

# Time Series Analysis

## Complete Final-Exam Preparation

MATH-342 – École Polytechnique Fédérale de Lausanne

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### What this document is

A self-contained revision companion covering *every* chapter of the lecture notes: all **definitions**, **theorems** / **propositions** / **lemmas** (statement + proof idea + why it matters), the **key techniques** you reproduce in the exam, and **worked exercises** (selected from the problem sheets and freshly created) with **full solutions**. It ends with an **Exam Strategy Checklist** and a list of **classic pitfalls**.

Coverage: Stationarity · ACVS/ACF/PACF · AR/MA/ARMA/ARIMA/SARIMA · Causality & Invertibility · Estimation (Yule–Walker, LS, MLE) · Forecasting · Fourier & Spectral Analysis · Periodogram & Tapering · Multivariate / VAR · Diagnostics & Model Selection · Long Memory · ARCH/GARCH · LTI Filters

## How to use this document

### Reading strategy

1. **First pass.** Read each section's *Definitions* and *Theorems* boxes. For every theorem, learn the *one-line statement*, the *proof idea*, and the “*why*” — examiners reward knowing *when* a result applies, not just *that* it is true.
2. **Second pass.** Work through the *Key Technique* boxes. These are the recurring computational moves (compute an ACVS, check causality via roots, derive an  $\text{MA}(\infty)$  representation, forecast recursively, get a spectral density). The exam is mostly these moves.
3. **Third pass.** Do the *Exercises* **with the solution hidden**. Exercises tagged [Sheet  $k$ ] come from the problem sheets; those tagged [New] were written to probe understanding from a different angle. *When you can do every exercise unaided, you are ready.*
4. **Night before.** Read only the [Exam Strategy Checklist](#) and the [Classic Pitfalls](#).

### Colour legend

|              |            |               |                   |                 |
|--------------|------------|---------------|-------------------|-----------------|
| ■ Definition | ■ Theorem  | ■ Proposition | ■ Lemma/Corollary | ■ Key Technique |
| ■ Exercise   | ■ Solution |               |                   |                 |

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# 1 Foundations: Stationarity, Autocovariance & Ergodicity

A *time series*  $\{X_t\}$  is both the data recorded in time order and the stochastic process  $\{X_t : t \in T\}$  on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  it is a realisation of. The index set  $T$  is a set of time points (here  $T = \mathbb{Z}$  unless stated otherwise; we take the sampling step  $\Delta = 1$ ). Time series differ from classical statistics because the observations are **dependent**: “there are many more ways to be dependent than to be independent”. Everything in this course is built on describing that dependence through second-order structure.

## 1.1 Definitions

### Definition 1: (Weak / second-order) stationarity

$\{X_t\}$  is **second-order** (equivalently **weakly** or **covariance**) **stationary** if all first- and second-order joint moments exist, are finite, and are invariant to time shifts. Concretely, for all  $t, s \in T$  and all admissible shifts  $\tau$ :

$$(i) \mathbb{E}[X_t] = \mu; \quad (ii) \text{Var}(X_t) = \sigma^2 < \infty; \quad (iii) \mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_s X_{s+\tau}].$$

Condition (iii) forces  $\mathbb{E}[X_t X_{t+\tau}]$  to be a function of the lag  $\tau$  only.

### Definition 2: Strong / strict stationarity

$\{X_t\}$  is **strictly** (completely) **stationary** if for every  $n \geq 1$ , every  $t_1, \dots, t_n$  and every shift  $\tau$ , the joint distribution of  $(X_{t_1}, \dots, X_{t_n})$  equals that of  $(X_{t_1+\tau}, \dots, X_{t_n+\tau})$ .

### Intuition / Why it matters

Neither notion implies the other in general. **Strict**  $\not\Rightarrow$  **weak**: i.i.d. Student- $t$  on 1 d.o.f. (Cauchy) is strictly stationary but has no finite moments, so it is not weakly stationary. **Weak**  $\not\Rightarrow$  **strict**: weak stationarity only constrains the first two moments and says nothing about higher ones. The two coincide for *Gaussian* processes (Definition 3).

### Definition 3: Gaussian process

$\{X_t\}$  is a **Gaussian process** if for all  $n$  and  $t_1, \dots, t_n$  the vector  $(X_{t_1}, \dots, X_{t_n})$  is multivariate normal. Such a process is completely characterised by its (finite) first two moments; hence for Gaussian processes **weak**  $\Leftrightarrow$  **strict stationarity**.

### Definition 4: Autocovariance sequence (ACVS)

For a discrete-time second-order stationary  $\{X_t\}$ , the **ACVS** is

$$\gamma_\tau = \text{Cov}(X_0, X_\tau) = \text{Cov}(X_t, X_{t+\tau}), \quad \tau \in \mathbb{Z}.$$

Immediate properties:  $\gamma_0 = \text{Var}(X_t) \geq 0$ , symmetry  $\gamma_{-\tau} = \gamma_\tau$ , and (Proposition 9) positive semi-definiteness.

### Definition 5: Autocorrelation sequence (ACF)

The **ACF** is  $\rho_\tau = \text{Corr}(X_t, X_{t+\tau}) = \gamma_\tau / \gamma_0$ , with  $\rho_0 = 1$  and  $|\rho_\tau| \leq 1$ .

### Definition 6: Sample autocovariance estimators

Given  $X_1, \dots, X_n$  with sample mean  $\bar{X}$ , two estimators are used for  $|\tau| \leq n-1$  (both 0 otherwise):

$$\tilde{\gamma}_\tau = \frac{1}{n-|\tau|} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}), \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}).$$

With known mean,  $\tilde{\gamma}_\tau$  is *unbiased* but need not be positive semi-definite;  $\hat{\gamma}_\tau$  is *biased* ( $\mathbb{E}[\hat{\gamma}_\tau] = \frac{n-|\tau|}{n} \gamma_\tau$ ) but *is* positive semi-definite, which is why it is the default. The plug-in ACF estimator is  $\hat{\rho}_\tau = \hat{\gamma}_\tau / \hat{\gamma}_0$ .

### Definition 7: Mean ergodicity

$\{X_t\}$  is **mean ergodic** if its first two moments are finite and  $\bar{X} \xrightarrow{P} \mathbb{E}[X_t]$  as  $n \rightarrow \infty$  (“temporal averages = population averages”). Ergodicity and stationarity are *not* equivalent.

## 1.2 Theorems & Propositions

### Theorem 8: Kolmogorov existence theorem

A family of finite-dimensional distribution functions  $\{F_{\mathbf{t}}\}$  are the f.d.d.’s of a stochastic process *iff* they are **consistent**: for every  $n$ , every  $\mathbf{t}$  and every  $1 \leq i \leq n$ ,

$$\lim_{x_i \rightarrow \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}(i)}(\mathbf{x}(i)),$$

where  $\mathbf{t}(i), \mathbf{x}(i)$  delete the  $i$ -th coordinate.

*Why*: it guarantees an infinite-index stochastic process actually exists given only its finite-dimensional “marginals”, justifying modelling  $\{X_t\}_{t \in \mathbb{Z}}$  even though we observe finitely many points.

### Proposition 9: The ACVS is positive semi-definite

For any  $n$ , any  $t_1, \dots, t_n \in \mathbb{Z}$  and any  $a_1, \dots, a_n \in \mathbb{R}$ ,

$$\sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k} \geq 0.$$

*Proof idea*: set  $W = \sum_j a_j X_j$ ; then  $0 \leq \text{Var}(W) = \sum_{j,k} a_j a_k \text{Cov}(X_j, X_k) = \sum_{j,k} a_j a_k \gamma_{j-k}$ .

*Why*: p.s.d. is the defining algebraic property of a valid ACVS; a candidate sequence that is *not* p.s.d. **cannot** be an autocovariance (used constantly to reject “fake” ACVS, see Exercise in Part 6).

### Proposition 10: Variance of the sample mean & consistency

If  $\sum_\tau |\gamma_\tau| < \infty$  then  $\bar{X}$  is unbiased and

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i,j=1}^n \gamma_{j-i} = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n}\right) \gamma_\tau, \quad n \text{Var}(\bar{X}) \xrightarrow{n \rightarrow \infty} \sum_{\tau=-\infty}^{\infty} \gamma_\tau.$$

Hence  $\text{Var}(\bar{X}) \rightarrow 0$  and  $\bar{X} \xrightarrow{P} \mu$  (consistency).

*Proof idea*: expand the double sum, collect by lag  $\tau$ , then apply the **Cesàro summability theorem** to replace  $\sum (1 - \frac{|\tau|}{n}) \gamma_\tau$  by  $\sum \gamma_\tau$ .

*Why:* this is the prototype “when can a sample average replace a population average?”. Strong positive correlation inflates  $\text{Var}(\bar{X})$  (e.g. AR(1) with  $\phi \rightarrow 1$ ): dependence, not just sample size, controls estimation accuracy.

### 1.3 Key Techniques

#### Key Technique 1: Proving (or disproving) weak stationarity

Check the three conditions *in order*, and stop at the first failure:

1. **Mean constant in  $t$ :** compute  $\mathbb{E}[X_t]$ ; it must not depend on  $t$ .
  2. **Variance finite & constant:** compute  $\text{Var}(X_t)$ ; must be finite and  $t$ -free.
  3. **Covariance shift-invariant:** compute  $\text{Cov}(X_t, X_{t+\tau})$  and check it depends on  $\tau$  only.
- Common disqualifiers: undefined moments (Cauchy), or a covariance that retains a  $t$ -dependence such as  $\cos(ct) \cos(c(t + \tau))$ .

#### Key Technique 2: ACVS of a finite linear filter of white noise

For  $X_t = \sum_k \theta_k \varepsilon_{t-k}$  with  $\varepsilon$  white noise of variance  $\sigma^2$ , use bilinearity and  $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ :

$$\gamma_\tau = \text{Cov}\left(\sum_k \theta_k \varepsilon_{t+\tau-k}, \sum_j \theta_j \varepsilon_{t-j}\right) = \sigma^2 \sum_j \theta_j \theta_{j+|\tau|}.$$

Only the terms whose noise indices coincide survive — this single move powers *all* MA covariance computations.

### 1.4 Exercises

#### Exercise 1: Student- $t$ noise [Sheet 1]

Let  $X_t$  be i.i.d. Student- $t$  on one degree of freedom. Is  $\{X_t\}$  weakly stationary?

#### Solution

A Student- $t$  on 1 d.o.f. is a standard Cauchy: its mean and variance are *undefined*. Condition (ii) of weak stationarity (finite, constant variance) fails, so  $\{X_t\}$  is **not** weakly stationary — even though it is strictly stationary (i.i.d.). A clean illustration that strict  $\nRightarrow$  weak.

#### Exercise 2: Four candidate processes [Sheet 1]

Let  $Y_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_Y^2)$  and  $a, b, c$  constants. Classify each as weakly / strongly stationary and give the mean and ACVS: (1)  $X_t = a + bY_t + cY_{t-1}$ ; (2)  $X_t = a + bY_0$ ; (3)  $X_t = Y_1 \cos(ct) + Y_2 \sin(ct)$ ; (4)  $X_t = Y_0 \cos(ct)$ .

#### Solution

(1)  $\mathbb{E}[X_t] = a$  and, using independence,

$$\gamma_\tau = \begin{cases} (b^2 + c^2)\sigma_Y^2 & \tau = 0, \\ bc\sigma_Y^2 & \tau = \pm 1, \\ 0 & \text{else.} \end{cases}$$

Depends on  $\tau$  only, finite  $\Rightarrow$  weakly stationary; a linear combination of Gaussians  $\Rightarrow$  Gaussian  $\Rightarrow$  also strongly stationary. (This is an MA(1).)

(2)  $\mathbb{E}[X_t] = a$ ,  $\gamma_\tau = \text{Cov}(bY_0, bY_0) = b^2\sigma_Y^2$  for all  $\tau$  (the process is constant in randomness across  $t$ ). Weakly and strongly stationary.

(3)  $\mathbb{E}[X_t] = 0$  and, using  $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$ ,

$$\gamma_\tau = \sigma_Y^2 [\cos(ct) \cos(c(t+\tau)) + \sin(ct) \sin(c(t+\tau))] = \sigma_Y^2 \cos(c\tau),$$

a function of  $\tau$  only  $\Rightarrow$  weakly (and, being Gaussian, strongly) stationary.

(4)  $\mathbb{E}[X_t] = 0$  but  $\gamma_\tau = \sigma_Y^2 \cos(ct) \cos(c(t+\tau))$  depends on  $t \Rightarrow$  **not** stationary. The contrast (3) vs (4) is the classic trap: two random coefficients (3) restore stationarity that a single one (4) destroys.

### Exercise 3: Sum of uncorrelated stationary series [Sheet 1]

Let  $\{X_t\}, \{Y_t\}$  be stationary and mutually uncorrelated ( $X_t, Y_s$  uncorrelated for all  $t, s$ ). Show  $\{X_t + Y_t\}$  is stationary with  $\gamma_\tau^{X+Y} = \gamma_\tau^X + \gamma_\tau^Y$ .

#### Solution

Mean:  $\mathbb{E}[X_t + Y_t] = \mathbb{E}[X_t] + \mathbb{E}[Y_t]$  is constant. Variance:  $\text{Var}(X_t + Y_t) = \text{Var}(X_t) + \text{Var}(Y_t) < \infty$  since  $\text{Cov}(X_t, Y_t) = 0$ . Covariance, by bilinearity and uncorrelatedness of the cross terms,

$$\text{Cov}(X_t + Y_t, X_{t+\tau} + Y_{t+\tau}) = \gamma_\tau^X + \underbrace{\text{Cov}(X_t, Y_{t+\tau})}_0 + \underbrace{\text{Cov}(Y_t, X_{t+\tau})}_0 + \gamma_\tau^Y = \gamma_\tau^X + \gamma_\tau^Y,$$

a function of  $\tau$  only. Hence stationary with additive ACVS.

### Exercise 4: Superposition of $m$ orthogonal series [Sheet 1]

Let  $\{X_{t,1}\}, \dots, \{X_{t,m}\}$  be zero-mean stationary with  $\gamma_j(\tau) = \text{Cov}(X_{t,j}, X_{t+\tau,j})$ , and  $\mathbb{E}[X_{t,j} X_{t+\tau,k}] = 0$  for  $j \neq k$ . Find the ACVS of  $X_t = \sum_{j=1}^m X_{t,j}$ .

#### Solution

By bilinearity,  $\gamma_X(\tau) = \sum_{j=1}^m \sum_{k=1}^m \text{Cov}(X_{t,j}, X_{t+\tau,k})$ . The cross terms ( $j \neq k$ ) vanish by orthogonality, leaving only the diagonal:  $\gamma_X(\tau) = \sum_{j=1}^m \gamma_j(\tau)$ . (Geometrically: sum a matrix that is zero off the diagonal.)

### Exercise 5: Signal plus independent noise [Sheet 1]

Let  $\{X_t\}$  be second-order stationary with ACVS  $\gamma_\tau$  and let  $\varepsilon_t$  be i.i.d. with variance  $\sigma^2$ , independent of  $\{X_t\}$ . Find the ACVS of  $Z_t = X_t + \varepsilon_t$ .

#### Solution

Independence kills all cross-covariances. For  $\tau \neq 0$ ,  $\gamma_Z(\tau) = \gamma_\tau$ ; at  $\tau = 0$ ,  $\gamma_Z(0) = \gamma_0 + \sigma^2$ :

$$\gamma_Z(\tau) = \gamma_\tau + \sigma^2 \mathbf{1}_{\{\tau=0\}}.$$

Adding white noise inflates only the variance (the  $\tau = 0$  “nugget”), leaving every other lag unchanged.

### Exercise 6: Bias, variance and MSE of the two ACVS estimators [Sheet 2]

For  $X_t$  i.i.d. Gaussian white noise (variance  $\sigma_X^2$ , known mean replaced for  $\bar{X}$ ), find the mean, variance and MSE of  $\tilde{\gamma}_\tau$  and  $\hat{\gamma}_\tau$ , and say which wins.

### Solution

Both are unbiased for white noise:  $\mathbb{E}[\tilde{\gamma}_\tau] = \mathbb{E}[\hat{\gamma}_\tau] = 0$  for  $\tau \neq 0$  (and  $\sigma_X^2$  at  $\tau = 0$ ). Using Isserlis' theorem for the fourth moments one finds, for  $\tau \neq 0$ ,

$$\text{Var}(\hat{\gamma}_\tau) = \frac{n - |\tau|}{n^2} \sigma_X^4, \quad \text{Var}(\tilde{\gamma}_\tau) = \frac{1}{n - |\tau|} \sigma_X^4.$$

Since the estimators are unbiased here,  $\text{MSE} = \text{Var}$ , and  $\text{Var}(\hat{\gamma}_\tau) = \left(\frac{n - |\tau|}{n}\right)^2 \text{Var}(\tilde{\gamma}_\tau) \leq \text{Var}(\tilde{\gamma}_\tau)$ . The **biased** divide-by- $n$  estimator  $\hat{\gamma}_\tau$  has the *smaller* MSE — a recurring lesson that unbiasedness is not the goal, low MSE is.

### Exercise 7: Sample autocovariances sum to zero [Sheet 2 / New angle]

Show that for any data  $\{x_1, \dots, x_n\}$ ,  $\sum_{|\tau| < n} \hat{\gamma}_\tau = 0$ .

### Solution

$$\sum_{\tau=-(n-1)}^{n-1} \hat{\gamma}_\tau = \frac{1}{n} \sum_{t_1=1}^n \sum_{t_2=1}^n (x_{t_1} - \bar{x})(x_{t_2} - \bar{x}) = \frac{1}{n} \left( \sum_{t=1}^n (x_t - \bar{x}) \right)^2 = 0,$$

because the double sum factorises and the centred data sum to zero. (The first equality just re-indexes the triangular lag sum as a full  $t_1, t_2$  square.)

### Exercise 8: ACVS of a two-tap filter [New]

Let  $\varepsilon_t$  be mean-zero white noise with variance  $\sigma^2$  and set  $X_t = \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1}$ . Compute the ACVS and the ACF, and state for which  $\theta_1, \theta_2$  the lag-1 correlation is maximised in magnitude.

### Solution

By the white-noise filter technique,

$$\gamma_\tau = \begin{cases} (\theta_1^2 + \theta_2^2) \sigma^2 & \tau = 0, \\ \theta_1 \theta_2 \sigma^2 & \tau = \pm 1, \\ 0 & \text{else,} \end{cases} \quad \rho_1 = \frac{\theta_1 \theta_2}{\theta_1^2 + \theta_2^2}.$$

Since  $|\theta_1 \theta_2| \leq \frac{1}{2}(\theta_1^2 + \theta_2^2)$  (AM–GM), we have  $|\rho_1| \leq \frac{1}{2}$ , with equality iff  $|\theta_1| = |\theta_2|$ . So **no MA(1) can have**  $|\rho_1| > \frac{1}{2}$  — a fact worth remembering as a sanity check on any claimed MA(1) ACF.

## 2 ARMA Models: White Noise, MA, AR, ARMA and the PACF

The ARMA family is the workhorse of linear time series modelling. Starting from **white noise** — the indivisible “atom” of randomness — we build **moving average** (MA), **autoregressive** (AR) and combined **ARMA** processes by feeding white noise through finite linear filters and linear recursions. Two diagnostic curves run through the whole story: the autocorrelation function (ACF,  $\rho_\tau$ ) and its less-obvious partner, the **partial autocorrelation function** (PACF,  $\alpha_\tau$ ). The punchline, encoded in one small table, is that the ACF identifies the MA order  $q$  while the PACF identifies the AR order  $p$ . We close with the **Durbin–Levinson** recursions that compute the PACF directly from the ACF.

## 2.1 Definitions

### Definition 1: White noise (purely random process)

$\{X_t\}$  is **white noise** (a purely random process) if it is a sequence of *uncorrelated*, mean-zero, finite-variance random variables: for all  $t \in \mathbb{Z}$ ,

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2 < \infty,$$

and  $\text{Cov}(X_s, X_t) = 0$  for  $s \neq t$ . Equivalently its second-order structure is

$$\gamma_\tau = \begin{cases} \sigma^2 & \tau = 0, \\ 0 & \tau \neq 0, \end{cases} \quad \rho_\tau = \begin{cases} 1 & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

White noise is automatically (weakly) stationary and is the **building block** of every model below. We write  $\varepsilon_t$  for a mean-zero white noise of variance  $\sigma_\varepsilon^2$ .

### Intuition / Why it matters

“Uncorrelated” is weaker than “independent”. White noise only constrains the first two moments, so a white-noise process need not be i.i.d. and need not be Gaussian. Calling something *white* refers to its flat spectral density (equal power at all frequencies), which mirrors the flat ACF above.

### Definition 2: Moving average process MA( $q$ )

Let  $\{\varepsilon_t\}$  be mean-zero white noise of variance  $\sigma_\varepsilon^2$ . The  $q$ -th order moving average process MA( $q$ ) is

$$X_t = \mu - \sum_{j=0}^q \theta_j \varepsilon_{t-j}, \quad \theta_0 = -1, \quad \theta_q \neq 0.$$

(The convention  $\theta_0 = -1$  makes the leading term  $-\theta_0 \varepsilon_t = \varepsilon_t$ .) Simple examples use the equivalent zero-mean one-sign convention  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$  for MA(1). An MA( $q$ ) is just a finite linear filter of white noise, hence always stationary; the only requirement is  $|\theta_j| < \infty$ .

### Definition 3: Moments of an MA( $q$ )

For the MA( $q$ ) of Definition 2, with  $\theta_j := 0$  for  $j > q$ ,

$$\mathbb{E}[X_t] = \mu, \quad \gamma_\tau = \text{Cov}(X_{t+\tau}, X_t) = \begin{cases} \sigma_\varepsilon^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|} & |\tau| \leq q, \\ 0 & |\tau| > q. \end{cases}$$

Hence the **ACVS of an MA( $q$ ) cuts off (is exactly 0) for lags  $|\tau| > q$**  — this is the signature used to read  $q$  off the ACF.

### Definition 4: Autoregressive process AR( $p$ )

Let  $\{\varepsilon_t\}$  be mean-zero white noise. The  $p$ -th order autoregressive process AR( $p$ ) is

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t, \quad \phi_p \neq 0.$$

Unlike the MA case there *are* constraints on  $\{\phi_j\}$  for stationarity (the roots of the AR polynomial must lie outside the unit circle), and there is no simple closed form for the ACVS in general.

### Definition 5: ARMA( $p, q$ ) process

$\{X_t\}$  is an **autoregressive moving average** process of orders  $p$  and  $q$ , ARMA( $p, q$ ), if

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k \varepsilon_{t-k}, \quad \theta_0 = -1,$$

with  $\{\varepsilon_t\}$  mean-zero white noise and  $\phi_j, \theta_k$  as in the AR and MA cases. Using the backshift operator  $B$  (with  $B^j X_t = X_{t-j}$ ) this is written compactly as

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j, \quad \Theta(z) = - \sum_{k=0}^q \theta_k z^k.$$

As for AR, conditions on the  $\phi_j$  are needed for the model to be valid, and closed-form ACVS is generally unavailable (though computable).

### Definition 6: Causal ARMA( $p, q$ )

An ARMA( $p, q$ ) process is **causal** (more precisely a causal function of  $\{\varepsilon_t\}$ ) if there exists a sequence  $\{\psi_j\}$  with  $\sum_{j=0}^{\infty} |\psi_j| < \infty$  such that

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad t = 0, \pm 1, \pm 2, \dots$$

i.e.  $X_t$  depends only on *present and past* noise. The  $\{\psi_j\}$  are the  **$\psi$ -weights** (the MA( $\infty$ ) representation). A mean-zero MA( $q$ ) is *always* causal (it already has this form with finitely many nonzero  $\psi_j$ ). For ARMA, causality holds iff all roots of  $\Phi(z)$  lie *outside* the unit circle.

### Roots convention

**Stationarity / causality:** all roots of the AR polynomial  $\Phi(z)$  lie *outside* the unit circle ( $|z| > 1$ ). **Invertibility:** all roots of the MA polynomial  $\Theta(z)$  lie *outside* the unit circle. For AR(1),  $\Phi(z) = 1 - \phi z$  has root  $z = 1/\phi$ , so  $|z| > 1 \iff |\phi| < 1$ .

### Definition 7: Best linear predictor

Given mean-zero random variables  $X_1, \dots, X_n, Y$ , the **best linear predictor** of  $Y$  from  $X_1, \dots, X_n$  is

$$P_{X_1, \dots, X_n}(Y) = \sum_{j=1}^n \beta_j X_j,$$

where  $\beta_1, \dots, \beta_n$  minimise the mean-squared error  $\mathbb{E}[(Y - P_{X_1, \dots, X_n}(Y))^2]$ . The minimiser is characterised by the **prediction (orthogonality) equations**: the prediction error is uncorrelated with every regressor,

$$\text{Cov}(Y - P_{X_1, \dots, X_n}(Y), X_k) = 0, \quad k = 1, \dots, n.$$

### Definition 8: Partial correlation

For random variables  $X, Y$  and confounders  $Z = (Z_1, \dots, Z_n)^\top$ , the **partial correlation** of  $X$  and  $Y$  given  $Z$  is

$$\rho_{XY \bullet Z} = \text{Corr}(X - P_Z(X), Y - P_Z(Y)),$$

i.e. the correlation of the residuals after removing the best linear effect of  $Z$  from each. If  $X, Y, Z$  are jointly Gaussian then  $\rho_{XY \bullet Z} = \mathbb{E}[\text{Corr}(X, Y \mid Z)]$ , and in fact  $\rho_{XY \bullet Z} \stackrel{\text{a.s.}}{=} \text{Corr}(X, Y \mid Z)$ .

### Definition 9: Partial autocorrelation function (PACF)

For a mean-zero stationary  $\{X_t\}$ , the **PACF** at lag  $\tau$  is the partial correlation of  $X_t$  and  $X_{t+\tau}$  given the intervening values  $Z = (X_{t+1}, \dots, X_{t+\tau-1})$ . Writing  $\hat{X}_t = P_{X_{t+1}, \dots, X_{t+\tau-1}}(X_t)$  and  $\hat{X}_{t+\tau} = P_{X_{t+1}, \dots, X_{t+\tau-1}}(X_{t+\tau})$ ,

$$\alpha_\tau = \text{Corr}(X_{t+\tau} - \hat{X}_{t+\tau}, X_t - \hat{X}_t).$$

Equivalently (regression form): if  $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$  is the best linear predictor of  $X_\tau$  from its  $\tau$  predecessors, then  $\alpha_\tau = \alpha_{\tau,\tau}$  (the *last* regression coefficient). Conventionally  $\alpha_1 = \rho_1$ .

### Intuition / Why it matters

The PACF answers “how much *new* linear information about  $X_{t+\tau}$  is in  $X_t$ , once everything in between is accounted for?”. In an AR(1),  $\rho_\tau = \phi^{|\tau|} \neq 0$  for all  $\tau$  even though  $X_t$  only directly depends on  $X_{t-1}$ : the correlation at lag 2 is entirely *transmitted* through  $X_{t-1}$ . The PACF strips out that transmitted part, so it vanishes for  $\tau > 1$  — exactly identifying the order  $p = 1$ .

## 2.2 Theorems & Propositions

### Proposition 10: Causal AR(1) expansion and its ACVS

For the AR(1)  $X_t = \phi X_{t-1} + \varepsilon_t$  with  $|\phi| < 1$ , iterating the recursion gives the causal (MA( $\infty$ )) representation

$$X_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}, \quad \psi_k = \phi^k, \quad \sum_k |\psi_k| = \frac{1}{1-|\phi|} < \infty,$$

with  $\mathbb{E}[X_t] = 0$  and autocovariance

$$\gamma_\tau = \frac{\sigma_\varepsilon^2}{1-\phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

*Proof idea:* substitute  $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$  repeatedly; the remainder  $\phi^n X_{t-n} \rightarrow 0$  in mean square. Then  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{k \geq 0} \phi^k \phi^{k+|\tau|} = \sigma_\varepsilon^2 \phi^{|\tau|} \sum_{k \geq 0} \phi^{2k} = \sigma_\varepsilon^2 \phi^{|\tau|} / (1 - \phi^2)$  by the geometric series.

*Why:* the AR(1) is the canonical example of a causal process and the template for “write an AR as an infinite MA, then use the white-noise filter trick to get the ACVS”. The ACF  $\phi^{|\tau|}$  *tails off* (never hits exactly 0), motivating the PACF.

### Proposition 11: ACF–PACF relation (prediction equations)

For a mean-zero stationary  $\{X_t\}$ , fix  $\tau > 0$  and write  $\hat{X}_\tau = P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$ . Then for each  $k \in \{1, \dots, \tau\}$ ,

$$\rho_k = \sum_{j=1}^{\tau} \alpha_{\tau,j} \rho_{k-j}.$$

*Proof idea:* decompose  $\gamma_k = \text{Cov}(X_{\tau-k}, X_\tau) = \text{Cov}(X_{\tau-k}, \hat{X}_\tau) + \text{Cov}(X_{\tau-k}, X_\tau - \hat{X}_\tau)$ ; the second term is 0 by the orthogonality of the prediction error, and expanding the first using  $\hat{X}_\tau = \sum_j \alpha_{\tau,j} X_{\tau-j}$  gives  $\gamma_k = \sum_j \alpha_{\tau,j} \gamma_{k-j}$ ; divide by  $\gamma_0$ .

*Why:* these are the **Yule–Walker-type** equations linking the ACF to the prediction coefficients  $\alpha_{\tau,j}$ . They are exactly what the Durbin–Levinson recursion solves to extract the PACF  $\alpha_\tau = \alpha_{\tau,\tau}$ .

### Theorem 12: ACF & PACF signatures of ARMA models

For *causal and invertible*  $\text{ARMA}(p, q)$  models the ACF and PACF behave as:

|             | AR( $p$ )              | MA( $q$ )              | ARMA( $p, q$ ) |
|-------------|------------------------|------------------------|----------------|
| <b>ACF</b>  | Tails off              | Cuts off after lag $q$ | Tails off      |
| <b>PACF</b> | Cuts off after lag $p$ | Tails off              | Tails off      |

*Proof idea:* the ACF cut-off for MA( $q$ ) is the moment computation of Definition 3; the PACF cut-off for AR( $p$ ) is Proposition 13; “tails off” means a geometric/damped-sine decay with no exact zero (as for the AR(1) ACF  $\phi^{|\tau|}$ ).

*Why:* this  $2 \times 3$  table is *the* model-identification tool: read  $q$  from where the ACF cuts,  $p$  from where the PACF cuts; if neither cuts (both tail off) the model is mixed ARMA.

### Proposition 13: PACF of an AR( $p$ ) vanishes beyond lag $p$

For a causal AR( $p$ ),

$$\alpha_\tau = 0 \quad \text{for all } \tau > p.$$

Moreover the *estimated* PACF at lags  $> p$  is, asymptotically, mean 0 with variance  $1/n$  ( $n$  the sample size), giving the  $\pm 1.96/\sqrt{n}$  significance bands.

*Proof idea:* for  $\tau > p$  write the prediction error  $X_\tau - \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$ . Because the process is AR( $p$ ), the true best predictor uses only  $\phi_1, \dots, \phi_p$  and the noise  $\varepsilon_\tau$ ; setting  $\tilde{\phi}_j = \phi_j$  for  $j \leq p$  and  $\tilde{\phi}_j = 0$  for  $j > p$ , the prediction equations  $\text{Cov}(X_\tau - \sum_j \alpha_{\tau,j} X_{\tau-j}, X_k) = 0$  are solved by  $\alpha_{\tau,j} = \tilde{\phi}_j$  (using  $\text{Cov}(\varepsilon_\tau, X_k) = 0$  for  $k < \tau$ , by causality). In particular  $\alpha_\tau = \alpha_{\tau,\tau} = \tilde{\phi}_\tau = 0$  for  $\tau > p$ .

*Why:* this is the theoretical backbone of the table: it is why a sharp cut-off in the (sample) PACF at lag  $p$  diagnoses an AR( $p$ ), and the  $1/n$  variance gives the threshold for “cut off”.

### Proposition 14: Non-identifiability of MA(1) from the ACVS

The MA(1) processes

$$X_t = \varepsilon_t - \theta \varepsilon_{t-1} \quad (\text{Var } \varepsilon_t = \sigma_\varepsilon^2) \quad \text{and} \quad Y_t = \eta_t - \frac{1}{\theta} \eta_{t-1} \quad (\text{Var } \eta_t = \sigma_\varepsilon^2 \theta^2)$$

have *identical* autocovariance sequences, hence are indistinguishable from second-order structure alone.

*Proof idea:* both give  $\gamma_0 = \sigma_\varepsilon^2(1 + \theta^2)$ ,  $\gamma_{\pm 1} = -\sigma_\varepsilon^2 \theta$ , and 0 elsewhere (direct computation, see Exercise 7).

*Why:* an MA(1) and its “root-reciprocal” twin share an ACVS; choosing the *invertible* representation ( $|\theta| < 1$ , root outside the unit circle) pins down a unique model. This is the reason invertibility is imposed.

## 2.3 Key Techniques

### Key Technique 1: Computing the ACVS of an MA( $q$ )

Use bilinearity of covariance and  $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma_\varepsilon^2 \mathbf{1}_{\{s=u\}}$ . For  $X_t = \mu - \sum_{j=0}^q \theta_j \varepsilon_{t-j}$  (constant  $\mu$  drops out of covariances),

$$\gamma_\tau = \text{Cov}\left(\sum_k \theta_k \varepsilon_{t+\tau-k}, \sum_j \theta_j \varepsilon_{t-j}\right) = \sigma_\varepsilon^2 \sum_j \theta_j \theta_{j+|\tau|},$$

the surviving terms being those with  $k = j + \tau$ . Steps: (1) write both sums; (2) keep only matching noise indices; (3) read off that  $\gamma_\tau = 0$  for  $|\tau| > q$ .

### Key Technique 2: Checking causality / stationarity via roots

To test an AR or ARMA: (1) form  $\Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j$ ; (2) find its roots; (3) the model is causal/stationary iff every root satisfies  $|z| > 1$  (outside the unit circle). For AR(1) this reduces to  $|\phi| < 1$ ; for AR(2)  $\Phi(z) = 1 - \phi_1 z - \phi_2 z^2$ , the conditions are  $\phi_1 + \phi_2 < 1$ ,  $\phi_2 - \phi_1 < 1$ ,  $|\phi_2| < 1$ . Invertibility uses the same test on  $\Theta(z)$ .

### Key Technique 3: Deriving the MA( $\infty$ ) ( $\psi$ -weight) representation

Match coefficients in  $\Phi(B)\psi(B) = \Theta(B)$  where  $\psi(z) = \sum_{j \geq 0} \psi_j z^j$ . Equivalently expand  $\psi(z) = \Theta(z)/\Phi(z)$  as a power series. For AR(1):  $\psi(z) = 1/(1 - \phi z) = \sum_{j \geq 0} \phi^j z^j$ , so  $\psi_j = \phi^j$ . For an ARMA(1,1) written  $X_t = \phi X_{t-1} + \varepsilon_t + \vartheta \varepsilon_{t-1}$  (so  $\Phi(z) = 1 - \phi z$ ,  $\Theta(z) = 1 + \vartheta z$ ):  $\psi_0 = 1$  and  $\psi_j = (\phi + \vartheta)\phi^{j-1}$  for  $j \geq 1$ . Once you have  $\{\psi_j\}$ , the ACVS is  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j \geq 0} \psi_j \psi_{j+|\tau|}$ .

### Key Technique 4: Durbin–Levinson recursion for the PACF

Given the ACF  $\{\rho_\tau\}$ , compute  $\alpha_\tau = \alpha_{\tau,\tau}$  by:

$$\alpha_{1,1} = \rho_1, \quad \alpha_{\tau,\tau} = \frac{\rho_\tau - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1} \alpha_{\tau-1,j} \rho_j},$$

then update the auxiliary coefficients

$$\alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau} \alpha_{\tau-1,\tau-j}, \quad j = 1, \dots, \tau-1.$$

Works with theoretical *or* sample ACF. A clean cut to 0 in  $\alpha_{\tau,\tau}$  at  $\tau = p+1$  flags an AR( $p$ ); the last diagonal coefficients then equal the AR coefficients.

### Key Technique 5: Reading ( $p, q$ ) off ACF & PACF plots

1. First confirm **stationarity**: stable mean/variance; an ACF that decays very slowly (almost linearly) signals non-stationarity — difference first.
2. **ACF cuts after lag  $q$ , PACF tails off**  $\Rightarrow$  MA( $q$ ).
3. **PACF cuts after lag  $p$ , ACF tails off**  $\Rightarrow$  AR( $p$ ).
4. **Both tail off**  $\Rightarrow$  mixed ARMA( $p, q$ ).
5. **Neither significant at any nonzero lag**  $\Rightarrow$  white noise.

Use the  $\pm 1.96/\sqrt{n}$  band to decide “significant”. Choices are rarely unique — justify with a sensible argument.

### Classic Pitfall

Two recurring sign/convention traps. (i) The notes’ MA/ARMA definition carries a *minus* sign in front of the  $\theta$ -sum with  $\theta_0 = -1$ , so  $X_t = \mu - \sum_{j=0}^q \theta_j \varepsilon_{t-j}$  starts as  $\varepsilon_t + \dots$ ; simple examples instead write  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$ . Track which convention a problem uses before plugging into the ACVS formula. (ii) “Roots outside the unit circle” is for the polynomials  $\Phi(z), \Theta(z)$  in  $z$ ; in terms of the coefficient it means  $|\phi| < 1$  for AR(1). Do not confuse it with “roots inside”.

## 2.4 Exercises

### Exercise 1: Why fix $\theta_0 = -1$ ? (MA identifiability) [Sheet 2]

When defining the MA( $q$ ) we set  $\theta_0 = -1$ . In terms of covariance structure, why do we gain no generality by letting  $\theta_0$  be arbitrary?

#### Solution

Rescaling the leading coefficient produces a *different* parametrisation with the *same* covariance, so  $\theta_0$  is not free — we fix it for identifiability. Take the zero-mean MA(1)

$$X_t = -\theta_0 \varepsilon_t - \theta_1 \varepsilon_{t-1} = \varepsilon_t - \theta_1 \varepsilon_{t-1}, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad \theta_0 = -1.$$

Suppose a second parametrisation uses  $\theta'_0 = c\theta_0 = c$  and  $\theta'_1$  with noise  $\varepsilon'_t \sim \mathcal{N}(0, \sigma_{\varepsilon'}^2)$ :  $X_t = -\theta'_0 \varepsilon'_t - \theta'_1 \varepsilon'_{t-1} = c\varepsilon'_t - \theta'_1 \varepsilon'_{t-1}$ . Rewriting the first form,

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} = c \frac{\varepsilon_t}{c} - (c\theta_1) \frac{\varepsilon_{t-1}}{c},$$

so  $\varepsilon'_t = c^{-1}\varepsilon_t$ ,  $\theta'_1 = c\theta_1$ , and  $\sigma_{\varepsilon'}^2 = c^{-2}\sigma_\varepsilon^2$ . The two representations share moments:

$$\text{Var}(X_t) = (1 + \theta_1^2)\sigma_\varepsilon^2, \quad \{(\theta'_0)^2 + (\theta'_1)^2\}\sigma_{\varepsilon'}^2 = c^2(1 + \theta_1^2)c^{-2}\sigma_\varepsilon^2 = (1 + \theta_1^2)\sigma_\varepsilon^2,$$

and  $\text{Cov}(X_t, X_{t-1}) = \theta_0\theta_1\sigma_\varepsilon^2$  is likewise unchanged. Thus many MA(1) representations give the same  $X_t$ ; fixing  $\theta_0 = -1$  makes the representation unique. The same scaling argument extends verbatim to MA( $q$ ).

### Exercise 2: Rewriting an MA(1) recursively [Sheet 2]

Show that the MA(1) process  $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$  can be written, for any positive integer  $p$ , as

$$X_t = \varepsilon_t - \sum_{j=1}^p \theta^j X_{t-j} - \theta^{p+1} \varepsilon_{t-p-1}.$$

#### Solution

For any  $p \geq 1$ ,  $X_{t-p} = \varepsilon_{t-p} - \theta\varepsilon_{t-p-1}$ , hence  $\varepsilon_{t-p} = X_{t-p} + \theta\varepsilon_{t-p-1}$ . Substitute this recursively:

$$\begin{aligned} X_t &= \varepsilon_t - \theta\varepsilon_{t-1} = \varepsilon_t - \theta(X_{t-1} + \theta\varepsilon_{t-2}) = \varepsilon_t - \theta X_{t-1} - \theta^2\varepsilon_{t-2} \\ &= \varepsilon_t - \theta X_{t-1} - \theta^2(X_{t-2} + \theta\varepsilon_{t-3}) = \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \theta^3\varepsilon_{t-3} \\ &= \dots = \varepsilon_t - \sum_{j=1}^p \theta^j X_{t-j} - \theta^{p+1} \varepsilon_{t-p-1}. \end{aligned}$$

If  $|\theta| < 1$  (invertible MA), the remainder  $\theta^{p+1}\varepsilon_{t-p-1} \rightarrow 0$  in mean square as  $p \rightarrow \infty$ , giving the AR( $\infty$ ) representation  $X_t = \varepsilon_t - \sum_{j \geq 1} \theta^j X_{t-j}$ .

### Exercise 3: Moments of a sum of two filtered noises [Sheet 2]

Let  $Y_t = \sum_{s=-p}^p g_{s-t}\varepsilon_s$  and  $Z_t = \sum_{s=-p}^p h_{s-t}\varepsilon_s$  with  $\{\varepsilon_t\}$  mean-zero white noise of variance  $\sigma^2$ , and define  $X_t = Y_t + Z_t$ . Determine the first and second moments of  $X_t$ .

### Solution

Combine the filters:  $X_t = \sum_{s=-p}^p (g_{s-t} + h_{s-t})\varepsilon_s$ . Since  $\mathbb{E}[\varepsilon_s] = 0$ ,

$$\mathbb{E}[X_t] = \sum_{s=-p}^p (g_{s-t} + h_{s-t}) \mathbb{E}[\varepsilon_s] = 0.$$

For the (auto)covariance, use  $\text{Cov}(\varepsilon_s, \varepsilon_l) = \sigma^2 \mathbf{1}_{\{s=l\}}$ :

$$\begin{aligned} \mathbb{E}[X_t X_{t+\tau}] &= \text{Cov}(X_t, X_{t+\tau}) = \text{Cov}\left(\sum_{s=-p}^p (g_{s-t} + h_{s-t})\varepsilon_s, \sum_{l=-p}^p (g_{l-t-\tau} + h_{l-t-\tau})\varepsilon_l\right) \\ &= \sum_{s=-p}^p \sum_{l=-p}^p (g_{s-t} + h_{s-t})(g_{l-t-\tau} + h_{l-t-\tau}) \text{Cov}(\varepsilon_s, \varepsilon_l) \\ &= \sigma^2 \sum_{s=-p}^p (g_{s-t} + h_{s-t})(g_{s-t-\tau} + h_{s-t-\tau}). \end{aligned}$$

Only the diagonal  $s = l$  survives, collapsing the double sum to a single one. (Note: because the filter is indexed by  $s - t$  over the fixed window  $\{-p, \dots, p\}$ , this is a covariance computation rather than a stationarity claim — the surviving sum can depend on  $t$  near the window edges.)

### Exercise 4: PACF of an AR( $p$ ) is zero beyond lag $p$ [Sheet 2]

Prove that for a causal AR( $p$ ),  $\alpha_\tau = 0$  for all  $\tau > p$ . Use the prediction equations  $\text{Cov}(Y - P_{X_1, \dots, X_n}(Y), X_k) = 0$  and the causal fact  $\text{Cov}(X_t, \varepsilon_s) = 0$  for  $t < s$ .

### Solution

Recall  $\alpha_\tau = \alpha_{\tau, \tau}$ , the last coefficient in  $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau, j} X_{\tau-j}$ . Take  $\tau > p$ . The prediction equations state, for all  $k \in \{0, \dots, \tau - 1\}$ ,

$$\text{Cov}\left(X_\tau - \sum_{j=1}^{\tau} \alpha_{\tau, j} X_{\tau-j}, X_k\right) = 0.$$

Substitute the AR( $p$ ) form  $X_\tau = \sum_{j=1}^p \phi_j X_{\tau-j} + \varepsilon_\tau$  and define  $\tilde{\phi}_j = \phi_j$  for  $j \leq p$ ,  $\tilde{\phi}_j = 0$  for  $j > p$ :

$$\text{Cov}\left(\sum_{j=1}^{\tau} (\tilde{\phi}_j - \alpha_{\tau, j}) X_{\tau-j} + \varepsilon_\tau, X_k\right) = \sum_{j=1}^{\tau} (\tilde{\phi}_j - \alpha_{\tau, j}) \text{Cov}(X_{\tau-j}, X_k) + \text{Cov}(\varepsilon_\tau, X_k).$$

By causality  $X_k$  depends only on  $\varepsilon_s$  with  $s \leq k < \tau$ , so  $\text{Cov}(\varepsilon_\tau, X_k) = 0$ . Hence choosing  $\alpha_{\tau, j} = \tilde{\phi}_j$  for all  $j$  satisfies every prediction equation; by uniqueness of the best linear predictor these are the coefficients. In particular  $\alpha_\tau = \alpha_{\tau, \tau} = \tilde{\phi}_\tau = 0$  for  $\tau > p$ . ■

### Exercise 5: Model identification from ACF & PACF [Sheet 2]

For six time series, each shown with its sample ACF and PACF, decide which of white noise, MA( $q$ ), AR( $p$ ), ARMA( $p, q$ ) is appropriate, with reasoning.

### Solution

The diagnostic is the signature table (Theorem 12). The (hidden) truths and sample arguments were:

1. Stationary-looking; *both* ACF and PACF decay without a sharp cut-off  $\Rightarrow$  **ARMA** (truth: ARMA(1,1); one could argue AR(2)).
2. Stationary; ACF cuts sharply after lag 1 while the PACF decays  $\Rightarrow$  **MA(1)**.
3. Stationary; ACF decays, PACF cuts after lag 1  $\Rightarrow$  **AR(1)** (an ARMA is also arguable).
4. Stationary; neither ACF nor PACF significant at nonzero lags  $\Rightarrow$  **white noise**.
5. Very strong lag-1 PACF, ACF decaying very slowly, apparent changing variance  $\Rightarrow$  likely **non-stationary** (could be a near-unit-root AR(1) with  $\phi \approx 1$ ).
6. Stationary; ACF decays, PACF cuts after lag 1  $\Rightarrow$  **AR(1)**.

The point is to make a defensible argument; model selection is not unique, and ACF/PACF are sanity checks, not proofs.

**Exercise 6: Durbin–Levinson:**  $\rho_1 = \frac{2}{5}$ ,  $\rho_2 = -\frac{1}{20}$ ,  $\rho_3 = -\frac{1}{8}$  [Lecture worked example]

A stationary process has  $\rho_1 = \frac{2}{5}$ ,  $\rho_2 = -\frac{1}{20}$ ,  $\rho_3 = -\frac{1}{8}$ . Find  $\alpha_1, \alpha_2, \alpha_3$ , and identify a plausible model.

### Solution

Apply the Durbin–Levinson recursion (Key Technique 4).

**Lag 1:**  $\alpha_{1,1} = \rho_1 = \frac{2}{5}$ .

**Lag 2:**

$$\alpha_{2,2} = \frac{\rho_2 - \alpha_{1,1}\rho_1}{1 - \alpha_{1,1}\rho_1} = \frac{-\frac{1}{20} - \frac{2}{5} \cdot \frac{2}{5}}{1 - \frac{2}{5} \cdot \frac{2}{5}} = \frac{-\frac{1}{20} - \frac{4}{25}}{1 - \frac{4}{25}} = \frac{-\frac{21}{100}}{\frac{84}{100}} = -\frac{21}{84} = -\frac{1}{4}.$$

Update:  $\alpha_{2,1} = \alpha_{1,1} - \alpha_{2,2}\alpha_{1,1} = \frac{2}{5} - (-\frac{1}{4})\frac{2}{5} = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}$ .

**Lag 3:**

$$\alpha_{3,3} = \frac{\rho_3 - \alpha_{2,1}\rho_2 - \alpha_{2,2}\rho_1}{1 - \alpha_{2,1}\rho_1 - \alpha_{2,2}\rho_2}.$$

Numerator:  $-\frac{1}{8} - \frac{1}{2} \cdot (-\frac{1}{20}) - (-\frac{1}{4}) \cdot \frac{2}{5} = -\frac{1}{8} + \frac{1}{40} + \frac{1}{10} = -\frac{5}{40} + \frac{1}{40} + \frac{4}{40} = 0$ . Hence  $\alpha_{3,3} = 0$ .  
So  $\alpha_1 = \frac{2}{5}$ ,  $\alpha_2 = -\frac{1}{4}$ ,  $\alpha_3 = 0$ . The PACF cutting to 0 at lag 3 suggests an **AR(2)**. Indeed

$$Y_t = \frac{1}{2}Y_{t-1} - \frac{1}{4}Y_{t-2} + \varepsilon_t$$

matches: its Yule–Walker ACF is  $\rho_1 = \phi_1/(1 - \phi_2) = \frac{1/2}{1+1/4} = \frac{2}{5}$  and  $\rho_2 = \phi_1\rho_1 + \phi_2 = \frac{1}{2} \cdot \frac{2}{5} - \frac{1}{4} = -\frac{1}{20}$ , exactly the given values; and the diagonal PACF coefficients  $\alpha_{2,1} = \frac{1}{2} = \phi_1$ ,  $\alpha_{2,2} = -\frac{1}{4} = \phi_2$  recover the AR coefficients.

**Exercise 7: MA(1) and its reciprocal twin share an ACVS** [New]

Let  $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$  with  $\text{Var } \varepsilon_t = \sigma_\varepsilon^2$  ( $\theta \neq 0$ ), and let  $Y_t = \eta_t - \frac{1}{\theta}\eta_{t-1}$  with  $\text{Var } \eta_t = \sigma_\varepsilon^2\theta^2$ . Compute both ACVSs and show they coincide. Which representation is invertible?

### Solution

For  $X_t$ , by the MA covariance technique (Key Technique 1) with taps  $(1, -\theta)$ ,

$$\gamma_0 = \sigma_\varepsilon^2(1 + \theta^2), \quad \gamma_{\pm 1} = -\sigma_\varepsilon^2\theta, \quad \gamma_\tau = 0 \quad (|\tau| \geq 2).$$

For  $Y_t$  the MA(1) coefficient is  $1/\theta$  and the noise variance is  $\sigma_\eta^2 = \sigma_\varepsilon^2 \theta^2$ :

$$\begin{aligned}\tilde{\gamma}_0 &= \sigma_\eta^2 \left(1 + \frac{1}{\theta^2}\right) = \sigma_\varepsilon^2 \theta^2 \cdot \frac{\theta^2 + 1}{\theta^2} = \sigma_\varepsilon^2 (1 + \theta^2), \\ \tilde{\gamma}_{\pm 1} &= -\sigma_\eta^2 \cdot \frac{1}{\theta} = -\sigma_\varepsilon^2 \theta^2 \cdot \frac{1}{\theta} = -\sigma_\varepsilon^2 \theta, \quad \tilde{\gamma}_\tau = 0 \quad (|\tau| \geq 2).\end{aligned}$$

Thus  $\tilde{\gamma}_\tau = \gamma_\tau$  for all  $\tau$ : the two MA(1)s are **indistinguishable** from second-order structure. The MA polynomial roots are  $z = 1/\theta$  (for  $X_t$ ) and  $z = \theta$  (for  $Y_t$ ); the **invertible** representation is the one with the root outside the unit circle, i.e. the one with the smaller coefficient in modulus ( $|\theta| < 1$  picks  $X_t$ ,  $|\theta| > 1$  picks  $Y_t$ ).

#### Exercise 8: ACVS and ACF of an MA(2) [New]

Let  $X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$  with white-noise variance  $\sigma_\varepsilon^2$ . Find  $\gamma_\tau$  for all  $\tau$  and write down  $\rho_1, \rho_2$ . Verify  $\gamma_\tau = 0$  for  $|\tau| \geq 3$ .

#### Solution

The filter taps are  $(1, -\theta_1, -\theta_2)$ . By Key Technique 1,  $\gamma_\tau = \sigma_\varepsilon^2 \sum_j (\text{tap}_j)(\text{tap}_{j+|\tau|})$ :

$$\begin{aligned}\gamma_0 &= \sigma_\varepsilon^2 (1 + \theta_1^2 + \theta_2^2), \\ \gamma_{\pm 1} &= \sigma_\varepsilon^2 [(1)(-\theta_1) + (-\theta_1)(-\theta_2)] = \sigma_\varepsilon^2 (-\theta_1 + \theta_1 \theta_2) = \sigma_\varepsilon^2 \theta_1 (\theta_2 - 1), \\ \gamma_{\pm 2} &= \sigma_\varepsilon^2 [(1)(-\theta_2)] = -\sigma_\varepsilon^2 \theta_2, \\ \gamma_\tau &= 0 \quad (|\tau| \geq 3),\end{aligned}$$

the last because for  $|\tau| \geq 3$  no pair of taps overlaps (the filter has length 3). The ACF cuts off after lag  $q = 2$ , as the table predicts:

$$\rho_1 = \frac{\theta_1(\theta_2 - 1)}{1 + \theta_1^2 + \theta_2^2}, \quad \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}.$$

#### Exercise 9: AR(2) causality region and ACF via Yule–Walker [New]

Consider  $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ . (a) State the causality (stationarity) conditions on  $(\phi_1, \phi_2)$ . (b) For  $\phi_1 = \frac{1}{2}$ ,  $\phi_2 = -\frac{1}{4}$ , compute  $\rho_1, \rho_2, \rho_3$  from the Yule–Walker equations, and predict the PACF pattern.

#### Solution

(a) With  $\Phi(z) = 1 - \phi_1 z - \phi_2 z^2$ , causality requires both roots outside the unit circle. Equivalently (the standard “stationarity triangle”):

$$\phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad |\phi_2| < 1.$$

(b) Here  $\phi_1 + \phi_2 = \frac{1}{4} < 1$ ,  $\phi_2 - \phi_1 = -\frac{3}{4} < 1$ ,  $|\phi_2| = \frac{1}{4} < 1$ : causal. The Yule–Walker recursion  $\rho_\tau = \phi_1 \rho_{\tau-1} + \phi_2 \rho_{\tau-2}$  ( $\tau \geq 1$ , with  $\rho_0 = 1$ ,  $\rho_{-1} = \rho_1$ ) gives

$$\rho_1 = \frac{\phi_1}{1 - \phi_2} = \frac{1/2}{1 + 1/4} = \frac{2}{5}, \quad \rho_2 = \phi_1 \rho_1 + \phi_2 = \frac{1}{2} \cdot \frac{2}{5} - \frac{1}{4} = -\frac{1}{20},$$

$$\rho_3 = \phi_1 \rho_2 + \phi_2 \rho_1 = \frac{1}{2} \cdot \left(-\frac{1}{20}\right) - \frac{1}{4} \cdot \frac{2}{5} = -\frac{1}{40} - \frac{1}{10} = -\frac{1}{8}.$$

These are exactly the values of Exercise 6: the PACF cuts off after lag 2 ( $\alpha_3 = 0$ ), the AR(2) signature, while the ACF tails off.

### Exercise 10: One-step recursive forecast for an AR(1) [New]

For a causal AR(1)  $X_t = \phi X_{t-1} + \varepsilon_t$  ( $|\phi| < 1$ ), find the best linear one-step-ahead predictor  $P_{X_1, \dots, X_n}(X_{n+1})$  and its mean-squared error.

#### Solution

The prediction equations require the error  $X_{n+1} - \hat{X}_{n+1}$  to be uncorrelated with  $X_1, \dots, X_n$ . Try  $\hat{X}_{n+1} = \phi X_n$ : the error is  $X_{n+1} - \phi X_n = \varepsilon_{n+1}$ , and by causality  $X_k$  depends only on  $\varepsilon_s$ ,  $s \leq k \leq n < n+1$ , so  $\text{Cov}(\varepsilon_{n+1}, X_k) = 0$  for all  $k \leq n$ . The orthogonality conditions hold, hence

$$P_{X_1, \dots, X_n}(X_{n+1}) = \phi X_n, \quad \text{MSE} = \mathbb{E}[\varepsilon_{n+1}^2] = \sigma_\varepsilon^2.$$

Only the most recent observation matters — a direct consequence of the AR(1) having PACF 0 beyond lag 1 (the older points carry no *new* linear information).

## 3 ARMA Revisited: Backshift Operator, Wold Decomposition, Causality & Invertibility

Having met AR, MA and ARMA models as explicit sums of white noise, we now repackage them with the **backshift operator**  $B$ , which turns the difference equations into a clean algebra of *polynomials*  $\Phi(z)$  and  $\Theta(z)$ . The payoff is conceptual: questions about a process (“is it stationary? is it invertible? what is its MA( $\infty$ ) representation?”) become questions about the *roots* of these polynomials and the analyticity of a transfer function  $G(z)$ . Along the way the **Wold decomposition theorem** tells us that essentially every stationary process is a one-sided moving average of white noise, which is exactly why the linear/ARMA world is so central. Throughout,  $\{\varepsilon_t\}$  is mean-zero white noise with  $\text{Var}(\varepsilon_t) = \sigma^2$ , and we use the lecture-note conventions: ARMA written  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ , stationarity from roots of  $\Phi$  *outside* the unit circle, invertibility from roots of  $\Theta$  *outside* the unit circle.

### 3.1 Definitions

#### Definition 1: Backshift operator $B$

The **backshift operator** maps a time series  $\{X_t\}$  to the series  $\{Y_t\} = B[\{X_t\}]$  shifted back one step in time,

$$Y_t = X_{t-1} \quad \text{for all } t \in \mathbb{Z}.$$

We write informally  $BX_t = X_{t-1}$ . Iterating,  $B^k X_t = X_{t-k}$  for  $k \geq 0$ , with  $B^0 = I$  the identity. The operator is linear, and because shifts commute, products of polynomials in  $B$  may be multiplied and reordered like ordinary polynomials.

#### Intuition / Why it matters

$B$  is just “ $X_t \mapsto X_{t-1}$ ”, but treating it as an algebraic symbol  $z$  is what lets us *factor*, *invert* and *expand* model equations. Every result in this part is a statement about the polynomial  $\Phi(z)$  or  $\Theta(z)$  obtained by replacing  $B$  with the complex variable  $z$ .

#### Definition 2: ARMA polynomial notation

An ARMA( $p, q$ ) process satisfies

$$\sum_{j=0}^p \phi_j X_{t-j} = \sum_{k=0}^q \theta_k \varepsilon_{t-k}, \quad \phi_0 = -1, \theta_0 = -1,$$

which, with the backshift operator, becomes the compact statement

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \Phi(z) = \sum_{j=0}^p \phi_j z^j, \quad \Theta(z) = \sum_{k=0}^q \theta_k z^k.$$

In the common “signed” form one writes  $X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k \varepsilon_{t-k}$ ; simple examples instead use  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$  etc. Pure cases: AR( $p$ ) is  $\Phi(B)X_t = \varepsilon_t$  and MA( $q$ ) is  $X_t = \Theta(B)\varepsilon_t$ .

### Definition 3: Characteristic polynomials

For  $\Phi(B)X_t = \Theta(B)\varepsilon_t$  we call

- $\Phi(z)$  the **characteristic polynomial of the autoregressive part**, and
- $\Theta(z)$  the **characteristic polynomial of the moving-average part**.

For a pure AR,  $\Phi(B)X_t = \varepsilon_t$ , “the characteristic polynomial” means  $\Phi(z)$ ; for a pure MA,  $X_t = \Theta(B)\varepsilon_t$ , it means  $\Theta(z)$ . The *roots* of these polynomials govern stationarity and invertibility.

### Definition 4: Two-sided linear process

A **(two-sided) linear process** is

$$X_t = \sum_{k=-\infty}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 = \sum_k g_k^2 < \infty,$$

with  $\{g_k\} \subset \mathbb{R}$  and  $\{\varepsilon_t\}$  mean-zero white noise. Its first two moments are, for all  $t, \tau \in \mathbb{Z}$ ,

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2 \|g\|_2^2 < \infty, \quad \text{Cov}(X_t, X_{t+\tau}) = \sigma^2 \sum_{k=-\infty}^{\infty} g_k g_{k+\tau}.$$

The summability  $\|g\|_2^2 < \infty$  is what makes these moments finite, so the process is (second-order) stationary.

### Definition 5: General linear process (one-sided)

If in the two-sided process we set  $g_{-1} = g_{-2} = \dots = 0$ , we obtain a **general linear process**

$$X_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 < \infty.$$

Now  $X_t$  depends only on the *past and present* of  $\{\varepsilon_t\}$ , making the process **causal** (in the linear sense). The same mean/autocovariance formulas apply, so it is stationary. Writing  $G(z) = \sum_{k \geq 0} g_k z^k$  gives the operator form  $X_t = G(B)\varepsilon_t$ . By convention  $g_0 = 1$ .

### Definition 6: Transfer function $G(z)$ , zeros and poles

For a linear process  $X_t = G(B)\varepsilon_t$ , the function

$$G(z) = \sum_{k=0}^{\infty} g_k z^k$$

is the **transfer function**. When  $G$  is a rational ratio  $G(z) = G_1(z)/G_2(z)$ :

- the roots of the numerator  $G_1(z)$  are the **zeros** of  $G$ ;
- the roots of the denominator  $G_2(z)$  are the **poles** of  $G$ .

For an ARMA model  $G(z) = \Theta(z)/\Phi(z)$ : the zeros of  $\Phi$  (roots of the AR polynomial) are the poles of  $G$ , and the zeros of  $\Theta$  (roots of the MA polynomial) are the zeros of  $G$ . Whether the expansion of  $G$  involves only past/present (versus future) noise is decided by where these roots sit relative to the unit circle  $|z| = 1$ .

#### Definition 7: Innovations outlier

If a single realised value of  $\varepsilon_t$  affects  $X_t$  and all subsequent  $X_{t+1}, X_{t+2}, \dots$ , that value is called an **innovations outlier**. This is exactly the behaviour of a one-sided (MA( $\infty$ )) process: each innovation  $\varepsilon_t$  propagates forward through the weights  $g_0, g_1, \dots$ .

### 3.2 Theorems & Propositions

#### Theorem 8: Wold decomposition theorem

Any (zero-mean) second-order stationary process  $\{X_t\}$  can be written as

$$X_t = U_t + V_t, \quad U_t \text{ and } V_t \text{ uncorrelated,}$$

where

- $U_t$  has a **one-sided linear** (MA( $\infty$ )) representation

$$U_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad g_0 = 1, \quad \|g\|_2^2 < \infty,$$

with  $\{\varepsilon_t\}$  mean-zero white noise uncorrelated with  $V$ , i.e.  $\mathbb{E}[\varepsilon_s V_t] = 0$  for all  $s, t$ ; the sequences  $\{g_k\}$  and  $\{\varepsilon_t\}$  are *uniquely* determined;

- $V_t$  is **singular** (deterministic): it can be predicted from its own past with zero error.

*Proof idea:* project  $X_t$  onto the closed linear span of its own past; the residual one-step innovations are the white noise  $\varepsilon_t$ , the  $g_k$  are the projection coefficients, and  $V_t$  is the part lying in the infinite-past intersection of these spans (the purely deterministic component).

*Why:* this is the structural justification for the whole linear-model programme: up to a deterministic piece, *every* stationary process is a one-sided moving average of white noise, and ARMA models are the finite, rational approximations to this MA( $\infty$ ).

#### Intuition / Why it matters

Wold says: the only “new information” arriving at time  $t$  is the innovation  $\varepsilon_t$ ; everything else is either an echo of past innovations ( $U_t$ ) or perfectly forecastable from the past ( $V_t$ , e.g. a fixed sinusoid). For the processes we study  $V_t = 0$ , leaving the pure MA( $\infty$ ) form  $X_t = G(B)\varepsilon_t$ .

#### Proposition 9: Laurent expansion of $1/G_2$ and the past/present criterion

Let  $G(z) = G_1(z)/G_2(z)$  and let  $z_1, \dots, z_p$  be the zeros of  $G_2$ , ordered so that  $z_1, \dots, z_k$  lie *inside* the unit circle and  $z_{k+1}, \dots, z_p$  lie *outside*. A partial-fraction plus geometric-series (Laurent) expansion, convergent on  $|z| = 1$ , gives after replacing  $z \mapsto B$

$$\frac{1}{G_2(B)} \varepsilon_t = \underbrace{\sum_{j=1}^k A_j \sum_{l=0}^{\infty} z_j^l \varepsilon_{t+1+l}}_{\text{future}} - \underbrace{\sum_{j=k+1}^p A_j \sum_{l=0}^{\infty} z_j^{-(l+1)} \varepsilon_{t-l}}_{\text{past+present}}.$$

Hence if *all* roots of  $G_2(z)$  are outside the unit circle ( $k = 0$ ), only past and present values appear, and the one-sided general linear process exists.

*Proof idea:* expand  $A_j/(z - z_j)$  as a geometric series, choosing the *inside* expansion  $\sum_l (z_j/z)^l$  for  $|z_j| < 1$  and the *outside* expansion for  $|z_j| > 1$ , so that each series converges on  $|z| = 1$ .  
*Why:* this is the precise mechanism behind “roots outside the unit circle  $\Rightarrow$  causal”. It also gives the equivalent analytic statement:  $|G(z)| < \infty$  for  $|z| \leq 1$ , i.e.  $G$  is **analytic inside and on the unit circle**.

### Proposition 10: Stationarity of the general linear process

A general linear process  $X_t = \sum_{k \geq 0} g_k \varepsilon_{t-k}$  with  $\|g\|_2^2 < \infty$  is second-order stationary, with

$$\mathbb{E}[X_t] = 0, \quad \gamma_\tau = \sigma^2 \sum_{k=0}^{\infty} g_k g_{k+|\tau|}.$$

*Proof idea:* the mean is zero by linearity; bilinearity of covariance together with  $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$  collapses the double sum to the single sum above, which is finite by Cauchy–Schwarz and  $\|g\|_2^2 < \infty$ .

*Why:* this is the engine for computing the ACVS of *any* causal ARMA once its  $\text{MA}(\infty)$  weights  $\{g_k\} = \{\psi_k\}$  are known.

### Proposition 11: Stationarity (causality) of $\text{AR}(p)$

For  $\text{AR}(p)$ ,  $\Phi(B)X_t = \varepsilon_t$ , the (unique) stationary causal solution is  $X_t = \Phi^{-1}(B)\varepsilon_t = G(B)\varepsilon_t$  with  $G(z) = \Phi^{-1}(z)$ . This power expansion is analytic on  $|z| \leq 1$  *iff* all roots of  $\Phi(z)$  lie *outside* the unit circle. Therefore

$$\text{AR}(p) \text{ is stationary} \iff \text{all roots of } \Phi(z) \text{ satisfy } |z| > 1.$$

*Proof idea:* the poles of  $G = 1/\Phi$  are the roots of  $\Phi$ ; by the Laurent criterion (Prop. 9) the one-sided expansion exists exactly when those poles are outside  $|z| = 1$ .

*Why:* this is *the* root test for AR stationarity; it reduces an infinite-order question to checking the moduli of  $p$  roots.

### Proposition 12: $\text{AR}(p)$ is always invertible; $\text{MA}(q)$ is always stationary

- For  $\text{AR}(p)$ ,  $\Phi(B)X_t = \varepsilon_t$  already expresses the white noise as a *finite* polynomial in  $X$ , so  $\text{AR}(p)$  is **always invertible** (the AR polynomial  $\Phi$  has finite order; there is no MA polynomial,  $\Theta \equiv 1$ ).
- For  $\text{MA}(q)$ ,  $X_t = \Theta(B)\varepsilon_t = G(B)\varepsilon_t$  with  $G(z) = \Theta(z)$  a *finite* polynomial, so  $|G(z)| < \infty$  everywhere; hence  $\text{MA}(q)$  is **always stationary**.

*Proof idea:* “finite polynomial  $\Rightarrow$  bounded on the unit disk”; only the *inverse* expansion can fail to converge, never the finite direct one.

*Why:* it tells you which check is *free*: never test an AR for invertibility or an MA for stationarity.

### Proposition 13: Invertibility via $G^{-1}$ and the $\text{MA}(q)$ root test

A linear process  $X_t = G(B)\varepsilon_t$  is **invertible** if it can be rewritten as an  $\text{AR}(\infty)$ ,

$$G^{-1}(B)X_t = \varepsilon_t, \quad G^{-1}(z) = \sum_{k=0}^{\infty} h_k z^k,$$

with  $G^{-1}$  analytic on  $|z| \leq 1$ , equivalently  $|G^{-1}(z)| < \infty$  for  $|z| \leq 1$ , equivalently *all poles of  $G^{-1}$  are outside* the unit circle (equivalently, all *zeros* of  $G$  are inside). For  $\text{MA}(q)$ ,  $G = \Theta$ , so

$$\text{MA}(q) \text{ is invertible} \iff \text{all roots of } \Theta(z) \text{ satisfy } |z| > 1.$$

*Proof idea:* inverting  $X_t = \Theta(B)\varepsilon_t$  needs the power series of  $1/\Theta$  to converge on  $|z| \leq 1$ ; by the Laurent criterion this requires the roots of  $\Theta$  (the poles of  $1/\Theta$ ) outside the unit circle.  
*Why:* invertibility is what makes the innovations  $\varepsilon_t$  recoverable from the observed past  $X_t, X_{t-1}, \dots$ , which is essential for forecasting and for parameter identifiability.

### Intuition / Why it matters

Stationarity (causality) and invertibility are the *same* test applied to *two different* polynomials: “all roots outside the unit circle” for  $\Phi$  controls whether  $X_t = G(B)\varepsilon_t$  uses only past/present noise (causal), while the identical test for  $\Theta$  controls whether  $\varepsilon_t = G^{-1}(B)X_t$  uses only past/present data (invertible). An AR has no  $\Theta$  to fail, so it is always invertible; an MA has no  $\Phi$  to fail, so it is always stationary.

### Corollary 14: Stationarity & invertibility summary for ARMA

For  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ :

**AR**( $p$ ) : stationary  $\iff$  roots of  $\Phi$  outside  $|z| = 1$ ; *always* invertible.

**MA**( $q$ ) : *always* stationary; invertible  $\iff$  roots of  $\Theta$  outside  $|z| = 1$ .

**ARMA**( $p, q$ ) : stationary  $\iff$  roots of  $\Phi$  outside; invertible  $\iff$  roots of  $\Theta$  outside.

*Why:* the single most-used reference fact of the chapter; combine the two independent root tests.

### Classic Pitfall

“Roots outside the unit circle” refers to the roots of the polynomial  $\Phi(z) = \sum_j \phi_j z^j$  in the variable  $z$ , i.e.  $|z| > 1$ . Equivalently, in terms of the *reciprocal* roots  $g = 1/z$ , one needs  $|g| < 1$  — exactly the AR(1) condition  $|\phi| < 1$  (since the root of  $1 - \phi z$  is  $z = 1/\phi$ ). Do not confuse the two conventions, and do not check *inside* the circle.

### Proposition 15: Causal AR( $p$ ) has $\alpha_\tau = 0$ for $\tau > p$

If  $\{X_t\}$  is a causal AR( $p$ ), its partial autocorrelation function satisfies  $\alpha_\tau = 0$  for all  $\tau > p$ .

*Proof idea:* the order- $\tau$  best linear predictor is  $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$  and  $\alpha_\tau = \alpha_{\tau,\tau}$ . Since  $X_\tau = \sum_{j=1}^p \phi_j X_{\tau-j} + \varepsilon_\tau$ , choosing  $\alpha_{\tau,j} = \tilde{\phi}_j$  (with  $\tilde{\phi}_j = \phi_j$  for  $j \leq p$ ,  $\tilde{\phi}_j = 0$  for  $j > p$ ) makes the prediction residual equal  $\varepsilon_\tau$ , which by causality satisfies  $\text{Cov}(\varepsilon_\tau, X_k) = 0$  for all  $k < \tau$  (here  $\text{Cov}(X_t, \varepsilon_s) = 0$  when  $t < s$ ). Thus the orthogonality (prediction) equations are met, so these are the true coefficients; in particular  $\alpha_\tau = \alpha_{\tau,\tau} = \tilde{\phi}_\tau = 0$  for  $\tau > p$ .

*Why:* the PACF “cuts off” after lag  $p$  for an AR( $p$ ) — the diagnostic mirror image of the ACF cutting off after lag  $q$  for an MA( $q$ ) (this proof is Sheet 2, Ex. 2.7).

## 3.3 Key Techniques

### Key Technique 1: Root test for stationarity / invertibility

Given  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ :

1. Write the polynomials in  $z$ :  $\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$  and  $\Theta(z) = 1 - \theta_1 z - \dots - \theta_q z^q$  (matching your sign convention).
2. **Stationarity:** solve  $\Phi(z) = 0$  and check *every* root has  $|z| > 1$ . (AR/ARMA only; an MA is automatically stationary.)
3. **Invertibility:** solve  $\Theta(z) = 0$  and check *every* root has  $|z| > 1$ . (MA/ARMA only; an AR

is automatically invertible.)

For a quadratic  $az^2 + bz + c$  use  $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  and report  $|z| = \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2}$ . A handy AR(1)/AR(2) shortcut:  $1 - \phi z$  has root  $1/\phi$ ;  $1 - \phi_1 z - \phi_2 z^2$  is stationary iff  $|\phi_2| < 1$ ,  $\phi_2 + \phi_1 < 1$  and  $\phi_2 - \phi_1 < 1$  (the AR(2) triangle).

### Key Technique 2: $\psi$ -weights by recurrence (matching coefficients)

To find the MA( $\infty$ ) weights in  $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$  for a causal  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ , equate  $\Phi(z)(\sum_{j \geq 0} \psi_j z^j) = \Theta(z)$  and match powers of  $z$ . With  $\Phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$  and  $\Theta(z) = 1 - \theta_1 z - \dots - \theta_q z^q$  this gives the recurrence

$$\psi_j = \theta_j^* + \sum_{i=1}^p \phi_i \psi_{j-i}, \quad \psi_0 = 1, \quad \psi_j = 0 \quad (j < 0),$$

where  $\theta_j^* = -\theta_j$  for  $1 \leq j \leq q$  and 0 otherwise (with the  $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$  sign convention). Solve forward; for a constant-coefficient linear recurrence the closed form is a sum  $\sum_i (\text{polynomial in } j) r_i^{-j}$  over the AR roots  $r_i$  (with the familiar  $(c_0 + c_1 j)r^{-j}$  shape for a repeated root).

### Key Technique 3: $\psi$ -weights by partial fractions

Alternatively, expand  $G(z) = \Theta(z)/\Phi(z)$  directly:

1. Factor  $\Phi(z) = \phi_p \prod_i (z - r_i)$  with roots  $r_i$  ( $|r_i| > 1$ ).
2. Partial-fraction  $\frac{\Theta(z)}{\Phi(z)} = (\text{poly}) + \sum_i \frac{A_i}{z - r_i}$ , then write each  $\frac{A_i}{z - r_i} = -\frac{A_i}{r_i} \frac{1}{1 - z/r_i} = -\frac{A_i}{r_i} \sum_{j \geq 0} (z/r_i)^j$  (valid for  $|z| < |r_i|$ , in particular on/inside the unit circle).
3. Collect the coefficient of  $z^j$  to read off  $\psi_j$ .

Because  $|r_i| > 1$  the geometric series converge inside the unit disk, giving a decaying  $\psi_j$ . This is the method of Sheet 3, Ex. 3.1.

### Key Technique 4: ACVS of a causal ARMA via the difference equation

To get  $\gamma_\tau$  for  $\Phi(B)X_t = \Theta(B)\varepsilon_t$  without summing  $\psi$ -weights:

1. Multiply the model equation by  $X_{t-\tau}$  and take expectations. For  $\tau$  larger than the MA order  $q$  the right side vanishes, giving a *homogeneous* recurrence in  $\gamma_\tau$  whose characteristic equation is  $\Phi$  (so  $\gamma_\tau$  is a combination of  $r_i^{-\tau}$ ).
2. For the small lags  $\tau = 0, 1, \dots, q$  the right side keeps cross-terms  $\mathbb{E}[\varepsilon_{t-k} X_{t-\tau}] = \sigma^2 \psi_{k-\tau}$  (only  $k \geq \tau$  contribute, since  $X_{t-\tau}$  is a one-sided MA in  $\varepsilon_{t-\tau}, \varepsilon_{t-\tau-1}, \dots$ ). These give the initial conditions.
3. Fit the general homogeneous solution to the initial conditions.

This is the route used in the lecture-note worked example (Exercise 6).

### Key Technique 5: Spectral density of an ARMA from its polynomials

For a causal, invertible ARMA  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ , the spectral density (with frequency  $f$ , sampling step 1) is

$$\mathcal{S}(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2}, \quad -\frac{1}{2} \leq f \leq \frac{1}{2}.$$

For the pure MA(1)  $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$  this gives  $\mathcal{S}(f) = \sigma^2 |1 - \theta e^{-2\pi i f}|^2 = \sigma^2(1 + \theta^2 - 2\theta \cos(2\pi f))$ , and one checks  $\int_{-1/2}^{1/2} \mathcal{S}(f) df = \gamma_0 = \sigma^2(1 + \theta^2)$  and  $\int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f} df = \gamma_1 = -\theta\sigma^2$ .

### 3.4 Exercises

#### Exercise 1: $\psi$ -weights via partial fractions [Sheet 3]

Find the coefficients  $\psi_j$ ,  $j = 0, 1, 2, \dots$ , in the representation  $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$  of the ARMA process

$$(1 - 0.5B + 0.04B^2)X_t = (1 + 0.25B)\varepsilon_t,$$

where  $\varepsilon_t$  is white noise.

#### Solution

Write  $X_t = G(B)\varepsilon_t$  with

$$G(z) = \frac{1 + 0.25z}{1 - 0.5z + 0.04z^2} = \frac{1 + 0.25z}{0.04(z - 2.5)(z - 10)},$$

since  $0.04z^2 - 0.5z + 1 = 0$  has roots  $z = 2.5$  and  $z = 10$  (both outside the unit circle, so the process is causal). Partial-fraction the rational part  $\frac{1}{(z-2.5)(z-10)} = \frac{1}{7.5} \left( \frac{-1}{z-2.5} + \frac{1}{z-10} \right)$ , hence

$$G(z) = \frac{1 + 0.25z}{0.04 \cdot 7.5} \left( \frac{-1}{z - 2.5} + \frac{1}{z - 10} \right) = \frac{1 + 0.25z}{0.3} \left( \frac{1}{2.5} \sum_{j \geq 0} 0.4^j z^j - \frac{1}{10} \sum_{j \geq 0} 0.1^j z^j \right),$$

using  $\frac{-1}{z-2.5} = \frac{1}{2.5} \frac{1}{1-z/2.5} = \frac{1}{2.5} \sum_{j \geq 0} (0.4)^j z^j$  and  $\frac{1}{z-10} = -\frac{1}{10} \sum_{j \geq 0} (0.1)^j z^j$ . Distributing the factor  $(1 + 0.25z)/0.3$  and collecting powers of  $z$ ,

$$G(z) = \sum_{j \geq 0} \left( \frac{0.4^j}{0.75} - \frac{0.1^j}{3} \right) z^j + \sum_{l \geq 1} \left( \frac{0.4^l}{1.2} - \frac{0.1^l}{1.2} \right) z^l,$$

so for  $j \geq 1$ ,

$$\psi_j = \frac{0.4^j}{0.75} - \frac{0.1^j}{3} + \frac{0.4^j}{1.2} - \frac{0.1^j}{1.2} = \left( \frac{1}{0.75} + \frac{1}{1.2} \right) 0.4^j - \left( \frac{1}{3} + \frac{1}{1.2} \right) 0.1^j = \frac{13}{6} 0.4^j - \frac{7}{6} 0.1^j,$$

and  $\psi_0 = 1$ . Hence

$$\psi_j = \begin{cases} 1 & j = 0, \\ \frac{13}{6} (0.4)^j - \frac{7}{6} (0.1)^j & j \geq 1. \end{cases}$$

*Check (coefficient-matching):* the recurrence  $\psi_0 = 1$ ,  $\psi_1 - 0.5\psi_0 = 0.25 \Rightarrow \psi_1 = 0.75$ ,  $\psi_j = 0.5\psi_{j-1} - 0.04\psi_{j-2}$  ( $j \geq 2$ ) gives  $\psi_0, \psi_1, \psi_2, \psi_3 = 1, 0.75, 0.335, 0.1375$ , matching  $\frac{13}{6} 0.4^j - \frac{7}{6} 0.1^j$  exactly.

#### Exercise 2: MA( $\infty$ ) representation of AR(1) and AR(2) [Sheet 3]

For a stationary causal AR(1)  $(1 - \phi B)Y_t = \varepsilon_t$  and a stationary AR(2)  $(1 - \phi_1 B - \phi_2 B^2)Y_t = \varepsilon_t$ , find the representation of  $Y_t$  in terms of  $\varepsilon_t$ .

### Solution

**AR(1).** By causality ( $|\phi| < 1$ ),

$$Y_t = \frac{1}{1 - \phi B} \varepsilon_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}.$$

**AR(2).** Factor  $\Phi(z) = 1 - \phi_1 z - \phi_2 z^2 = (1 - g_1 z)(1 - g_2 z)$ , where  $g_1^{-1}, g_2^{-1}$  are the roots (outside the unit circle, so  $|g_1|, |g_2| < 1$ ). Then

$$Y_t = \frac{1}{(1 - g_1 B)(1 - g_2 B)} \varepsilon_t = \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} g_1^{k_1} g_2^{k_2} \varepsilon_{t-(k_1+k_2)} = \sum_{n=0}^{\infty} \left( \sum_{k=0}^n g_1^k g_2^{n-k} \right) \varepsilon_{t-n},$$

collecting terms with  $n = k_1 + k_2$  (Cauchy product, justified by  $|g_1|, |g_2| < 1$ ). For  $g_1 \neq g_2$  the inner sum is a geometric series,

$$\sum_{k=0}^n g_1^k g_2^{n-k} = \frac{g_1^{n+1} - g_2^{n+1}}{g_1 - g_2}, \quad \text{so} \quad Y_t = \frac{g_1}{g_1 - g_2} \sum_{n \geq 0} g_1^n \varepsilon_{t-n} - \frac{g_2}{g_1 - g_2} \sum_{n \geq 0} g_2^n \varepsilon_{t-n}.$$

The reciprocal roots are  $g_{1,2}^{-1} = \frac{-\phi_1 \mp \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2}$ . When  $g_1 = g_2 = g$  the inner sum becomes  $\sum_{k=0}^n g^n = (n+1)g^n$ , so  $\psi_n = (n+1)g^n$ .

### Exercise 3: AR(2) stationarity via roots [Sheet 3]

For the AR(2) process  $\frac{15}{16}Y_t = \frac{1}{4}Y_{t-1} - \frac{1}{16}Y_{t-2} + \varepsilon_t$  (with  $\varepsilon_t$  Gaussian), write the characteristic AR polynomial. Is it stationary? Invertible?

### Solution

Move all  $Y$  terms to the left and use the backshift operator:

$$\left( \frac{15}{16} - \frac{1}{4}B + \frac{1}{16}B^2 \right) Y_t = \varepsilon_t, \quad \Phi(z) = \frac{15}{16} - \frac{1}{4}z + \frac{1}{16}z^2.$$

Solve  $\frac{1}{16}z^2 - \frac{1}{4}z + \frac{15}{16} = 0$ , i.e.  $z^2 - 4z + 15 = 0$ :

$$z = \frac{4 \pm \sqrt{16 - 60}}{2} = 2 \pm \sqrt{-11} = 2 \pm \sqrt{11}i \approx 2 \pm 3.3166i, \quad |z| = \sqrt{4 + 11} = \sqrt{15} \approx 3.873.$$

Both roots have modulus  $\sqrt{15} > 1$ , i.e. they lie outside the unit circle, so the process is **stationary**. An AR process is **always invertible**.

### Exercise 4: ARMA(2, 1) stationary & invertible [Sheet 3]

For the ARMA(2, 1) process  $Y_t - \frac{1}{4}Y_{t-1} + \frac{1}{16}Y_{t-2} = \varepsilon_t - 0.5\varepsilon_{t-1}$ , is it stationary? Invertible?

### Solution

In backshift form,  $(1 - \frac{1}{4}B + \frac{1}{16}B^2)Y_t = (1 - 0.5B)\varepsilon_t$ , so

$$\Phi(z) = 1 - \frac{1}{4}z + \frac{1}{16}z^2, \quad \Theta(z) = 1 - \frac{1}{2}z.$$

**Stationarity.**  $\Phi(z) = 0 \iff z^2 - 4z + 16 = 0$  (multiply by 16), giving

$$z = \frac{4 \pm \sqrt{16 - 64}}{2} = 2 \pm 2\sqrt{3}i, \quad |z| = \sqrt{4 + 12} = 4 > 1,$$

both outside the unit circle  $\Rightarrow$  **stationary**.

**Invertibility.**  $\Theta(z) = 0 \iff z = 2$ , and  $|2| = 2 > 1$  is outside the unit circle  $\Rightarrow$  **invertible**.  
The process is both stationary and invertible.

### Exercise 5: Same ACF; inverting a non-invertible MA [Sheet 3]

Let  $\{U_t\}$  be stationary zero-mean. Define  $X_t = (1 - 0.4B)U_t$  and  $W_t = (1 - 2.5B)U_t$ . **(a)** Express the ACFs of  $\{X_t\}$  and  $\{W_t\}$  via that of  $\{U_t\}$ . **(b)** Show  $\{X_t\}$  and  $\{W_t\}$  have the *same* ACF. **(c)** Show  $\varepsilon_t = -\sum_{j \geq 1} (0.4)^j X_{t+j}$  satisfies  $\varepsilon_t - 2.5\varepsilon_{t-1} = X_t$ .

### Solution

**(a)** With  $X_t = U_t - 0.4U_{t-1}$ , bilinearity gives

$\gamma_X(\tau) = \text{Cov}(U_t - 0.4U_{t-1}, U_{t+\tau} - 0.4U_{t+\tau-1}) = 1.16\gamma_U(\tau) - 0.4\gamma_U(\tau-1) - 0.4\gamma_U(\tau+1)$ ,  
using  $1 + 0.4^2 = 1.16$ . In particular, with symmetry  $\gamma_U(-1) = \gamma_U(1)$ ,

$$\gamma_X(0) = 1.16\gamma_U(0) - 0.8\gamma_U(1) = \gamma_U(0)\{1.16 - 0.8\rho_U(1)\}.$$

Dividing, and writing  $\gamma_U(\tau) = \rho_U(\tau)\gamma_U(0)$ ,

$$\rho_X(\tau) = \frac{1.16\rho_U(\tau) - 0.4\rho_U(\tau-1) - 0.4\rho_U(\tau+1)}{1.16 - 0.8\rho_U(1)}.$$

Similarly, with  $W_t = U_t - 2.5U_{t-1}$  ( $1 + 2.5^2 = 7.25$ ),

$$\rho_W(\tau) = \frac{7.25\rho_U(\tau) - 2.5\rho_U(\tau-1) - 2.5\rho_U(\tau+1)}{7.25 - 5\rho_U(1)}.$$

**(b)** Note  $7.25 = 6.25 \times 1.16$ ,  $2.5 = 6.25 \times 0.4$  and  $5 = 6.25 \times 0.8$ . Factoring 6.25 from numerator and denominator of  $\rho_W$ ,

$$\rho_W(\tau) = \frac{6.25\{1.16\rho_U(\tau) - 0.4\rho_U(\tau-1) - 0.4\rho_U(\tau+1)\}}{6.25\{1.16 - 0.8\rho_U(1)\}} = \rho_X(\tau).$$

So the two have identical ACFs. (This is the MA non-uniqueness: the root  $z = 2.5$  for  $X$  and  $z = 0.4 = 1/2.5$  for  $W$  give the same correlation structure; only  $X$ , with root outside the circle, is invertible.)

**(c)** With  $\varepsilon_t = -\sum_{j \geq 1} (0.4)^j X_{t+j}$ ,

$$\begin{aligned} \varepsilon_t - 2.5\varepsilon_{t-1} &= -\sum_{j \geq 1} 0.4^j X_{t+j} + 2.5 \sum_{j \geq 1} 0.4^j X_{t+j-1} \\ &= -\sum_{j \geq 1} 0.4^j X_{t+j} + 2.5 \sum_{l \geq 0} 0.4^{l+1} X_{t+l} \quad (l = j-1) \\ &= -\sum_{j \geq 1} 0.4^j X_{t+j} + \sum_{l \geq 0} 0.4^l X_{t+l} \quad (2.5 \cdot 0.4 = 1) \\ &= X_t + \sum_{l \geq 1} 0.4^l X_{t+l} - \sum_{j \geq 1} 0.4^j X_{t+j} = X_t. \end{aligned}$$

Thus the non-invertible  $W$ -type representation can still be inverted, but using *future* values  $X_{t+j}$  — exactly the “future-noise” branch of the Laurent expansion (Prop. 9).

### Exercise 6: Worked lecture example: ACVS of an ARMA(2, 1) [Notes]

For  $\{I - B + \frac{1}{4}B^2\}X_t = \{I + B\}\varepsilon_t$  with  $\varepsilon_t$  white noise of variance  $\sigma^2$ , determine the autocovariance sequence  $\gamma_\tau$ .

#### Solution

The model is  $X_t - X_{t-1} + \frac{1}{4}X_{t-2} = \varepsilon_t + \varepsilon_{t-1}$ , with  $\Phi(z) = 1 - z + \frac{1}{4}z^2 = \frac{1}{4}(z - 2)^2$  (double root  $z = 2$  outside the unit circle  $\Rightarrow$  stationary) and  $\Theta(z) = 1 + z$  (root  $z = -1$  on the circle  $\Rightarrow$  not invertible, but stationarity is all we need here).

**Recurrence for  $\tau \geq 2$ .** Multiply by  $X_{t-\tau}$  and take expectations. For  $\tau \geq 2$  the right side is  $\mathbb{E}[(\varepsilon_t + \varepsilon_{t-1})X_{t-\tau}] = 0$  (since  $X_{t-\tau}$  involves only  $\varepsilon_{t-\tau}, \varepsilon_{t-\tau-1}, \dots$ , all uncorrelated with  $\varepsilon_t, \varepsilon_{t-1}$ ), so

$$\gamma_\tau - \gamma_{\tau-1} + \frac{1}{4}\gamma_{\tau-2} = 0, \quad \tau \geq 2.$$

The characteristic equation is  $\Phi$  with a double root, so the general solution is

$$\gamma_\tau = (\beta_0 + \beta_1\tau)2^{-\tau}, \quad \tau \geq 0.$$

**Initial conditions ( $\tau = 0, 1$ ).** First find the leading  $\psi$ -weights from  $\Phi(B)\psi(B) = \Theta(B)$ :  $\psi_0 = 1$  and  $\psi_1 - \psi_0 = 1 \Rightarrow \psi_1 = 2$ . Then the cross terms are  $\mathbb{E}[\varepsilon_t X_t] = \psi_0 \sigma^2 = \sigma^2$  and  $\mathbb{E}[\varepsilon_{t-1} X_t] = \psi_1 \sigma^2 = 2\sigma^2$ . Hence

$$\tau = 0: \quad \gamma_0 - \gamma_1 + \frac{1}{4}\gamma_2 = \mathbb{E}[(\varepsilon_t + \varepsilon_{t-1})X_t] = \sigma^2 + 2\sigma^2 = 3\sigma^2,$$

$$\tau = 1: \quad \gamma_1 - \gamma_0 + \frac{1}{4}\gamma_1 = \mathbb{E}[(\varepsilon_t + \varepsilon_{t-1})X_{t-1}] = \sigma^2.$$

(The second uses  $\mathbb{E}[\varepsilon_t X_{t-1}] = 0$  and  $\mathbb{E}[\varepsilon_{t-1} X_{t-1}] = \psi_0 \sigma^2 = \sigma^2$ .) Substituting  $\gamma_\tau = (\beta_0 + \beta_1\tau)2^{-\tau}$  gives the linear system  $\beta_0 = \frac{32}{3}\sigma^2$ ,  $\beta_1 = 8\sigma^2$ . Therefore

$$\gamma_\tau = \left(\frac{32}{3} + 8\tau\right)2^{-\tau} \sigma^2, \quad \tau \geq 0$$

(and  $\gamma_{-\tau} = \gamma_\tau$ ). In particular  $\gamma_0 = \frac{32}{3}\sigma^2$ ,  $\gamma_1 = \frac{28}{3}\sigma^2$ ,  $\gamma_2 = \frac{20}{3}\sigma^2$ .

*Check (via  $\psi$ -weights):* the recurrence  $\psi_j = \psi_{j-1} - \frac{1}{4}\psi_{j-2}$  has closed form  $\psi_j = (1 + 3j)2^{-j}$ , and  $\sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$  reproduces  $\frac{32}{3}, \frac{28}{3}, \frac{20}{3}, \dots$  exactly.

### Exercise 7: Prove the PACF cut-off for a causal AR( $p$ ) [Sheet 2, Ex. 2.7]

Prove that for a causal AR( $p$ ),  $\alpha_\tau = 0$  for all  $\tau > p$ . Use the prediction equations  $\text{Cov}(Y - P_{X_1, \dots, X_n}(Y), X_k) = 0$  for all  $k$ , and the fact that for causal processes  $\text{Cov}(X_t, \varepsilon_s) = 0$  when  $t < s$ .

#### Solution

Fix  $\tau > p$ . By definition  $\alpha_\tau = \alpha_{\tau, \tau}$ , the last coefficient in the order- $\tau$  best linear predictor

$$P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau, j} X_{\tau-j}.$$

We show the AR coefficients themselves solve the prediction equations. Set  $\tilde{\phi}_j = \phi_j$  for  $1 \leq j \leq p$  and  $\tilde{\phi}_j = 0$  for  $j > p$ , so that  $X_\tau = \sum_{j=1}^{\tau} \tilde{\phi}_j X_{\tau-j} + \varepsilon_\tau$ . Then for every  $k \in \{0, \dots, \tau-1\}$ ,

$$\text{Cov}\left(X_\tau - \sum_{j=1}^{\tau} \tilde{\phi}_j X_{\tau-j}, X_k\right) = \text{Cov}(\varepsilon_\tau, X_k) = 0,$$

where the last equality is causality ( $X_k$  depends only on  $\varepsilon_k, \varepsilon_{k-1}, \dots$ , all with index  $< \tau$ , so uncorrelated with  $\varepsilon_\tau$ ). Hence  $\{\tilde{\phi}_j\}$  satisfies the orthogonality (prediction) equations, and by uniqueness of the best linear predictor  $\alpha_{\tau,j} = \tilde{\phi}_j$ . In particular  $\alpha_\tau = \alpha_{\tau,\tau} = \tilde{\phi}_\tau = 0$  because  $\tau > p$ .  $\square$

### Exercise 8: Causality & invertibility of an ARMA(1,1), and its $\psi$ -weights [New]

Consider  $(1 - 0.5B)X_t = (1 + 0.3B)\varepsilon_t$ . (a) Is it stationary? Invertible? (b) Find the MA( $\infty$ ) weights  $\psi_j$ . (c) Compute  $\gamma_0$  and  $\gamma_1$ .

#### Solution

(a)  $\Phi(z) = 1 - 0.5z$  has root  $z = 2$  ( $|z| > 1$ )  $\Rightarrow$  **stationary**.  $\Theta(z) = 1 + 0.3z$  has root  $z = -1/0.3 \approx -3.33$  ( $|z| > 1$ )  $\Rightarrow$  **invertible**.

(b) Match coefficients in  $(1 - 0.5z) \sum_{j \geq 0} \psi_j z^j = 1 + 0.3z$ :  $\psi_0 = 1$ ;  $\psi_1 - 0.5\psi_0 = 0.3 \Rightarrow \psi_1 = 0.8$ ;  $\psi_j - 0.5\psi_{j-1} = 0 \Rightarrow \psi_j = 0.5\psi_{j-1}$  for  $j \geq 2$ . Closed form (general ARMA(1,1) with  $X_t = \phi X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$ ):  $\psi_j = (\phi + \theta)\phi^{j-1}$  for  $j \geq 1$  with  $\phi = 0.5$ ,  $\theta = 0.3$ , i.e.

$$\psi_0 = 1, \quad \psi_j = 0.8(0.5)^{j-1} = 1.6(0.5)^j, \quad j \geq 1.$$

(c) Using  $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$  with  $\sum_{j \geq 1} \psi_j^2 = 0.8^2 \sum_{j \geq 1} 0.25^{j-1} = 0.64/0.75 = \frac{64}{75}$ ,

$$\gamma_0 = \sigma^2 \left(1 + \frac{64}{75}\right) = \frac{139}{75} \sigma^2 \approx 1.853 \sigma^2.$$

For  $\gamma_1 = \sigma^2(\psi_0\psi_1 + \sum_{j \geq 1} \psi_j \psi_{j+1})$  with  $\sum_{j \geq 1} \psi_j \psi_{j+1} = 0.5 \sum_{j \geq 1} \psi_j^2 = \frac{1}{2} \cdot \frac{64}{75} = \frac{32}{75}$ ,

$$\gamma_1 = \sigma^2 \left(0.8 + \frac{32}{75}\right) = \frac{92}{75} \sigma^2 \approx 1.227 \sigma^2.$$

(Quick check:  $\rho_1 = 92/139 \approx 0.662$ , sensibly large for a positively correlated ARMA(1,1).)

### Exercise 9: Invertibility of an MA(2) and the unit-circle convention [New]

Let  $X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$  with  $(\theta_1, \theta_2) = (0.2, 0.48)$ , i.e.  $X_t = \varepsilon_t - 0.2\varepsilon_{t-1} - 0.48\varepsilon_{t-2}$ . Is it stationary? Invertible?

#### Solution

**Stationarity.** Every finite MA is stationary (Prop. 12); no computation needed.

**Invertibility.** With  $X_t = \Theta(B)\varepsilon_t$  and  $\Theta(z) = 1 - 0.2z - 0.48z^2$ , solve  $0.48z^2 + 0.2z - 1 = 0$ :

$$z = \frac{-0.2 \pm \sqrt{0.04 + 1.92}}{0.96} = \frac{-0.2 \pm 1.4}{0.96} = \left\{ \frac{5}{4}, -\frac{5}{3} \right\}.$$

Both roots satisfy  $|z| > 1$  (1.25 and  $-1.667$ ), so the process is **invertible**. Equivalently,  $\Theta(z) = (1 - 0.8z)(1 + 0.6z)$  has reciprocal roots 0.8 and  $-0.6$ , both of modulus  $< 1$ , the mirror statement of “roots outside”. (Had we instead picked coefficients whose root had  $|z| < 1$ , the process would be non-invertible despite being perfectly stationary.)

### Exercise 10: Spectral density of an ARMA(1,1) [New]

Find the spectral density  $\mathcal{S}(f)$  of the causal, invertible process  $(1 - \phi B)X_t = (1 - \theta B)\varepsilon_t$  with  $|\phi| < 1$ ,  $|\theta| < 1$ , and use it to confirm  $\gamma_0$  for the model  $(1 - 0.5B)X_t = (1 + 0.3B)\varepsilon_t$  of Exercise 8.

### Solution

By the ARMA spectral formula (Key Technique 5),

$$\mathcal{S}(f) = \sigma^2 \frac{|1 - \theta e^{-2\pi i f}|^2}{|1 - \phi e^{-2\pi i f}|^2} = \sigma^2 \frac{1 + \theta^2 - 2\theta \cos(2\pi f)}{1 + \phi^2 - 2\phi \cos(2\pi f)}, \quad -\frac{1}{2} \leq f \leq \frac{1}{2}.$$

Integrating over one period recovers the variance,  $\gamma_0 = \int_{-1/2}^{1/2} \mathcal{S}(f) df$ ; doing the integral (or summing  $\sigma^2 \sum_j \psi_j^2$  with  $\psi_0 = 1$ ,  $\psi_j = (\phi - \theta)\phi^{j-1}$ ) gives the standard ARMA(1,1) variance

$$\gamma_0 = \sigma^2 \frac{1 + \theta^2 - 2\phi\theta}{1 - \phi^2}.$$

The model  $(1 - 0.5B)X_t = (1 + 0.3B)\varepsilon_t$  corresponds to  $\phi = 0.5$  and  $\theta = -0.3$  in the  $(1 - \theta B)$  convention. Then numerator  $= 1 + 0.09 - 2(0.5)(-0.3) = 1.39$ , denominator  $= 1 - 0.25 = 0.75$ , so

$$\gamma_0 = \frac{1.39}{0.75} \sigma^2 = \frac{139}{75} \sigma^2 \approx 1.853 \sigma^2,$$

agreeing with Exercise 8. (Sanity check on the density: since  $\phi > 0$  and the effective MA root is positive,  $\mathcal{S}$  peaks at  $f = 0$ , the hallmark of a positively correlated, low-frequency-dominated process.)

## 4 Parametric Estimation: Yule–Walker and Least Squares

Having modelled  $\{X_t\}$  as an ARMA process, we now *fit* it: given data  $X_1, \dots, X_n$ , estimate the autoregressive coefficients  $\phi_1, \dots, \phi_p$ , the moving-average coefficients  $\theta_1, \dots, \theta_q$  and the innovation variance  $\sigma_\varepsilon^2$ . This part develops the two workhorse estimators of the course. The **Yule–Walker** (method-of-moments) estimator turns the AR recursion into a linear system in the autocovariances and is solved by a single matrix inversion (accelerated by the Levinson–Durbin algorithm). The **least-squares** estimator minimises the sum of squared one-step prediction errors; for pure AR models it is an ordinary linear regression (forward, backward, or forward–backward), and for general ARMA models it becomes a nonlinear minimisation in which the unobserved innovations  $\{\varepsilon_t\}$  are reconstructed recursively. Throughout,  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$  is Gaussian white noise and the AR part is assumed causal (roots of  $\Phi(z)$  outside the unit circle), so that  $\varepsilon_t$  is uncorrelated with the past  $X_{t-k}$ ,  $k > 0$  — the single fact that makes the whole derivation work.

### 4.1 Definitions

#### Definition 1: Yule–Walker estimators (AR( $p$ ))

Let  $\{X_t\}$  be a mean-zero causal AR( $p$ ) process  $X_t = \sum_{k=1}^p \phi_k X_{t-k} + \varepsilon_t$ , observed at  $X_1, \dots, X_n$ . Form the sample ACVS (mean known to be zero)

$$\hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} X_t X_{t+|\tau|},$$

collect  $\hat{\gamma}_p = (\hat{\gamma}_1, \dots, \hat{\gamma}_p)^\top$  and the symmetric Toeplitz matrix

$$\hat{\Gamma}_p = \begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 & \cdots & \hat{\gamma}_{p-1} \\ \hat{\gamma}_1 & \hat{\gamma}_0 & \cdots & \hat{\gamma}_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\gamma}_{p-1} & \hat{\gamma}_{p-2} & \cdots & \hat{\gamma}_0 \end{pmatrix}.$$

The **Yule–Walker estimators** are

$$\hat{\phi}_p = \hat{\Gamma}_p^{-1} \hat{\gamma}_p, \quad \hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \hat{\phi}_j \hat{\gamma}_j.$$

They are the empirical analogue of the population identities  $\gamma = \Gamma\phi$  and  $\gamma_0 = \sum_j \phi_j \gamma_j + \sigma_\varepsilon^2$ .

### Definition 2: Forward least-squares estimators (AR(p))

Write the AR equations for the observable times  $t = p+1, \dots, n$  as the linear model  $\mathbf{X}_F = F\phi + \varepsilon$ , where

$$\mathbf{X}_F = \begin{pmatrix} X_{p+1} \\ \vdots \\ X_n \end{pmatrix}, \quad F = \begin{pmatrix} X_p & X_{p-1} & \cdots & X_1 \\ X_{p+1} & X_p & \cdots & X_2 \\ \vdots & \vdots & & \vdots \\ X_{n-1} & X_{n-2} & \cdots & X_{n-p} \end{pmatrix}, \quad \phi = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix}.$$

The **forward least-squares estimator** minimises

$$SS_F(\phi) = \|\mathbf{X}_F - F\phi\|^2 = \sum_{t=p+1}^n \left( X_t - \sum_{k=1}^p \phi_k X_{t-k} \right)^2 = \sum_{t=p+1}^n \varepsilon_t^2,$$

and equals the ordinary least-squares solution

$$\hat{\phi}_F = (F^\top F)^{-1} F^\top \mathbf{X}_F, \quad \hat{\sigma}_F^2 = \frac{\|\mathbf{X}_F - F\hat{\phi}_F\|^2}{n - 2p}.$$

The first  $p$  observations  $X_1, \dots, X_p$  are discarded as conditioning values.

### Definition 3: Backward least-squares estimators (AR(p))

Exploiting time reversibility of a Gaussian AR process, regress the *first*  $n - p$  values on their *future* lags: for  $t = 1, \dots, n - p$ ,  $X_t = \sum_{j=1}^p \phi_j X_{t+j} + \nu_t$ . In matrix form  $\mathbf{X}_B = B\phi + \nu$  with

$$\mathbf{X}_B = \begin{pmatrix} X_1 \\ \vdots \\ X_{n-p} \end{pmatrix}, \quad B = \begin{pmatrix} X_2 & X_3 & \cdots & X_{p+1} \\ X_3 & X_4 & \cdots & X_{p+2} \\ \vdots & \vdots & & \vdots \\ X_{n-p+1} & X_{n-p+2} & \cdots & X_n \end{pmatrix}.$$

The **backward least-squares estimator** minimises  $SS_B(\phi) = \|\mathbf{X}_B - B\phi\|^2 = \sum_{t=1}^{n-p} (X_t - \sum_{k=1}^p \phi_k X_{t+k})^2$  and is

$$\hat{\phi}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B, \quad \hat{\sigma}_B^2 = \frac{\|\mathbf{X}_B - B\hat{\phi}_B\|^2}{n - 2p}.$$

The same  $\phi$ -parameters appear because  $Y_t = X_{-t}$  is an AR(p) with identical coefficients and an innovation  $\tilde{\nu}_t \stackrel{d}{=} \varepsilon_t$ .

### Definition 4: Forward–backward least-squares estimator

The **forward–backward estimator**  $\hat{\phi}_{FB}$  is the minimiser of the *combined* criterion

$$SS_F(\phi) + SS_B(\phi).$$

Equivalently it is the OLS solution of the stacked system obtained by piling  $\mathbf{X}_F, \mathbf{X}_B$  and  $F, B$  on top of each other. Simulation studies show it typically outperforms both the forward and the backward estimator.

#### Intuition / Why it matters

The three least-squares variants only differ in *which*  $n - p$  one-step residuals they square. Forward LS predicts each point from its past; backward LS predicts it from its future (legitimate because a Gaussian AR run backwards in time is the *same* AR model); forward-backward LS uses both directions at once, roughly doubling the effective data and stabilising the estimate near the stationarity boundary.

#### Definition 5: Least squares for general ARMA via residual reconstruction

For an ARMA( $p, q$ ) model  $\Phi(B)X_t = \Theta(B)\varepsilon_t$  the innovations  $\{\varepsilon_t\}$  are unobserved. The least-squares estimator chooses the parameters minimising  $S(\phi, \theta) = \sum_t \hat{\varepsilon}_t^2$ , where for each candidate parameter value the residuals are **reconstructed recursively** from the model equation after fixing the  $q$  pre-sample innovations (and any pre-sample data) to zero, e.g.  $\varepsilon_0 = \varepsilon_{-1} = \dots = 0$ . The minimisation is nonlinear (no closed form) except in the pure-AR case, where it reduces to the linear forward least squares above.

## 4.2 Theorems & Propositions

#### Proposition 6: Yule–Walker equations

Let  $\{X_t\}$  be a mean-zero causal AR( $p$ ) process with ACVS  $\gamma_\tau$ . Then for every  $k \geq 1$ ,

$$\gamma_k = \sum_{j=1}^p \phi_j \gamma_{k-j},$$

and in particular, taking  $k = 1, \dots, p$  and using  $\gamma_{-\tau} = \gamma_\tau$ ,

$$\gamma = \Gamma \phi, \quad \gamma = (\gamma_1, \dots, \gamma_p)^\top, \quad \Gamma = (\gamma_{i-j})_{i,j=1}^p.$$

Moreover  $\gamma_0 = \sum_{j=1}^p \phi_j \gamma_j + \sigma_\varepsilon^2$ .

*Proof idea:* multiply  $X_t = \sum_j \phi_j X_{t-j} + \varepsilon_t$  by  $X_{t-k}$  and take expectations. Because the process is *causal*,  $X_{t-k}$  depends only on  $\varepsilon_{t-k}, \varepsilon_{t-k-1}, \dots$ , so for  $k > 0$  we have  $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$ ; with mean zero  $\mathbb{E}[X_{t-j} X_{t-k}] = \gamma_{k-j}$ , giving the recursion. For the variance, multiply by  $X_t$  instead: the cross term becomes  $\mathbb{E}[\varepsilon_t X_t] = \mathbb{E}[\varepsilon_t (\sum_j \phi_j X_{t-j} + \varepsilon_t)] = \sigma_\varepsilon^2$  (all  $\mathbb{E}[\varepsilon_t X_{t-j}] = 0$ ,  $j \geq 1$ ).

*Why:* these  $p + 1$  equations express the  $p + 1$  unknowns  $(\phi_1, \dots, \phi_p, \sigma_\varepsilon^2)$  purely in terms of  $\gamma_0, \dots, \gamma_p$ ; replacing the population ACVS by its sample version yields the Yule–Walker estimators. It is the canonical method-of-moments fit for AR models.

#### Intuition / Why it matters

The phrase “causality  $\Rightarrow \mathbb{E}[\varepsilon_t X_{t-k}] = 0$ ” is the heart of the derivation. Causality means  $X_{t-k}$  is a function of innovations *strictly earlier* than  $\varepsilon_t$ ; uncorrelated white noise then kills the cross term, leaving a clean linear system in the autocovariances. Without causality the noise leaks into the past values and the Yule–Walker equations no longer hold.

### Proposition 7: Minimising the prediction MSE gives Yule–Walker

Let  $\{Y_t\}$  be mean-zero stationary with ACVS  $\gamma$ . The coefficients  $\phi_1, \dots, \phi_k$  minimising the one-step mean-square prediction error

$$\mathbb{E}\left[\left(Y_t - \sum_{i=1}^k \phi_i Y_{t-i}\right)^2\right]$$

satisfy the Yule–Walker equations  $\gamma_j = \sum_{i=1}^k \phi_i \gamma_{j-i}$  for  $j = 1, \dots, k$ .

*Proof idea:* since  $\{Y_t\}$  is mean-zero, the objective equals the variance  $\gamma_0 - 2 \sum_i \phi_i \gamma_i + \sum_{i,j} \phi_i \phi_j \gamma_{i-j}$ , a positive quadratic in  $\phi$ . Setting  $\partial/\partial \phi_i = 0$  gives  $-2\gamma_i + 2 \sum_j \phi_j \gamma_{i-j} = 0$ , i.e. the Yule–Walker equations.

*Why:* it shows Yule–Walker is not merely a moment trick but the *best linear predictor*: the AR coefficients are exactly the optimal linear-prediction weights, which is why YW and least squares agree asymptotically for AR models.

### Proposition 8: Forward / backward least squares are the OLS estimators

With the design matrices of Definitions 2–3, the minimisers of  $SS_F$  and  $SS_B$  are

$$\hat{\phi}_F = (F^\top F)^{-1} F^\top \mathbf{X}_F, \quad \hat{\phi}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B,$$

with residual-variance estimators  $\hat{\sigma}^2 = \|\text{residuals}\|^2 / (n - 2p)$ .

*Proof idea:* each is a standard linear-regression sum of squares  $\|\mathbf{y} - A\phi\|^2$ ; the normal equations  $A^\top A \phi = A^\top \mathbf{y}$  give the stated solution, and the mean-square residual divides by the degrees of freedom  $(n - p) - p$ .

*Why:* it places AR estimation inside the familiar OLS framework, so all regression machinery (existence, uniqueness when  $A^\top A$  is invertible,  $\hat{\sigma}^2$ ) applies directly.

### Proposition 9: Time reversibility of a Gaussian AR process

If  $\{X_t\}$  is a stationary Gaussian  $\text{AR}(p)$  process with coefficients  $\phi_1, \dots, \phi_p$  and innovation variance  $\sigma_\varepsilon^2$ , then the time-reversed process  $Y_t = X_{-t}$  is an  $\text{AR}(p)$  process with the *same* coefficients and innovation variance:

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \tilde{\nu}_t, \quad \tilde{\nu}_t \stackrel{d}{=} \varepsilon_t,$$

equivalently  $X_t = \sum_{j=1}^p \phi_j X_{t+j} + \nu_t$  with  $\nu_t = \tilde{\nu}_{-t}$ .

*Proof idea:* a stationary Gaussian process is determined by its ACVS, and the ACVS is symmetric,  $\gamma_{-\tau} = \gamma_\tau$ ; hence the forward and reversed processes have identical second-order structure and (being Gaussian) identical distributions. The full argument is given later via frequency-domain methods (the spectral density is even in  $f$ ).

*Why:* it justifies the backward least-squares regression — predicting the present from the future uses the very same parameters — and underpins the forward–backward estimator.

### Proposition 10: Levinson–Durbin: solving Yule–Walker in $O(p^2)$

The Yule–Walker system  $\hat{\Gamma}_p \hat{\phi}_p = \hat{\gamma}_p$  has a symmetric Toeplitz coefficient matrix. The **Levinson–Durbin algorithm** exploits this structure to solve it recursively in  $k = 1, \dots, p$ , costing  $O(p^2)$  operations rather than the  $O(p^3)$  of a naive matrix inversion. As a by-product it produces the partial autocorrelations  $\hat{\alpha}_k = \hat{\phi}_{k,k}$  and the prediction-error variances at every order.

*Proof idea:* the order- $k$  solution is obtained from the order- $(k-1)$  solution by one rank-one update governed by the reflection coefficient  $\hat{\phi}_{k,k} = (\hat{\gamma}_k - \sum_{j=1}^{k-1} \hat{\phi}_{k-1,j} \hat{\gamma}_{k-j}) / v_{k-1}$ , where  $v_{k-1}$

is the running error variance; the remaining coefficients update as  $\hat{\phi}_{k,j} = \hat{\phi}_{k-1,j} - \hat{\phi}_{k,k}\hat{\phi}_{k-1,k-j}$ . *Why:* for high-order AR fits, spectral estimation, or repeatedly solving nested systems, the quadratic cost (and the free PACF) makes Levinson–Durbin the standard solver.

### Proposition 11: Least-squares fits need not be stationary

The forward, backward and forward–backward least-squares estimators  $\hat{\phi}_F, \hat{\phi}_B, \hat{\phi}_{FB}$  are *not* constrained to yield a stationary (causal) model: the implied polynomial  $\hat{\Phi}(z)$  may have roots inside the unit circle.

*Proof idea:* the OLS objective places no constraint on the root moduli; it only minimises the residual sum of squares, which can be smaller for a non-causal parameter vector.

*Why:* non-stationary estimates are a problem for *prediction*, but acceptable for some uses such as *spectral estimation*, where they still produce a valid spectral density estimate. The Yule–Walker estimator, by contrast, always yields a stationary fit.

### Yule–Walker vs. least squares

For large  $n$  the two approaches give very similar AR estimates: the least-squares score equations reduce, to leading order, to  $\hat{\mu} \approx \bar{Y}$  and  $\hat{\phi} \approx \hat{\rho}_1$ , which is precisely what method-of-moments delivers. The practical differences are at the margins: Yule–Walker is a single linear solve and always stationary; least squares extends cleanly to ARMA and to including a mean, but can stray outside the stationary region.

## 4.3 Key Techniques

### Key Technique 1: Yule–Walker from a sample ACVS / ACF

To fit  $\text{AR}(p)$  by Yule–Walker:

1. **Estimate the ACVS**  $\hat{\gamma}_0, \dots, \hat{\gamma}_p$  (divide-by- $n$  estimator on the mean-corrected series).
2. **Build** the  $p \times p$  Toeplitz matrix  $\hat{\Gamma}_p = (\hat{\gamma}_{i-j})$  and the vector  $\hat{\gamma}_p = (\hat{\gamma}_1, \dots, \hat{\gamma}_p)^\top$ .
3. **Solve**  $\hat{\phi}_p = \hat{\Gamma}_p^{-1} \hat{\gamma}_p$  (by hand for  $p \leq 3$ ; otherwise Levinson–Durbin).
4. **Variance:**  $\hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \hat{\phi}_j \hat{\gamma}_j$ .

**ACF shortcut:** if only the autocorrelations  $\hat{\rho}_\tau$  are given, divide every entry by  $\hat{\gamma}_0$ . Writing  $V = \hat{\gamma}_0^{-1} I$ , the system  $\hat{\gamma}_p = \hat{\Gamma}_p \hat{\phi}_p$  becomes  $V \hat{\gamma}_p = V \hat{\Gamma}_p \hat{\phi}_p$ , i.e.

$$\hat{\phi}_p = \bar{\Gamma}_p^{-1} \hat{\rho}_p, \quad \bar{\Gamma}_p = (\hat{\rho}_{i-j}), \quad \hat{\rho}_p = (\hat{\rho}_1, \dots, \hat{\rho}_p)^\top,$$

so the *same* solve works with the correlation matrix and the correlation vector —  $\hat{\gamma}_0$  cancels. (It is needed only for  $\hat{\sigma}_\varepsilon^2$ , via  $\hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0(1 - \sum_j \hat{\phi}_j \hat{\rho}_j)$ .)

### Key Technique 2: Recursive innovation reconstruction for MA / ARMA least squares

To evaluate  $S(\cdot) = \sum_t \hat{\varepsilon}_t^2$  for an MA or ARMA model at a candidate parameter value, rebuild the unobserved innovations one step at a time:

1. **Solve the model equation for  $\varepsilon_t$ .** MA(1)  $X_t = \varepsilon_t - \theta \varepsilon_{t-1} \Rightarrow \varepsilon_t = X_t + \theta \varepsilon_{t-1}$ ; ARMA(1,1)  $X_t = \phi X_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1} \Rightarrow \varepsilon_t = X_t - \phi X_{t-1} + \theta \varepsilon_{t-1}$ .
2. **Initialise the  $q$  pre-sample innovations (and pre-sample data) to zero:**  $\varepsilon_0 = 0$  (and  $X_0 = 0$  for the ARMA case).
3. **Recurse forward**  $t = 1, 2, \dots, n$  to obtain  $\hat{\varepsilon}_1, \hat{\varepsilon}_2, \dots, \hat{\varepsilon}_n$ , each a function of the unknown parameters.

4. **Form  $S$  and minimise** over the parameters (closed form only if  $S$  is quadratic, as in the toy MA(1) of Exercise 3; otherwise numerically).

**Why it is needed:** an MA/ARMA model is linear in the data but the innovations are latent, so least squares is no longer a one-shot regression — the recursion supplies the residuals before squaring them. The method works for any ARMA( $p, q$ ); one always assumes the  $q$  initial innovations vanish.

### Key Technique 3: Forward vs. backward design matrices (AR(2) template)

For AR(2) ( $p = 2$ ) the two regressions are

$$\underbrace{\begin{pmatrix} X_3 \\ X_4 \\ \vdots \\ X_n \end{pmatrix}}_{\mathbf{X}_F} = \underbrace{\begin{pmatrix} X_2 & X_1 \\ X_3 & X_2 \\ \vdots & \vdots \\ X_{n-1} & X_{n-2} \end{pmatrix}}_F \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} + \varepsilon, \quad \underbrace{\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_{n-2} \end{pmatrix}}_{\mathbf{X}_B} = \underbrace{\begin{pmatrix} X_2 & X_3 \\ X_3 & X_4 \\ \vdots & \vdots \\ X_{n-1} & X_n \end{pmatrix}}_B \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} + \nu.$$

Forward regresses the last  $n - p$  points on their past lags; backward regresses the first  $n - p$  points on their future lags. Then  $\hat{\phi} = (A^\top A)^{-1} A^\top \mathbf{X}$  with  $A = F$  or  $A = B$ . Remove the sample mean first if it is significantly non-zero.

## 4.4 Exercises

### Exercise 1: Yule–Walker on the sunspot series [Sheet 4]

For the Wolfer sunspot numbers the sample autocovariances are  $\hat{\gamma}_0 = 1382.185$ ,  $\hat{\gamma}_1 = 1114.378$ ,  $\hat{\gamma}_2 = 591.721$ ,  $\hat{\gamma}_3 = 96.215$ . Fit the AR(2) model  $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$  ( $Y_t$  mean-corrected) by Yule–Walker, estimating  $\phi_1, \phi_2$  and  $\sigma_\varepsilon^2$ .

### Solution

With  $p = 2$  the Yule–Walker system  $\hat{\Gamma}_2 \hat{\phi} = \hat{\gamma}_2$  is

$$\begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 \\ \hat{\gamma}_1 & \hat{\gamma}_0 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} \implies \begin{pmatrix} 1382.185 & 1114.378 \\ 1114.378 & 1382.185 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} 1114.378 \\ 591.721 \end{pmatrix}.$$

The inverse of a  $2 \times 2$  symmetric matrix  $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$  is  $\frac{1}{a^2 - b^2} \begin{pmatrix} a & -b \\ -b & a \end{pmatrix}$ , with here  $a = 1382.185$ ,  $b = 1114.378$  and determinant  $a^2 - b^2 = 1.9104 \times 10^6 - 1.2418 \times 10^6 = 6.686 \times 10^5$ . Hence

$$\hat{\phi} = \frac{1}{6.686 \times 10^5} \begin{pmatrix} 1382.185 & -1114.378 \\ -1114.378 & 1382.185 \end{pmatrix} \begin{pmatrix} 1114.378 \\ 591.721 \end{pmatrix} \approx \begin{pmatrix} 1.3175 \\ -0.6341 \end{pmatrix}.$$

The innovation variance is

$$\hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \hat{\phi}_1 \hat{\gamma}_1 - \hat{\phi}_2 \hat{\gamma}_2 = 1382.185 - 1.3175(1114.378) - (-0.6341)(591.721) \approx 289.21.$$

So  $\hat{\phi}_1 \approx 1.3175$ ,  $\hat{\phi}_2 \approx -0.6341$ ,  $\hat{\sigma}_\varepsilon^2 \approx 289.21$ . (One checks the fit is causal: the roots of  $1 - 1.3175z + 0.6341z^2$  are complex with modulus  $1/\sqrt{0.6341} \approx 1.256 > 1$ .)

### Exercise 2: Deriving Yule–Walker by minimising the MSE [Sheet 4]

Let  $\{Y_t\}$  be mean-zero stationary with ACVS  $\gamma$ . Show that the  $\phi_1, \dots, \phi_k$  minimising  $\mathbb{E}[(Y_t - \sum_{i=1}^k \phi_i Y_{t-i})^2]$  satisfy  $\gamma_j = \sum_{i=1}^k \phi_i \gamma_{j-i}$ ,  $j = 1, \dots, k$ .

### Solution

Write  $W = Y_t - \sum_{i=1}^k \phi_i Y_{t-i}$ . Decompose  $\mathbb{E}[W^2] = \text{Var}(W) + (\mathbb{E}[W])^2$ . Because  $\{Y_t\}$  is stationary,  $\mathbb{E}[W] = (1 - \sum_i \phi_i) \mathbb{E}[Y_t]$ , and since the process is mean-zero this term vanishes, so  $\mathbb{E}[W^2] = \text{Var}(W)$ . Expanding the variance by bilinearity,

$$\begin{aligned} \mathbb{E}[W^2] &= \text{Cov} \left( Y_t - \sum_{i=1}^k \phi_i Y_{t-i}, Y_t - \sum_{j=1}^k \phi_j Y_{t-j} \right) \\ &= \text{Cov}(Y_t, Y_t) - 2 \sum_{i=1}^k \phi_i \text{Cov}(Y_t, Y_{t-i}) + \sum_{i=1}^k \sum_{j=1}^k \phi_i \phi_j \text{Cov}(Y_{t-i}, Y_{t-j}) \\ &= \gamma_0 - 2 \sum_{i=1}^k \phi_i \gamma_i + \sum_{i=1}^k \sum_{j=1}^k \phi_i \phi_j \gamma_{i-j}. \end{aligned}$$

This is a positive (semi-)definite quadratic in  $\phi$ , so its unique stationary point is the global minimum. Differentiating with respect to the  $i$ -th coefficient (using  $\gamma_{i-j} = \gamma_{j-i}$  to combine the two double-sum terms) and equating to zero,

$$\frac{\partial \mathbb{E}[W^2]}{\partial \phi_i} = -2\gamma_i + 2 \sum_{j=1}^k \phi_j \gamma_{i-j} = 0 \quad \implies \quad \gamma_i = \sum_{j=1}^k \phi_j \gamma_{i-j}, \quad i = 1, \dots, k,$$

which are exactly the Yule–Walker equations.

### Exercise 3: Estimating $\theta$ of an MA(1) by recursive least squares [Sheet 4]

A mean-zero MA(1)  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$  is observed at  $X_1 = 0$ ,  $X_2 = 1$ ,  $X_3 = 0.5$ . Estimate  $\theta$  by least squares.

### Solution

The innovations are latent, so reconstruct them recursively. Solving the model equation for  $\varepsilon_t$  gives  $\varepsilon_t = X_t + \theta \varepsilon_{t-1}$ . Take  $t = 0$  as the start and set  $\varepsilon_0 = 0$ :

$$\varepsilon_1 = X_1 + \theta \varepsilon_0 = 0, \quad \varepsilon_2 = X_2 + \theta \varepsilon_1 = 1, \quad \varepsilon_3 = X_3 + \theta \varepsilon_2 = 0.5 + \theta.$$

The least-squares criterion is

$$S(\theta) = \sum_{t=1}^3 \varepsilon_t^2 = 0^2 + 1^2 + (0.5 + \theta)^2 = 1 + (0.5 + \theta)^2.$$

This is minimised when  $(0.5 + \theta)^2$  is smallest, i.e. at  $\theta = -0.5$ . Hence  $\hat{\theta} = -0.5$ . (Note  $|\hat{\theta}| < 1$ , so the fit is invertible.)

### Exercise 4: AR(3) Yule–Walker from a given ACF [Sheet 4]

A series of length 100 has sample ACF  $\hat{\rho}_1 = 0.8$ ,  $\hat{\rho}_2 = 0.5$ ,  $\hat{\rho}_3 = 0.4$ . Assuming a zero-mean AR(3) model, estimate  $\phi_1, \phi_2, \phi_3$ .

### Solution

Use the ACF shortcut: dividing the Yule–Walker system by  $\gamma_0$  replaces covariances by correlations,  $\hat{\phi} = \bar{\Gamma}_3^{-1} \hat{\rho}_3$ , where  $\bar{\Gamma}_3 = (\hat{\rho}_{i-j})$  uses  $\hat{\rho}_0 = 1$ :

$$\begin{pmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.8 \\ 0.5 & 0.8 & 1 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \\ \hat{\phi}_3 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.5 \\ 0.4 \end{pmatrix}.$$

Solving this  $3 \times 3$  system (e.g. by Gaussian elimination) gives

$$\hat{\phi} = (1.30909, -0.95455, 0.50909)^\top.$$

*Check (row 1):*  $1(1.30909) + 0.8(-0.95455) + 0.5(0.50909) = 1.30909 - 0.76364 + 0.25455 = 0.8$ .  
*✓ Row 3:*  $0.5(1.30909) + 0.8(-0.95455) + 1(0.50909) = 0.65455 - 0.76364 + 0.50909 = 0.4$ . *✓*  
 So  $\hat{\phi}_1 \approx 1.309$ ,  $\hat{\phi}_2 \approx -0.955$ ,  $\hat{\phi}_3 \approx 0.509$ . (The series length  $n = 100$  does not enter; only the ACF values determine the point estimates.)

### Exercise 5: Forward and backward least squares (conceptual) [Sheet 4]

Describe how to compute the forward and backward least-squares AR(2) fits to the sunspot series, and interpret the reported output  $\hat{\phi}_F \approx (1.4046, -0.7113)^\top$ ,  $\hat{\sigma}_F^2 \approx 232.26$ ,  $\hat{\phi}_B \approx (1.4061, -0.7076)^\top$ ,  $\hat{\sigma}_B^2 \approx 231.03$ .

### Solution

**Procedure.** Centre the series,  $x_t \mapsto x_t - \bar{x}$  (the sunspot mean is clearly non-zero). For the forward fit build  $\mathbf{X}_F = (x_3, \dots, x_n)^\top$  and the  $(n-2) \times 2$  matrix  $F$  with rows  $(x_{t-1}, x_{t-2})$ , then  $\hat{\phi}_F = (F^\top F)^{-1} F^\top \mathbf{X}_F$  and  $\hat{\sigma}_F^2 = \|\mathbf{X}_F - F \hat{\phi}_F\|^2 / (n-2p)$  with  $p = 2$ . For the backward fit build  $\mathbf{X}_B = (x_1, \dots, x_{n-2})^\top$  and  $B$  with rows  $(x_{t+1}, x_{t+2})$ , then  $\hat{\phi}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B$  with the analogous  $\hat{\sigma}_B^2$ . In R these are two short functions building the lagged design columns and calling `solve(t(A)%*%A)%*%t(A)%*%X`.

**Interpretation.** The forward and backward estimates are almost identical ( $\hat{\phi}_F \approx \hat{\phi}_B$ ), a direct illustration of *time reversibility*: a Gaussian AR run backwards has the same coefficients, so both regressions estimate the same  $\phi$ . The small discrepancy is sampling noise from using different  $n-p$  residuals. Both  $\hat{\phi}_1 \approx 1.40$ ,  $\hat{\phi}_2 \approx -0.71$  are close to (but not equal to) the Yule–Walker values  $(1.3175, -0.6341)$  of Exercise 1; the two methods agree only asymptotically, and least squares does not constrain the fit to be stationary. A forward–backward fit would minimise  $SS_F + SS_B$  jointly, typically landing between the two and with slightly lower variance.

### Exercise 6: AR(1) Yule–Walker, by hand [New]

For a mean-zero AR(1)  $X_t = \phi X_{t-1} + \varepsilon_t$ , a series of length  $n = 200$  gives  $\hat{\gamma}_0 = 4.0$  and  $\hat{\gamma}_1 = 2.4$ . Find the Yule–Walker estimates  $\hat{\phi}$  and  $\hat{\sigma}_\varepsilon^2$ , and state the implied stationarity check.

### Solution

For  $p = 1$  the Yule–Walker system collapses to a scalar equation  $\hat{\gamma}_1 = \hat{\phi} \hat{\gamma}_0$ , so

$$\hat{\phi} = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{2.4}{4.0} = 0.6 = \hat{\rho}_1.$$

(For AR(1) the YW estimate is just the lag-1 sample autocorrelation.) The variance estimator is

$$\hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \hat{\phi} \hat{\gamma}_1 = 4.0 - 0.6 \times 2.4 = 4.0 - 1.44 = 2.56.$$

This is consistent with the population identity  $\gamma_0 = \sigma_\varepsilon^2 / (1 - \phi^2)$ : indeed  $\hat{\gamma}_0(1 - \hat{\phi}^2) = 4.0(1 - 0.36) = 2.56 = \hat{\sigma}_\varepsilon^2$ . ✓ **Stationarity:**  $\Phi(z) = 1 - 0.6z$  has root  $z = 1/0.6 \approx 1.667$ , of modulus  $> 1$ , so the fit is causal/stationary.

### Exercise 7: ARMA(1,1) residual recursion [New]

For a mean-zero ARMA(1,1)  $X_t = \phi X_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$ , write out the recursion that reconstructs  $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_4$  as functions of  $(\phi, \theta)$  from data  $X_1, X_2, X_3, X_4$ , stating the initialisation, and exhibit the least-squares objective.

### Solution

Solve the model equation for the current innovation:  $\varepsilon_t = X_t - \phi X_{t-1} + \theta \varepsilon_{t-1}$ . Initialise the pre-sample data and innovation to zero,  $X_0 = 0$  and  $\varepsilon_0 = 0$ . Then

$$\begin{aligned}\hat{\varepsilon}_1 &= X_1 - \phi X_0 + \theta \varepsilon_0 = X_1, \\ \hat{\varepsilon}_2 &= X_2 - \phi X_1 + \theta \hat{\varepsilon}_1 = X_2 - \phi X_1 + \theta X_1, \\ \hat{\varepsilon}_3 &= X_3 - \phi X_2 + \theta \hat{\varepsilon}_2 = X_3 - \phi X_2 + \theta(X_2 - \phi X_1 + \theta X_1), \\ \hat{\varepsilon}_4 &= X_4 - \phi X_3 + \theta \hat{\varepsilon}_3.\end{aligned}$$

Each  $\hat{\varepsilon}_t$  is an explicit (nonlinear in  $\theta$ ) function of the parameters. The least-squares estimator minimises

$$S(\phi, \theta) = \sum_{t=1}^4 \hat{\varepsilon}_t^2 = \sum_{t=1}^4 (X_t - \phi X_{t-1} + \theta \hat{\varepsilon}_{t-1})^2,$$

which, because of the  $\theta \hat{\varepsilon}_{t-1}$  feedback, is *not* quadratic in  $\theta$  and must be minimised numerically. This is the general ARMA( $p, q$ ) recipe with  $q = 1$  pre-sample innovation set to zero.

### Exercise 8: Why an MA(1) least-squares fit can be quadratic — or not [New]

For the MA(1)  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$  with  $\varepsilon_0 = 0$  and data  $X_1, \dots, X_n$ , show that the reconstructed residual  $\hat{\varepsilon}_t$  is a polynomial of degree  $t - 1$  in  $\theta$ , and explain why the toy three-point problem of Exercise 3 nevertheless had a clean quadratic minimisation.

### Solution

The recursion  $\hat{\varepsilon}_t = X_t + \theta \hat{\varepsilon}_{t-1}$  with  $\hat{\varepsilon}_0 = 0$  unrolls to the geometric-type sum

$$\hat{\varepsilon}_t = \sum_{k=0}^{t-1} \theta^k X_{t-k} = X_t + \theta X_{t-1} + \theta^2 X_{t-2} + \dots + \theta^{t-1} X_1,$$

a polynomial in  $\theta$  of degree  $t - 1$ . Hence  $S(\theta) = \sum_{t=1}^n \hat{\varepsilon}_t^2$  is a polynomial of degree  $2(n - 1)$  in  $\theta$ , generally with several stationary points — so in general the MA(1) least-squares fit requires numerical optimisation, not a single derivative.

In Exercise 3 the data were special:  $X_1 = 0$  forces  $\hat{\varepsilon}_1 = 0$  and  $\hat{\varepsilon}_2 = X_2 = 1$  (free of  $\theta$ ), so only the last residual  $\hat{\varepsilon}_3 = X_3 + \theta \hat{\varepsilon}_2 = 0.5 + \theta$  carries any  $\theta$ -dependence, and it is *linear*. Therefore  $S(\theta) = 1 + (0.5 + \theta)^2$  is exactly quadratic with the unique minimiser  $\hat{\theta} = -0.5$ . The clean answer was an artefact of  $X_1 = 0$ ; with generic data the degree- $2(n - 1)$  surface would need a numerical solver.

### Classic Pitfall

Three traps in this part. (1) **Indexing of the Toeplitz system:**  $\Gamma_p$  uses  $\gamma_{i-j}$  with  $\gamma_0$  on the diagonal, and the *right-hand side* is  $(\gamma_1, \dots, \gamma_p)$  — do not put  $\gamma_0$  in the vector. (2) **Degrees**

**of freedom in  $\hat{\sigma}^2$ :** the least-squares variance divides by  $n - 2p$  (there are  $n - p$  residuals and  $p$  fitted coefficients), not by  $n$ . (3) **Forgetting to initialise the recursion:** in MA/ARMA least squares you must set the  $q$  pre-sample innovations (and any pre-sample data) to zero before recursing, otherwise  $\hat{\varepsilon}_1$  is undefined.

## 5 Trends, Seasonality and Differencing: SARIMA

Real signals rarely look stationary: they drift (a *trend*) and they repeat (a *season*). The working model of this chapter is  $X_t = \mu_t + Y_t$  with  $\mu_t$  a deterministic time-dependent mean and  $\{Y_t\}$  stationary. There are two ways to peel off  $\mu_t$ : *estimate and subtract* (regression on  $t$  and on seasonal dummies), or *difference it away* with the operator  $\nabla = I - B$ . Differencing of order  $d$  annihilates a polynomial trend of degree  $d - 1$ ; a *seasonal* difference  $\nabla_{(s)} = I - B^s$  kills a period- $s$  season. Layering autoregressive and moving-average structure onto *both* the ordinary lags and the seasonal lags gives the SARMA family, and folding the differencing in front gives the (S)ARIMA model  $\Phi(B)(1 - B)^d X_t = \Theta(B)\varepsilon_t$  — the standard Box–Jenkins workhorse. Throughout,  $\{\varepsilon_t\}$  is mean-zero white noise of variance  $\sigma_\varepsilon^2$ , AR/seasonal-AR coefficients are  $\phi_j, \phi_j^{(s)}$  and MA/seasonal-MA coefficients are  $\theta_k, \theta_k^{(s)}$ .

### 5.1 Definitions

#### Definition 1: Mean-trend (signal-plus-noise) model

A series with a slowly varying level is modelled as

$$X_t = \mu_t + Y_t,$$

where  $\mu_t$  is a deterministic, time-dependent mean (the **trend**) and  $\{Y_t\}$  is a zero-mean second-order stationary process. The canonical linear case is  $\mu_t = a + bt$ ; more generally  $\mu_t$  may be a degree- $(q - 1)$  polynomial in  $t$ . When the level instead repeats,  $\mu_t = s_t$  with  $s_{t+s} = s_t$  is a **periodic** (seasonal) trend of period  $s$ ; the two can coexist,  $X_t = a + bt + s_t + Y_t$ .

#### Definition 2: Difference operator $\nabla$

The (first-)difference operator is

$$\nabla = I - B, \quad \nabla X_t = X_t - X_{t-1},$$

where  $B$  is the backshift operator  $BX_t = X_{t-1}$ . It is *linear*. Its  $d$ -th power is, by the binomial theorem,

$$\nabla^d = (I - B)^d = \sum_{k=0}^d \binom{d}{k} (-1)^k B^k, \quad \nabla^d X_t = \sum_{k=0}^d \binom{d}{k} (-1)^k X_{t-k}.$$

For example  $\nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}$ .

#### Definition 3: Seasonal difference operator $\nabla_{(s)}$

The **seasonal difference** of period  $s$  is

$$\nabla_{(s)} = I - B^s, \quad \nabla_{(s)} X_t = X_t - X_{t-s}.$$

It removes a deterministic component of period  $s$  (since  $\nabla_{(s)} s_t = s_t - s_{t-s} = 0$ ) and, applied to a linear trend, leaves a constant:  $\nabla_{(s)}(bt) = bs$ . The ordinary difference is the case  $s = 1$ ,

$\nabla = \nabla_{(1)}$ . To kill a linear-plus-seasonal trend one composes them,  $\nabla\nabla_{(s)} = (I - B)(I - B^s)$ .

**Definition 4: Seasonal moving average  $\text{SMA}(Q)_s$**

A **seasonal moving average** of order  $Q$  at period  $s$  is

$$X_t = \varepsilon_t - \sum_{j=1}^Q \theta_j^{(s)} \varepsilon_{t-js} = \Theta^{(s)}(B) \varepsilon_t, \quad \Theta^{(s)}(z) = 1 - \sum_{j=1}^Q \theta_j^{(s)} z^{sj}.$$

Only lags that are *multiples of  $s$*  appear. The basic seasonal MA(1) is  $X_t = \varepsilon_t - \theta \varepsilon_{t-s}$ .

**Definition 5: Seasonal autoregression  $\text{SAR}(P)_s$**

A **seasonal autoregression** of order  $P$  at period  $s$  is

$$X_t = \varepsilon_t + \sum_{j=1}^P \phi_j^{(s)} X_{t-js}, \quad \Phi^{(s)}(z) = 1 - \sum_{j=1}^P \phi_j^{(s)} z^{sj}, \quad \Phi^{(s)}(B) X_t = \varepsilon_t.$$

Again only the seasonal lags  $s, 2s, \dots, Ps$  enter.

**Definition 6:  $\text{SARMA}(p, q) \times (P, Q)_s$**

Combining an ordinary  $\text{ARMA}(p, q)$  with a seasonal  $\text{ARMA}(P, Q)_s$  gives the **multiplicative seasonal ARMA**

$$\Phi(B) \Phi^{(s)}(B) X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t,$$

with the four polynomials  $\Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j$ ,  $\Theta(z) = 1 - \sum_{k=1}^q \theta_k z^k$ ,  $\Phi^{(s)}(z) = 1 - \sum_{j=1}^P \phi_j^{(s)} z^{sj}$ ,  $\Theta^{(s)}(z) = 1 - \sum_{k=1}^Q \theta_k^{(s)} z^{sk}$ . Special cases:  $\text{SMA}(Q)_s$  is  $\text{SARMA}(0, 0) \times (0, Q)_s$  and  $\text{SAR}(P)_s$  is  $\text{SARMA}(0, 0) \times (P, 0)_s$ . Example, a  $\text{SARMA}(0, 1) \times (0, 1)_{12}$ :

$$X_t = (1 - \theta B)(1 - \theta^{(12)} B^{12}) \varepsilon_t.$$

**Definition 7:  $\text{ARIMA}(p, d, q)$**

$\{X_t\}$  is an **ARIMA**( $p, d, q$ ) (“integrated ARMA”) process if its  $d$ -th difference is a (causal, invertible)  $\text{ARMA}(p, q)$ :

$$\Phi(B) (1 - B)^d X_t = \Theta(B) \varepsilon_t.$$

Equivalently, writing  $Z_t = \nabla^d X_t$ ,

$$Z_t = \sum_{j=1}^p \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}.$$

The “I” stands for *integrated*:  $X_t$  is recovered from  $Z_t$  by summing (the inverse of differencing)  $d$  times. ARMA is the case  $d = 0$ .

**Definition 8:  $\text{SARIMA}(p, d, q) \times (P, D, Q)_s$**

The full **seasonal ARIMA** folds both an ordinary difference of order  $d$  and a seasonal difference of order  $D$  in front of a SARMA:

$$\Phi(B) \Phi^{(s)}(B) (1 - B)^d (1 - B^s)^D X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t.$$

Here  $d$  removes a polynomial trend,  $D$  removes a (possibly stochastic) seasonal pattern, and the four polynomials capture the remaining short-range and seasonal dependence.

### Definition 9: Random walk and the IMA(1,1)/ARI(1,1) models

The **random walk** is the AR(1) at the boundary  $\phi = 1$ ,

$$X_t = X_{t-1} + \varepsilon_t,$$

which is *not* stationary, but whose first difference  $\nabla X_t = \varepsilon_t$  is white noise: it is an ARIMA(0, 1, 0). Two minimal one-step-integrated models:

- **IMA**(1, 1) = ARIMA(0, 1, 1):  $Z_t = \nabla Y_t = \varepsilon_t - \theta \varepsilon_{t-1}$ , i.e.  $Y_t = Y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$ ;
- **ARI**(1, 1) = ARIMA(1, 1, 0):  $Z_t = \nabla Y_t = \phi Z_{t-1} + \varepsilon_t$  with  $|\phi| < 1$ , i.e.  $Y_t - Y_{t-1} = \phi(Y_{t-1} - Y_{t-2}) + \varepsilon_t$ .

Neither  $\{Y_t\}$  is stationary.

### Intuition / Why it matters

“Regress-and-subtract” versus “difference” is the central practical choice. Regression on  $a + bt$  (or on  $\sin / \cos$  harmonics) gives an interpretable, deterministic trend but assumes you know its functional form. Differencing is model-free and robust —  $\nabla$  knocks out any linear drift,  $\nabla^q$  any degree- $(q - 1)$  polynomial,  $\nabla_{(s)}$  any period- $s$  season — at the price of slightly distorting the noise ( $\nabla Y_t$  replaces  $Y_t$ ) and risking *over*-differencing (injecting a spurious unit root into the MA part).

## 5.2 Theorems & Propositions

### Proposition 10: $\nabla$ removes a linear trend; $\nabla Y$ stays stationary

Let  $X_t = a + bt + Y_t$  with  $\{Y_t\}$  stationary. Then

$$\nabla X_t = X_t - X_{t-1} = b + \nabla Y_t,$$

so the trend  $a + bt$  is reduced to the constant  $b$ , and the random part  $\nabla Y_t = Y_t - Y_{t-1}$  is again stationary. Differencing once more removes the constant too:  $\nabla^2 X_t = \nabla^2 Y_t = Y_t - 2Y_{t-1} + Y_{t-2}$ .

*Proof idea:* substitute  $a + bt + Y_t$  and cancel:  $a + bt - (a + b(t - 1)) = b$ . Linearity of  $\nabla$  gives  $\nabla Y_t$ ; a finite linear combination of a stationary process is stationary, hence so is  $\nabla Y_t$ .

*Why:* the formal justification for first-differencing a series that “looks linear”; it converts a non-stationary level into a stationary increment, up to a known constant that the mean estimate then absorbs.

### Proposition 11: $\nabla^q$ annihilates a degree- $(q - 1)$ polynomial trend

Let  $X_t = \mu_t + Y_t$  with  $\mu_t$  a polynomial of degree  $q - 1$  in  $t$ . Then  $\nabla^q \mu_t \equiv 0$ , so

$$\nabla^q X_t = \nabla^q Y_t = \sum_{k=0}^q \binom{q}{k} (-1)^k Y_{t-k}.$$

*Proof idea:*  $\nabla$  lowers the degree of a polynomial in  $t$  by exactly one (the leading  $t^m$  becomes  $t^m - (t - 1)^m = m t^{m-1} + \dots$ ); applying  $\nabla$  a total of  $q$  times to a degree- $(q - 1)$  polynomial drives it to the zero sequence. Linearity then leaves  $\nabla^q Y_t$ , expanded by the binomial theorem. (See Exercise 1 for the careful induction that  $\nabla^k t^{k-1} = 0$ .)

*Why:* tells you exactly how many differences  $d = q$  are needed to flatten a polynomial trend of a given degree, which is how one chooses  $d$  in ARIMA( $p, d, q$ ).

### Theorem 12: Combined $\nabla\nabla_{(12)}$ yields a stationary process

Let  $X_t = a + bt + s_t + Y_t$ , where  $\{Y_t\}$  is zero-mean stationary with ACVS  $\gamma_\tau$ ,  $a, b$  are constants and  $s_{t+12} = s_t$  has period 12. Then

$$\nabla\nabla_{(12)}X_t = \nabla\nabla_{(12)}Y_t = Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$$

is weakly stationary, with mean 0 and

$$\begin{aligned} \text{Cov}(\nabla\nabla_{(12)}X_t, \nabla\nabla_{(12)}X_{t-\tau}) &= \gamma_\tau - \gamma_{\tau+1} - \gamma_{\tau+12} + \gamma_{\tau+13} \\ &\quad - \gamma_{\tau-1} + \gamma_\tau + \gamma_{\tau+11} - \gamma_{\tau+12} \\ &\quad - \gamma_{\tau-12} + \gamma_{\tau-11} + \gamma_\tau - \gamma_{\tau+1} \\ &\quad + \gamma_{\tau-13} - \gamma_{\tau-12} - \gamma_{\tau-1} + \gamma_\tau. \end{aligned}$$

*Proof idea:* first  $\nabla_{(12)}X_t = 12b + \nabla_{(12)}Y_t$  kills  $s_t$  and turns  $bt$  into the constant  $12b$ ; then  $\nabla$  kills the constant, leaving the fixed linear combination  $Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$ . Its mean is 0 and its covariance, expanded by bilinearity, is a function of  $\tau$  *only* (the four-by-four table above). *Why:* the theoretical backing for the standard “ $\nabla_{12}$  then  $\nabla$ ” pre-processing of monthly data (e.g. the Mauna Loa CO<sub>2</sub> series); it shows no  $t$ -dependence survives.

### Proposition 13: Multiplicative-trend variant: $\nabla_{(12)}^2$

If instead  $X_t = (a + bt)s_t + Y_t$  with  $s_{t+12} = s_t$ , then

$$\nabla_{(12)}X_t = 12b s_{t-12} + Y_t - Y_{t-12}, \quad \nabla_{(12)}^2X_t = Y_t - 2Y_{t-12} + Y_{t-24},$$

which has zero mean and a covariance depending only on  $|\tau|$ , hence is stationary.

*Proof idea:*  $\nabla_{(12)}$  once turns  $(a + bt)s_t$  into  $12b s_{t-12}$  (using  $s_t = s_{t-12}$ ); a second  $\nabla_{(12)}$  removes that periodic term as well, leaving a fixed combination of  $Y$ 's.

*Why:* warns that when trend and season *multiply* (amplitude grows with level), a *single seasonal* double-difference — not  $\nabla\nabla_{(12)}$  — is what flattens the series.

### Proposition 14: ACVS of the seasonal MA(1)

For  $X_t = \varepsilon_t - \theta\varepsilon_{t-s}$  with white-noise variance  $\sigma_\varepsilon^2$ ,

$$\gamma_\tau = \sigma_\varepsilon^2 \left[ (1 + \theta^2)\delta_\tau - \theta(\delta_{\tau-s} + \delta_{\tau+s}) \right] = \begin{cases} (1 + \theta^2)\sigma_\varepsilon^2, & \tau = 0, \\ -\theta\sigma_\varepsilon^2, & \tau = \pm s, \\ 0, & \text{otherwise,} \end{cases}$$

so the ACF is  $\rho_{\pm s} = -\theta/(1 + \theta^2)$  and zero at every other lag.

*Proof idea:* apply the white-noise filter rule  $\gamma_\tau = \sigma_\varepsilon^2 \sum_j a_j a_{j+|\tau|}$  to the taps  $a_0 = 1$ ,  $a_s = -\theta$ ; only  $\tau \in \{0, \pm s\}$  pair surviving noise indices.

*Why:* a single nonzero ACF spike at the seasonal lag  $s$  is the fingerprint that flags an  $\text{SMA}(1)_s$  component in a correlogram.

### Proposition 15: Random walk is non-stationary; its difference is white noise

For the random walk  $X_t = \sum_{j=1}^t \varepsilon_j$  (with  $X_0 = 0$ ),

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = t\sigma_\varepsilon^2, \quad \text{Cov}(X_t, X_{t+\tau}) = t\sigma_\varepsilon^2 \quad (\tau \geq 0),$$

all of which depend on  $t$ , so  $\{X_t\}$  is **not** stationary; but  $\nabla X_t = \varepsilon_t$  is white noise.

*Proof idea:* the variance of a sum of  $t$  uncorrelated unit pieces grows linearly in  $t$ ; the overlap

of  $X_t$  and  $X_{t+\tau}$  is the first  $t$  shared terms.

*Why:* the prototype unit-root process — exactly the “ $\phi = 1$  boundary” that ARIMA differencing is designed to handle.

#### Theorem 16: Second-order growth of IMA(1, 1); correlation $\rightarrow 1$

Let  $Y_t = Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}$  with  $Y_0 = 0$ . Then for  $t \geq 1$

$$Y_t = \varepsilon_t + (1 - \theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta\varepsilon_0,$$

hence

$$\text{Var}(Y_t) = \sigma_\varepsilon^2 [1 + (1 - \theta)^2(t - 1) + \theta^2],$$

and for  $\tau > 0$

$$\text{Cov}(Y_t, Y_{t+\tau}) = \sigma_\varepsilon^2 [1 - \theta + (1 - \theta)^2(t - 1) + \theta^2].$$

As  $\tau/t \rightarrow 0$  the correlation  $\text{Corr}(Y_t, Y_{t+\tau}) \rightarrow 1$ .

*Proof idea:* unroll the recursion from  $Y_0 = 0$  to get the explicit linear combination of  $\varepsilon_0, \dots, \varepsilon_t$ ; read off Var and Cov from the shared coefficients (the bulk  $(1 - \theta)$  terms dominate). Both grow like  $(1 - \theta)^2(t - 1)\sigma_\varepsilon^2$ , so their ratio  $\rightarrow 1$ .

*Why:* shows quantitatively that an integrated series has *variance growing linearly in  $t$*  and *nearby values almost perfectly correlated* — the diagnostic signature of needing a difference.

#### Theorem 17: Second-order growth of ARI(1, 1); correlation near unity

Let  $Y_t - Y_{t-1} = \phi(Y_{t-1} - Y_{t-2}) + \varepsilon_t$  with  $|\phi| < 1$  and  $Y_0 = 0$ . Then

$$Y_t = \sum_{j=1}^t \sum_{i=0}^{t-j} \phi^i \varepsilon_j = \sum_{j=1}^t \frac{1 - \phi^{t-j+1}}{1 - \phi} \varepsilon_j,$$

so that

$$\begin{aligned} \text{Var}(Y_t) &= \sigma_\varepsilon^2 \sum_{j=1}^t \left( \frac{1 - \phi^{t-j+1}}{1 - \phi} \right)^2, \\ \text{Cov}(Y_t, Y_{t+\tau}) &= \sigma_\varepsilon^2 \sum_{j=1}^t \frac{1 - \phi^{t-j+1}}{1 - \phi} \cdot \frac{1 - \phi^{t+\tau-j+1}}{1 - \phi} \quad (\tau > 0). \end{aligned}$$

Again the correlation between nearby times is close to unity.

*Proof idea:* the increments  $Z_t = \nabla Y_t$  form a stationary AR(1),  $Z_t = \sum_{i \geq 0} \phi^i \varepsilon_{t-i}$ ; integrating ( $Y_t = \sum_{m \leq t} Z_m$  from 0) and summing the inner geometric series gives the closed coefficients  $(1 - \phi^{t-j+1})/(1 - \phi)$ .

*Why:* confirms that even when the increments are a well-behaved stationary AR(1), *integrating* them recreates the random-walk-like variance growth and near-unit correlation, so the level  $\{Y_t\}$  must be differenced before fitting.

#### Classic Pitfall

$\nabla_{(s)} = I - B^s$  is *not*  $\nabla^s = (I - B)^s$ . The seasonal difference subtracts the value  $s$  steps back ( $X_t - X_{t-s}$ , two terms); the  $s$ -th ordinary difference is an  $(s + 1)$ -term binomial combination. They remove different things:  $\nabla_{(s)}$  a period- $s$  season,  $\nabla^s$  a degree- $(s - 1)$  polynomial. Mixing them up is a classic exam error.

### 5.3 Key Techniques

#### Key Technique 1: Choosing the differencing orders $d$ and $D$

1. **Trend.** If the level looks like a degree- $(q-1)$  polynomial, take  $d = q$  ordinary differences (linear drift  $\Rightarrow d = 1$ ; quadratic  $\Rightarrow d = 2$ ). Symptoms of a remaining unit root: a slowly decaying sample ACF and near-linear variance growth.
2. **Season.** If a fixed pattern of period  $s$  persists, take one seasonal difference  $\nabla_{(s)}$  ( $D = 1$ ); rarely  $D = 2$ .
3. **Order.** Apply  $\nabla_{(s)}$  first to remove the season, then  $\nabla^d$ ; for monthly data with linear drift this is  $\nabla\nabla_{(12)}$ .
4. **Stop** when the differenced series has a stable mean, constant variance and a quickly decaying ACF. **Do not over-difference:** an extra  $\nabla$  injects a spurious MA unit root ( $\rho_1 \rightarrow -\frac{1}{2}$ ) and inflates variance.

#### Key Technique 2: Expanding a SARMA into explicit lag form

Multiply the polynomials out, treating  $B$  as a formal variable:

1. Write the four factors  $\Phi(B), \Phi^{(s)}(B)$  (left) and  $\Theta(B), \Theta^{(s)}(B)$  (right).
2. Expand each product, e.g.  $(1 - \theta B)(1 - \theta^{(12)}B^{12}) = 1 - \theta B - \theta^{(12)}B^{12} + \theta\theta^{(12)}B^{13}$ .
3. Read off the surviving lags. The seasonal factor only contributes lags that are multiples of  $s$ ; cross terms give lags like  $s \pm 1$ .
4. Translate  $B^k X_t = X_{t-k}$ ,  $B^k \varepsilon_t = \varepsilon_{t-k}$  to get the difference equation, then solve for  $X_t$  (move all but  $X_t$  to the right) for a one-step recursion.

#### Key Technique 3: Deriving the $\text{MA}(\infty)$ ( $\psi$ -weights) of a SARMA/ARMA

To get  $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$  from  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ :

1. Invert the AR factor as a geometric series:  $\frac{1}{1 - \phi B} = \sum_{l \geq 0} \phi^l B^l$  (valid when  $|\phi| < 1$ ).
2. Multiply by the full MA numerator and collect the coefficient of  $B^j$ .
3. **Match powers of  $B$ :**  $\Phi(B) \sum_j \psi_j B^j = \Theta(B)$  gives the recursion  $\psi_j = \theta_j^{(\text{num})} + \sum_{i=1}^p \phi_i \psi_{j-i}$ ,  $\psi_0 = 1$ , with  $\theta_j^{(\text{num})}$  the coefficient of  $B^j$  in the (expanded) MA numerator. Beyond the highest numerator lag the pure recursion  $\psi_j = \sum_i \phi_i \psi_{j-i}$  takes over. Spot the recursion early (e.g.  $\psi_3 = \phi\psi_2$ ) — it collapses the algebra.

#### Key Technique 4: ACVS/ACF of a seasonal $\text{MA}(\infty)$ via even/odd lags

For a process with only even (or only seasonal) taps,  $X_t = \sum_j a_j \varepsilon_{t-j}$ :

1. Use  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j \geq 0} a_j a_{j+\tau}$  and  $\rho_\tau = \gamma_\tau / \gamma_0$ .
2. If the taps vanish at odd lags ( $a_{\text{odd}} = 0$ ), then so do  $\gamma_\tau$  and  $\rho_\tau$  at odd  $\tau$  — only even (seasonal) lags carry correlation.
3. Sum the resulting geometric series in  $\phi^2$  for the closed form;  $\text{Var}(X_t) = \gamma_0 = \sigma_\varepsilon^2 \sum_j a_j^2$ .

## 5.4 Exercises

### Exercise 1: ARIMA invariance to an added polynomial [Sheet 5]

Suppose  $\{X_t\}$  is an ARIMA( $p, d, q$ ) process satisfying  $\Phi(B)(I - B)^d X_t = \Theta(B)Z_t$  with  $Z_t$  white noise. Show that the same equations are satisfied by  $W_t = X_t + A_0 + A_1 t + \dots + A_{d-1} t^{d-1}$ , where the  $A_j$  are independent random coefficients.

#### Solution

Because  $\Phi(B)$  and  $\Theta(B)$  act on whatever  $(I - B)^d X_t$  produces, it suffices to show that the added polynomial is annihilated by  $(I - B)^d$ , i.e.  $(I - B)^d W_t = (I - B)^d X_t$ . First,  $(I - B)\mathbf{0} = \mathbf{0}$  on the zero sequence. By linearity of  $(I - B)^d$ ,

$$(I - B)^d W_t = (I - B)^d X_t + \sum_{k=1}^d A_{k-1} (I - B)^d t^{k-1}.$$

So we only need  $(I - B)^d t^{k-1} = \mathbf{0}$  for  $k = 1, \dots, d$ ; it is enough to prove the stronger statement

$$(I - B)^k t^{k-1} = \mathbf{0} \quad \text{for all } k \in \mathbb{N},$$

by induction on  $k$ .

*Base case*  $k = 1$ :  $(I - B)t^0 = (I - B)\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0}$  (here  $\mathbf{1}$  is the constant-one sequence).

*Inductive step*: assume  $(I - B)^r t^{r-1} = \mathbf{0}$  for all  $r \leq k$ . Then

$$(I - B)^{k+1} t^k = (I - B)^k (I - B)t^k = (I - B)^k (t^k - (t - 1)^k).$$

By the binomial theorem the top power cancels,

$$t^k - (t - 1)^k = - \sum_{r=0}^{k-1} \binom{k}{r} (-1)^{k-r} t^r,$$

a polynomial of degree  $k - 1$ . Applying  $(I - B)^k$  and using linearity together with the induction hypothesis (each  $(I - B)^{r+1} t^r = \mathbf{0}$  for  $r \leq k - 1$ , and one extra difference of  $\mathbf{0}$  is still  $\mathbf{0}$ ), every term vanishes, so  $(I - B)^{k+1} t^k = \mathbf{0}$ .

By induction the claim holds for all  $k$ , hence  $(I - B)^d W_t = (I - B)^d X_t$  and  $\{W_t\}$  satisfies the identical ARIMA equations. *Interpretation*: the  $d$ -fold integration in an ARIMA( $p, d, q$ ) leaves the degree- $(d - 1)$  polynomial “initial conditions” completely undetermined by the model.

### Exercise 2: $\psi$ -weights of a SARMA( $1, 2$ ) $\times$ $(0, 1)_4$ [Sheet 5]

$U_t$  follows a SARMA( $1, 2$ )  $\times$   $(0, 1)_4$  model. Write the defining equation and find the weights  $\psi_j$  in the MA( $\infty$ ) representation  $U_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ .

#### Solution

With one seasonal MA factor at  $s = 4$  the model is

$$(1 - \phi B) U_t = (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4) \varepsilon_t \iff U_t = \phi U_{t-1} + (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4) \varepsilon_t.$$

Invert the AR factor as a geometric series,  $1/(1 - \phi B) = \sum_{l \geq 0} \phi^l B^l$ :

$$\begin{aligned} U_t &= \sum_{l \geq 0} (\phi B)^l (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4) \varepsilon_t \\ &= \sum_{l \geq 0} (\phi^l B^l - \theta_1 \phi^l B^{l+1} - \theta_2 \phi^l B^{l+2} - \Theta \{\phi^l B^{l+4} - \theta_1 \phi^l B^{l+5} - \theta_2 \phi^l B^{l+6}\}) \varepsilon_t. \end{aligned}$$

Collecting the coefficient of  $B^j$  gives  $\psi_j$ . For the low lags (before the seasonal term enters):

$$\psi_0 = 1, \quad \psi_1 = -\theta_1 + \phi, \quad \psi_2 = -\theta_2 - \theta_1\phi + \phi^2, \quad \psi_3 = -\theta_2\phi - \theta_1\phi^2 + \phi^3 = \phi\psi_2.$$

From  $j = 4$  the seasonal factor contributes:

$$\begin{aligned}\psi_4 &= -\Theta - \theta_2\phi^2 - \theta_1\phi^3 + \phi^4 = -\Theta + \phi\psi_3, \\ \psi_5 &= -\Theta(\phi - \theta_1) - \theta_2\phi^3 - \theta_1\phi^4 + \phi^5, \\ \psi_6 &= -\Theta(\phi^2 - \theta_1\phi - \theta_2) + (\phi^6 - \theta_1\phi^5 - \theta_2\phi^4), \\ \psi_7 &= -\Theta(\phi^3 - \theta_1\phi^2 - \theta_2\phi) + (\phi^7 - \theta_1\phi^6 - \theta_2\phi^5) = \phi\psi_6.\end{aligned}$$

The pattern stabilises: for all  $l \geq 6$ ,

$$\psi_l = (\phi^l - \theta_1\phi^{l-1} - \theta_2\phi^{l-2}) - \Theta(\phi^{l-4} - \theta_1\phi^{l-5} - \theta_2\phi^{l-6}).$$

(The recursion  $\psi_l = \phi\psi_{l-1}$  holds once  $l$  is past every fixed numerator lag, here  $l \geq 7$ , which is the labour-saving shortcut.)

### Exercise 3: Seasonal ARMA $(1 - 0.7B^2)X = (1 - 0.3B^2)\varepsilon$ [Sheet 5]

Study  $(1 - 0.7B^2)X_t = (1 - 0.3B^2)\varepsilon_t$ ,  $\varepsilon_t$  white noise with unit variance. (1) Find  $\{a_j\}$  in  $X_t = \sum_{j \geq 0} a_j \varepsilon_{t-j}$ . (2) Find  $\{b_j\}$  in  $\varepsilon_t = \sum_{j \geq 0} b_j X_{t-j}$ . (3) Find and describe the autocorrelation  $\rho_\tau$ .

### Solution

Write  $\phi = 0.7$ ,  $\theta = 0.3$ , so  $(1 - \phi B^2)X_t = (1 - \theta B^2)\varepsilon_t$ . The AR polynomial  $\Phi(z) = 1 - \phi z^2$  has roots  $z = \pm\phi^{-1/2}$  with modulus  $> 1$ , so the process is stationary (and causal) and we may invert.

(1) **MA( $\infty$ ).** Geometric expansion in  $B^2$ :

$$X_t = \frac{1 - \theta B^2}{1 - \phi B^2} \varepsilon_t = \sum_{k \geq 0} (\phi B^2)^k (1 - \theta B^2) \varepsilon_t.$$

The coefficient of  $B^{2k}$  is  $\phi^k$  and of  $B^{2k+2}$  is  $-\phi^k\theta$ , so only even lags survive:

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = \phi - \theta, \quad a_3 = 0, \quad a_4 = \phi^2 - \phi\theta, \dots$$

and in general  $a_j = 0$  for odd  $j$ , while for even  $j \geq 2$

$$a_j = \phi^{j/2} - \phi^{(j-2)/2}\theta = \phi^{(j-2)/2}(\phi - \theta).$$

(2) **AR( $\infty$ ) (invertibility).** The MA polynomial  $1 - \theta z^2$  has roots  $\pm\theta^{-1/2}$ , modulus  $> 1$ , so the process is invertible. By symmetry, swap the roles of  $\phi$  and  $\theta$ :

$$\varepsilon_t = \frac{1 - \phi B^2}{1 - \theta B^2} X_t = \sum_{k \geq 0} (\theta B^2)^k (1 - \phi B^2) X_t,$$

hence  $b_0 = 1$ ,  $b_j = 0$  for odd  $j$ , and for even  $j \geq 2$

$$b_j = \theta^{j/2} - \theta^{(j-2)/2}\phi.$$

(3) **Autocorrelation.** Using  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j \geq 0} a_j a_{j+\tau}$  (white noise) and  $\rho_\tau = \gamma_\tau / \gamma_0$ . Since the  $a_j$  vanish at odd  $j$ ,  $\rho_\tau = 0$  for all odd  $\tau$ , and  $\rho_0 = 1$ .

First the variance. With  $a_0 = 1$  and  $a_{2m} = \phi^{m-1}(\phi - \theta)$  for  $m \geq 1$ ,

$$\text{Var}(X_t) = \gamma_0 = \sigma_\varepsilon^2 \left( 1 + (\phi - \theta)^2 \sum_{m \geq 1} \phi^{2(m-1)} \right) = \sigma_\varepsilon^2 \left( 1 + \frac{(\phi - \theta)^2}{1 - \phi^2} \right).$$

Writing  $(\phi - \theta)^2 = \phi^2(1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2)$  recovers the notes' form

$$\text{Var}(X_t) = \sigma_\varepsilon^2 \left\{ 1 + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1 - \phi^2} \right\}.$$

For even  $\tau = 2m$  ( $m \geq 1$ ),

$$\gamma_\tau = \sigma_\varepsilon^2 \left[ a_0 a_\tau + \sum_{k \geq 1} a_{2k} a_{2k+\tau} \right] = \sigma_\varepsilon^2 \left[ (\phi^m - \phi^{m-1}\theta) + (\phi - \theta)^2 \phi^m \sum_{k \geq 1} \phi^{2(k-1)} \right],$$

where the cross term used  $a_{2k} a_{2k+2m} = \phi^{k-1}(\phi - \theta) \cdot \phi^{k+m-1}(\phi - \theta)$ . Summing the geometric series and dividing by  $\gamma_0$  (with  $m = \frac{\tau}{2}$ ),

$$\begin{aligned} \rho_\tau &= \frac{\phi^m - \phi^{m-1}\theta + (\phi - \theta)^2 \frac{\phi^m}{1 - \phi^2}}{1 + \frac{(\phi - \theta)^2}{1 - \phi^2}} \\ &= \phi^m \cdot \frac{1 - \phi^{-1}\theta + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1 - \phi^2}}{1 + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1 - \phi^2}}. \end{aligned}$$

With  $\phi = 0.7$ ,  $\theta = 0.3$ :  $\text{Var}(X_t) = 1 + \frac{0.16}{0.51} \approx 1.3137$ , and  $\rho_2 \approx 0.472$ ,  $\rho_4 \approx 0.330$ ,  $\rho_6 \approx 0.231$ ,  $\rho_8 \approx 0.162$  — a clean *geometric decay at rate  $\phi$  along the even lags*, with all odd-lag correlations zero. (Numerically verified.) The correlogram therefore shows spikes only at  $\tau = 2, 4, 6, \dots$  decaying like  $0.7^{\tau/2}$ .

#### Exercise 4: Order of $\nabla\nabla_{(s)}$ on trend-plus-season [New]

Let  $X_t = a + bt + s_t + Y_t$  with  $s_{t+4} = s_t$  (quarterly season) and  $\{Y_t\}$  zero-mean stationary. Compute  $\nabla\nabla_{(4)}X_t$  explicitly and verify it is a fixed linear combination of  $\{Y_t\}$  with zero mean.

#### Solution

Apply the seasonal difference first:

$$\nabla_{(4)}X_t = X_t - X_{t-4} = (a + bt + s_t + Y_t) - (a + b(t-4) + s_{t-4} + Y_{t-4}) = 4b + (Y_t - Y_{t-4}),$$

using  $s_t = s_{t-4}$ . The trend has collapsed to the constant  $4b$  and the season is gone. Now apply  $\nabla = I - B$ :

$$\nabla\nabla_{(4)}X_t = (4b + Y_t - Y_{t-4}) - (4b + Y_{t-1} - Y_{t-5}) = Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}.$$

This is a fixed linear combination of the stationary  $\{Y_t\}$ , so its mean is  $\mathbb{E}[Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}] = 0$  and, by bilinearity, its covariance depends on the lag only; hence  $\nabla\nabla_{(4)}X_t$  is weakly stationary. (Same shape as the  $\nabla\nabla_{(12)}$  result with  $12 \rightarrow 4$ .)

**Exercise 5: Expand and recurse a SARMA(0, 1) × (1, 0)<sub>12</sub> [New]**

Write the explicit one-step recursion for a SARMA(0, 1) × (1, 0)<sub>12</sub> model  $(1 - \phi^{(12)}B^{12})X_t = (1 - \theta B)\varepsilon_t$ , give its MA(∞) weights, and find its ACVS at lags 0, 1, 11, 12, 13.

**Solution**

Expanding the seasonal AR factor and solving for  $X_t$ :

$$X_t = \phi^{(12)}X_{t-12} + \varepsilon_t - \theta\varepsilon_{t-1}.$$

This is an AR(1) acting only on the seasonal lag 12, driven by an MA(1) input. Its MA(∞) comes from inverting  $1/(1 - \phi^{(12)}B^{12}) = \sum_{l \geq 0} (\phi^{(12)})^l B^{12l}$ :

$$X_t = \sum_{l \geq 0} (\phi^{(12)})^l (\varepsilon_{t-12l} - \theta\varepsilon_{t-12l-1}), \quad \psi_{12l} = (\phi^{(12)})^l, \quad \psi_{12l+1} = -\theta(\phi^{(12)})^l,$$

all other  $\psi_j = 0$ . By the white-noise filter rule  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$ :

$$\begin{aligned} \gamma_0 &= \sigma_\varepsilon^2 \frac{1 + \theta^2}{1 - (\phi^{(12)})^2}, & \gamma_1 &= \sigma_\varepsilon^2 \sum_{l \geq 0} \psi_{12l} \psi_{12l+1} = \frac{-\theta \sigma_\varepsilon^2}{1 - (\phi^{(12)})^2}, \\ \gamma_{11} &= \sigma_\varepsilon^2 \sum_{l \geq 0} \psi_{12l+1} \psi_{12(l+1)} = \frac{-\theta \phi^{(12)} \sigma_\varepsilon^2}{1 - (\phi^{(12)})^2}, & \gamma_{12} &= \sigma_\varepsilon^2 \sum_{l \geq 0} \psi_{12l} \psi_{12l+12} = \phi^{(12)} \gamma_0, \\ \gamma_{13} &= \sigma_\varepsilon^2 \sum_{l \geq 0} \psi_{12l} \psi_{12(l+1)+1} = \frac{-\theta \phi^{(12)} \sigma_\varepsilon^2}{1 - (\phi^{(12)})^2} = \gamma_{11}. \end{aligned}$$

So correlation appears in clusters around the seasonal lags 0, 12, 24, ... (each carrying MA(1) side-lobes at ±1), decaying geometrically in  $\phi^{(12)}$  — the textbook shape of a seasonal AR with a non-seasonal MA.

**Exercise 6: IMA(1, 1) variance growth and the over-differencing trap [New]**

For  $Y_t = Y_{t-1} + \varepsilon_t - \theta\varepsilon_{t-1}$  with  $Y_0 = 0$ : (a) confirm  $\text{Var}(Y_t)$  grows linearly in  $t$  for  $\theta \neq 1$ ; (b) what happens at the boundary  $\theta = 1$ , and why is that the over-differencing warning?

**Solution**

(a) From  $Y_t = \varepsilon_t + (1 - \theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta\varepsilon_0$  and uncorrelated noise,

$$\text{Var}(Y_t) = \sigma_\varepsilon^2 [1 + (1 - \theta)^2(t - 1) + \theta^2],$$

which is *linear* in  $t$  with slope  $(1 - \theta)^2 \sigma_\varepsilon^2 > 0$  whenever  $\theta \neq 1$ : the series wanders like a (damped) random walk, the hallmark of an integrated process. The covariance  $\text{Cov}(Y_t, Y_{t+\tau}) = \sigma_\varepsilon^2 [1 - \theta + (1 - \theta)^2(t - 1) + \theta^2]$  shares the same leading  $(1 - \theta)^2(t - 1)$  term, so the correlation of nearby points  $\rightarrow 1$ .

(b) At  $\theta = 1$  the slope vanishes and  $\text{Var}(Y_t) = \sigma_\varepsilon^2 [1 + 1] = 2\sigma_\varepsilon^2$  is *bounded*. In fact  $Y_t = Y_{t-1} + \varepsilon_t - \varepsilon_{t-1}$  telescopes to  $Y_t = \varepsilon_t - \varepsilon_0$ , so  $\nabla Y_t = \varepsilon_t - \varepsilon_{t-1}$  is a (non-invertible) MA(1) with an MA unit root and lag-1 ACF  $\rho_1 = -\theta/(1 + \theta^2) = -1/2$ . This is exactly the signature of *over-differencing*: if you difference a series that was already stationary, you create a  $(1 - B)$  factor in the MA polynomial, i.e. a non-invertible MA(1) with  $\rho_1 = -\frac{1}{2}$ . Seeing  $\rho_1 \approx -\frac{1}{2}$  after differencing is the cue to difference one time fewer.

## 6 Fourier Theory, Spectral Density and the Periodogram

The frequency domain is the second face of a stationary process. Every second-order property carried by the autocovariance sequence (ACVS)  $\{\gamma_\tau\}$  in the time domain is mirrored, via the Fourier transform, by the **spectral density function** (SDF)  $\mathcal{S}(f)$  in the frequency domain. This part builds that bridge in three stages. First, pure Fourier theory: continuous-time, discrete-time and finite-data transforms, the convolution theorem, aliasing and the Nyquist frequency. Second, the spectral representation of a stationary process:  $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau}$ , Herglotz's theorem, the integrated spectrum and its Lebesgue decomposition, and the spectral representation theorem with its orthogonal-increment process  $\{Z(f)\}$ . Third, estimation: the periodogram  $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$ , why it is (asymptotically) unbiased but *inconsistent*, and how tapering cures bias while multitapering cures variance. Throughout we take the sampling interval  $\Delta = 1$  once we reach the spectral chapters, and we always work with ordinary frequency  $f$  (not angular frequency  $\omega = 2\pi f$ ).

### 6.1 Definitions

#### Definition 1: Continuous-time Fourier transform and its inverse

For  $g : \mathbb{R} \rightarrow \mathbb{C}$  with  $g \in L^1$ , the **Fourier transform**  $G$  (upper case denotes the transform; some authors write  $\hat{g}$ ) is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i f t} dt, \quad f \in \mathbb{R}.$$

If in addition  $G \in L^1$ , the **inverse transform** recovers  $g$  for almost all  $t$ :

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df.$$

We write  $g \longleftrightarrow G$  and call  $(g, G)$  a **Fourier pair**.

#### Definition 2: Properties of the continuous-time transform

For  $g, h \in L^1$  with transforms  $G, H$  and scalar  $\alpha$ , the basic pairs are

$$\begin{aligned} \overline{g(t)} &\longleftrightarrow \overline{G(-f)}, & g(\alpha t) \ (\alpha \neq 0) &\longleftrightarrow \frac{1}{|\alpha|} G(f/\alpha), \\ g(t + \tau) &\longleftrightarrow G(f) e^{2\pi i \tau f}, & \alpha g(t) + h(t) &\longleftrightarrow \alpha G(f) + H(f). \end{aligned}$$

(Conjugation reflects, scaling in time inversely scales frequency, a time shift multiplies by a phase, and the transform is linear.)

#### Definition 3: Continuous-time convolution

For  $g, h \in L^1$ , the **convolution**  $u = g * h$  is

$$u(t) = \int_{-\infty}^{\infty} g(t - u) h(u) du, \quad t \in \mathbb{R}.$$

Convolution at a point is a weighted average of one function by the other, centred at that point.

#### Definition 4: Discrete-time Fourier transform and its inverse

For a sequence  $\{g_t\}_{t \in \Delta\mathbb{Z}}$  in  $\ell^1$ , the **discrete-time Fourier transform** is

$$G(f) = \Delta \sum_{t \in \Delta\mathbb{Z}} g_t e^{-2\pi i t f}, \quad f \in \mathbb{R},$$

with inverse (a Fourier series with the roles of domains swapped)

$$g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i t f} df, \quad t \in \Delta\mathbb{Z}.$$

A defining feature is **periodicity**:  $G(f) = G(f + k/\Delta)$  for every  $k \in \mathbb{Z}$ , so  $G$  is determined by its values on one period of length  $1/\Delta$ .

#### Definition 5: Discrete-time and periodic convolution

For  $g, h \in \ell^1$  the **discrete convolution**  $u = g * h$  is

$$u_t = \Delta \sum_{u \in \Delta\mathbb{Z}} g_{t-u} h_u, \quad t \in \Delta\mathbb{Z},$$

(with  $\Delta = 1$  the prefactor disappears). The dual object, the **periodic convolution**  $U = G * H$  of two periodic spectra, is

$$U(f) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f - f') H(f') df'.$$

#### Definition 6: Finite-data Fourier transform and the Dirichlet kernel

Given observations  $g_0, \dots, g_{(n-1)\Delta}$  at times  $T = \{0, \Delta, \dots, (n-1)\Delta\}$ , the **finite Fourier transform** is

$$G_n(f) = \sum_{t \in T} g_t e^{-2\pi i t f}, \quad f \in \mathbb{R}.$$

It equals the full transform of  $g_t$  multiplied by the indicator  $d_t = \frac{1}{\Delta} \mathbf{1}_T(t)$ , whose own transform is the **Dirichlet kernel**

$$\mathcal{D}_T(f) = \sum_{t \in T} e^{-2\pi i t f} = \frac{\sin(n\pi f \Delta)}{\sin(\pi f \Delta)} e^{-\pi i (n-1) f \Delta}.$$

Thus  $G_n = \mathcal{D}_T * G$ : observing a finite window *blurs* the true transform.

#### Definition 7: Fourier frequencies

The FFT evaluates  $G_n$  at the **Fourier frequencies**

$$f_k = \frac{k}{\Delta n}, \quad k \in \mathbb{Z}, \quad -\frac{n}{2} \leq k \leq \frac{n-1}{2}.$$

With  $\Delta = 1$  these are  $f_k = k/n$ . They are equally spaced over one period and are the frequencies at which distinct periodogram ordinates are (approximately) uncorrelated.

#### Definition 8: Spectral density function (discrete and continuous)

For a discrete-time stationary process  $\{X_t\}_{t \in \mathbb{Z}}$  with ACVS  $\{\gamma_\tau\} \in \ell^1$ , the **spectral density function** is the discrete-time Fourier transform of the ACVS (with  $\Delta = 1$ ):

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau}, \quad f \in \mathbb{R}.$$

For a continuous-time process with  $\gamma \in L^1$ ,  $\mathcal{S}(f) = \int_{-\infty}^{\infty} \gamma(\tau) e^{-2\pi i f \tau} d\tau$ . By Fourier inversion the ACVS is recovered as  $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$ ; in particular  $\gamma_0 = \text{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f) df$ ,

so the SDF distributes the total “power”  $\text{Var}(X_t)$  across frequencies. A valid SDF is real, even  $(\mathcal{S}(-f) = \mathcal{S}(f))$ , since  $\gamma_{-\tau} = \gamma_\tau$ , non-negative, and periodic with period 1.

#### Definition 9: Harmonic (sinusoidal) process

Let  $A > 0$  and  $\Theta \sim \text{Unif}(-\pi, \pi)$  be independent. The **harmonic process** is

$$X_t = A \cos(\nu t + \Theta), \quad t \in \mathbb{Z},$$

with  $\nu \in \mathbb{R}$  the oscillation frequency. Then  $\mathbb{E}[X_t] = 0$  and

$$\text{Cov}(X_{t+\tau}, X_t) = \frac{1}{2} \mathbb{E}[A^2] \cos(\nu\tau),$$

so it is stationary but its ACVS does *not* decay; in particular  $\{\gamma_\tau\} \notin \ell^1$  and no ordinary SDF exists. Its spectrum is a pair of lines (jumps) at  $\pm\nu/(2\pi)$ .

#### Definition 10: Integrated spectrum (spectral distribution function)

By Herglotz’s theorem, every ACVS admits the representation

$$\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i \tau f} dS^{(I)}(f),$$

where  $S^{(I)}$ , the **integrated spectrum**, is right-continuous, bounded, non-decreasing, with  $S^{(I)}(-\frac{1}{2}) = 0$  and  $S^{(I)}(\frac{1}{2}) = \sigma^2 = \gamma_0$ . It behaves like a (scaled) cumulative distribution function over frequency. When  $\{\gamma_\tau\} \in \ell^1$  there is a density  $\mathcal{S}$  with  $S^{(I)}(f) = \int_{-1/2}^f \mathcal{S}(\lambda) d\lambda$ , i.e.  $\mathcal{S} = dS^{(I)}/df$ , recovering Definition 8.

#### Definition 11: Periodogram

For data  $X_1, \dots, X_n$  ( $T = \{1, \dots, n\}$ ,  $\Delta = 1$ ), the **periodogram** is the plug-in estimator obtained by Fourier transforming the biased sample ACVS:

$$\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau \in \mathbb{Z}} \hat{\gamma}_\tau e^{-2\pi i f \tau}, \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X})$$

for  $|\tau| \leq n-1$  (0 otherwise). Equivalently (Proposition 24)  $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$  with  $J(f) = \sqrt{1/n} \sum_{t \in T} (X_t - \bar{X}) e^{-2\pi i t f}$ .

#### Definition 12: Data taper and tapered periodogram

A **data taper** is a sequence  $\{h_t\}_{t \in T}$  with  $\|h\|_2^2 = \sum_t h_t^2 = 1$ . The **tapered Fourier transform** and **tapered periodogram** are

$$J_h(f) = \sum_{t \in T} h_t (X_t - \bar{X}) e^{-2\pi i t f}, \quad \hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2.$$

Choosing  $h_t = \sqrt{1/n}$  (the constant taper) recovers the ordinary periodogram. The normalisation  $\|h\|_2^2 = 1$  keeps the estimator unbiased for white noise (Exercise 11).

### Definition 13: Multitaper spectral estimator

Given  $K$  **orthonormal** tapers  $h_k = \{h_{t,k}\}_{t \in T}$  with  $\sum_{t \in T} h_{t,k} h_{t,k'} = \delta_{k,k'}$ , the **multitaper estimator** averages the individual tapered periodograms,

$$\hat{\mathcal{S}}^{(mt)}(f) = \frac{1}{K} \sum_{k=1}^K \hat{\mathcal{S}}_{h_k}^{(p)}(f).$$

Its expectation is a convolution with the **average kernel**  $\bar{H}(f) = \frac{1}{K} \sum_{k=1}^K |H_k(f)|^2$ , and its variance is  $\approx \mathcal{S}(f)^2/K$ .

## 6.2 Theorems & Propositions

### Theorem 14: Fourier inversion (continuous and discrete time)

(Continuous time.) If  $g, G \in L^1$  then the inverse transform recovers  $g$  for almost all  $t$ :  $g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df$ . (Discrete time.) If  $\{g_t\} \in \ell^1$  then  $G$  is recovered on one period as a Fourier series with the domains swapped:

$$g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t} df, \quad t \in \Delta\mathbb{Z}.$$

*Proof idea:* substitute the forward transform into the right-hand side and use the Fourier orthogonality relation ( $\int e^{2\pi i f(t-s)} df$  acts as a delta), the integrability hypotheses justifying the interchange of integration/summation.

*Why:* inversion is what makes  $(g, G)$  a genuine *Fourier pair* — no information is lost. It is the engine behind recovering the ACVS from the SDF ( $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$ ) and behind every “time  $\leftrightarrow$  frequency” duality used below.

### Theorem 15: Convolution theorem (continuous time)

For  $g, h \in L^1$ , if  $u = g * h$  then  $U = G \cdot H$ : the transform of a convolution is the product of the transforms. If also  $G, H \in L^1$  and  $v = g \cdot h$ , then dually  $V = G * H$ .

*Proof idea:* substitute the convolution integral into the definition of  $U$ , swap the order of integration (Fubini, justified by  $\iint |g(x-y)h(y)| dx dy = \|g\|_1 \|h\|_1 < \infty$ ), and recognise the shift factor  $e^{-2\pi i s f}$  as producing  $G(f)$ .

*Why:* the central computational identity of Fourier analysis — “multiplication in one domain is convolution in the other”. It underlies filtering, the SDF of a filtered process, and the periodogram bias formula.

### Theorem 16: Convolution theorem (discrete time)

For  $g, h \in \ell^1$  both pairs hold:

$$h \cdot g \longleftrightarrow H * G, \quad h * g \longleftrightarrow H \cdot G.$$

*Proof idea:* for  $u = h * g$ , expand  $U(f) = \sum_t u_t e^{-2\pi i f t}$ , insert  $e^{-2\pi i(t-s)f} e^{-2\pi i s f}$ , factor the double sum into  $(\sum_s h_s e^{-2\pi i s f})(\sum_x g_x e^{-2\pi i x f})$ ; interchange is legal because  $g, h \in \ell^1$ . For  $v = h \cdot g$ , replace one factor by its inverse transform and integrate.

*Why:* the discrete analogue used to derive the finite-data transform ( $G_n = \mathcal{D}_T * G$ ) and the periodogram expectation (a frequency convolution with the Fejér kernel).

### Theorem 17: Aliasing theorem and the Nyquist frequency

Let  $g \in L^1$  be continuous with transform  $G$ , and let  $G$  denote the transform of the sampled sequence  $\{g(t)\}_{t \in \Delta\mathbb{Z}}$ . Under the summability hypotheses, for every  $f$

$$G(f) = \sum_{k \in \mathbb{Z}} G(f + k/\Delta).$$

Sampling *folds* all the continuous-frequency content at  $\{f + k/\Delta\}_k$  onto the single frequency  $f$ . For an SDF this reads  $\mathcal{S}(f) = \sum_{k \in \mathbb{Z}} \mathcal{S}(f + k)$  (with  $\Delta = 1$ ). The **Nyquist frequency** is  $1/(2\Delta)$  (i.e.  $\frac{1}{2}$  when  $\Delta = 1$ ): content above it cannot be recovered.

*Proof idea:* write  $g(t)$  at integer times by both inversion formulas; split  $\int_{-\infty}^{\infty} G(f)e^{2\pi ift}df$  into period-length blocks, shift each block by  $k/\Delta$  (the phase  $e^{2\pi itk/\Delta} = 1$  for  $t \in \Delta\mathbb{Z}$ ), swap sum and integral (summability), and match the two representations of  $g(t)$ .

*Why:* explains why we sample fast enough; if  $\mathcal{S}$  is negligible beyond  $1/(2\Delta)$  the discrete spectrum faithfully matches the continuous one, otherwise it is corrupted (poor agreement).

### Why aliasing matters

A high-frequency sinusoid sampled too slowly looks identical to a low-frequency one — the wagon-wheel effect. Frequencies  $f$  and  $f + k/\Delta$  are indistinguishable from the samples; they are “aliases”. The cure is to sample finely (small  $\Delta$ , large Nyquist) or to low-pass filter before sampling.

### Proposition 18: FFT complexity

The finite Fourier transform at the  $n$  Fourier frequencies  $\{k/(\Delta n)\}$  can be computed in  $O(n \log n)$  time via the **Fast Fourier Transform**, versus  $O(n^2)$  for the naive sum.

*Proof idea:* divide-and-conquer — a transform of length  $n$  splits into two of length  $n/2$  (even/odd indices), recursing  $\log_2 n$  levels with  $O(n)$  work each.

*Why:* makes periodogram/spectral computation feasible at scale; one can also invert the periodogram by FFT to get the sample ACVS cheaply.

### Theorem 19: Herglotz’s theorem

A complex sequence  $\{g_t\}_{t \in \mathbb{Z}}$  is **non-negative definite** if and only if there exists a right-continuous, bounded, non-decreasing function  $G^{(I)}$  with  $G^{(I)}(-\frac{1}{2}) = 0$  such that

$$g_t = \int_{-1/2}^{1/2} e^{2\pi itf} dG^{(I)}(f), \quad t \in \mathbb{Z}.$$

*Proof idea:* ( $\Leftarrow$ ) any such integral representation yields a non-negative-definite sequence because  $G^{(I)}$  is non-decreasing. ( $\Rightarrow$ ) construct  $G^{(I)}$  as the (weak) limit of the non-negative Cesàro/Fejér averages of the partial sums  $\sum_{|t| \leq N} g_t e^{-2\pi itf}$ , which form a tight family of measures.

*Why:* it is the existence theorem for the spectrum. Since every ACVS is non-negative definite (a foundational property of any stationary process), *every* stationary process has an integrated spectrum  $S^{(I)}$  — even those, like harmonic processes, with no ordinary SDF.

### Corollary 20: Integrated spectrum exists for every stationary process

Applying Herglotz’s theorem to the non-negative-definite ACVS gives  $\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i\tau f} dS^{(I)}(f)$  with  $S^{(I)}(-\frac{1}{2}) = 0$ ,  $S^{(I)}(\frac{1}{2}) = \sigma^2$ , and  $S^{(I)}$  non-decreasing.

*Why:* guarantees a frequency-domain description always exists; the SDF is merely the special (absolutely continuous) case  $dS^{(I)} = \mathcal{S} df$ .

### Theorem 21: Lebesgue decomposition of the integrated spectrum

Any integrated spectrum decomposes uniquely as

$$S^{(I)}(\cdot) = S_1^{(I)}(\cdot) + S_2^{(I)}(\cdot) + S_3^{(I)}(\cdot),$$

where each  $S_j^{(I)}$  is non-negative, non-decreasing with  $S_j^{(I)}(-\frac{1}{2}) = 0$ , and

1.  $S_1^{(I)}$  is **absolutely continuous**:  $S_1^{(I)}(f) = \int_{-1/2}^f \mathcal{S}(\lambda) d\lambda$  (the ordinary SDF  $\mathcal{S}$ );
2.  $S_2^{(I)}$  is a **step function** with jumps  $\{p_\ell\}$  at frequencies  $\{f_\ell\}$  of pure sinusoids (a *line spectrum*);
3.  $S_3^{(I)}$  is a **continuous singular** function (pathological, of no practical use).

*Proof idea*: the Lebesgue decomposition theorem for monotone functions / measures, specialised to the finite interval  $[-\frac{1}{2}, \frac{1}{2}]$ .

*Why*: classifies the possible spectral behaviours: purely continuous ( $S_1^{(I)}$  only), purely discrete/line ( $S_2^{(I)}$  only), or mixed.

### Reading the three components

**Absolutely continuous** ( $S_1^{(I)}$ ): the ACVS damps to zero,  $\gamma_\tau \rightarrow 0$  as  $\tau \rightarrow \infty$  (since for integrable  $\mathcal{S}$ ,  $\int \cos(2\pi f\tau) \mathcal{S} df \rightarrow 0$ ). Example: any ARMA process. **Discrete/line** ( $S_2^{(I)}$ ): the ACVS does *not* damp to zero (a sinusoid keeps oscillating). Example: a sinusoid with random phase and amplitude (the harmonic process). **Mixed**: a signal-plus-noise sum of the two, e.g. a periodic component buried in coloured noise.

### Example 22: Spectral characterisation of common processes

- **Purely continuous spectrum** ( $S_1^{(I)} \geq 0$ ,  $S_2^{(I)} \equiv 0$ ): ARMA processes; the SDF  $\mathcal{S}$  exists and  $\gamma_\tau \rightarrow 0$ .
- **Purely discrete (line) spectrum** ( $S_1^{(I)} \equiv 0$ ,  $S_2^{(I)} \geq 0$ ): random-phase/amplitude sinusoid; the ACVS persists.
- **Mixed spectrum** ( $S_1^{(I)} \geq 0$ ,  $S_2^{(I)} \geq 0$ ): pointwise aggregation, e.g. harmonic signal plus an ARMA background.

### Theorem 23: Spectral representation theorem

Let  $\{X_t\}$  be a real-valued discrete-time stationary process with mean  $\mu$ . There exists an **orthogonal-increment process**  $\{Z(f)\}$  on  $[-\frac{1}{2}, \frac{1}{2}]$  with

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f) \quad (\text{equality in mean square}),$$

where for  $f \neq f'$ ,

$$\mathbb{E}[dZ(f)] = 0, \quad \text{Var}(dZ(f)) = dS^{(I)}(f), \quad \text{Cov}(dZ(f), dZ(f')) = 0.$$

*Proof idea*: build  $Z$  as the  $L^2$  (mean-square) limit of finite Fourier sums of the  $X_t$ 's; orthogonality of increments mirrors the non-decrease of  $S^{(I)}$ , and the variance of an increment is exactly the mass  $S^{(I)}$  places there.

*Why*: the deep structural result: *every* stationary process is a superposition of uncorrelated random sinusoids  $e^{2\pi i f t} dZ(f)$ , with “amount of variance”  $dS^{(I)}(f)$  at each frequency. It makes precise the slogan “ $\mathcal{S}(f) df$  is the power in  $[f, f + df]$ ”.

### Recall: covariance of complex random variables

For complex  $Z_1, Z_2$  the covariance is  $\text{Cov}(Z_1, Z_2) = \mathbb{E}[(Z_1 - \mathbb{E}Z_1)(\overline{Z_2 - \mathbb{E}Z_2})]$  (note the conjugate on the second factor). The *complementary covariance* (relation) is  $\text{Rel}\{Z_1, Z_2\} = \mathbb{E}[(Z_1 - \mathbb{E}Z_1)(Z_2 - \mathbb{E}Z_2)]$  (no conjugate).

### Proposition 24: Periodogram as squared modulus of $J$

With  $T = \{1, \dots, n\}$  and  $Y_t = X_t - \bar{X}$ ,

$$\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2, \quad J(f) = \sqrt{\frac{1}{n}} \sum_{t \in T} Y_t e^{-2\pi i t f}.$$

*Proof idea:* starting from  $\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau} \frac{1}{n} \sum_t Y_t Y_{t+|\tau|} e^{-2\pi i \tau f}$ , re-index the double lag sum as a free double sum over  $s, t \in T$ ,

$$\hat{\mathcal{S}}^{(p)}(f) = \frac{1}{n} \sum_{s=1}^n \sum_{t=1}^n Y_t Y_s e^{-2\pi i s f} e^{2\pi i t f} = \left( \sqrt{\frac{1}{n}} \sum_t Y_t e^{-2\pi i t f} \right) \overline{\left( \sqrt{\frac{1}{n}} \sum_s Y_s e^{-2\pi i s f} \right)} = |J(f)|^2.$$

*Why:* lets the periodogram be computed by a single FFT in  $O(n \log n)$ , and makes manifest that  $\hat{\mathcal{S}}^{(p)} \geq 0$ .

### Theorem 25: Expectation of the periodogram: the Fejér kernel

For a zero-mean process,

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \sum_{\tau \in \mathbb{Z}} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f}, \quad w_{\tau} = \left(1 - \frac{|\tau|}{n}\right) \mathbf{1}_{[-n, n]}(\tau),$$

equivalently the frequency convolution

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') \mathcal{F}_n(f - f') df', \quad \mathcal{F}_n(f) = \begin{cases} \frac{1}{n} \frac{\sin^2(n\pi f)}{\sin^2(\pi f)}, & f \neq 0, \\ n, & f = 0, \end{cases}$$

the **Fejér kernel**.

*Proof idea:*  $\mathbb{E}[\hat{\gamma}_{\tau}] = (1 - \frac{|\tau|}{n})\gamma_{\tau}$  for  $|\tau| < n$ ; transform term by term. The triangular weight  $w_{\tau} = \frac{1}{n}(d * \tilde{d})_{\tau}$  is the autocorrelation of the rectangular window, whose transform is  $|\mathcal{D}_T(f)|^2 / n = \mathcal{F}_n(f)$ , so multiplication by  $w_{\tau}$  in time becomes convolution with  $\mathcal{F}_n$  in frequency.

*Why:* quantifies the **bias**. Because  $\mathcal{F}_n$  has wide side lobes that decay slowly,  $\mathbb{E}[\hat{\mathcal{S}}^{(p)}]$  is a *blurred* (leaked) version of  $\mathcal{S}$ ; peaks are smeared and power leaks into neighbouring frequencies. As  $n \rightarrow \infty$ ,  $\mathcal{F}_n$  concentrates at 0, so the periodogram is asymptotically unbiased.

### Theorem 26: Variance of the periodogram: inconsistency

For  $f \neq 0 \bmod \frac{1}{2}$ ,

$$\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \xrightarrow{n \rightarrow \infty} \mathcal{S}(f)^2 > 0,$$

and at distinct Fourier frequencies the ordinates are asymptotically uncorrelated.

*Proof idea:* for a Gaussian process  $J(f)$  is asymptotically complex Gaussian, so  $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$  is asymptotically  $\frac{1}{2}\mathcal{S}(f)\chi_2^2 = \mathcal{S}(f) \cdot \text{Exp}(1)$ , whose variance is  $\mathcal{S}(f)^2$ .

*Why:* the periodogram is **not consistent** — its variance does *not* vanish as  $n$  grows; more data merely gives more (equally noisy) ordinates. This is the motivation for smoothing/averaging (multitapering, or averaging neighbouring ordinates).

### 🔍 Classic Pitfall

The periodogram is *not* a consistent estimator of the SDF. Doubling  $n$  does **not** halve the variance at a fixed  $f$ ; it stays  $\approx \mathcal{S}(f)^2$ . “It got noisier as I added data” is expected behaviour, not a bug — you see finer frequency resolution but the same per-ordinate scatter. To reduce variance you must *average* (multitaper or smooth).

### Theorem 27: Expectation of the tapered periodogram

With a taper  $\{h_t\}$  (set  $h_t = 0$  off  $T$ ) and mean replaced by  $\mu$ ,

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df',$$

where  $H$  is the discrete Fourier transform of  $h$ .

*Proof idea:* expand  $\hat{\mathcal{S}}_h^{(p)}(f) = \sum_{t,s} h_t h_s Y_t Y_s e^{-2\pi i f(t-s)}$ , take expectations to get  $\sum_{\tau} w_{\tau} \gamma_{\tau} e^{-2\pi i f \tau}$  with  $w_{\tau} = \sum_t h_t h_{t+\tau} = (\tilde{h} * \tilde{h})_{\tau}$  (autocorrelation of the taper). By the convolution theorem its transform is  $\tilde{H}(f) \tilde{H}(f) = |H(f)|^2$ , giving the frequency convolution.

*Why:* the kernel is now  $|H(f)|^2$  instead of the Fejér kernel  $\mathcal{F}_n$ . A well-chosen taper (e.g. a dpss / Slepian taper) makes  $|H(f)|^2$  far more concentrated near 0 with much lower side lobes, so **tapering reduces bias / leakage**.

### Corollary 28: Unbiasedness of the tapered periodogram for white noise

If  $\{X_t\}$  is white noise with variance  $\sigma^2$  (so  $\mathcal{S} \equiv \sigma^2$ ) and  $\|h\|_2^2 = 1$ , then  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 = \mathcal{S}(f)$  for all  $n$  and all  $f$ .

*Proof idea:* the kernel integrates out,  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 \int_{-1/2}^{1/2} |H(f')|^2 df' = \sigma^2 \sum_t h_t^2 = \sigma^2 \|h\|_2^2$  (Parseval), which equals  $\sigma^2$  exactly when  $\|h\|_2^2 = 1$ .

*Why:* pins down the normalisation  $\|h\|_2^2 = 1$  in the definition of a taper, and shows tapering costs nothing in bias for flat spectra.

### Theorem 29: Variance reduction by multitapering

For  $K$  orthonormal tapers, asymptotically

$$\text{Cov}(\hat{\mathcal{S}}_{h_k}^{(p)}(f), \hat{\mathcal{S}}_{h_{k'}}^{(p)}(f)) \rightarrow \mathcal{S}(f)^2 \delta_{k,k'}, \quad \text{hence} \quad \text{Var}(\hat{\mathcal{S}}^{(mt)}(f)) \approx \frac{\mathcal{S}(f)^2}{K}.$$

*Proof idea:* orthogonality of the tapers makes the  $K$  individual tapered periodograms asymptotically uncorrelated, each with variance  $\mathcal{S}(f)^2$ ; averaging  $K$  uncorrelated terms divides the variance by  $K$ .

*Why:* **multitapering restores consistency**: as  $n \rightarrow \infty$  one may let  $K \rightarrow \infty$  (slowly) so  $\text{Var} \rightarrow 0$ . It is the standard fix for the periodogram’s non-vanishing variance, with only a modest bias increase.

### Bias vs. variance, the whole story in one line

The raw periodogram has tolerable (vanishing) *bias* but stubborn *variance*. **Tapering** attacks bias (sharper kernel  $|H|^2$  vs.  $\mathcal{F}_n$ ); **multitapering** attacks variance (average  $K$  near-independent estimates,  $\text{Var} \approx \mathcal{S}^2/K$ ). Together they give a usable spectral estimate.

### 6.3 Key Techniques

#### Key Technique 1: Compute an SDF from an ACVS

To get  $\mathcal{S}(f) = \sum_{\tau} \gamma_{\tau} e^{-2\pi i f \tau}$ :

1. Write down the ACVS (only finitely many lags for MA; geometric for AR).
2. Pair  $\tau$  with  $-\tau$  and use  $e^{-2\pi i f \tau} + e^{2\pi i f \tau} = 2 \cos(2\pi f \tau)$ , giving  $\mathcal{S}(f) = \gamma_0 + 2 \sum_{\tau \geq 1} \gamma_{\tau} \cos(2\pi f \tau)$  (real, even).
3. For a finite linear filter, prefer the factored form: if  $X_t = \sum_j \psi_j \varepsilon_{t-j}$  with  $\varepsilon$  white noise of variance  $\sigma^2$ , then  $\mathcal{S}(f) = \sigma^2 \left| \sum_j \psi_j e^{-2\pi i j f} \right|^2$ .

For an AR/ARMA  $\Phi(B)X_t = \Theta(B)\varepsilon_t$ ,  $\mathcal{S}(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2}$ . **Sanity checks:**  $\mathcal{S}$  must be real, even, non-negative, and integrate to  $\gamma_0$  over  $[-\frac{1}{2}, \frac{1}{2}]$ .

#### Key Technique 2: Test whether a candidate $\gamma$ is a valid ACVS

A sequence  $\{\gamma_{\tau}\}$  is a valid ACVS *iff* it is non-negative definite, which (in the summable case) is equivalent to its Fourier transform being non-negative:

$$\mathcal{S}(f) = \sum_{\tau} \gamma_{\tau} e^{-2\pi i f \tau} \geq 0 \quad \text{for all } f.$$

Procedure: (i) check  $\gamma_{-\tau} = \gamma_{\tau}$  and  $\gamma_0 \geq |\gamma_{\tau}|$  (necessary); (ii) compute  $\mathcal{S}(f)$ ; (iii) hunt for a frequency where  $\mathcal{S}(f) < 0$  — often the endpoints  $f = 0$  (gives  $\sum_{\tau} \gamma_{\tau}$ ) or  $f = \frac{1}{2}$  (gives  $\sum_{\tau} (-1)^{\tau} \gamma_{\tau}$ ) are the culprits. A single negative value **disqualifies** the candidate.

#### Key Technique 3: Recover an ACVS from a given SDF

Use  $\gamma_{\tau} = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$ . Practical moves:

- Drop the imaginary part: since  $\gamma_{\tau}$  is real and  $\mathcal{S}$  even,  $\gamma_{\tau} = \int_{-1/2}^{1/2} \mathcal{S}(f) \cos(2\pi f \tau) df = 2 \int_0^{1/2} \mathcal{S}(f) \cos(2\pi f \tau) df$ .
- For polynomial  $\mathcal{S}$  integrate by parts; the building blocks are  $\int_0^{1/2} \cos(2\pi f \tau) df = \frac{\sin(\pi \tau)}{2\pi \tau}$  and  $\int_0^{1/2} f \cos(2\pi f \tau) df = \frac{\sin(\pi \tau)}{4\pi \tau} + \frac{\cos(\pi \tau) - 1}{4\pi^2 \tau^2}$ .
- For a rational  $\mathcal{S}$  (ARMA), match it to  $\sigma^2 |\Theta|^2 / |\Phi|^2$  and read off the recursion, or do a partial-fraction / contour integral.
- Treat  $\tau = 0$  separately (the  $\sin(\pi \tau)/\tau$  terms need a limit).

#### Key Technique 4: Read off periodogram bias and variance

Given a periodogram problem, recall the three facts:

- **Bias / blurring:**  $\mathbb{E}[\hat{\mathcal{S}}^{(p)}] = \mathcal{S} * \mathcal{F}_n$ ; the Fejér kernel's slow-decaying side lobes leak power between frequencies. Sharp spectral peaks are the worst affected.
- **Variance:**  $\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \rightarrow \mathcal{S}(f)^2 \neq 0$  — not consistent.
- **Fixes:** tapering swaps  $\mathcal{F}_n$  for the sharper  $|H|^2$  (less bias); multitapering averages  $K$  tapers ( $\text{Var} \approx \mathcal{S}^2/K$ , less variance).

## 6.4 Exercises

### Exercise 1: Fourier transform property table [Sheet 6]

For  $g \in L^1$  with transform  $G$ , prove: (1) if  $h(t) = \overline{g(t)}$  then  $H(f) = \overline{G(-f)}$ ; (2) if  $h(t) = g(\alpha t)$ ,  $\alpha \neq 0$ , then  $H(f) = G(f/\alpha)/|\alpha|$ ; (3) if  $h(t) = g(t + \tau)$  then  $H(f) = G(f)e^{2\pi i f \tau}$ ; (4) if  $h(t) = \alpha g(t) + p(t)$  then  $H(f) = \alpha G(f) + P(f)$ .

#### Solution

(1) Conjugation: pull the conjugate outside (it flips the sign of the exponent),

$$H(f) = \int_{-\infty}^{\infty} \overline{g(t)} e^{-2\pi i f t} dt = \overline{\int_{-\infty}^{\infty} g(t) e^{2\pi i f t} dt} = \overline{G(-f)}.$$

(2) Scaling: substitute  $t' = \alpha t$ ,  $dt = dt'/|\alpha|$  (the  $|\alpha|$  comes from reversing limits when  $\alpha < 0$ ):

$$H(f) = \int_{-\infty}^{\infty} g(\alpha t) e^{-2\pi i f t} dt = \frac{1}{|\alpha|} \int_{-\infty}^{\infty} g(t') e^{-2\pi i (f/\alpha) t'} dt' = \frac{1}{|\alpha|} G(f/\alpha).$$

(3) Shift: insert  $1 = e^{-2\pi i f \tau} e^{2\pi i f \tau}$ ,

$$H(f) = \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f t} dt = \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f (t + \tau)} dt e^{2\pi i f \tau} = G(f) e^{2\pi i f \tau}.$$

(4) Linearity of the integral gives  $H = \alpha G + P$  immediately.

### Exercise 2: Continuous-time convolution theorem [Sheet 6]

Let  $g, h \in L^1$  and  $u = g * h$ . Prove  $U = G \cdot H$ .

#### Solution

$$\begin{aligned} U(f) &= \int_{-\infty}^{\infty} u(x) e^{-2\pi i x f} dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x - y) h(y) dy e^{-2\pi i x f} dx \\ &= \int_{-\infty}^{\infty} \left( \int_{-\infty}^{\infty} g(x - y) e^{-2\pi i x f} dx \right) h(y) dy \quad (\text{Fubini}) \\ &= \int_{-\infty}^{\infty} \left( \int_{-\infty}^{\infty} g(s) e^{-2\pi i s f} ds \right) e^{-2\pi i y f} h(y) dy \quad (s = x - y) \\ &= G(f) \int_{-\infty}^{\infty} h(y) e^{-2\pi i y f} dy = G(f) H(f). \end{aligned}$$

The interchange is justified because  $\iint |g(x - y) h(y)| dx dy = \|g\|_1 \|h\|_1 < \infty$ , so Fubini applies.

### Exercise 3: Aliasing via Poisson summation [Sheet 6]

For continuous  $g \in L^1$  with transform  $G$ , and  $G$  the transform of the sampled sequence  $\{g(t)\}_{t \in \Delta \mathbb{Z}}$ , show  $G(f) = \sum_{k \in \mathbb{Z}} G(f + k/\Delta)$  (under the stated summability hypotheses).

#### Solution

For  $t \in \Delta \mathbb{Z}$  both inversion formulas hold:

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df, \quad g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t} df.$$

Split the real line into period blocks centred at  $k/\Delta$  and shift each:

$$\begin{aligned} g(t) &= \sum_{k \in \mathbb{Z}} \int_{k/\Delta - 1/(2\Delta)}^{k/\Delta + 1/(2\Delta)} G(f) e^{2\pi i f t} df = \sum_{k \in \mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f + k/\Delta) e^{2\pi i (f + k/\Delta)t} df \\ &= \sum_{k \in \mathbb{Z}} \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f + k/\Delta) e^{2\pi i f t} df, \end{aligned}$$

because  $t = z\Delta$  for some  $z \in \mathbb{Z}$  makes  $e^{2\pi i (k/\Delta)t} = e^{2\pi i k z} = 1$ . By the summability hypothesis  $\sum_k \int |G(f + k/\Delta)| df < \infty$  we may swap sum and integral:

$$\int_{-1/(2\Delta)}^{1/(2\Delta)} \left( \sum_{k \in \mathbb{Z}} G(f + k/\Delta) \right) e^{2\pi i f t} df = g(t) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i f t} df.$$

Equality of these Fourier-series coefficients (for all  $t \in \Delta\mathbb{Z}$ ) forces  $\sum_k G(f + k/\Delta) = G(f)$  for a.e.  $f$ ; both sides are continuous (by the  $\ell^1$ /summability hypotheses), so equality holds for all  $f$ .

#### Exercise 4: SDF of white noise [Sheet 7]

Let  $\{\varepsilon_t\}$  be white noise with variance  $\sigma_\varepsilon^2$ . Find its SDF.

#### Solution

The ACVS is  $\gamma_\tau = \sigma_\varepsilon^2$  for  $\tau = 0$  and 0 otherwise, so only the  $\tau = 0$  term survives:

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau} = \sigma_\varepsilon^2, \quad \text{for all } f.$$

The spectrum is **flat** — equal power at every frequency, which is why it is called “white”.

#### Exercise 5: Is a truncated ACVS a valid ACVS? [Sheet 7]

Let  $\gamma_\tau = 1$  for  $|\tau| \leq K$  and 0 otherwise,  $K$  a positive integer. Is  $\{\gamma_\tau\}$  a valid ACVS? What would the SDF be?

#### Solution

It is summable and symmetric, so we may form

$$\mathcal{S}(f) = \sum_{\tau=-K}^K e^{-2\pi i \tau f} = 1 + 2 \sum_{\tau=1}^K \cos(2\pi \tau f) = \frac{\sin((2K+1)\pi f)}{\sin(\pi f)}$$

(the Dirichlet kernel). This is *not* non-negative everywhere. Evaluate at the Nyquist endpoint  $f = \frac{1}{2}$ : using  $\cos(\pi \tau) = (-1)^\tau$ ,

$$\mathcal{S}\left(\frac{1}{2}\right) = 1 + 2 \sum_{\tau=1}^K (-1)^\tau = (-1)^K.$$

So  $\mathcal{S}(\frac{1}{2}) = -1 < 0$  whenever  $K$  is **odd** (write  $K = 2q + 1$ ). Since the SDF goes negative,  $\{\gamma_\tau\}$  is **not** a valid ACVS for any stationary process. (It fails to be non-negative definite.)

### Intuition / Why it matters

Truncating a genuine ACVS to a finite window in general breaks non-negative-definiteness — exactly why one cannot just chop the ACVS and hope. The clean way to “truncate” is to taper the spectrum, not the ACVS.

### Exercise 6: ACVS from a triangular SDF [Sheet 7]

Find the ACVS of  $\{X_t\}$  with  $\mathcal{S}(f) = \frac{\sigma^2}{\pi} (1 - 2|f|) \mathbf{1}_{\{|f| \leq 1/2\}}$ .

### Solution

For  $\tau \neq 0$ , since  $\mathcal{S}$  is even the sin part drops out:

$$\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i f \tau} \mathcal{S}(f) df = \frac{2\sigma^2}{\pi} \int_0^{1/2} \cos(2\pi f \tau) (1 - 2f) df = \frac{2\sigma^2}{\pi} (I_1 - 2I_2),$$

with  $I_1 = \int_0^{1/2} \cos(2\pi f \tau) df$  and  $I_2 = \int_0^{1/2} f \cos(2\pi f \tau) df$ . Computing,

$$I_1 = \frac{\sin(\pi\tau)}{2\pi\tau}, \quad I_2 = \frac{\sin(\pi\tau)}{4\pi\tau} + \frac{\cos(\pi\tau) - 1}{4\pi^2\tau^2}.$$

Then  $I_1 - 2I_2 = -\frac{\cos(\pi\tau) - 1}{2\pi^2\tau^2} = \frac{1 - \cos(\pi\tau)}{2\pi^2\tau^2}$ , so

$$\boxed{\gamma_\tau = \frac{\sigma^2}{\pi^3} \frac{1 - \cos(\pi\tau)}{\tau^2}} \quad (\tau \neq 0).$$

For  $\tau = 0$ ,  $\gamma_0 = \frac{2\sigma^2}{\pi} \int_0^{1/2} (1 - 2f) df = \frac{2\sigma^2}{\pi} \cdot \frac{1}{4} = \frac{\sigma^2}{2\pi}$ . (Note  $1 - \cos(\pi\tau) = 0$  for even  $\tau$ ,  $= 2$  for odd  $\tau$ , so the ACVS is supported on odd lags plus  $\tau = 0$ . Numerically:  $\gamma_0 = \sigma^2/(2\pi)$ ,  $\gamma_1 = 2\sigma^2/\pi^3$ ,  $\gamma_2 = 0$ ,  $\gamma_3 = 2\sigma^2/(9\pi^3)$ , all verified.)

### Exercise 7: SDF of an MA(q) process [Sheet 7]

What is the SDF of an MA(q) process  $X_t = \sum_{j=0}^q \theta_j \varepsilon_{t-j}$  (here writing  $\theta_0 = 1$  for concreteness,  $\varepsilon$  white noise of variance  $\sigma_\varepsilon^2$ )?

### Solution

Its ACVS is  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|}$  for  $|\tau| \leq q$  (zero otherwise), which we write as the double sum  $\gamma_\tau = \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \mathbf{1}_{\{\tau=k-j\}}$ . Since  $\gamma \in \ell^1$  the SDF exists:

$$\begin{aligned} \mathcal{S}(f) &= \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i \tau f} = \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k \sum_{\tau} \mathbf{1}_{\{\tau=k-j\}} e^{-2\pi i \tau f} \\ &= \sigma_\varepsilon^2 \sum_{j=0}^q \sum_{k=0}^q \theta_j \theta_k e^{-2\pi i (k-j)f} = \sigma_\varepsilon^2 \left( \sum_{j=0}^q \theta_j e^{-2\pi i j f} \right) \overline{\left( \sum_{k=0}^q \theta_k e^{-2\pi i k f} \right)} \\ &= \sigma_\varepsilon^2 \left| \sum_{j=0}^q \theta_j e^{-2\pi i j f} \right|^2 = \sigma_\varepsilon^2 \left| \Theta(e^{-2\pi i f}) \right|^2. \end{aligned}$$

The SDF is the squared modulus of the MA transfer function — automatically non-negative, as any SDF must be.

### Exercise 8: Time-varying-amplitude harmonic process [Sheet 7]

Let  $X_t = \varepsilon_t \cos(\nu t + \Theta)$  with  $\nu$  fixed,  $\varepsilon_t$  mean-zero white noise of variance  $\sigma^2$  (not constrained positive), independent of the random phase  $\Theta$  whose pdf is  $f_\Theta(\theta) = \frac{1}{2\pi}(1 + \cos \theta)$  on  $|\theta| \leq \pi$ . Find the first two moments of  $X_t$ ; are they  $t$ -free?

#### Solution

**Mean.** By independence,  $\mathbb{E}[X_t] = \mathbb{E}[\varepsilon_t] \mathbb{E}[\cos(\nu t + \Theta)] = 0 \cdot (\cdot) = 0$ , constant in  $t$  (the second factor is finite since  $|\cos| \leq 1$ ).

**Autocovariance.** By independence,  $\mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[\varepsilon_t \varepsilon_{t+\tau}] \mathbb{E}[\cos(\nu t + \Theta) \cos(\nu(t + \tau) + \Theta)]$ . White noise gives  $\mathbb{E}[\varepsilon_t \varepsilon_{t+\tau}] = \sigma^2 \mathbf{1}_{\{\tau=0\}}$ . Using  $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ ,

$$\mathbb{E}[\cos(\nu t + \Theta) \cos(\nu(t + \tau) + \Theta)] = \frac{1}{2}[\cos(\nu\tau) + \mathbb{E}[\cos(\nu(2t + \tau) + 2\Theta)]].$$

The remaining expectation vanishes for this  $\Theta$ :

$$\begin{aligned} \mathbb{E}[\cos(\nu(2t + \tau) + 2\Theta)] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\nu(2t + \tau) + 2\theta) d\theta \\ &\quad + \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\nu(2t + \tau) + 2\theta) \cos \theta d\theta. \end{aligned}$$

The first integral is  $\frac{1}{4\pi} [\sin(\nu(2t + \tau) + 2\theta)]_{-\pi}^{\pi} = 0$ . For the second, write  $\cos(\cdots + 2\theta) \cos \theta = \frac{1}{2}[\cos(\cdots + \theta) + \cos(\cdots + 3\theta)]$ ; both pieces integrate to 0 over the full period. Hence  $\mathbb{E}[\cos(\nu(2t + \tau) + 2\Theta)] = 0$ , and

$$\mathbb{E}[X_t X_{t+\tau}] = \sigma^2 \mathbf{1}_{\{\tau=0\}} \cdot \frac{1}{2} \cos(\nu\tau) = \begin{cases} \sigma^2/2, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases}$$

Both moments are independent of  $t$ , so  $\{X_t\}$  is stationary; in fact it is white noise with variance  $\sigma^2/2$ . (The non-uniform phase did not spoil stationarity precisely because its  $2\theta$ -harmonics integrate out.)

### Exercise 9: Periodogram as $|J(f)|^2$ [Sheet 8]

With  $T = \{1, \dots, n\}$ , show  $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$  for  $J(f) = \sqrt{1/n} \sum_{t \in T} (X_t - \bar{X}) e^{-2\pi i t f}$ .

#### Solution

Put  $Y_t = X_t - \bar{X}$ . Starting from the definition and re-indexing the lag sum as a free double sum over  $s, t \in \{1, \dots, n\}$  (the substitution  $\tau = t - s$  sweeps all  $|\tau| \leq n - 1$ ),

$$\begin{aligned} \hat{\mathcal{S}}^{(p)}(f) &= \sum_{\tau=-n+1}^{n-1} \frac{1}{n} \sum_{t=1}^{n-|\tau|} Y_t Y_{t+|\tau|} e^{-2\pi i \tau f} = \frac{1}{n} \sum_{s=1}^n \sum_{t=1}^n Y_t Y_s e^{-2\pi i s f} e^{2\pi i t f} \\ &= \left( \sqrt{\frac{1}{n}} \sum_{t=1}^n Y_t e^{-2\pi i t f} \right) \overline{\left( \sqrt{\frac{1}{n}} \sum_{s=1}^n Y_s e^{-2\pi i s f} \right)} = J(f) \overline{J(f)} = |J(f)|^2. \end{aligned}$$

(In the second equality we relabel  $s = t + |\tau|$  so the lag  $\tau = t - s$  runs freely, and  $e^{-2\pi i \tau f} = e^{2\pi i t f} e^{-2\pi i s f}$  separates the two conjugate sums.)

### Exercise 10: Expectation of the tapered periodogram [Sheet 8]

Replacing  $\bar{X}$  by  $\mu$ , show  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df'$ , with  $H$  the transform of  $h$ .

### Solution

With  $Y_t = X_t - \mu$  and  $g_t = h_t$  ( $g_t = 0$  off  $T$ ),

$$\hat{\mathcal{S}}_h^{(p)}(f) = \left| \sum_{t=1}^n g_t Y_t e^{-2\pi i f t} \right|^2 = \sum_{\tau=-n+1}^{n-1} \sum_t g_t g_{t+|\tau|} Y_t Y_{t+\tau} e^{-2\pi i f \tau}.$$

Taking expectations ( $\mathbb{E}[Y_t Y_{t+\tau}] = \gamma_\tau$ ),

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sum_{\tau \in \mathbb{Z}} w_\tau \gamma_\tau e^{-2\pi i f \tau}, \quad w_\tau = \sum_{t \in \mathbb{Z}} h_t h_{t+\tau}.$$

By the convolution theorem  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') W(f - f') df'$  where  $W$  is the transform of  $w$ . Setting  $\tilde{h}_t = h_{-t}$  and  $\bar{h}_t = h_t$ , we recognise  $w = \tilde{h} * \bar{h}$ , so by the convolution theorem again  $W(f) = \tilde{H}(f) \bar{H}(f) = |H(f)|^2$ . Therefore  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df'$ .

### Exercise 11: Unbiased tapered periodogram for white noise [Sheet 8]

If  $\{X_t\}$  is white noise with variance  $\sigma^2$ , prove  $\hat{\mathcal{S}}_h^{(p)}(f)$  is unbiased for all  $n$  provided  $\|h\|_2^2 = 1$ .

### Solution

White noise has  $\mathcal{S} \equiv \sigma^2$ . Using the tapered-expectation formula and relabelling (periodicity),

$$\begin{aligned} \mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] &= \int_{-1/2}^{1/2} \mathcal{S}(f - f') |H(f')|^2 df' = \sigma^2 \int_{-1/2}^{1/2} |H(f')|^2 df' \\ &= \sigma^2 \sum_{t \in \mathbb{Z}} \sum_{s \in \mathbb{Z}} h_t h_s \int_{-1/2}^{1/2} e^{-2\pi i(t-s)f'} df' = \sigma^2 \sum_t h_t^2 = \sigma^2 \|h\|_2^2, \end{aligned}$$

where  $\int_{-1/2}^{1/2} e^{-2\pi i \tau f} df = \mathbf{1}_{\{\tau=0\}}$  collapses the double sum (Parseval). Hence if  $\|h\|_2^2 = 1$  then  $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 = \mathcal{S}(f)$ , i.e. exactly unbiased.

### Exercise 12: SDF of an MA(1) process [New]

Let  $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$  with  $\varepsilon$  white noise of variance  $\sigma^2$ . Find  $\mathcal{S}(f)$  two ways (from the ACVS and from the transfer function), and locate its maximum and minimum.

### Solution

**From the ACVS.**  $\gamma_0 = \sigma^2(1 + \theta^2)$ ,  $\gamma_{\pm 1} = -\sigma^2\theta$ , 0 otherwise. Then

$$\mathcal{S}(f) = \gamma_0 + 2\gamma_1 \cos(2\pi f) = \sigma^2(1 + \theta^2 - 2\theta \cos(2\pi f)).$$

**From the transfer function.**  $\Theta(z) = 1 - \theta z$ , so

$$\mathcal{S}(f) = \sigma^2 \left| 1 - \theta e^{-2\pi i f} \right|^2 = \sigma^2 (1 - \theta e^{-2\pi i f})(1 - \theta e^{2\pi i f}) = \sigma^2(1 + \theta^2 - 2\theta \cos(2\pi f)),$$

the same answer. Since  $\cos(2\pi f)$  ranges over  $[-1, 1]$ : for  $\theta > 0$  the **minimum** is at  $f = 0$ ,  $\mathcal{S}(0) = \sigma^2(1 - \theta)^2$ , and the **maximum** at  $f = \frac{1}{2}$ ,  $\mathcal{S}(\frac{1}{2}) = \sigma^2(1 + \theta)^2$  (high-pass shape); for  $\theta < 0$  these swap (low-pass). The SDF is non-negative everywhere, as required (it touches 0 only on the unit circle when  $|\theta| = 1$ ).

### Exercise 13: SDF of an AR(1) process [New]

Let  $X_t = \phi X_{t-1} + \varepsilon_t$ ,  $|\phi| < 1$ ,  $\varepsilon$  white noise of variance  $\sigma^2$ . Find  $\mathcal{S}(f)$  from the ACVS and confirm it via the transfer function. Describe its shape for  $\phi > 0$  and  $\phi < 0$ .

#### Solution

The ACVS is  $\gamma_\tau = \frac{\sigma^2}{1 - \phi^2} \phi^{|\tau|}$ . Summing the two geometric tails,

$$\begin{aligned} \mathcal{S}(f) &= \frac{\sigma^2}{1 - \phi^2} \sum_{\tau \in \mathbb{Z}} \phi^{|\tau|} e^{-2\pi i f \tau} = \frac{\sigma^2}{1 - \phi^2} \left( 1 + \sum_{\tau \geq 1} \phi^\tau (e^{-2\pi i f \tau} + e^{2\pi i f \tau}) \right) \\ &= \frac{\sigma^2}{1 - \phi^2} \left( 1 + \frac{\phi e^{-2\pi i f}}{1 - \phi e^{-2\pi i f}} + \frac{\phi e^{2\pi i f}}{1 - \phi e^{2\pi i f}} \right) = \frac{\sigma^2}{1 - \phi^2} \cdot \frac{1 - \phi^2}{1 - 2\phi \cos(2\pi f) + \phi^2} \\ &= \frac{\sigma^2}{1 - 2\phi \cos(2\pi f) + \phi^2}. \end{aligned}$$

**Transfer-function check.**  $\Phi(z) = 1 - \phi z$ , so  $\mathcal{S}(f) = \sigma^2 / |1 - \phi e^{-2\pi i f}|^2 = \sigma^2 / (1 - 2\phi \cos(2\pi f) + \phi^2)$  — identical. For  $\phi > 0$  the denominator is smallest at  $f = 0$ , so  $\mathcal{S}$  peaks at low frequency (slowly varying, “red” spectrum); for  $\phi < 0$  it peaks at  $f = \frac{1}{2}$  (rapidly alternating, “blue” spectrum). As  $\phi \rightarrow 1$  the peak at 0 blows up (near-unit-root, very long-range dependence).

### Exercise 14: Endpoint test for a candidate ACVS [New]

Decide whether  $\gamma_0 = 1$ ,  $\gamma_{\pm 1} = \rho$ ,  $\gamma_\tau = 0$  ( $|\tau| \geq 2$ ) is a valid ACVS, as a function of  $\rho \in \mathbb{R}$ .

#### Solution

This is the ACVS shape of an MA(1). Compute the SDF:

$$\mathcal{S}(f) = 1 + 2\rho \cos(2\pi f).$$

Since  $\cos(2\pi f) \in [-1, 1]$ , the minimum of  $\mathcal{S}$  is  $1 - 2|\rho|$ , attained at  $f = 0$  (if  $\rho < 0$ ) or  $f = \frac{1}{2}$  (if  $\rho > 0$ ). Non-negativity  $\mathcal{S}(f) \geq 0$  for all  $f$  holds iff  $1 - 2|\rho| \geq 0$ , i.e.

$$|\rho| \leq \frac{1}{2}$$

For  $|\rho| > \frac{1}{2}$  the SDF dips below zero and the sequence is **not** a valid ACVS. (Consistent with the fact that an MA(1) cannot have  $|\rho_1| > \frac{1}{2}$ .)

### Exercise 15: Variance from the SDF; Parseval [New]

For a stationary process with SDF  $\mathcal{S}$ , show  $\text{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f) df$ , and use it to read off  $\gamma_0$  for the AR(1) SDF  $\mathcal{S}(f) = \sigma^2 / (1 - 2\phi \cos(2\pi f) + \phi^2)$ .

#### Solution

By the inversion formula at lag  $\tau = 0$ ,

$$\gamma_0 = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \cdot 0} df = \int_{-1/2}^{1/2} \mathcal{S}(f) df = \text{Var}(X_t),$$

the spectral statement of Parseval: total variance equals total spectral mass. For the AR(1), rather than integrate directly we use the known  $\gamma_0 = \sigma^2 / (1 - \phi^2)$ ; consistency is the standard

integral

$$\int_{-1/2}^{1/2} \frac{\sigma^2 df}{1 - 2\phi \cos(2\pi f) + \phi^2} = \frac{\sigma^2}{1 - \phi^2},$$

which one can verify by residues (substitute  $z = e^{2\pi i f}$ ; the integrand has poles at  $z = \phi$  and  $z = 1/\phi$ , only  $z = \phi$  lying inside the unit circle). Thus  $\text{Var}(X_t) = \sigma^2/(1 - \phi^2)$ , matching the time-domain computation.

## 7 Multivariate Time Series and Vector Autoregression (VAR)

So far each  $X_t$  has been a scalar. Many real systems are  $d$ -dimensional: a basket of normalised stock prices (Apple, Nasdaq-100, S&P 500), or the longitude/latitude/temperature of an ocean drifter. We now let  $\{X_t\}$  take values in  $\mathbb{R}^d$  and ask two questions: (i) how do we *describe* the second-order structure — now a matrix-valued autocovariance and a matrix-valued spectrum, with genuine **cross**-terms between components, and (ii) how do we *model* it. The canonical model is the vector autoregression  $\text{VAR}(p)$ , and the key trick is that *every*  $\text{VAR}(p)$  is a  $\text{VAR}(1)$  in disguise (companion form), so its theory reduces to that of a single matrix  $\Phi_1$  (or  $F$ ).

### 7.1 Definitions

#### Definition 1: Multivariate time series

A **multivariate time series** is a real  $d$ -vector-valued discrete-time stochastic process

$$X_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(d)})^\top, \quad t \in \mathbb{Z},$$

whose marginal processes  $\{X_t^{(1)}\}, \dots, \{X_t^{(d)}\}$  are each univariate time series. When  $d = 2$  we call it **bivariate**.

#### Definition 2: Multivariate second-order stationarity

$\{X_t\}$  is **multivariate second-order (weakly) stationary** if for all  $t, s, \tau \in \mathbb{Z}$ ,

$$\mathbb{E}[X_t] = \mathbb{E}[X_s], \quad \text{Cov}(X_{t+\tau}, X_t) = \text{Cov}(X_{s+\tau}, X_s), \quad \text{tr}(\text{Var}(X_t)) < \infty.$$

Equivalently: each marginal  $\{X_t^{(j)}\}$  is second-order stationary **and** every pair has a shift-invariant cross-covariance,  $\text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)}) = \text{Cov}(X_{s+\tau}^{(j)}, X_s^{(k)})$  for all  $1 \leq j, k \leq d$ . (Here  $\text{Cov}(U, V)$  of two vectors is the matrix with  $(j, k)$  entry  $\text{Cov}(U^{(j)}, V^{(k)})$ .)

#### Definition 3: Matrix autocovariance sequence (ACVS)

For a multivariate second-order stationary  $\{X_t\}$ , the **(matrix) autocovariance sequence** is the  $d \times d$  matrix-valued sequence

$$\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t), \quad \tau \in \mathbb{Z},$$

with  $(j, k)$  entry  $\gamma_\tau^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)})$ .

- **The lag  $\tau$  sits in the FIRST argument.** In the scalar case this is immaterial, but here it determines the convention for  $\Gamma_{-\tau} = \Gamma_\tau^\top$  below.
- When  $j = k$ ,  $\gamma_\tau^{(j,j)}$  is the ordinary ACVS of the marginal  $\{X_t^{(j)}\}$ .
- When  $j \neq k$ ,  $\gamma_\tau^{(j,k)}$  is called the **cross-covariance sequence** (CCS) between components  $j$  and  $k$ .

#### Definition 4: Cross-correlation sequence (CCS)

The **cross-correlation sequence** between components  $j$  and  $k$  is

$$\rho_{\tau}^{(j,k)} = \frac{\gamma_{\tau}^{(j,k)}}{\sqrt{\gamma_0^{(j,j)} \gamma_0^{(k,k)}}}.$$

For  $j = k$  this is the ordinary ACF (so  $\rho_0^{(j,j)} = 1$ ). For  $j \neq k$  it need *not* equal 1 at  $\tau = 0$  and need *not* be symmetric in  $\tau$ ; it satisfies  $\rho_{\tau}^{(j,k)} = \rho_{-\tau}^{(k,j)}$ .

#### Definition 5: Multivariate white noise $\text{WN}(0, \Sigma)$

A  $d$ -variate series  $\{\varepsilon_t\}$  is **white noise with mean 0 and covariance matrix  $\Sigma$** , written  $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$ , iff it is stationary with mean vector 0 and

$$\Gamma_{\tau}^{(\varepsilon)} = \begin{cases} \Sigma & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

- There *can* be correlation between different components at lag 0 (off-diagonal entries of  $\Sigma$ ): the components are uncorrelated **across time** but possibly correlated **contemporaneously**.
- Often one further assumes  $\Sigma$  *diagonal* (componentwise independent noise), but this is an extra assumption, not part of the definition.

#### Definition 6: Uncorrelated processes

Two jointly second-order stationary processes  $\{X_t^{(1)}\}$  and  $\{X_t^{(2)}\}$  are **uncorrelated** if their cross-covariance vanishes at *every* lag:

$$\gamma_{\tau}^{(1,2)} = 0 \quad \text{for all } \tau \in \mathbb{Z}.$$

#### Definition 7: Spectral density matrix function

Assume  $\{\gamma_{\tau}^{(j,k)}\}_{\tau \in \mathbb{Z}} \in \ell^1$  for all  $1 \leq j, k \leq d$  (absolutely summable cross-covariances). The **spectral density matrix** is the  $d \times d$  matrix-valued function

$$S(f) = \sum_{\tau \in \mathbb{Z}} \Gamma_{\tau} e^{-2\pi i \tau f}, \quad f \in [-\tfrac{1}{2}, \tfrac{1}{2}],$$

with inverse  $\Gamma_{\tau} = \int_{-1/2}^{1/2} S(f) e^{2\pi i f \tau} df$ .

- The  $(j, k)$  entry is  $S_{j,k}(f)$ . For  $j = k$  it is the ordinary (real-valued) spectral density  $\mathcal{S}_j(f)$ .
- For  $j \neq k$ ,  $S_{j,k}(f)$  is the **cross-spectral density**, which is *complex-valued* in general.

#### Definition 8: Coherence

The **coherence** between components  $j$  and  $k$  is

$$r_{j,k}(f) = \frac{S_{j,k}(f)}{\sqrt{S_{j,j}(f) S_{k,k}(f)}} \in \mathbb{C},$$

the frequency-domain analogue of a correlation between the increments  $dZ_j$  and  $dZ_k$ . One reports:

- its **magnitude**  $|r_{j,k}(f)|$  (or magnitude-squared  $|r_{j,k}(f)|^2 \in [0, 1]$ ), the strength of frequency-by-frequency association;

- its **phase**  $\arg r_{j,k}(f)$ , the “sign” / lead-lag angle in the complex plane.

#### Definition 9: Vector autoregression VAR( $p$ )

Let  $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$  be  $d$ -variate. A process  $\{X_t\}$  is a **vector autoregressive process of order  $p$** , VAR( $p$ ), if

$$X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \cdots + \Phi_p X_{t-p} + \varepsilon_t, \quad \Phi_j \in \mathbb{R}^{d \times d}, \Phi_p \neq 0.$$

Defining the matrix polynomial  $\Phi(z) = I_d - \sum_{j=1}^p \Phi_j z^j$ , this is written compactly as

$$\Phi(B)X_t = \varepsilon_t,$$

where  $B$  is the backshift operator. The same regressors (past values of the *whole* vector  $X_t$ ) appear in every component equation; the off-diagonal entries of the  $\Phi_j$  are exactly the cross-component dynamics.

#### Definition 10: Companion form of a VAR( $p$ )

Stack the last  $p$  vectors into  $Y_t = (X_t^\top, X_{t-1}^\top, \dots, X_{t-p+1}^\top)^\top \in \mathbb{R}^{dp}$ . Then a VAR( $p$ ) is equivalently the VAR(1)

$$\underbrace{Y_t}_{dp \times 1} = \underbrace{F}_{dp \times dp} \underbrace{Y_{t-1}}_{dp \times 1} + \underbrace{U_t}_{dp \times 1}, \quad F = \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \cdots & 0 & 0 \\ 0 & I_d & & \vdots & \vdots \\ \vdots & & \ddots & 0 & 0 \\ 0 & \cdots & 0 & I_d & 0 \end{pmatrix}, \quad U_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The top block row reproduces the VAR( $p$ ) equation; the lower rows are identities  $X_{t-i} = X_{t-i}$  that shift the stack. Recover the original process via  $X_t = GY_t$  with  $G = (I_d \ 0 \ \cdots \ 0) \in \mathbb{R}^{d \times dp}$ .

#### Definition 11: Stability of a VAR( $p$ )

A VAR( $p$ ) is **stable** if the roots of the **reverse characteristic polynomial**

$$\det(I_d - \Phi_1 z - \Phi_2 z^2 - \cdots - \Phi_p z^p) = 0$$

all lie *strictly outside* the complex unit circle ( $|z| > 1$ ). Equivalently, the companion matrix  $F$  has all eigenvalues of modulus *strictly less than one* ( $|\lambda| < 1$ ). Stability  $\Rightarrow$  stationarity (but not conversely in general).

#### Two equivalent stability tests — and why they are reciprocal

The reverse polynomial  $\det(I_d - \sum_j \Phi_j z^j)$  uses  $z = 1/\lambda$ , so “roots  $z$  outside the unit circle” is exactly “eigenvalues  $\lambda$  of  $F$  inside the unit circle”. For a VAR(1),  $\det(I_d - \Phi_1 z) = 0$  means  $1/z$  is an eigenvalue of  $\Phi_1$ , i.e.  $z = 1/\lambda_i(\Phi_1)$ ; demanding  $|z| > 1$  is demanding  $|\lambda_i(\Phi_1)| < 1$ . This mirrors the scalar AR causality condition “roots of  $\phi(z)$  outside the unit circle”.

## 7.2 Theorems & Propositions

### Proposition 12: Symmetry of the matrix ACVS

For a multivariate second-order stationary process,

$$\gamma_{\tau}^{(j,k)} = \gamma_{-\tau}^{(k,j)} \iff \Gamma_{-\tau} = \Gamma_{\tau}^{\top}.$$

*Proof idea:*  $\gamma_{\tau}^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)}) = \text{Cov}(X_t^{(k)}, X_{t+\tau}^{(j)})$  (covariance is symmetric in its arguments)  $= \text{Cov}(X_{s-\tau}^{(k)}, X_s^{(j)})$  (stationarity, set  $s = t + \tau$ )  $= \gamma_{-\tau}^{(k,j)}$ .

*Why:* the cross-covariance need *not* be even in  $\tau$  ( $\gamma_{\tau}^{(j,k)} \neq \gamma_{-\tau}^{(j,k)}$  in general); only the *transpose* symmetry  $\Gamma_{-\tau} = \Gamma_{\tau}^{\top}$  holds. This asymmetry is precisely what encodes lead-lag relationships between channels.

### Classic Pitfall

In the univariate case  $\gamma_{-\tau} = \gamma_{\tau}$  (even). Do *not* carry this over: in the multivariate case the diagonal entries are still even, but the off-diagonal cross-covariances generally satisfy only  $\gamma_{\tau}^{(j,k)} = \gamma_{-\tau}^{(k,j)}$ . Forgetting to swap the indices  $(j, k) \rightarrow (k, j)$  when flipping the sign of  $\tau$  is the classic error.

### Proposition 13: Positive semi-definiteness of the matrix ACVS

The matrix sequence  $\{\Gamma_{\tau}\}$  is positive semi-definite: for every  $n \in \mathbb{N}$  and every  $a_1, \dots, a_n \in \mathbb{R}^d$ ,

$$\sum_{j=1}^n \sum_{k=1}^n a_j^{\top} \Gamma_{k-j} a_k \geq 0.$$

*Proof idea:* the left side equals  $\text{Var}(\sum_j a_j^{\top} X_{t_j}) \geq 0$  for suitable times (the scalar random variable  $\sum_j a_j^{\top} X_{t_j}$  has non-negative variance).

*Why:* this is the defining algebraic constraint a valid matrix ACVS must satisfy; a candidate matrix sequence violating it cannot be the autocovariance of any process.

### Proposition 14: Contemporaneous correlation induced by shared noise

Let  $X_t = (X_t^{(1)}, X_t^{(2)})^{\top}$  be second-order stationary with ACVS  $\Gamma_{\tau}^{(X)}$ , and let  $\{\phi_t\}$  be a univariate mean-zero white noise (variance  $\sigma_{\phi}^2$ ) independent of  $\{X_t\}$ . Define  $Y_t = X_t + \phi_t \mathbf{1}$  (the *same* scalar noise added to both components), where  $\mathbf{1} = (1, 1)^{\top}$ . Then

$$\Gamma_{\tau}^{(Y)} = \Gamma_{\tau}^{(X)} + \sigma_{\phi}^2 \delta_{\tau,0} \mathbf{1} \mathbf{1}^{\top}, \quad \mathbf{1} \mathbf{1}^{\top} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

*Proof idea:*  $\Gamma_{\tau}^{(Y)} = \text{Cov}(X_{t+\tau} + \phi_{t+\tau} \mathbf{1}, X_t + \phi_t \mathbf{1})$ ; independence kills the  $X$ - $\phi$  cross terms, and  $\text{Cov}(\phi_{t+\tau}, \phi_t) = \sigma_{\phi}^2 \delta_{\tau,0}$  contributes  $\sigma_{\phi}^2 \delta_{\tau,0}$  to *every* entry.

*Why:* adding a common shock creates lag-0 correlation between  $Y^{(1)}$  and  $Y^{(2)}$  *regardless* of the correlation in  $X$ . This is the prototypical mechanism behind “everything moves together” contemporaneous correlation.

### Proposition 15: Hermitian symmetry of the spectral matrix

For all  $f \in [-\frac{1}{2}, \frac{1}{2}]$  and all  $1 \leq j, k \leq d$ ,

$$S_{j,k}(f) = \overline{S_{k,j}(f)}, \quad S_{j,k}(f) = \overline{S_{j,k}(-f)},$$

equivalently in matrix form

$$S(f) = S(f)^H, \quad S(f) = S(-f)^T,$$

and  $S(f)$  is positive semi-definite for every  $f$ . *Proof idea:* substitute  $\Gamma_{-\tau} = \Gamma_{\tau}^T$  into  $S(f) = \sum_{\tau} \Gamma_{\tau} e^{-2\pi i \tau f}$  and relabel  $\tau \rightarrow -\tau$ ; the conjugate of  $e^{-2\pi i \tau f}$  flips the sign in the exponent. *Why:*  $S(f)^H = S(f)$  (Hermitian) guarantees real diagonal spectra and makes coherence well defined; positive semi-definiteness is the frequency-domain counterpart of Proposition 13.

### Theorem 16: Multivariate spectral representation theorem

Let  $\{X_t\}$  be a vector-valued discrete-time stationary process with mean  $\mu$ . There exists a vector-valued orthogonal-increment process  $\{Z(f)\}$  on  $[-\frac{1}{2}, \frac{1}{2}]$  such that

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f).$$

If moreover  $\{\gamma_{\tau}^{(j,k)}\} \in \ell^1$ , then for  $f, f' \in [-\frac{1}{2}, \frac{1}{2}]$ ,

$$\mathbb{E}[dZ(f)] = 0, \quad \text{Var}(dZ(f)) = S(f) df, \quad \text{Cov}(dZ(f), dZ(f')) = 0 \quad (f \neq f'),$$

where  $S(f)$  is the spectral density *matrix* and the 0 in the last identity is the zero matrix.

*Proof idea:* the multivariate analogue of the scalar Cramér/spectral representation; the increment process  $Z(f)$  carries the random “Fourier content” of each component, with covariance structure given by the spectral matrix.

*Why:* it justifies thinking of a multivariate stationary process as a superposition of uncorrelated random sinusoids whose (matrix) variance is  $S(f) df$ ; coherence is then literally the correlation of the increments  $dZ_j(f), dZ_k(f)$ .

### Proposition 17: Cross-spectrum of a pure delay process

Let  $\{Y_t\}$  be stationary and  $X_t = (Y_t, Y_{t-\nu})^T$  for fixed  $\nu \in \mathbb{Z}$ . Then

$$S_{1,1}(f) = S_{2,2}(f) = \mathcal{S}_Y(f), \quad S_{1,2}(f) = \mathcal{S}_Y(f) e^{2\pi i f \nu},$$

so the coherence is  $r_{1,2}(f) = e^{2\pi i f \nu}$ , of magnitude 1 and phase  $2\pi \nu f$ . *Proof idea:*  $\gamma_{\tau}^{(1,2)} = \text{Cov}(Y_{t+\tau}, Y_{t-\nu}) = \gamma_{\tau+\nu}^{(Y)}$ ; Fourier-transforming and shifting the summation index pulls out  $e^{2\pi i f \nu}$ .

*Why:* a pure time shift between two channels produces *perfect* coherence (magnitude 1) with a phase that is *linear in f* with slope  $2\pi \nu$  — the textbook signature of a lead/lag of  $\nu$  samples.

### Theorem 18: Everything is a VAR(1) (companion reduction)

Any VAR( $p$ ) process  $\{X_t\}$  admits the VAR(1) representation  $Y_t = F Y_{t-1} + U_t$  of Definition 10, with  $\{U_t\}$  a  $dp$ -variate white noise whose only nonzero covariance block is the top-left  $\Sigma$ . Hence all VAR( $p$ ) properties (stability, MA( $\infty$ ) expansion, covariances) follow from the VAR(1) theory applied to  $F$  and then projected back by  $X_t = G Y_t$ .

*Proof idea:* write out the stacked vector  $Y_t$ ; its top block equals the VAR( $p$ ) recursion and its remaining blocks are trivial identities, giving exactly  $F$  and  $U_t$ .

*Why:* it lets us prove one clean VAR(1) result and reuse it for every order  $p$ , e.g. the stability test becomes “eigenvalues of  $F$  inside the unit circle”.

### Proposition 19: MA( $\infty$ ) representation of a stable VAR(1)

If the VAR(1)  $X_t = \Phi_1 X_{t-1} + \varepsilon_t$  is stable, then

$$X_t = \sum_{j=0}^{\infty} \Phi_1^j \varepsilon_{t-j}.$$

*Proof idea:* back-substitute repeatedly,  $X_t = \Phi_1(\Phi_1 X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \Phi_1^2 X_{t-2} + \Phi_1 \varepsilon_{t-1} + \varepsilon_t = \dots$ ; stability ( $\Phi_1^j \rightarrow 0$ ) makes the remainder term vanish and the series converge (in mean square).

*Why:* the matrix geometric series is the multivariate causal linear filter of the noise — it underlies the covariance formula below and all forecasting.

### Proposition 20: Covariance of a stable VAR(1)

For a stable VAR(1) with  $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$  and  $\tau \geq 0$ ,

$$\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t) = \sum_{k=0}^{\infty} \Phi_1^{k+\tau} \Sigma (\Phi_1^k)^\top.$$

*Proof idea:* substitute the MA( $\infty$ ) form into  $\text{Cov}(\sum_j \Phi_1^j \varepsilon_{t+\tau-j}, \sum_k \Phi_1^k \varepsilon_{t-k})$ ; white noise makes only the terms with  $t + \tau - j = t - k$ , i.e.  $j = k + \tau$ , survive, each contributing  $\Phi_1^{k+\tau} \Sigma (\Phi_1^k)^\top$ .

*Why:* closed-form second-order structure of a VAR(1); for  $\tau = 0$  it gives  $\Gamma_0 = \sum_k \Phi_1^k \Sigma (\Phi_1^k)^\top$ , the stationary covariance, which also solves the discrete Lyapunov equation  $\Gamma_0 = \Phi_1 \Gamma_0 \Phi_1^\top + \Sigma$ .

### Corollary 21: Covariance of a VAR( $p$ ) via the companion form

For a stable VAR( $p$ ) with companion matrix  $F$  and  $\Sigma_U = \text{Var}(U_t)$  (top-left block  $\Sigma$ , all else 0),

$$\Gamma_\tau^{(X)} = \text{Cov}(X_{t+\tau}, X_t) = G \left( \sum_{k=0}^{\infty} F^{k+\tau} \Sigma_U (F^k)^\top \right) G^\top, \quad G = (I_d \ 0 \ \dots \ 0).$$

*Proof idea:* apply Proposition 20 to the VAR(1)  $Y_t = F Y_{t-1} + U_t$  to get  $\Gamma_\tau^{(Y)}$ , then use  $X_t = G Y_t$  so  $\Gamma_\tau^{(X)} = G \Gamma_\tau^{(Y)} G^\top$ .

*Why:* reduces the order- $p$  covariance to one VAR(1) computation plus a projection.

### Theorem 22: Yule–Walker equations for a VAR( $p$ )

For a stable VAR( $p$ ) and  $\tau \geq 0$ ,

$$\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \Phi_2 \Gamma_{\tau-2} + \dots + \Phi_p \Gamma_{\tau-p} + \delta_{\tau,0} \Sigma.$$

In particular for VAR(1):  $\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \delta_{\tau,0} \Sigma$ . *Proof idea:* multiply the VAR equation on the right by  $X_t^\top$  and take expectations:  $\text{Cov}(X_{t+\tau}, X_t) = \text{Cov}(\sum_i \Phi_i X_{t+\tau-i} + \varepsilon_{t+\tau}, X_t)$ ; the noise term  $\text{Cov}(\varepsilon_{t+\tau}, X_t)$  contributes  $\Sigma$  only when  $\tau = 0$  (since  $X_t$  depends only on  $\varepsilon_s$  for  $s \leq t$ ).

*Why:* given  $\Sigma$  and the  $\Phi_j$ , solving the first  $p + 1$  equations yields  $\Gamma_0, \dots, \Gamma_p$ , and the recursion then generates all  $\Gamma_\tau$  for  $\tau > p$ . Run in reverse, it estimates the  $\Phi_j$  from sample covariances (VAR fitting).

### Why the noise term only shows up at $\tau = 0$

$X_t$  is built from  $\varepsilon_t, \varepsilon_{t-1}, \dots$  (the causal MA( $\infty$ )). For  $\tau > 0$ ,  $\varepsilon_{t+\tau}$  is in the *future* of  $X_t$ , hence uncorrelated with it, so  $\text{Cov}(\varepsilon_{t+\tau}, X_t) = 0$ . Only at  $\tau = 0$  does  $\text{Cov}(\varepsilon_t, X_t) = \text{Cov}(\varepsilon_t, \varepsilon_t) = \Sigma$  appear. This is the exact multivariate analogue of the scalar Yule–Walker derivation.

### 7.3 Key Techniques

#### Key Technique 1: Test VAR stability via the reverse characteristic polynomial

Given  $\Phi_1, \dots, \Phi_p$ :

1. Form  $M(z) = I_d - \Phi_1 z - \dots - \Phi_p z^p$  (a  $d \times d$  matrix whose entries are polynomials in  $z$ ).
2. Compute  $\det M(z)$  — a scalar polynomial of degree at most  $dp$ . *Factor it* if possible (block/triangular structure often factors it cleanly).
3. Find all roots  $z_i$ . The VAR is **stable** iff  $|z_i| > 1$  for *every* root.

**Shortcut for VAR(1):**  $\det(I_d - \Phi_1 z) = 0 \iff z = 1/\lambda_i(\Phi_1)$ , so just check  $|\lambda_i(\Phi_1)| < 1$  (spectral radius of  $\Phi_1$  below one). For a triangular  $\Phi_1$ , the eigenvalues are the diagonal entries.  
*Sanity check:*  $\det M(0) = \det I_d = 1$  always (the constant term is 1).

#### Key Technique 2: Compute a cross-spectrum from a cross-covariance

To get  $S_{j,k}(f)$ :

1. Find the cross-covariance  $\gamma_\tau^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)})$  in closed form, usually a *shift* of a known (auto)covariance, e.g.  $\gamma_\tau^{(j,k)} = \alpha \gamma_{\tau-\nu}^{(k,k)}$ .
2. Fourier transform:  $S_{j,k}(f) = \sum_\tau \gamma_\tau^{(j,k)} e^{-2\pi i f \tau}$ , then *re-index* the sum to extract any  $e^{\pm 2\pi i f \nu}$  factor. A lag-shift  $\tau \rightarrow \tau - \nu$  produces a phase factor  $e^{-2\pi i f \nu}$ .
3. Read off amplitude  $|S_{j,k}(f)|$  and phase  $\arg S_{j,k}(f)$ ; divide by  $\sqrt{S_{j,j} S_{k,k}}$  for coherence.

Mnemonic: *a delay of  $\nu$  samples in the time domain = a linear phase  $\mp 2\pi \nu f$  in the frequency domain.*

#### Key Technique 3: Derive the VAR(1) MA( $\infty$ ) and its covariance

For a stable VAR(1)  $X_t = \Phi_1 X_{t-1} + \varepsilon_t$ :

1. Write  $X_t = \sum_{j \geq 0} \Phi_1^j \varepsilon_{t-j}$  (matrix geometric filter).
2. Plug into  $\text{Cov}(X_{t+\tau}, X_t)$ ; keep only index pairs with matching noise times ( $j = k + \tau$ ).  
Result:  $\Gamma_\tau = \sum_{k \geq 0} \Phi_1^{k+\tau} \Sigma (\Phi_1^k)^\top$ .
3. For a *numerical*  $\Gamma_0$ , solve the Lyapunov equation  $\Gamma_0 = \Phi_1 \Gamma_0 \Phi_1^\top + \Sigma$  (vectorise:  $\text{vec } \Gamma_0 = (I - \Phi_1 \otimes \Phi_1)^{-1} \text{vec } \Sigma$ ), or sum the series until  $\Phi_1^k$  is negligible.

If  $\Phi_1$  and  $\Sigma$  are diagonal, everything decouples into  $d$  scalar AR(1)'s:  $\Gamma_0^{(j,j)} = \sigma_j^2 / (1 - \phi_j^2)$ .

#### Key Technique 4: Run the VAR Yule–Walker recursion

Given  $\Phi_1, \dots, \Phi_p, \Sigma$  and known  $\Gamma_0, \dots, \Gamma_{p-1}$ :

- For  $\tau > 0$ :  $\Gamma_\tau = \sum_{i=1}^p \Phi_i \Gamma_{\tau-i}$ , using  $\Gamma_{-s} = \Gamma_s^\top$  whenever a negative-lag matrix appears.
- To *initialise*, write the  $\tau = 0, \dots, p$  equations as a linear system in  $\Gamma_0, \dots, \Gamma_p$  (the  $\tau = 0$  equation carries the  $+\Sigma$  term) and solve once.

Watch the transpose: a term like  $\Phi_2 \Gamma_{-1}$  means  $\Phi_2 \Gamma_1^\top$ .

### 7.4 Exercises

#### Exercise 1: Common-noise cross-correlation [Sheet 9]

Let  $Y_t = (Y_t^{(1)}, Y_t^{(2)})^\top$  with  $Y_t^{(j)} = X_t^{(j)} + \varepsilon_t$ ,  $j = 1, 2$ , where  $X^{(1)}, X^{(2)}, \varepsilon$  are mutually independent and  $\{\varepsilon_t\}$  is white noise. Write  $\sigma_j^2 = \text{Var}(X_t^{(j)})$ ,  $\sigma_\varepsilon^2 = \text{Var}(\varepsilon_t)$  and  $\tilde{\gamma}_\tau^{(1,2)} = \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)})$ . Find the cross-correlation  $\rho_\tau^{(1,2)}$  between  $\{Y^{(1)}\}$  and  $\{Y^{(2)}\}$  at all lags.

### Solution

The *same* scalar noise  $\varepsilon_t$  is added to both channels, so expand

$$\gamma_\tau^{(1,2)} = \text{Cov}(Y_{t+\tau}^{(1)}, Y_t^{(2)}) = \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) + \underbrace{\text{Cov}(X_{t+\tau}^{(1)}, \varepsilon_t)}_0 + \underbrace{\text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)})}_0 + \text{Cov}(\varepsilon_{t+\tau}, \varepsilon_t).$$

The two middle terms vanish by independence of  $X$  and  $\varepsilon$ , and  $\text{Cov}(\varepsilon_{t+\tau}, \varepsilon_t) = \sigma_\varepsilon^2 \delta_{\tau,0}$ . Hence

$$\gamma_\tau^{(1,2)} = \begin{cases} \tilde{\gamma}_0^{(1,2)} + \sigma_\varepsilon^2 & \tau = 0, \\ \tilde{\gamma}_\tau^{(1,2)} & \tau \neq 0. \end{cases}$$

For the marginal variances,  $\gamma_0^{(1,1)} = \text{Var}(X_t^{(1)}) + \text{Var}(\varepsilon_t) = \sigma_1^2 + \sigma_\varepsilon^2$  and likewise  $\gamma_0^{(2,2)} = \sigma_2^2 + \sigma_\varepsilon^2$ . Therefore

$$\rho_\tau^{(1,2)} = \frac{\gamma_\tau^{(1,2)}}{\sqrt{(\sigma_1^2 + \sigma_\varepsilon^2)(\sigma_2^2 + \sigma_\varepsilon^2)}} = \begin{cases} \frac{\tilde{\gamma}_0^{(1,2)} + \sigma_\varepsilon^2}{\sqrt{(\sigma_1^2 + \sigma_\varepsilon^2)(\sigma_2^2 + \sigma_\varepsilon^2)}} & \tau = 0, \\ \frac{\tilde{\gamma}_\tau^{(1,2)}}{\sqrt{(\sigma_1^2 + \sigma_\varepsilon^2)(\sigma_2^2 + \sigma_\varepsilon^2)}} & \tau \neq 0. \end{cases}$$

The shared noise injects extra correlation *only* at lag 0 (the  $+\sigma_\varepsilon^2$ ), exactly Proposition 14 in disguise.

### Exercise 2: Delay process: cross-spectrum amplitude & phase [Sheet 9]

For the delay process with  $\alpha > 0$ ,  $\nu \in \mathbb{Z}$ ,

$$X_t^{(1)} = \alpha X_{t-\nu}^{(2)} + \varepsilon_t, \quad t \in \mathbb{Z},$$

where  $\{\varepsilon_t\}$  is white noise (variance  $\sigma_\varepsilon^2$ ) and  $\{X_t^{(2)}\}$  is stationary and uncorrelated with  $\{\varepsilon_t\}$  at all lags. **(a)** Find the cross-correlation between  $\{X^{(1)}\}$  and  $\{X^{(2)}\}$  at all lags. **(b)** Find the cross-spectrum  $S_{1,2}(f)$  and its amplitude and phase.

### Solution

**(a)** Using independence of  $X^{(2)}$  and  $\varepsilon$ ,

$$\gamma_\tau^{(1,2)} = \text{Cov}(X_{t+\tau}^{(1)}, X_t^{(2)}) = \alpha \text{Cov}(X_{t+\tau-\nu}^{(2)}, X_t^{(2)}) + \underbrace{\text{Cov}(\varepsilon_{t+\tau}, X_t^{(2)})}_0 = \alpha \gamma_{\tau-\nu}^{(2,2)}.$$

The marginal variances are  $\gamma_0^{(1,1)} = \text{Var}(\alpha X_{t-\nu}^{(2)} + \varepsilon_t) = \alpha^2 \sigma_2^2 + \sigma_\varepsilon^2$  (with  $\sigma_2^2 = \gamma_0^{(2,2)}$ ) and  $\gamma_0^{(2,2)} = \sigma_2^2$ . Hence

$$\rho_\tau^{(1,2)} = \frac{\alpha \gamma_{\tau-\nu}^{(2,2)}}{\sqrt{\alpha^2 \sigma_2^2 + \sigma_\varepsilon^2} \sqrt{\sigma_2^2}} = \frac{\rho_{\tau-\nu}^{(2,2)}}{\sqrt{1 + \sigma_\varepsilon^2 / (\alpha \sigma_2)^2}},$$

i.e. the autocorrelation of  $X^{(2)}$  shifted by  $\nu$  and shrunk by the noise factor.

**(b)** Fourier-transform the cross-covariance and shift the index  $\tau \rightarrow \tau + \nu$ :

$$\begin{aligned} S_{1,2}(f) &= \sum_{\tau=-\infty}^{\infty} \gamma_\tau^{(1,2)} e^{-2\pi i f \tau} = \alpha \sum_{\tau} \gamma_{\tau-\nu}^{(2,2)} e^{-2\pi i f \tau} \\ &= \alpha e^{-2\pi i f \nu} \sum_{\tau} \gamma_\tau^{(2,2)} e^{-2\pi i f \tau} = \alpha S_{2,2}(f) e^{-2\pi i f \nu}. \end{aligned}$$

Since  $S_{2,2}(f) \geq 0$  is real (the spectral density of  $X^{(2)}$ ), the **amplitude** is  $|S_{1,2}(f)| = \alpha S_{2,2}(f)$  (the spectrum of  $X^{(2)}$  scaled by  $\alpha$ ) and the **phase** is  $\arg S_{1,2}(f) = -2\pi \nu f$ , linear in  $f$  with

slope  $-2\pi\nu$  — the delay signature.

### Exercise 3: Expectation & variance of the sample cross-covariance [Sheet 9]

Let  $Y_t = (Y_t^{(1)}, Y_t^{(2)})^\top \stackrel{\text{ind}}{\sim} \mathcal{N}(0, \Sigma)$  in  $t$ , with  $\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$ . With  $\hat{\Gamma}_\tau = \frac{1}{N} \sum_{t=1}^{N-|\tau|} Y_{t+\tau} Y_t^\top$ , determine  $\mathbb{E}[\hat{\gamma}_\tau^{(1,2)}]$  and  $\text{Var}(\hat{\gamma}_\tau^{(1,2)})$ .

#### Solution

**Expectation.** By linearity,

$$\mathbb{E}[\hat{\gamma}_\tau^{(1,2)}] = \frac{1}{N} \sum_{t=1}^{N-|\tau|} \mathbb{E}[Y_{t+\tau}^{(1)} Y_t^{(2)}].$$

Since the  $Y_t$  are independent across  $t$ ,  $\mathbb{E}[Y_{t+\tau}^{(1)} Y_t^{(2)}] = 0$  for  $\tau \neq 0$ , and  $= \sigma_{12}$  for  $\tau = 0$ . With  $N - |\tau| = N$  terms at  $\tau = 0$ ,

$$\mathbb{E}[\hat{\gamma}_\tau^{(1,2)}] = \sigma_{12} \delta_{\tau,0}.$$

**Variance.** Write  $\text{Var}(\hat{\gamma}_\tau^{(1,2)}) = \mathbb{E}[(\hat{\gamma}_\tau^{(1,2)})^2] - \mathbb{E}[\hat{\gamma}_\tau^{(1,2)}]^2$  and expand the square:

$$\mathbb{E}[(\hat{\gamma}_\tau^{(1,2)})^2] = \frac{1}{N^2} \left( \sum_{t=1}^{N-|\tau|} \mathbb{E}[(Y_{t+\tau}^{(1)})^2 (Y_t^{(2)})^2] + \sum_{t \neq s} \mathbb{E}[Y_{t+\tau}^{(1)} Y_t^{(2)} Y_{s+\tau}^{(1)} Y_s^{(2)}] \right).$$

For jointly Gaussian zero-mean variables, Isserlis' (Wick's) theorem gives the key fourth moment

$$\mathbb{E}[(Y_t^{(1)})^2 (Y_t^{(2)})^2] = \sigma_{11}\sigma_{22} + 2\sigma_{12}^2.$$

*Case  $\tau = 0$ .* The diagonal sum has  $N$  identical terms  $\sigma_{11}\sigma_{22} + 2\sigma_{12}^2$ . For the off-diagonal ( $t \neq s$ ), independence across  $t$  factorises the expectation into  $\mathbb{E}[Y_t^{(1)} Y_t^{(2)}] \mathbb{E}[Y_s^{(1)} Y_s^{(2)}] = \sigma_{12}^2$ , and there are  $N(N-1)$  such pairs. Thus

$$\mathbb{E}[(\hat{\gamma}_0^{(1,2)})^2] = \frac{1}{N^2} [N(\sigma_{11}\sigma_{22} + 2\sigma_{12}^2) + N(N-1)\sigma_{12}^2] = \frac{\sigma_{11}\sigma_{22} + 2\sigma_{12}^2}{N} + \frac{N-1}{N} \sigma_{12}^2.$$

Subtracting  $\mathbb{E}[\hat{\gamma}_0^{(1,2)}]^2 = \sigma_{12}^2$ ,

$$\text{Var}(\hat{\gamma}_0^{(1,2)}) = \frac{\sigma_{11}\sigma_{22} + 2\sigma_{12}^2}{N} + \frac{N-1}{N} \sigma_{12}^2 - \sigma_{12}^2 = \frac{1}{N} (\sigma_{11}\sigma_{22} + \sigma_{12}^2).$$

*Case  $\tau \neq 0$ .* Now  $\mathbb{E}[\hat{\gamma}_\tau^{(1,2)}] = 0$ . The diagonal sum has  $N - |\tau|$  terms, but because  $Y_{t+\tau}$  and  $Y_t$  are *independent* (different time indices), the Gaussian fourth moment splits as  $\mathbb{E}[(Y_{t+\tau}^{(1)})^2] \mathbb{E}[(Y_t^{(2)})^2] = \sigma_{11}\sigma_{22}$ . Every off-diagonal term contains a factor with a unique time index appearing once and so vanishes. Hence

$$\text{Var}(\hat{\gamma}_\tau^{(1,2)}) = \frac{1}{N^2} [(N - |\tau|)\sigma_{11}\sigma_{22} + 0] = \frac{N - |\tau|}{N^2} \sigma_{11}\sigma_{22}.$$

**Summary.**

$$\mathbb{E}[\hat{\gamma}_\tau^{(1,2)}] = \sigma_{12} \delta_{\tau,0}, \quad \text{Var}(\hat{\gamma}_\tau^{(1,2)}) = \frac{N - |\tau|}{N^2} (\sigma_{11}\sigma_{22} + \sigma_{12}^2 \delta_{\tau,0}).$$

(At  $\tau = 0$  this reads  $\frac{1}{N} (\sigma_{11}\sigma_{22} + \sigma_{12}^2)$ , matching the case above.)

**Exercise 4: Stability of a bivariate VAR(2)** [New]

Consider  $X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \varepsilon_t$  with  $\Phi_1 = \begin{pmatrix} 0.5 & 0.1 \\ 0.4 & 0.5 \end{pmatrix}$ ,  $\Phi_2 = \begin{pmatrix} 0 & 0 \\ 0.25 & 0 \end{pmatrix}$ . Form the reverse characteristic polynomial, find its roots, and decide whether the VAR(2) is stable.

**Solution**

We need  $\det(I_2 - \Phi_1 z - \Phi_2 z^2)$ . The inner matrix is

$$M(z) = \begin{pmatrix} 1 - 0.5z & -0.1z \\ -0.4z - 0.25z^2 & 1 - 0.5z \end{pmatrix}.$$

Its determinant is

$$\det M(z) = (1 - 0.5z)^2 - (-0.1z)(-0.4z - 0.25z^2) = (1 - z + 0.25z^2) - (0.04z^2 + 0.025z^3),$$

$$\det M(z) = 1 - z + 0.21z^2 - 0.025z^3.$$

Finding the roots of  $1 - z + 0.21z^2 - 0.025z^3 = 0$  gives

$$z_1 = 1.3, \quad z_2 = 3.55 + 4.26i, \quad z_3 = 3.55 - 4.26i.$$

The moduli are  $|z_1| = 1.3 > 1$  and  $|z_2| = |z_3| = \sqrt{3.55^2 + 4.26^2} = \sqrt{12.60 + 18.15} = \sqrt{30.75} \approx 5.545 > 1$ . All three roots lie strictly outside the unit circle, so the VAR(2) is **stable** (hence stationary). *Check:*  $\det M(0) = 1$ , as it must be (the constant term equals  $\det I_2 = 1$ ).

**Exercise 5: Stability of a triangular trivariate VAR(1)** [New]

Let  $X_t = \Phi_1 X_{t-1} + \varepsilon_t$  with  $\Phi_1 = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix}$ . Factor the reverse characteristic polynomial and decide stability.

**Solution**

Because the first row of  $\Phi_1$  has only its  $(1, 1)$  entry nonzero, expand  $\det(I_3 - \Phi_1 z)$  along the first row of

$$I_3 - \Phi_1 z = \begin{pmatrix} 1 - 0.5z & 0 & 0 \\ -0.1z & 1 - 0.1z & -0.3z \\ 0 & -0.2z & 1 - 0.3z \end{pmatrix}.$$

This gives

$$\det(I_3 - \Phi_1 z) = (1 - 0.5z) \det \begin{pmatrix} 1 - 0.1z & -0.3z \\ -0.2z & 1 - 0.3z \end{pmatrix} = (1 - 0.5z) [(1 - 0.1z)(1 - 0.3z) - 0.06z^2].$$

The  $2 \times 2$  determinant is  $1 - 0.4z + 0.03z^2 - 0.06z^2 = 1 - 0.4z - 0.03z^2$ , so

$$\det(I_3 - \Phi_1 z) = (1 - 0.5z)(1 - 0.4z - 0.03z^2).$$

The first factor gives  $z_1 = 2$ . The quadratic  $1 - 0.4z - 0.03z^2 = 0$ , i.e.  $0.03z^2 + 0.4z - 1 = 0$ , has roots  $z = \frac{-0.4 \pm \sqrt{0.16 + 0.12}}{0.06} = \frac{-0.4 \pm \sqrt{0.28}}{0.06}$ , giving  $z_2 \approx 2.153$  and  $z_3 \approx -15.486$ . All three roots satisfy  $|z| > 1$ , so the VAR(1) is **stable**. (Equivalently:  $\Phi_1$  is block lower-triangular with eigenvalues 0.5 and those of  $\begin{pmatrix} 0.1 & 0.3 \\ 0.2 & 0.3 \end{pmatrix}$ , namely  $\approx 0.465$  and  $\approx -0.065$ ; all have modulus  $< 1$ .)

### Exercise 6: MA( $\infty$ ) covariance of a diagonal VAR(1) [New]

Let  $X_t = \Phi_1 X_{t-1} + \varepsilon_t$  with  $\Phi_1 = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.3 \end{pmatrix}$  and  $\{\varepsilon_t\} \sim \text{WN}(0, I_2)$ . (a) Write the MA( $\infty$ ) form. (b) Compute the stationary covariance  $\Gamma_0$  and the lag-1 matrix  $\Gamma_1$ . (c) What is the coherence between the two channels?

#### Solution

(a) Since  $\Phi_1 = \text{diag}(0.5, 0.3)$  has spectral radius  $0.5 < 1$ , the VAR is stable and

$$X_t = \sum_{j=0}^{\infty} \Phi_1^j \varepsilon_{t-j}, \quad \Phi_1^j = \text{diag}(0.5^j, 0.3^j).$$

(b) By the VAR(1) covariance formula with  $\Sigma = I_2$ ,

$$\Gamma_0 = \sum_{k=0}^{\infty} \Phi_1^k (\Phi_1^k)^T = \sum_{k=0}^{\infty} \text{diag}(0.25^k, 0.09^k) = \text{diag}\left(\frac{1}{1-0.25}, \frac{1}{1-0.09}\right) = \text{diag}\left(\frac{4}{3}, \frac{100}{91}\right),$$

i.e.  $\Gamma_0 \approx \text{diag}(1.333, 1.099)$ . (Equivalently solve  $\Gamma_0 = \Phi_1 \Gamma_0 \Phi_1^T + I_2$  componentwise:  $g_{jj} = \phi_j^2 g_{jj} + 1$ .) The lag-1 matrix is, by Yule–Walker  $\Gamma_1 = \Phi_1 \Gamma_0$ ,

$$\Gamma_1 = \text{diag}(0.5, 0.3) \text{diag}\left(\frac{4}{3}, \frac{100}{91}\right) = \text{diag}\left(\frac{2}{3}, \frac{30}{91}\right) \approx \text{diag}(0.667, 0.330).$$

(c) All off-diagonal entries of  $\Gamma_\tau$  are 0 for every  $\tau$  (the channels are independent AR(1)'s), so  $S_{1,2}(f) \equiv 0$  and the coherence  $r_{1,2}(f) = 0$  at all frequencies: the two series are completely uncoupled. The off-diagonal entries of  $\Phi_1$  (here zero) are exactly the cross-channel dynamics, confirming the decoupling.

### Exercise 7: Reading off a VAR(1) from its Yule–Walker recursion [New]

A stable bivariate VAR(1)  $X_t = \Phi_1 X_{t-1} + \varepsilon_t$  has  $\Phi_1 = \begin{pmatrix} 0.6 & 0.2 \\ 0.1 & 0.5 \end{pmatrix}$  and  $\{\varepsilon_t\} \sim \text{WN}(0, I_2)$ . (a) Verify stability via the reverse characteristic polynomial. (b) Express  $\Gamma_2$  in terms of  $\Gamma_0$  using Yule–Walker. (c) Give  $\Gamma_{-1}$  in terms of  $\Gamma_1$ .

#### Solution

(a)  $\det(I_2 - \Phi_1 z) = (1 - 0.6z)(1 - 0.5z) - (0.2z)(0.1z) = 1 - 1.1z + 0.30z^2 - 0.02z^2 = 1 - 1.1z + 0.28z^2$ . Solving  $0.28z^2 - 1.1z + 1 = 0$ ,

$$z = \frac{1.1 \pm \sqrt{1.21 - 1.12}}{0.56} = \frac{1.1 \pm 0.3}{0.56} \Rightarrow z_1 = \frac{1.4}{0.56} = 2.5, \quad z_2 = \frac{0.8}{0.56} = \frac{10}{7} \approx 1.43.$$

Both have modulus  $> 1$ , so the VAR(1) is stable. (Equivalently  $\Phi_1$  has eigenvalues 0.7 and 0.4, both inside the unit circle.)

(b) Yule–Walker for VAR(1) gives  $\Gamma_\tau = \Phi_1 \Gamma_{\tau-1}$  for  $\tau \geq 1$ . Iterating,

$$\Gamma_1 = \Phi_1 \Gamma_0, \quad \Gamma_2 = \Phi_1 \Gamma_1 = \Phi_1^2 \Gamma_0, \quad \Phi_1^2 = \begin{pmatrix} 0.38 & 0.22 \\ 0.11 & 0.27 \end{pmatrix}.$$

(c) By the transpose symmetry  $\Gamma_{-\tau} = \Gamma_\tau^T$ ,

$$\Gamma_{-1} = \Gamma_1^T = (\Phi_1 \Gamma_0)^T = \Gamma_0 \Phi_1^T$$

(using  $\Gamma_0 = \Gamma_0^T$ ). Note  $\Gamma_{-1} \neq \Gamma_1$  in general, since  $\Phi_1$  is not symmetric — the multivariate ACVS is *not* even in  $\tau$ .

## 8 Forecasting

The purpose of time series analysis is often to *forecast* future values from the data observed up to now. The recipe has three steps: (1) assume the form of the model; (2) estimate its parameters; (3) use the assumed structure and the estimated parameters to form predictions. In this part we take the model structure and the parameters as **known** and predict  $h$  lags ahead, i.e. we want  $X_{n+h}$  from  $X_1, \dots, X_n$ . The integer  $h > 0$  is the **lead** (or **horizon**) and  $n$  is the **origin**. The central theme is that, for the second-order theory we use, the optimal predictor depends *only* on the covariance structure  $\{\gamma_\tau\}$ , and for causal ARMA models it is computed cleanly by setting future innovations to zero.

### 8.1 Definitions

#### Definition 1: Prediction mean square error

Let  $g(X_1, \dots, X_n)$  be any estimator (predictor) of  $X_{n+h}$  based on the observations  $X_1, \dots, X_n$ . Its performance is measured by the **prediction mean square error**

$$P_{n+h}^n := \mathbb{E} \left[ (X_{n+h} - g(X_1, \dots, X_n))^2 \right].$$

Smaller  $P_{n+h}^n$  means a better forecast; minimising it over a class of predictors defines the “best” predictor in that class.

#### Definition 2: Best linear predictor

The **best linear predictor** of  $X_{n+h}$  based on  $X_1, \dots, X_n$  is the linear function

$$X_{n+h}^n = P_{X_1, \dots, X_n}(X_{n+h}) = \sum_{j=1}^n \beta_j X_j$$

that minimises the prediction mean square error among all linear combinations of  $X_1, \dots, X_n$ . We write  $P_{X_1, \dots, X_n}(\cdot)$  for the projection onto the linear span of the data.

- For **Gaussian** time series the conditional expectation  $\mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$  is itself linear in the data, so the best linear predictor coincides with the (overall) best predictor.
- In general the conditional expectation is a *nonlinear* function of the data; for tractability one restricts attention to linear predictors, which need only second-order information.

#### Definition 3: $h$ -step forecast error and its variance

For the best linear predictor  $X_{n+h}^n$  define the  **$h$ -step ahead forecast error**

$$e_n(h) := X_{n+h} - X_{n+h}^n.$$

It is unbiased,  $\mathbb{E}[e_n(h)] = 0$ , and its variance equals the prediction MSE,  $\text{Var}(e_n(h)) = P_{n+h}^n$ . For a causal ARMA process with  $MA(\infty)$  weights  $\{\psi_j\}$  one has  $e_n(h) = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$  and  $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ .

#### Definition 4: Sources of prediction uncertainty

There are **three** components of uncertainty in a forecast:

1. **model uncertainty** — the assumed model form may be wrong;
2. **estimation uncertainty** — the parameters are estimated, not known;
3. **innovation uncertainty** — the inherent randomness of the future innovations

$\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ .

Throughout we assume the model is *known* and the parameters are estimated *without error*, which removes the first two sources and leaves only innovation uncertainty. (Bootstrap methods can incorporate the first two, but are not covered here.)

#### Definition 5: Gaussian prediction interval

For a **Gaussian** process the  $h$ -step error  $e_n(h)$  is normal with mean 0 and variance  $\text{Var}(e_n(h))$ , so a  $(1 - \alpha)$  **prediction interval** for  $X_{n+h}$  based on  $X_1, \dots, X_n$  is

$$X_{n+h}^n \pm z_{1-\alpha/2} \sqrt{\text{Var}(e_n(h))},$$

where  $z_{1-\alpha/2}$  is the  $(1 - \alpha/2)$  quantile of the standard normal distribution (e.g.  $z_{0.975} \approx 1.96$  for a 95% interval).

#### Intuition / Why it matters

A point forecast alone is almost useless without a measure of its spread. The interval widens with  $h$  because more unknown innovations  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  accumulate in the error. For a stationary ARMA model the width saturates: as  $h \rightarrow \infty$ ,  $\text{Var}(e_n(h)) \rightarrow \gamma_0$ , i.e. the interval converges to “mean  $\pm$  a multiple of the marginal standard deviation” — forecasting far ahead is no better than quoting the unconditional distribution. For integrated (ARIMA) processes the interval grows *without bound*.

## 8.2 Theorems & Propositions

#### Lemma 6: Conditional expectation minimises the prediction MSE

The conditional expectation

$$g(X_1, \dots, X_n) = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$$

minimises the prediction mean square error for  $X_{n+h}$  based on  $X_1, \dots, X_n$ .

*Proof idea:* For any real random variable  $Y$  with  $\mu = \mathbb{E}[Y]$  and any constant  $c$ ,

$$\text{MSE}(c) = \mathbb{E}[(Y - c)^2] = \mathbb{E}[(Y - \mu + \mu - c)^2] = \mathbb{E}[(Y - \mu)^2] + (\mu - c)^2 = \text{Var}(Y) + (\mu - c)^2,$$

which is minimised at  $c = \mu = \mathbb{E}[Y]$  (the cross term vanishes because  $\mathbb{E}[Y - \mu] = 0$ ). Applying this *conditionally on*  $X_1, \dots, X_n$  and using the tower property  $\mathbb{E}[(X_{n+h} - g)^2] = \mathbb{E}[\mathbb{E}[(X_{n+h} - g)^2 \mid X_1, \dots, X_n]]$ , the inner MSE is minimised pointwise by taking  $g = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$ .

*Why:* this is the theoretical gold standard for forecasting. For Gaussian processes the conditional expectation is linear, so the best predictor *is* the best linear predictor; in general it motivates restricting to linear predictors, which only need the ACVS.

#### Example 7: Gaussian AR(1) predictor — mean reversion

Let  $X_t = \phi X_{t-1} + \varepsilon_t$  with  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$  and  $|\phi| < 1$  (stationary). Because the process is Gaussian, the best linear predictor equals the conditional expectation, and iterating the recursion,

$$X_{n+h}^n = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n] = \mathbb{E}[\phi X_{n+h-1} + \varepsilon_{n+h} \mid X_1, \dots, X_n] = \phi^h X_n.$$

Since  $|\phi| < 1$ ,  $X_{n+h}^n = \phi^h X_n \rightarrow 0$  (the process mean) as  $h \rightarrow \infty$ : the forecast **reverts to the mean**. This is a general property of stationary ARMA models. (A worked figure in the notes

uses  $\phi = 0.8$ ,  $\sigma^2 = 1$ .)

#### Example 8: Gaussian MA(1) predictor

Let  $X_t = \mu + \varepsilon_t - \theta \varepsilon_{t-1}$  with  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ . For  $h > 0$ ,

$$X_{n+h}^n = \mu + \mathbb{E}[\varepsilon_{n+h} \mid X_1, \dots, X_n] - \theta \mathbb{E}[\varepsilon_{n+h-1} \mid X_1, \dots, X_n] = \mu - \theta \mathbb{E}[\varepsilon_{n+h-1} \mid X_1, \dots, X_n],$$

because  $\varepsilon_t$  for  $t > n$  is uncorrelated with  $X_1, \dots, X_n$  (so its conditional expectation is 0). Hence:

- $h = 1$ :  $X_{n+1}^n = \mu - \theta \hat{\varepsilon}_n$  (the one observable innovation  $\varepsilon_n$  remains);
- $h > 1$ :  $X_{n+h}^n = \mu$  — the forecast hits the mean immediately, consistent with an MA(1) having memory only of length 1.

#### Theorem 9: Prediction equations (zero-mean stationary process)

For a zero-mean stationary process  $\{X_t\}$ , the best linear predictor  $X_{n+h}^n = \sum_{j=1}^n \beta_j X_j$  is found by solving for  $\beta_1, \dots, \beta_n$  the **prediction equations**

$$\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0, \quad k = 1, \dots, n,$$

i.e. the forecast error is orthogonal to every observation. Equivalently, in matrix form  $\Gamma_n \beta = \gamma_{[h]}$ :

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \cdots & \gamma_0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \begin{pmatrix} \gamma_{n+h-1} \\ \gamma_{n+h-2} \\ \vdots \\ \gamma_h \end{pmatrix}.$$

*Proof idea:* the best linear predictor is the orthogonal projection of  $X_{n+h}$  onto  $\text{span}\{X_1, \dots, X_n\}$ ; the residual  $X_{n+h} - X_{n+h}^n$  must be orthogonal (uncorrelated) to each  $X_k$ . Expanding  $\mathbb{E}[(X_{n+h} - \sum_j \beta_j X_j) X_k] = 0$  gives  $\gamma_{n+h-k} = \sum_{j=1}^n \beta_j \gamma_{j-k}$ , which is row  $k$  of  $\Gamma_n \beta = \gamma_{[h]}$ .

*Why:* this is the master equation for linear prediction of *any* stationary process — it shows the optimal forecast uses only the ACVS.

#### Corollary 10: Matrix solution and prediction MSE

If  $\Gamma_n$  is invertible, then with  $X = (X_1, \dots, X_n)^\top$  the best linear predictor and its MSE are

$$X_{n+h}^n = \gamma_{[h]}^\top \Gamma_n^{-1} X, \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

*Proof idea:* solve  $\Gamma_n \beta = \gamma_{[h]}$  for  $\beta = \Gamma_n^{-1} \gamma_{[h]}$  and substitute into  $X_{n+h}^n = \beta^\top X$  and into  $P_{n+h}^n = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta$ ; the last two terms collapse using  $\Gamma_n \beta = \gamma_{[h]}$ .

*Why:* gives the closed-form forecast for short series.

- $\Gamma_n$  is invertible for *invertible* ARMA models (and not in all other cases), but the best prediction  $X_{n+h}^n$  is **always unique** even when  $\Gamma_n$  is singular.
- Computing  $\gamma_{[h]}^\top \Gamma_n^{-1} X$  is inefficient for large  $n$  because the  $n \times n$  matrix must be inverted; the Durbin–Levinson and innovations algorithms avoid this (Shumway & Stoffer, §3.5).

#### Theorem 11: Prediction equations with non-zero mean (and intercept $\beta_0$ )

For a stationary process with  $\mu = \mathbb{E}[X_t]$ , write  $X_{n+h}^n = \beta_0 + \sum_{j=1}^n \beta_j X_j$ . The coefficients solve the prediction equations  $\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0$  for  $k = 0, 1, \dots, n$  (with the convention

$X_0 \equiv 1$ ), which give

$$\Gamma_n \beta = \gamma_{[h]}, \quad \beta_0 = \mu(1 - \beta^\top \mathbf{1}_n),$$

and, when  $\Gamma_n$  is invertible,

$$X_{n+h}^n = \mu + \gamma_{[h]}^\top \Gamma_n^{-1} (X - \mu \mathbf{1}_n), \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

*Proof idea:* the predictor has the form  $X_{n+h}^n = \mu + \beta^\top (X - \mu \mathbf{1}_n)$ , so centring removes the mean and reduces the problem to the zero-mean case — there is no loss of generality in assuming  $\mu = 0$ , where  $\beta_0 = 0$ . The MSE is *identical* to the zero-mean case because the constant shift does not affect the error variance.

*Why:* real data are rarely centred; this is the version actually used in practice. (Proved as Exercise 2.)

### Theorem 12: Best linear predictor of a causal ARMA via $\psi$ -weights

For a causal ARMA process with  $MA(\infty)$  representation  $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ , the best linear predictor of  $X_{n+h}$  based on  $X_1, \dots, X_n$  is

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j} = \psi_h \varepsilon_n + \psi_{h+1} \varepsilon_{n-1} + \dots,$$

with prediction mean square error

$$P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

*Proof idea:* write the predictor as a linear filter of the past innovations,  $X_{n+h}^n = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}$ . Then

$$\mathbb{E}[(X_{n+h} - X_{n+h}^n)^2] = \sigma^2 \left( \sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right),$$

which is minimised by matching  $c_j = \psi_{j+h}$ , killing the second sum and leaving  $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ . Practically: **set the unrealised future innovations  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  to zero** in  $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$ .

*Why:* this is the engine of all ARMA forecasting. It also shows that for a causal ARMA,  $\psi_j \rightarrow 0$  exponentially so  $X_{n+h}^n \rightarrow \mu$ , and  $P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \uparrow \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 = \gamma_0$  as  $h \rightarrow \infty$ . (Proved as Exercise 3.)

### Intuition / Why it matters

The two ARMA forecasting theorems (prediction equations vs.  $\psi$ -weights) are the same projection viewed differently. The prediction equations project onto  $\text{span}\{X_1, \dots, X_n\}$  using the ACVS; the  $\psi$ -weight formula projects onto  $\text{span}\{\varepsilon_s : s \leq n\}$ , the *innovation* space. Because for an invertible ARMA these two spans coincide, the answers agree — but the innovation view is far easier to compute: “write the future, erase the future innovations.”

### Proposition 13: Limiting behaviour of ARMA forecasts

For a causal ARMA model with mean  $\mu$ ,  $X_{n+h}^n = \mu + \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}$  with  $\psi_j \rightarrow 0$  exponentially fast, hence

$$X_{n+h}^n \rightarrow \mu, \quad \text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \rightarrow \gamma_0, \quad h \rightarrow \infty.$$

*Proof idea:* immediate from the  $\psi$ -weight theorem and exponential decay of  $\psi_j$  (a consequence of causality, roots of  $\Phi$  outside the unit circle).

*Why:* quantifies “the distant future is the unconditional distribution” — the point forecast converges to the mean and the error variance converges to the marginal variance  $\gamma_0$ .

### Proposition 14: Forecasts for integrated (ARIMA) processes

Let  $\{X_t\}$  be ARIMA( $p, d, q$ ),  $\Phi(B)(1-B)^d X_t = \Theta(B)\varepsilon_t$ . Then  $Z_t = \nabla^d X_t$  is the ARMA( $p, q$ ) process  $Z_t = \sum_{j=1}^p \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}$ , and one forecasts  $X$  by undoing the differencing,

$$\nabla^d X_{n+h}^n = Z_{n+h}^n.$$

For  $d = 1$  (so  $X_{n+h} - X_{n+h-1} = Z_{n+h}$ ),

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n, \quad h \geq 1.$$

If  $\{Z_t\}$  has mean  $\alpha \neq 0$  then  $Z_{n+h}^n \rightarrow \alpha$  and  $X_{n+h}^n \approx X_n + \alpha h$  for large  $h$  (a deterministic drift line).

*Proof idea:* apply the linear projection operator  $P_{X_1, \dots, X_n}$ , which is linear, to  $\nabla^d X_{n+h} = Z_{n+h}$ ; since differencing is a fixed linear filter it commutes with the projection, giving  $\nabla^d X_{n+h}^n = Z_{n+h}^n$ . Telescoping the  $d = 1$  identity  $X_{n+h} = X_{n+h-1} + Z_{n+h}$  down to the last observation  $X_n$  yields  $X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n$ .

*Why:* reduces ARIMA forecasting to ARMA forecasting on the differenced series, then “re-integrates”.

- For ARIMA models the prediction MSE **diverges**:  $P_{n+h}^n \rightarrow \infty$  as  $h \rightarrow \infty$  (no mean reversion — the process is non-stationary).
- For SARIMA models with  $d = D = 1$  and  $\alpha = 0$  the forecast is linear and seasonal in  $h$ .

### Classic Pitfall

After expanding an ARIMA( $p, 1, q$ ) into the form  $X_t = (1 + \phi_1)X_{t-1} - (\phi_1 - \phi_2)X_{t-2} - \dots$ , the equation *looks* like an ARMA( $p+1, 0, q$ ), but it is **not** a stationary ARMA: its AR coefficients do not satisfy the causality/stationarity conditions (there is a unit root). Treat it only as a convenient recursion for forecasting, never as a stationary model. This is exactly why  $P_{n+h}^n \rightarrow \infty$ .

## 8.3 Key Techniques

### Key Technique 1: Solving the prediction equations

To get the best linear predictor of a zero-mean stationary process:

1. Build  $\Gamma_n = [\gamma_{i-j}]_{i,j=1}^n$  (Toeplitz, from the ACVS).
2. Build the right-hand side  $\gamma_{[h]} = (\gamma_{n+h-1}, \gamma_{n+h-2}, \dots, \gamma_h)^\top$ .

3. Solve  $\Gamma_n \beta = \gamma_{[h]}$  for  $\beta$ , set  $X_{n+h}^n = \sum_j \beta_j X_j = \gamma_{[h]}^T \Gamma_n^{-1} X$ .

4. Report the MSE  $P_{n+h}^n = \gamma_0 - \gamma_{[h]}^T \Gamma_n^{-1} \gamma_{[h]}$ .

For **non-zero mean**  $\mu$ , centre first: use  $X - \mu \mathbf{1}_n$  in place of  $X$  and add  $\mu$  back,  $X_{n+h}^n = \mu + \gamma_{[h]}^T \Gamma_n^{-1} (X - \mu \mathbf{1}_n)$ ; the MSE is unchanged. *Shortcut for AR(p)*: for  $h = 1$  and  $n \geq p$  the answer is just  $X_{n+1}^n = \sum_{i=1}^p \phi_i X_{n+1-i}$  — no inversion needed (the next technique explains why).

### Key Technique 2: Causal-ARMA forecast by zeroing future innovations

The fast route for any causal ARMA:

1. Write  $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$  (the  $MA(\infty)$  form), or just the ARMA equation rearranged for  $X_{n+h}$ .
2. Take conditional expectation given the data: every future innovation  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  becomes 0; every past/present innovation  $\varepsilon_s$  ( $s \leq n$ ) stays (replace by its residual  $\hat{\varepsilon}_s$ ); every past observation  $X_s$  ( $s \leq n$ ) stays; every future observation  $X_s$  ( $s > n$ ) is replaced by its forecast  $X_s^n$ .
3. The MSE is  $P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$  — obtain the first  $h$   $\psi$ -weights from  $\Theta(B)/\Phi(B) = \sum_j \psi_j B^j$  by matching powers of  $B$ .

### Key Technique 3: Recursive truncated forecasting recipe (ARMA & ARIMA)

The general-purpose hand-computation method. **Three steps:**

1. **Isolate**  $y_t$ : expand the (ARI)MA equation so that  $y_t$  is alone on the left and *everything else* (past  $y$ 's, present/past  $\varepsilon$ 's) is on the right.
2. **Shift to  $n + h$** : replace  $t$  by  $n + h$  throughout.
3. **Substitute**: on the right-hand side replace
  - future observations  $y_s$  ( $s > n$ ) by their forecasts  $\hat{y}_{s|n} = X_s^n$ ;
  - future errors  $\varepsilon_s$  ( $s > n$ ) by 0;
  - past errors  $\varepsilon_s$  ( $s \leq n$ ) by the corresponding residuals  $\hat{\varepsilon}_s$  (and any  $\varepsilon_s$  with  $s \leq 0$  by 0).

Run  $h = 1, 2, 3, \dots$  in order; each forecast feeds the next. The residuals are themselves built recursively (for  $t = 1, \dots, n$ , and 0 otherwise) via

$$\hat{\varepsilon}_t = \left( \tilde{X}_t^n - \sum_{i=1}^p \phi_i \tilde{X}_{t-i}^n \right) + \sum_{k=1}^q \theta_k \hat{\varepsilon}_{t-k}.$$

The truncation (setting pre-sample terms to 0) makes  $\tilde{X}_t^n$  an *approximation*; the forecast error is estimated by  $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ .

### Key Technique 4: Re-integrating ARIMA forecasts

For ARIMA( $p, d, q$ ): difference  $d$  times to get the ARMA series  $Z_t = \nabla^d X_t$ , forecast  $Z_{n+h}^n$  by the ARMA technique, then undo the differencing. For  $d = 1$ ,

$$X_{n+1}^n = X_n + Z_{n+1}^n, \quad X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n \quad (h \geq 1).$$

Equivalently, expand  $\Phi(B)(1-B)^d$  as a single (unit-root) AR polynomial in  $X$  and apply the recursive recipe directly (often easier in practice). Remember the MSE *grows without bound*.

### Key Technique 5: Building Gaussian prediction intervals

1. Get the point forecast  $X_{n+h}^n$ .
2. Get the  $\psi$ -weights  $\psi_0 = 1, \psi_1, \dots, \psi_{h-1}$  from  $\Theta(B)/\Phi(B)$ .
3. Forecast-error variance:  $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ .
4. Interval:  $X_{n+h}^n \pm z_{1-\alpha/2} \sqrt{\text{Var}(e_n(h))}$  (use 1.96 for 95%).

Sanity check: for stationary ARMA the half-width increases in  $h$  toward  $z_{1-\alpha/2} \sqrt{\gamma_0}$ ; for ARIMA it keeps growing.

## 8.4 Exercises

### Exercise 1: One-step AR(2) predictors from 1 and 2 observations [Sheet 10]

Let  $\varepsilon_t$  be white noise. For the causal AR(2) process  $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ , show that the best one-step-ahead linear predictors based on one observation  $X_1$  and on two observations  $X_1, X_2$  are

$$X_2^1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1, \quad X_3^2 = \phi_1 X_2 + \phi_2 X_1.$$

### Solution

**Based on  $X_1$  (so  $n = 1, h = 1$ ).** The prediction equation  $\Gamma_1 \beta = \gamma_{[1]}$  is the scalar equation  $\gamma_0 \beta_1 = \gamma_1$ , so  $\beta_1 = \gamma_1 / \gamma_0 = \rho_1$  and

$$X_2^1 = \beta_1 X_1 = \frac{\gamma_1}{\gamma_0} X_1 = \rho_1 X_1.$$

**Based on  $X_1, X_2$  (so  $n = 2, h = 1$ ).** We could build  $\Gamma_2$  and solve, but it is quicker to use the model directly. Since  $X_3 = \phi_1 X_2 + \phi_2 X_1 + \varepsilon_3$  we have  $X_3 - (\phi_1 X_2 + \phi_2 X_1) = \varepsilon_3$ , and  $\varepsilon_3$  (white noise, future) is uncorrelated with  $X_1$  and  $X_2$ :

$$\mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1)X_1] = \mathbb{E}[\varepsilon_3 X_1] = 0, \quad \mathbb{E}[(X_3 - \phi_1 X_2 - \phi_2 X_1)X_2] = \mathbb{E}[\varepsilon_3 X_2] = 0.$$

Thus the candidate  $\phi_1 X_2 + \phi_2 X_1$  has error orthogonal to both observations, so by the prediction-equations theorem it is the best linear predictor:  $X_3^2 = \phi_1 X_2 + \phi_2 X_1$ . (This is the general fact: for an AR( $p$ ) with  $n \geq p$  observations, the one-step forecast is simply  $\sum_{i=1}^p \phi_i X_{n+1-i}$ .)

### Exercise 2: Prediction equations with non-zero mean [Sheet 10]

For a stationary process with  $\mu = \mathbb{E}[X_t]$  and predictor  $X_{n+h}^n = \beta_0 + \sum_{j=1}^n \beta_j X_j$  (set  $X_0 \equiv 1$ ), prove that the prediction equations  $\mathbb{E}[(X_{n+h} - X_{n+h}^n)X_k] = 0, k = 0, \dots, n$ , give  $\Gamma_n \beta = \gamma_{[h]}$  and  $\beta_0 = \mu(1 - \beta^\top \mathbf{1}_n)$ , and that when  $\Gamma_n$  is invertible  $X_{n+h}^n = \mu + \gamma_{[h]}^\top \Gamma_n^{-1} (X - \mu \mathbf{1}_n)$  with MSE  $\gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}$ .

### Solution

**(a) MSE in matrix form.** With  $X = (X_1, \dots, X_n)^\top$  and  $\mathbf{1}_n$  the all-ones vector,

$$\begin{aligned} S(\beta_0, \dots, \beta_n) &= \mathbb{E}[(X_{n+h} - \beta_0 - \beta^\top X)^2] \\ &= \text{Var}(X_{n+h} - \beta^\top X) + (\mathbb{E}[X_{n+h} - \beta_0 - \beta^\top X])^2 \\ &= \gamma_0 - 2\beta^\top \text{Cov}(X_{n+h}, X) + \beta^\top \text{Var}(X)\beta + (\mu - \beta_0 - \mu\beta^\top \mathbf{1}_n)^2, \end{aligned}$$

a quadratic bounded below by 0, hence minimised where all  $\partial S / \partial \beta_r = 0$ .

**(b) Solve for  $\beta_0$ .** Only the last term depends on  $\beta_0$ :

$$\frac{\partial S}{\partial \beta_0} = 2(\mu - \beta_0 - \mu \beta^\top \mathbf{1}_n)(-1) = 0 \implies \beta_0 = \mu(1 - \beta^\top \mathbf{1}_n).$$

**(c) Reduce to zero mean.** Substituting  $\beta_0$  back,

$$X_{n+h}^n = \beta_0 + \beta^\top X = \mu + \beta^\top (X - \mu \mathbf{1}_n),$$

so the predictor depends on the data only through the *centred* vector  $X - \mu \mathbf{1}_n$ . Hence there is no loss of generality in taking  $\mu = 0$ , in which case  $\beta_0 = 0$  and  $X_{n+h}^n = \beta^\top X$ .

**(d) Normal equations.** For  $\mu = 0$ ,  $S(\beta) = \gamma_0 - 2\beta^\top \text{Cov}(X_{n+h}, X) + \beta^\top \text{Var}(X)\beta$ ; setting  $\nabla_\beta S = 0$  gives

$$\text{Var}(X) \beta = \text{Cov}(X_{n+h}, X), \quad \text{i.e.} \quad \Gamma_n \beta = \gamma_{[h]}.$$

**(e) Equivalence to the orthogonality form.** Row  $k$  of  $\Gamma_n \beta = \gamma_{[h]}$  reads  $\sum_{j=1}^n \beta_j \gamma_{j-k} = \gamma_{n+h-k}$ , which is exactly  $\mathbb{E}[(X_{n+h} - \sum_j \beta_j X_j) X_k] = 0$  for  $k = 1, \dots, n$ . The  $k = 0$  equation (with  $X_0 = 1$ ) gives  $\mu - \beta_0 - \sum_j \beta_j \mu = 0$ , recovering  $\beta_0 = \mu(1 - \beta^\top \mathbf{1}_n)$  from part (b).

**(2a) Closed form.** If  $\Gamma_n$  is invertible,  $\beta = \Gamma_n^{-1} \gamma_{[h]}$ , so

$$X_{n+h}^n = \mu + \gamma_{[h]}^\top \Gamma_n^{-1} (X - \mu \mathbf{1}_n),$$

differing from the lecture's zero-mean formula only by the centring/shift by  $\mu$ .

**(2b) MSE.** Since  $\mathbb{E}[X_{n+h} - X_{n+h}^n] = 0$ ,

$$P_{n+h}^n = \text{Var}(X_{n+h} - \beta^\top X) = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]},$$

identical to the zero-mean case (the mean does not affect forecast accuracy).

### Exercise 3: Causal-ARMA predictor via $\psi$ -weights [Sheet 10]

Prove that the best linear predictor  $X_{n+h}^n$  for  $X_{n+h}$  in a causal ARMA process with linear representation  $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$  is  $X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}$ , with MSE  $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ .

#### Solution

Consider a linear predictor  $X_{n+h}^n = \sum_{i=1}^n \beta_i X_i$ . Substituting the linear representation  $X_i = \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j}$ ,

$$X_{n+h}^n = \sum_{i=1}^n \beta_i \sum_{j=0}^{\infty} \psi_j \varepsilon_{i-j} = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}$$

for some coefficients  $c_j$ ; i.e. *any* linear predictor based on  $X_1, \dots, X_n$  is a linear combination of innovations up to time  $n$  (it “starts at  $\varepsilon_n$ ”). Now compute the MSE, using  $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$  and splitting the future innovations  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$  (indices  $j = 0, \dots, h-1$ ) from the rest:

$$\begin{aligned} \mathbb{E}[(X_{n+h} - X_{n+h}^n)^2] &= \mathbb{E}\left[\left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j} - \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}\right)^2\right] \\ &= \sigma^2 \left( \sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right), \end{aligned}$$

where the first sum collects the future innovations (which no past-based predictor can touch) and the second matches  $\psi_j$  against  $c_{j-h}$  for the observable innovations (cross terms vanish)

because the  $\varepsilon$ 's are uncorrelated). The first sum is fixed; the second is minimised — to zero — by choosing  $c_{j-h} = \psi_j$ , i.e.  $c_j = \psi_{j+h}$  for  $j = 0, 1, \dots$ . The predictor is then

$$X_{n+h}^n = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j} = \sum_{j=0}^{\infty} \psi_{j+h} \varepsilon_{n-j} = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j},$$

and the residual is  $X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$ , giving

$$P_{n+h}^n = \mathbb{E} \left[ (X_{n+h} - X_{n+h}^n)^2 \right] = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

#### Exercise 4: ARIMA(3,1,1) worked truncated forecast [Lecture]

For the fitted model  $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3)(1 - B)y_t = (1 + \theta_1 B)\varepsilon_t$  (the convention used in the lecture's worked example, with the MA polynomial written  $\Theta(B) = 1 + \theta_1 B$ ), derive the one-, two- and general  $h$ -step truncated forecasts. Then plug in the fitted values  $\hat{\phi}_1 = 0.0044$ ,  $\hat{\phi}_2 = 0.0916$ ,  $\hat{\phi}_3 = 0.3698$ ,  $\hat{\theta}_1 = -0.3921$  and write the explicit  $\hat{y}_{T+1|T}$  and  $\hat{y}_{T+2|T}$ .

#### Solution

**Step 1 — isolate  $y_t$ .** Multiply out the AR×difference factor  $(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3)(1 - B)$ :

$$1 - (1 + \phi_1)B + (\phi_1 - \phi_2)B^2 + (\phi_2 - \phi_3)B^3 + \phi_3 B^4,$$

so the model is

$$y_t - (1 + \phi_1)y_{t-1} + (\phi_1 - \phi_2)y_{t-2} + (\phi_2 - \phi_3)y_{t-3} + \phi_3 y_{t-4} = \varepsilon_t + \theta_1 \varepsilon_{t-1}.$$

Moving everything except  $y_t$  to the right,

$$y_t = \underbrace{(1 + \phi_1)y_{t-1} - (\phi_1 - \phi_2)y_{t-2} - (\phi_2 - \phi_3)y_{t-3} - \phi_3 y_{t-4}}_{\text{AR part}} + \underbrace{\varepsilon_t + \theta_1 \varepsilon_{t-1}}_{\text{MA part}}.$$

(We use the lecture's convention  $\Theta(B) = 1 + \theta_1 B$ , so the MA term is  $\varepsilon_t + \theta_1 \varepsilon_{t-1}$  and the forecast picks up  $+\theta_1 \hat{\varepsilon}$ . Beware: in the earlier general theory the MA polynomial was written  $\Theta(B) = 1 - \theta_1 B$ ; that sign convention would replace every  $+\theta_1$  below by  $-\theta_1$ . We follow the worked example exactly so the numbers match.)

**Step 2–3 — shift and substitute.** For  $h = 1$  (replace  $t$  by  $n + 1$ ): the only error terms are  $\varepsilon_{n+1} \rightarrow 0$  and  $\varepsilon_n \rightarrow \hat{\varepsilon}_n$  (last residual). Hence

$$y_{n+1}^n = (1 + \phi_1)y_n - (\phi_1 - \phi_2)y_{n-1} - (\phi_2 - \phi_3)y_{n-2} - \phi_3 y_{n-3} + \theta_1 \hat{\varepsilon}_n.$$

For  $h > 1$  all error terms are future ( $\rightarrow 0$ ) and the future observations are replaced by forecasts:

$$y_{n+h}^n = (1 + \phi_1)y_{n+h-1}^n - (\phi_1 - \phi_2)y_{n+h-2}^n - (\phi_2 - \phi_3)y_{n+h-3}^n - \phi_3 y_{n+h-4}^n,$$

with the convention  $y_s^n = y_s$  for  $s \leq n$ .

**Numbers.** Using  $1 + \hat{\phi}_1 = 1.0044$ ,  $\hat{\phi}_1 - \hat{\phi}_2 = 0.0044 - 0.0916 = -0.0872$ ,  $\hat{\phi}_2 - \hat{\phi}_3 = 0.0916 - 0.3698 = -0.2782$ ,  $\hat{\phi}_3 = 0.3698$ ,  $\hat{\theta}_1 = -0.3921$ , the one-step forecast (writing  $e_T$  for the last residual) is

$$\hat{y}_{T+1|T} = 1.0044 y_T + 0.0872 y_{T-1} + 0.2782 y_{T-2} - 0.3698 y_{T-3} + \hat{\theta}_1 e_T,$$

i.e. with  $\hat{\theta}_1 = -0.3921$ ,

$$\hat{y}_{T+1|T} = 1.0044 y_T + 0.0872 y_{T-1} + 0.2782 y_{T-2} - 0.3698 y_{T-3} - 0.3921 e_T.$$

(The two middle signs are positive because  $-(\hat{\phi}_1 - \hat{\phi}_2) = +0.0872$  and  $-(\hat{\phi}_2 - \hat{\phi}_3) = +0.2782$ .) The two-step forecast replaces  $y_{T+1}$  by  $\hat{y}_{T+1|T}$  and sets all errors to 0:

$$\hat{y}_{T+2|T} = 1.0044 \hat{y}_{T+1|T} + 0.0872 y_T + 0.2782 y_{T-1} - 0.3698 y_{T-2}.$$

Continue likewise for  $h = 3, 4, \dots$ ; each step uses previously computed forecasts. Because  $d = 1$ , the prediction MSE grows without bound as  $h$  increases.

#### Exercise 5: One- and two-step forecast of an MA(2) with interval [New]

Let  $X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$  with  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$  (invertible, zero mean). Give  $X_{n+1}^n$ ,  $X_{n+2}^n$ ,  $X_{n+3}^n$ , their forecast-error variances, and a 95% interval for  $X_{n+2}$ .

#### Solution

Here  $\psi_0 = 1$ ,  $\psi_1 = -\theta_1$ ,  $\psi_2 = -\theta_2$  and  $\psi_j = 0$  for  $j \geq 3$ . By the  $\psi$ -weight theorem, set future innovations to 0 in  $X_{n+h} = \varepsilon_{n+h} - \theta_1 \varepsilon_{n+h-1} - \theta_2 \varepsilon_{n+h-2}$ :

$$X_{n+1}^n = -\theta_1 \hat{\varepsilon}_n - \theta_2 \hat{\varepsilon}_{n-1},$$

$$X_{n+2}^n = -\theta_2 \hat{\varepsilon}_n,$$

$$X_{n+3}^n = 0 \quad (= \mu, \text{ the mean; an MA(2) is forecast as the mean for } h > 2).$$

Forecast-error variances  $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ :

$$\text{Var}(e_n(1)) = \sigma^2, \quad \text{Var}(e_n(2)) = \sigma^2(1 + \theta_1^2), \quad \text{Var}(e_n(3)) = \sigma^2(1 + \theta_1^2 + \theta_2^2) = \gamma_0.$$

For  $h \geq 3$  the variance saturates at  $\gamma_0$ , as it must. A 95% interval for  $X_{n+2}$  is

$$-\theta_2 \hat{\varepsilon}_n \pm 1.96 \sigma \sqrt{1 + \theta_1^2}.$$

#### Exercise 6: Two-step AR(1) forecast variance and interval [New]

For the Gaussian AR(1)  $X_t = \phi X_{t-1} + \varepsilon_t$  ( $|\phi| < 1$ ,  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ ), compute  $X_{n+1}^n$ ,  $X_{n+2}^n$ , their error variances, and verify  $\text{Var}(e_n(h)) \rightarrow \gamma_0$ .

#### Solution

The MA( $\infty$ ) weights of a causal AR(1) are  $\psi_j = \phi^j$ . Point forecasts (Example 7):  $X_{n+1}^n = \phi X_n$ ,  $X_{n+2}^n = \phi^2 X_n$ , and generally  $X_{n+h}^n = \phi^h X_n$ . Forecast-error variances:

$$\text{Var}(e_n(1)) = \sigma^2 \psi_0^2 = \sigma^2, \quad \text{Var}(e_n(2)) = \sigma^2(\psi_0^2 + \psi_1^2) = \sigma^2(1 + \phi^2),$$

and in general  $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}$ . As  $h \rightarrow \infty$  this tends to  $\frac{\sigma^2}{1 - \phi^2} = \gamma_0$ , the marginal variance of an AR(1) — confirming the saturation. A 95% interval for  $X_{n+2}$  is  $\phi^2 X_n \pm 1.96 \sigma \sqrt{1 + \phi^2}$ .

### Exercise 7: Re-integrating a random walk with drift [New]

Let  $X_t$  be ARIMA(0, 1, 0) with drift:  $Z_t = X_t - X_{t-1} = \alpha + \varepsilon_t$ ,  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ . Find  $X_{n+h}^n$  and  $\text{Var}(e_n(h))$ , and show the MSE diverges.

#### Solution

The differenced series  $Z_t$  is white noise plus a constant mean  $\mathbb{E}[Z_t] = \alpha$ , so  $Z_{n+h}^n = \alpha$  for all  $h$ . Re-integrating (Proposition on integrated processes),

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n = X_n + \alpha h,$$

a straight drift line from the last observation. The forecast error is  $e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=1}^h \varepsilon_{n+j}$  (the accumulated future innovations), so

$$\text{Var}(e_n(h)) = h\sigma^2 \xrightarrow{h \rightarrow \infty} \infty.$$

Thus  $P_{n+h}^n = h\sigma^2$  grows linearly and the prediction interval  $X_n + \alpha h \pm 1.96\sigma\sqrt{h}$  fans out without bound — the hallmark of an integrated process (no mean reversion).

## 9 Diagnostics, Model Selection and Long Memory

Once a candidate ARIMA model has been identified and fitted, two questions remain: *does it fit?* (diagnostics, based on the residuals) and *is it the right model among several?* (model selection, based on information criteria). This part collects the standard residual checks (mean, variance, Gaussianity, autocorrelation), the Box–Pierce and Ljung–Box portmanteau tests, the AIC/AICc/BIC criteria, and the Box–Jenkins methodology that ties them together. It then steps outside the ARMA world entirely: ARMA covariances decay exponentially fast, but many real series have covariances that decay only *polynomially*. This is **long memory**, modelled by fractional differencing  $(1 - B)^d$  and the FARIMA class, and intimately related to self-similarity and fractional Brownian motion.

### 9.1 Definitions

#### Definition 1: Residuals and standardized residuals

For a fitted model with fitted value  $\hat{X}_t$  at time  $t$ , the **residuals** are

$$e_t = X_t - \hat{X}_t,$$

and the **standardized residuals** are

$$\tilde{e}_t = \frac{e_t}{\sqrt{\text{Var}(e_t)}}.$$

Model checking is built almost entirely on the  $\{e_t\}$  (or  $\{\tilde{e}_t\}$ ): if the model is correct, the residuals should behave like the driving white noise. The four questions to ask are:

1. Do the residuals have constant zero mean?
2. Is their variance constant in  $t$  (*homoscedasticity*)?
3. Are they uncorrelated in  $t$ ?
4. Are they Gaussian?

## Why residuals should be the noise

A correctly specified model has captured all the predictable (linearly dependent) structure of  $\{X_t\}$ . What is left over,  $e_t = X_t - \hat{X}_t$ , should be the *unpredictable* part — the innovation  $\varepsilon_t$ . So every property we demanded of the white-noise input (zero mean, constant variance, no correlation, optionally Gaussianity) is exactly what we now *test for* in the residuals. Failure of any of the four points the modeller back to a deficiency: a trend left in (mean), volatility clustering (variance), an unmodelled AR/MA term (correlation), or non-normal innovations (Gaussianity).

### Definition 2: Checking mean, variance and Gaussianity

Three of the four checks are visual:

- **Mean:** plot  $\{(t, \tilde{e}_t)\}$ . There should be no trend and the points should hover around zero.
- **Variance (homoscedasticity):** plot  $\{(t, \tilde{e}_t)\}$  and look for constant spread; a fanning-out or fanning-in band signals non-constant variance.
- **Gaussianity:** a **Q–Q plot** of the residual quantiles against normal quantiles; Gaussian residuals lie on a straight line, whereas heavy tails (e.g. Student) curve away at the ends. Note this checks only *marginal* Gaussianity.

### Definition 3: Sample residual autocorrelation & the correlogram band

The residual sample autocorrelation at lag  $\tau$  is

$$\hat{r}_\tau = \hat{\rho}_\tau^{(e)} = \frac{\sum_{t=1}^{n-\tau} (e_{t+\tau} - \bar{e})(e_t - \bar{e})}{\sum_{t=1}^n (e_t - \bar{e})^2}.$$

Plotting  $\hat{r}_\tau$  against  $\tau$  gives the residual **correlogram**. Under the null hypothesis that the residuals are white noise, each  $\hat{r}_\tau$  is approximately  $\mathcal{N}(0, 1/n)$ , so individual values are compared against the band

$$\left( -\frac{1.96}{\sqrt{n}}, \frac{1.96}{\sqrt{n}} \right),$$

the (pointwise) 95% acceptance band for “ $\hat{r}_\tau$  is zero”.

### Definition 4: $R^2$ statistic

With  $s_e^2$  the residual variance and  $s_y^2$  the sample variance of the data,

$$R^2 = 1 - \frac{s_e^2}{s_y^2}.$$

It measures the fraction of variance “explained” by the fit. In time-series model selection it is a **dangerous** criterion (see Proposition 13): adding parameters can only increase  $R^2$ , and even the *true* model has a bounded  $R^2$  because a stochastic process is not perfectly predictable.

### Definition 5: Information criteria: AIC, AICc, BIC

For a model with  $k$  estimated parameters fitted to  $n$  observations, with maximized likelihood  $L(\theta | y)$ :

$$\text{AIC}(\theta) = -2 \log L(\theta | y) + 2k,$$

$$\text{AICc}(\theta) = -2 \log L(\theta | y) + 2k \frac{n}{n - k - 1} = \text{AIC}(\theta) + \frac{2k(k + 1)}{n - k - 1},$$

$$\text{BIC}(\theta) = -2 \log L(\theta | y) + k \log(n).$$

**Smaller is better** for all three. For an ARMA( $p, q$ ) the parameter count is

$$k = p + q + 1$$

( $p$  AR coefficients,  $q$  MA coefficients, plus the noise variance  $\sigma^2$ ). AIC is known to *overestimate*  $p$ ; AICc (Hurvich & Tsai, 1989) corrects the small-sample bias and coincides with AIC as  $n \rightarrow \infty$ ; BIC penalizes parameters more heavily (for  $n > e^2 \approx 7.4$ ) and so favours more parsimonious models.

#### Definition 6: Long memory (covariance definition)

A second-order stationary process  $\{X_t\}$  with ACVS  $\{\gamma_\tau\}$  is said to exhibit **long memory** (long-range dependence) if its covariances decay only polynomially:

$$\gamma_\tau \sim L_\gamma(\tau) |\tau|^{-\alpha}, \quad |\tau| \rightarrow \infty,$$

where  $L_\gamma$  is a *slowly varying* function and the exponent satisfies  $0 < \alpha < 1$ . Contrast this with ARMA models, whose covariances are either finite-support ( $\gamma_\tau = 0$  for  $|\tau| > q$ ) or exponentially decaying ( $\gamma_\tau \sim |\phi|^\tau$ ). The full admissible range  $0 < \alpha < 2$  yields three regimes:

$$\underbrace{0 < \alpha < 1}_{\text{long memory}}, \quad \underbrace{\alpha = 1}_{\text{short-range dependence}}, \quad \underbrace{1 < \alpha < 2}_{\text{anti-persistence}}.$$

#### Definition 7: Long memory (spectral definition)

Equivalently, long memory is characterized by a spectral density that blows up (or vanishes) like a power of the frequency near the origin:

$$\mathcal{S}(f) \sim L_S(f) |f|^{\alpha-1}, \quad f \rightarrow 0,$$

with  $L_S$  slowly varying. For  $0 < \alpha < 1$  the exponent  $\alpha - 1 \in (-1, 0)$ , so  $\mathcal{S}(f) \rightarrow \infty$  as  $f \rightarrow 0$  (a spectral pole at zero): the process concentrates power at the lowest frequencies, the spectral signature of persistence. The slowly varying parts of the two definitions are linked by

$$L_S(f) = 2\gamma(0)\Gamma(1-\alpha)\sin\frac{\pi\alpha}{2}, \quad L_\gamma(\tau) = 2\Gamma(\alpha)\sin\frac{\pi(1-\alpha)}{2}.$$

The case  $\alpha = 1$  (short-range dependence) gives  $\mathcal{S}(0)$  finite and nonzero, exactly the ARMA situation  $\mathcal{S}(0) = \sum_\tau \gamma_\tau < \infty$ ; the anti-persistent case  $1 < \alpha < 2$  forces  $\sum_k \gamma_k = 0$  (and  $\mathcal{S}(0) = 0$ ).

#### Definition 8: Fractional differencing operator & FARIMA

The **fractional differencing operator** is defined for real  $d$  via the binomial expansion of  $(1 - B)^d$  in the backshift operator  $B$ :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k B^k = \sum_{k=0}^{\infty} \frac{\Gamma(d+1)}{\Gamma(k+1)\Gamma(d-k+1)} (-1)^k B^k = \sum_{j=0}^{\infty} \frac{\Gamma(j-d)}{\Gamma(-d)\Gamma(j+1)} B^j.$$

The simplest long-memory process is the **fractionally differenced** (pure) process  $(1 - B)^d X_t = \varepsilon_t$ . The general **FARIMA** (fractional ARIMA) class is the stationary solution of

$$(1 - B)^d \phi(B) X_t = \psi(B) \varepsilon_t, \quad d = \frac{1 - \alpha}{2} \in (-0.5, 0.5),$$

where  $\varepsilon_t$  are i.i.d. zero-mean with variance  $\sigma_\varepsilon^2 > 0$ , and  $\phi(z)$ ,  $\psi(z)$  are degree- $p$  and degree- $q$  polynomials with all roots outside the unit circle. Integer  $d$  recovers ordinary ARIMA; fractional

$d \in (0, 0.5)$  gives long memory.

### Definition 9: Self-similarity & fractional Brownian motion

A process  $\{Y_t\}$  is **self-similar** with self-similarity parameter (Hurst exponent)  $H$  if, for every  $c > 0$ ,

$$Y_{ct} \stackrel{d}{=} c^H Y_t$$

(equality in distribution, jointly over  $t$ ). Self-similarity with stationary increments forces the covariance structure

$$\text{Cov}(Y_t, Y_s) = \frac{\sigma^2}{2} \left( |t|^{2H} + |s|^{2H} - |t - s|^{2H} \right). \quad (\star)$$

A *Gaussian* process with covariance  $(\star)$  is **fractional Brownian motion**  $B_H(t)$ ; it is the *only* self-similar Gaussian process. Its increment process  $X_t = B_H(t) - B_H(t - 1)$  is **fractional Gaussian noise**, with

$$\gamma_X(\tau) = \frac{\sigma^2}{2} \left( |\tau + 1|^{2H} - 2|\tau|^{2H} + |\tau - 1|^{2H} \right) \sim H(2H - 1) |\tau|^{2H-2}, \quad |\tau| \rightarrow \infty.$$

The graph  $(t, B_H(t))$  has Hausdorff (fractal) dimension  $D = 2 - H$  in one dimension. Fractal dimension is a *local* property; long-range dependence is a *global* one — the two can be made completely independent.

## 9.2 Theorems & Propositions

### Proposition 10: Residuals of a correctly specified AR(1) are the noise

For an AR(1) model  $X_t = \phi X_{t-1} + \varepsilon_t$  with known true parameters, the fitted one-step value is  $\hat{X}_t = \phi X_{t-1}$ , so the residual is

$$e_t = X_t - \phi X_{t-1} = \varepsilon_t.$$

*Proof idea:* substitute the model equation; the AR structure cancels exactly, leaving the innovation.

*Why:* this is the template for *all* residual diagnostics. If the model and parameters are right, residuals reproduce the white noise: constant zero mean, constant variance, uncorrelated, and Gaussian whenever  $\varepsilon_t$  is. Any departure from these is evidence *against* the model.

### Proposition 11: Box–Pierce portmanteau statistic

To test  $H_0 : \rho_1 = \rho_2 = \dots = \rho_m = 0$  (residuals white up to lag  $m$ ) against the alternative that some  $\rho_\tau \neq 0$ , the **Box–Pierce** statistic is

$$Q_m = n \sum_{\tau=1}^m \hat{r}_\tau^2.$$

Under  $H_0$ , for an ARMA( $p, q$ ) fit,  $Q_m$  is approximately  $\chi_{m-p-q}^2$ .

*Proof idea:* under white noise each  $\sqrt{n} \hat{r}_\tau \xrightarrow{d} \mathcal{N}(0, 1)$  and the  $\hat{r}_\tau$  are asymptotically independent, so  $\sum_\tau (\sqrt{n} \hat{r}_\tau)^2$  is a sum of squared standard normals; fitting  $p + q$  parameters removes  $p + q$  degrees of freedom.

*Why:* it tests *many* lags at once (a “portmanteau”), avoiding the multiple-testing problem of eyeballing the correlogram lag by lag.

### Proposition 12: Ljung–Box statistic (modified Box–Pierce)

The small-sample-corrected (**Ljung–Box**) statistic is

$$Q_m = n(n+2) \sum_{\tau=1}^m \frac{\hat{r}_\tau^2}{n-\tau},$$

again approximately  $\chi_{m-p-q}^2$  under  $H_0$  for an ARMA( $p, q$ ).

*Proof idea:* the factor  $\frac{n+2}{n-\tau}$  rescales each term so its finite- $n$  variance matches the  $\chi^2$  target more closely than the plain  $n\hat{r}_\tau^2$ ; as  $n \rightarrow \infty$ ,  $\frac{n(n+2)}{n-\tau} \rightarrow n$  and the two statistics agree.

*Why:* the Box–Pierce  $\chi^2$  approximation is poor at realistic sample sizes, so Ljung–Box is the version used in practice (e.g. R's `Box.test(type="Ljung-Box")`). One usually computes  $Q_m$  over a range of  $m$  and plots the resulting  $p$ -values.

### Classic Pitfall

The degrees of freedom are  $m - p - q$ , **not**  $m$ . You must subtract the number of *estimated* ARMA parameters ( $p + q$ ; the noise variance is *not* subtracted). Using  $\text{df} = m$  inflates the critical value and makes you reject far too rarely. Also require  $m > p + q$  so the df is positive, and  $m$  should be large enough to catch the lags of interest but not so large that the high-lag  $\hat{r}_\tau$  (estimated from few pairs) add only noise.

### Proposition 13: The danger of $R^2$ : an AR(1) caps at $\phi^2$

For an AR(1) with innovation variance  $\sigma^2$ , the process variance is  $s_y^2 = \sigma^2/(1 - \phi^2)$  and the one-step residual variance is  $s_e^2 = \sigma^2$ , so

$$R^2 = 1 - \frac{s_e^2}{s_y^2} = 1 - \frac{\sigma^2(1 - \phi^2)}{\sigma^2} = \phi^2.$$

*Proof idea:* substitute the AR(1) variance formula (geometric sum of the MA( $\infty$ ) coefficients  $\phi^{2j}$ ) and the fact that the best one-step prediction error is  $\varepsilon_t$ .

*Why:* even the *true* model has bounded  $R^2$  ( $\phi = 0.4 \Rightarrow R^2 = 0.16$ , which “looks bad” but is optimal). A stochastic process is not perfectly predictable, so a low  $R^2$  is no indictment, and a high one (from piling on parameters) is no endorsement. **Use information criteria, not  $R^2$ .**

### Proposition 14: AIC vs. AICc gap

The corrected and uncorrected AIC differ by exactly the parameter-count penalty

$$\text{AICc}(\theta) - \text{AIC}(\theta) = 2k \left( \frac{n}{n-k-1} - 1 \right) = \frac{2k(k+1)}{n-k-1} (> 0).$$

*Proof idea:* algebra — put  $\frac{n}{n-k-1} - 1 = \frac{k+1}{n-k-1}$  over a common denominator.

*Why:* the correction is always positive (it penalizes complexity more), vanishes as  $n \rightarrow \infty$  with  $k$  fixed, and stays negligible whenever  $k^2/n \rightarrow 0$ . AICc is therefore the safe default in small samples or when  $k$  is comparable to  $n$ .

### Proposition 15: BIC penalty exceeds AIC penalty for $n > e^2$

The per-parameter penalties are 2 (AIC) and  $\log n$  (BIC). Hence the BIC penalty is the larger of the two iff

$$\log n > 2 \iff n > e^2 \approx 7.39.$$

*Proof idea:* compare the per-parameter coefficients 2 and  $\log n$  directly.

*Why:* for any realistic  $n$  ( $\geq 8$ ) BIC charges more per parameter, hence is more conservative and

consistent (selects the true order with probability  $\rightarrow 1$ ), whereas AIC is more permissive and tends to over-fit  $p$ . Choose BIC when you want parsimony / consistency, AIC(c) for predictive performance.

### Proposition 16: Long-memory inflation of $\text{Var}(\bar{X})$

For a stationary process with absolutely continuous spectrum,  $\text{Var}(\bar{X}) \sim \mathcal{S}(0)/n$  (the classical  $1/n$  rate, since  $\mathcal{S}(0) = \sum_{\tau} \gamma_{\tau} < \infty$ ). Under long memory with  $\gamma_{\tau} \sim |\tau|^{-\alpha}$  ( $0 < \alpha < 1$ ), however,

$$\text{Var}(\bar{X}) \sim \frac{c}{n^{\alpha}}, \quad 0 < \alpha < 1,$$

i.e. the variance decays *slower* than  $1/n$ .

*Proof idea:*  $\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i,j} \gamma_{j-i}$ ; aggregating  $\sum_{j=-n}^n \gamma_j$  with  $\gamma_j \sim |j|^{-\alpha}$  recovers a power  $n^{1-\alpha}$ , so dividing by  $n$  gives  $n^{-\alpha}$ .

*Why:* this is the empirical fingerprint of long memory (Smith, 1938): sample means converge much more slowly than i.i.d. or short-range theory predicts, so naive standard errors  $\sqrt{\mathcal{S}(0)/n}$  are badly too small.

### Theorem 17: Spectral density of a FARIMA / fractionally differenced process

The spectral density of the FARIMA process  $(1 - B)^d \phi(B)X_t = \psi(B)\varepsilon_t$  is

$$\mathcal{S}(f) = \sigma_{\varepsilon}^2 \frac{|\psi(e^{-2\pi i f})|^2}{|\phi(e^{-2\pi i f})|^2} |1 - e^{-2\pi i f}|^{-2d} \xrightarrow{f \rightarrow 0} c_f |f|^{-2d}, \quad c_f = \sigma_{\varepsilon}^2 \frac{|\psi(1)|^2}{|\phi(1)|^2}.$$

*Proof idea:* the spectral density of a filtered process multiplies the input spectrum by the squared transfer function; the fractional difference contributes  $|1 - e^{-2\pi i f}|^{-2d}$ , and near  $f = 0$ ,  $|1 - e^{-2\pi i f}| \sim 2\pi |f|$ , leaving the pole  $|f|^{-2d}$ .

*Why:* it shows the long-memory parameter  $\alpha$  ( $d = (1 - \alpha)/2$ , so  $-2d = \alpha - 1$ ) is read off the slope of  $\log \mathcal{S}$  vs.  $\log |f|$  near the origin — the basis of log-periodogram (GPH) estimation of  $d$ .

### Where the spectral pole comes from

Differencing once,  $(1 - B)$ , has transfer function  $1 - e^{-2\pi i f}$ , which is *zero* at  $f = 0$ : it kills the trend / mean (a high-pass filter). Taking a *fractional* negative power,  $(1 - B)^{-|d|}$ , inverts a fraction of that, producing a mild *pole* at  $f = 0$  instead. So a long-memory series is, loosely, “a little bit non-stationary”: it has more low-frequency power than any ARMA but not so much that it fails to be stationary (as long as  $d < 1/2$ ).

### Proposition 18: Estimating $d$ from the periodogram (log-periodogram idea)

Since  $\mathbb{E} I(f) \approx \mathcal{S}(f) \sim L_{\mathcal{S}}(f) |f|^{\alpha-1}$  near  $f = 0$ , taking logarithms gives, for low Fourier frequencies  $f_j$ ,

$$\log I(f_j) \approx \text{const} + (\alpha - 1) \log |f_j| + \text{error} = \text{const} - 2d \log |f_j| + \text{error}.$$

*Proof idea:* the periodogram is an (asymptotically unbiased but inconsistent) estimate of the spectral density; regressing  $\log I(f_j)$  on  $\log |f_j|$  over the lowest frequencies estimates the slope  $\alpha - 1 = -2d$ .

*Why:* it turns long-memory estimation into a simple linear regression on the log-log periodogram; one must take  $\log$  *before* taking expectations, because  $\mathbb{E} \log I \neq \log \mathbb{E} I$ .

### 9.3 Key Techniques

#### Key Technique 1: Running a Ljung–Box test ( $df = m - p - q$ )

Given residuals from a fitted ARMA( $p, q$ ) on  $n$  observations:

1. Compute the residual sample autocorrelations  $\hat{r}_1, \dots, \hat{r}_m$  (choose  $m > p + q$ , often  $m \approx 10$  or  $\approx \log n$ ).
2. Form the Ljung–Box statistic  $Q_m = n(n+2) \sum_{\tau=1}^m \frac{\hat{r}_\tau^2}{n-\tau}$ .
3. Degrees of freedom:  $df = m - p - q$ .
4. Critical value at level  $\alpha$ : the upper- $\alpha$  quantile of  $\chi_{df}^2$  (in R: `qchisq(1- $\alpha$ , df)`); reject  $H_0$  if  $Q_m$  exceeds it (equivalently if the  $p$ -value = `1-pchisq(Q, df)` is  $< \alpha$ ).
5. **Reject**  $\Rightarrow$  residual autocorrelation remains  $\Rightarrow$  the model is inadequate. **Fail to reject**  $\Rightarrow$  no evidence against whiteness  $\Rightarrow$  model adequate.

#### Key Technique 2: Choosing between AIC, AICc and BIC

1. Fit every candidate model and record  $-2 \log L$  and  $k = p + q + 1$ .
2. Compute the desired criterion; **pick the smallest**.
3. **Which one?**
  - **AICc** when  $n$  is small or  $k$  is comparable to  $n$  (it corrects AIC's tendency to over-fit  $p$ ); always safe since  $AICc \rightarrow AIC$  for large  $n$ .
  - **BIC** when you want a *parsimonious*, consistent choice (its  $\log n$  penalty dominates for  $n > e^2$ ).
  - **AIC** for predictive/forecasting goals where slight over-fitting is tolerable.
4. For an extra parameter (going from  $k-1$  to  $k$  estimated parameters) to be worth it, the drop in  $-2 \log L$  must beat the increase in the penalty: 2 (AIC),  $\log n$  (BIC), or the (larger) AICc increment

$$\Delta_{AICc} = 2 + \frac{2k(k+1)}{n-k-1} - \frac{2(k-1)k}{n-k} \quad (> 2).$$

#### Key Technique 3: Box–Jenkins model-building loop

The Box–Jenkins methodology iterates four steps:

1. **Identify** reasonable  $p, d, q$ :
  - *Choosing  $d$* : plot the data; if non-stationary, plot differences; use the smallest order of differencing that stabilizes the series.
  - *Choosing  $p, q$  (on the differenced data)*: read the ACF and PACF — a sharp drop in the ACF at lag  $q$  suggests MA( $q$ ); a sharp drop in the PACF at lag  $p$  suggests AR( $p$ ); no sharp drop suggests a mixed ARMA; very slow decay suggests non-stationarity (re-difference).
2. **Estimate** the parameters of the proposed ARIMA model.
3. **Diagnose** via residuals (mean/variance/QQ/correlogram + Ljung–Box); if inadequate, return to step 1.
4. **Use** the validated model (forecasting or other inference).

#### Key Technique 4: Identifying $p, q$ from ACF/PACF cut-offs

Memorize the dual table (computed on the appropriately differenced series):

| Model          | ACF                                      | PACF                                     |
|----------------|------------------------------------------|------------------------------------------|
| AR( $p$ )      | tails off (decays)                       | <b>cuts off after lag <math>p</math></b> |
| MA( $q$ )      | <b>cuts off after lag <math>q</math></b> | tails off (decays)                       |
| ARMA( $p, q$ ) | tails off                                | tails off                                |
| non-stationary | very slow decay                          | —                                        |

“Cuts off” means the values fall inside the  $\pm 1.96/\sqrt{n}$  band beyond the order; “tails off” means a geometric/sinusoidal decay that never abruptly stops.

#### Key Technique 5: Reading the long-memory regime from $\alpha$ (or $d$ , or $H$ )

The same phenomenon has three equivalent parameters. Convert and classify:

$$d = \frac{1 - \alpha}{2}, \quad H = d + \frac{1}{2} = 1 - \frac{\alpha}{2}.$$

| Regime                   | $\alpha$         | $d$                    | $H$                   |
|--------------------------|------------------|------------------------|-----------------------|
| Long memory (persistent) | $0 < \alpha < 1$ | $0 < d < \frac{1}{2}$  | $\frac{1}{2} < H < 1$ |
| Short-range dependence   | $\alpha = 1$     | $d = 0$                | $H = \frac{1}{2}$     |
| Anti-persistence         | $1 < \alpha < 2$ | $-\frac{1}{2} < d < 0$ | $0 < H < \frac{1}{2}$ |

Signatures: long memory  $\Rightarrow \mathcal{S}(0) = \infty$  (spectral pole),  $\sum_{\tau} \gamma_{\tau} = \infty$ ; anti-persistence  $\Rightarrow \mathcal{S}(0) = 0$  and  $\sum_k \gamma_k = 0$ ; short-range  $\Rightarrow \mathcal{S}(0)$  finite, nonzero (ARMA).

## 9.4 Exercises

#### Exercise 1: When are AIC and AICc equivalent? [Sheet 11]

The criteria are  $\text{AIC} = -2 \ln L + 2k$  and  $\text{AICc} = -2 \ln L + 2k \frac{n}{n-k-1}$ . Explain under what conditions on  $n$  and  $k$  the two become approximately equivalent.

#### Solution

Subtract:

$$\text{AICc} - \text{AIC} = 2k \left( \frac{n}{n-k-1} - 1 \right) = 2k \cdot \frac{n - (n-k-1)}{n-k-1} = 2k \cdot \frac{k+1}{n-k-1} = \frac{2k(k+1)}{n-k-1}.$$

The two criteria are approximately equivalent exactly when this correction term is small. In particular:

- If  $k$  is **fixed** and  $n \rightarrow \infty$ , then  $\frac{2k(k+1)}{n-k-1} \rightarrow 0$ , so  $\text{AICc} \rightarrow \text{AIC}$ .
- If  $k = k_n$  grows with  $n$ , the correction is still small provided  $n$  is large relative to  $k_n^2$ , e.g.  $k_n^2/n \rightarrow 0$ . If  $k$  grows too fast ( $k_n^2/n \not\rightarrow 0$ ) the gap need not vanish and AICc stays meaningfully larger than AIC.

So: *large samples relative to the number of parameters* make the two agree; small samples or richly parameterized models make AICc strictly more conservative.

**Exercise 2: AICc with  $m$  coefficients fixed to zero** [Sheet 11]

Determine the form of AICc for a mean-zero ARMA( $p, q$ ) model when exactly  $m$  of the AR and MA coefficients are known in advance and fixed to zero.

**Solution**

If  $m$  of the  $p + q$  AR/MA coefficients are fixed (not estimated), only  $p + q - m$  of them are estimated. Adding the innovation variance  $\sigma^2$ , the estimated-parameter count is

$$k = p + q + 1 - m.$$

Substituting into the AICc formula,

$$\text{AICc}(\theta) = -2 \ln L(\theta | y) + 2(p + q + 1 - m) \frac{n}{n - (p + q + 1 - m) - 1}.$$

(For the mean-zero model no separate mean parameter is counted.) Fixing coefficients to zero *lowers*  $k$ , which both reduces the penalty and *increases* the residual degrees of freedom in any subsequent Ljung–Box test.

**Exercise 3: Compute  $Q_4$  and test (Ljung–Box)** [Sheet 11]

For  $n = 150$  observations a mean-zero ARMA(1, 1) is fitted; the residual autocorrelations at lags 1–4 are

$$\hat{r}_1 = 0.03, \quad \hat{r}_2 = -0.09, \quad \hat{r}_3 = 0.11, \quad \hat{r}_4 = -0.05.$$

(a) Compute the Ljung–Box statistic  $Q_4$ . (b) Test  $H_0 : \rho_1 = \dots = \rho_4 = 0$  at the 5% level (state the df, the critical value, and the conclusion). (c) Would the conclusion change for an AR(1) fit?

**Solution**

(a) With  $n = 150$ ,  $n(n + 2) = 150 \cdot 152 = 22800$ , and  $m = 4$ :

$$Q_4 = 22800 \left( \frac{0.03^2}{149} + \frac{(-0.09)^2}{148} + \frac{0.11^2}{147} + \frac{(-0.05)^2}{146} \right).$$

Term by term,

$$\frac{0.0009}{149} = 6.04 \times 10^{-6}, \quad \frac{0.0081}{148} = 5.47 \times 10^{-5}, \quad \frac{0.0121}{147} = 8.23 \times 10^{-5}, \quad \frac{0.0025}{146} = 1.71 \times 10^{-5}.$$

Summing:  $6.04 \times 10^{-6} + 5.47 \times 10^{-5} + 8.23 \times 10^{-5} + 1.71 \times 10^{-5} = 1.60 \times 10^{-4}$ . Hence

$$Q_4 \approx 22800 \times 1.60 \times 10^{-4} \approx 3.65.$$

(b) For ARMA(1, 1),  $p = q = 1$ , so  $\text{df} = m - p - q = 4 - 1 - 1 = 2$ . The 5% critical value is the upper-5% point of  $\chi^2_2$ , i.e.  $\text{qchisq}(0.95, \text{df}=2) = 5.991$ . Since  $Q_4 \approx 3.65 < 5.991$ , we **do not reject**  $H_0$ : there is no significant evidence of residual autocorrelation, and the ARMA(1, 1) appears adequate.

(c) For AR(1),  $p = 1, q = 0$ , so  $\text{df} = 4 - 1 - 0 = 3$  and the critical value rises to  $\text{qchisq}(0.95, \text{df}=3) = 7.815$ . Since  $3.65 < 7.815$  the conclusion is unchanged (do not reject). In general, fewer fitted parameters leave *more* degrees of freedom, widening the acceptance region.

#### Exercise 4: AIC vs. BIC penalty comparison [Sheet 11]

With  $\text{AIC} = -2\log L + 2k$  and  $\text{BIC} = -2\log L + k\log n$ : (a) Show the BIC penalty exceeds the AIC penalty iff  $n > e^2 \approx 7.4$ . (b) For fixed  $\log L$ , a  $k = 3$  model competes with a  $k = 2$  model at  $n = 50$ . Which criterion is more likely to select the larger model?

#### Solution

(a) Per parameter the penalties are 2 (AIC) and  $\log n$  (BIC), so

$$\log n > 2 \iff n > e^2 \approx 7.39.$$

For any realistic  $n$  ( $\geq 8$ ) BIC charges the larger penalty per parameter and hence favours more parsimonious models.

(b) Going from  $k = 2$  to  $k = 3$  adds one parameter. The extra penalty is

$$\text{AIC: } 2 \times 1 = 2, \quad \text{BIC: } \log(50) \approx 3.91.$$

So AIC selects the larger model whenever  $-2\log L$  drops by more than 2 (i.e. the log-likelihood improves by  $> 1$ ), while BIC requires the drop to exceed 3.91 (log-likelihood improvement  $> 1.96$ ). **AIC is more willing to select the larger model**; BIC is harder to satisfy and protects against over-fitting.

#### Exercise 5: Box–Pierce vs. Ljung–Box on the same data [New]

Using the data of Exercise 3 ( $n = 150$ ,  $\hat{r}_{1:4} = 0.03, -0.09, 0.11, -0.05$ , ARMA(1, 1) fit), compute the *Box–Pierce* statistic  $Q_4^{\text{BP}}$  and compare it with the Ljung–Box value  $Q_4 \approx 3.65$ . Does the test decision change?

#### Solution

The Box–Pierce statistic is  $Q_m^{\text{BP}} = n \sum_{\tau=1}^m \hat{r}_\tau^2$ :

$$\sum_{\tau=1}^4 \hat{r}_\tau^2 = 0.03^2 + 0.09^2 + 0.11^2 + 0.05^2 = 0.0009 + 0.0081 + 0.0121 + 0.0025 = 0.0236.$$

Hence

$$Q_4^{\text{BP}} = 150 \times 0.0236 = 3.54.$$

This is slightly smaller than the Ljung–Box  $Q_4 \approx 3.65$  (Ljung–Box inflates each term by  $\frac{n+2}{n-2} > 1$ ). Both use  $\text{df} = 4 - 1 - 1 = 2$  and the same critical value 5.991; since  $3.54 < 5.991$  and  $3.65 < 5.991$ , **both fail to reject  $H_0$** . With  $n = 150$  the two agree closely; the difference matters only at small  $n$ , where the Box–Pierce  $\chi^2$  approximation is unreliable and Ljung–Box is preferred.

#### Exercise 6: $R^2$ of a fitted AR(1) [New]

An AR(1) model is fitted to a long stationary series and the estimated coefficient is  $\hat{\phi} = 0.7$ . (a) What value of  $R^2$  should you expect? (b) A colleague adds three more lags (fitting AR(4)) and reports  $R^2 = 0.55$ . Has the model genuinely improved? How would you decide?

#### Solution

(a) For an AR(1) the (population)  $R^2$  equals  $\phi^2$ :

$$R^2 = 1 - \frac{s_e^2}{s_y^2} = 1 - \frac{\sigma^2}{\sigma^2/(1 - \phi^2)} = \phi^2 = 0.7^2 = 0.49.$$

So roughly  $R^2 \approx 0.49$  even for the correct, optimal model — a “modest” value that is in fact the best attainable.

(b) A rise from 0.49 to 0.55 is *not* evidence of a better model: adding parameters can only increase the in-sample  $R^2$ . To decide, compare *information criteria*: compute AICc (or BIC) for the AR(1) and AR(4) and pick the smaller. Equivalently, test whether the extra coefficients are significant and whether the AR(1) residuals already pass a Ljung–Box test — if they do, the extra lags are spurious and the parsimonious AR(1) wins.

### Exercise 7: Classify a process by its long-memory parameter [New]

A stationary series has ACVS  $\gamma_\tau \sim c |\tau|^{-0.4}$  as  $|\tau| \rightarrow \infty$ . (a) Find  $\alpha$ , the fractional difference parameter  $d$ , and the Hurst exponent  $H$ . (b) Classify the dependence regime. (c) Is  $\sum_\tau \gamma_\tau$  finite? What does  $\mathcal{S}(f)$  do near  $f = 0$ ?

### Solution

(a) Matching  $\gamma_\tau \sim L_\gamma(\tau) |\tau|^{-\alpha}$  gives  $\alpha = 0.4$ . Then

$$d = \frac{1 - \alpha}{2} = \frac{1 - 0.4}{2} = 0.3, \quad H = d + \frac{1}{2} = 0.8 \quad (= 1 - \frac{\alpha}{2} = 1 - 0.2).$$

(b) Since  $0 < \alpha = 0.4 < 1$  (equivalently  $0 < d = 0.3 < \frac{1}{2}$ , or  $\frac{1}{2} < H = 0.8 < 1$ ), the process exhibits **long memory** / **positive persistence**.

(c) The tail  $|\tau|^{-0.4}$  is *not* summable ( $\sum |\tau|^{-\alpha}$  diverges for  $\alpha \leq 1$ ), so  $\sum_\tau \gamma_\tau = \infty$ . Correspondingly  $\mathcal{S}(f) \sim L_S(f) |f|^{\alpha-1} = L_S(f) |f|^{-0.6} \rightarrow \infty$  as  $f \rightarrow 0$ : a spectral pole at the origin, the hallmark of long memory.

### Exercise 8: Variance of $B_H$ and the self-similarity scaling [New]

For fractional Brownian motion  $B_H(t)$  with covariance  $\text{Cov}(B_H(t), B_H(s)) = \frac{\sigma^2}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H})$ : (a) compute  $\text{Var}(B_H(t))$ ; (b) verify the self-similarity relation  $B_H(ct) \stackrel{d}{=} c^H B_H(t)$  at the level of variances; (c) for  $H = \frac{1}{2}$  identify the process.

### Solution

(a) Set  $s = t$  in the covariance:

$$\text{Var}(B_H(t)) = \text{Cov}(B_H(t), B_H(t)) = \frac{\sigma^2}{2}(|t|^{2H} + |t|^{2H} - 0) = \sigma^2 |t|^{2H}.$$

(b) Then  $\text{Var}(B_H(ct)) = \sigma^2 |ct|^{2H} = c^{2H} \sigma^2 |t|^{2H} = \text{Var}(c^H B_H(t))$ , consistent with  $B_H(ct) \stackrel{d}{=} c^H B_H(t)$ . (For a centred Gaussian process, matching the full covariance structure — which scales by  $c^{2H}$  in every entry — gives equality in distribution.)

(c) For  $H = \frac{1}{2}$ ,  $\text{Var}(B_{1/2}(t)) = \sigma^2 |t|$  and  $\text{Cov}(B_{1/2}(t), B_{1/2}(s)) = \frac{\sigma^2}{2}(|t| + |s| - |t - s|) = \sigma^2 \min(t, s)$  for  $t, s \geq 0$  — exactly **standard Brownian motion**. Its increments are ordinary (uncorrelated) white noise:  $H = \frac{1}{2}$  is the short-range / memoryless boundary ( $d = 0$ ,  $\alpha = 1$ ).

## 10 Non-Linear Time Series: ARCH and GARCH

Linear models (AR, MA, ARMA) describe the conditional *mean* of a series and assume a constant innovation variance. Financial log-returns violate this: they are nearly uncorrelated (so a linear conditional-mean model is almost useless), yet their *squares* are strongly dependent — large moves cluster with large moves and quiet periods with quiet periods. This **volatility clustering**, together

with heavy (**fat**) tails, motivates models for the *conditional variance*. **ARCH** and **GARCH** keep the series mean-zero and uncorrelated (white noise) while letting the conditional variance  $\sigma_t^2$  evolve in time as a function of past squared returns (and, for GARCH, past variances). Throughout,  $\mathcal{F}_t$  denotes the information available up to and including time  $t$ , and  $\{\varepsilon_t\}$  is white noise with  $\mathbb{E}[\varepsilon_t] = 0$ ,  $\text{Var}(\varepsilon_t) = 1$ , independent of  $\mathcal{F}_{t-1}$ .

## 10.1 Definitions

### Definition 1: Log-returns & volatility clustering

For a price series  $\{X_t\}$  (e.g. a stock index), the **log-returns** are

$$r_t = \log\left(\frac{X_t}{X_{t-1}}\right) = \log(X_t) - \log(X_{t-1}),$$

measuring relative changes in price. Empirically log-returns show:

- **little linear dependence:** the sample ACF/PACF of  $r_t$  is essentially that of white noise;
- **dependent squares:** the sample ACF/PACF of  $r_t^2$  decays slowly  $\Rightarrow$  the conditional variance changes over time;
- **volatility clustering:** large changes tend to be followed by large changes, small by small;
- **fat tails:** kurtosis  $> 3$  (heavier tails than the normal), visible as deviations in a QQ-plot against  $\mathcal{N}(0, 1)$ .

### Definition 2: ARCH(1) model

$\{r_t\}$  is an **AutoRegressive Conditionally Heteroscedastic** model of order 1, ARCH(1), if for  $t \in \mathbb{Z}$

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2,$$

with  $\{\varepsilon_t\}$  white noise of mean 0 and variance 1, independent of  $\mathcal{F}_{t-1}$ , and parameters  $\alpha_0 > 0$ ,  $\alpha_1 \geq 0$ . The volatility  $\sigma_t$  is  $\mathcal{F}_{t-1}$ -measurable (known one step ahead). Immediate properties (Proposition 7):

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0, \quad \text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2.$$

### Definition 3: ARCH(p) model

$\{r_t\}$  is an ARCH( $p$ ) model if  $r_t = \sigma_t \varepsilon_t$  with

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \cdots + \alpha_p r_{t-p}^2 = \alpha_0 + \sum_{j=1}^p \alpha_j r_{t-j}^2,$$

$\{\varepsilon_t\}$  white noise (0,1) independent of  $\mathcal{F}_{t-1}$ , and  $\alpha_0 > 0$ ,  $\alpha_p > 0$ ,  $\alpha_j \geq 0$  for  $1 \leq j < p$ . Volatility may now depend on several past squared shocks, so as before  $\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$ .

### Definition 4: GARCH(p,q) model

$\{r_t\}$  is a **Generalised ARCH** model GARCH( $p, q$ ) if  $r_t = \sigma_t \varepsilon_t$  with

$$\sigma_t^2 = \alpha_0 + \sum_{j=1}^p \alpha_j r_{t-j}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

$\{\varepsilon_t\}$  white noise (0,1) independent of  $\mathcal{F}_{t-1}$ , and parameters  $\alpha_0 > 0$ ,  $\alpha_p > 0$ ,  $\beta_q > 0$ , with  $\alpha_j, \beta_j \geq 0$  otherwise. Feeding back past *variances*  $\sigma_{t-j}^2$  lets volatility persist over time far more

parsimoniously than an ARCH( $p$ ) with large  $p$ . A GARCH( $p, 0$ ) is exactly an ARCH( $p$ ). As before  $\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$ .

### Why feed back $\sigma_{t-1}^2$ ?

In ARCH(1) the conditional variance depends only on the *single* most recent squared return  $r_{t-1}^2$ . One quiet day ( $r_{t-1}^2$  small) instantly collapses  $\sigma_t^2$  back toward  $\alpha_0$ , so the model cannot sustain long high-volatility regimes. GARCH adds the smooth, slowly varying term  $\beta_1 \sigma_{t-1}^2$ : yesterday's *variance estimate* is mostly carried forward, so volatility is sticky. In practice fitted GARCH(1, 1) models have  $\alpha_1 + \beta_1$  close to 1, giving long memory in volatility (still geometric decay, but very slow).

### Definition 5: Standardised residuals (diagnostics)

After fitting, the **standardised residuals** are  $\tilde{e}_t = r_t / \hat{\sigma}_t$ . If the model is correct these should look like i.i.d. (0, 1) noise: their ACF/PACF *and* the ACF/PACF of their *squares*  $\tilde{e}_t^2$  should resemble white noise, and a QQ-plot against the assumed innovation law should be roughly straight. Under Gaussian innovations one also overlays in-sample bands  $\pm 1.96 \hat{\sigma}_t$  on  $r_t$ .

### Definition 6: ARMA–GARCH model

Because a GARCH process is itself white noise, it can serve as the innovation of an ARMA model: with  $r_t$  a GARCH( $\tilde{p}, \tilde{q}$ ) process,

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k r_{t-k}, \quad \theta_0 = -1,$$

where the  $\phi_j, \theta_k$  satisfy the usual ARMA( $p, q$ ) stationarity and invertibility conditions. This captures a conditional mean (ARMA) and a time-varying conditional variance (GARCH) at once; one must only be careful with the likelihood, as the errors are no longer i.i.d. Gaussian.

## 10.2 Theorems & Propositions

### Proposition 7: Conditional and unconditional moments of ARCH(1)

For the ARCH(1) model with  $\alpha_1 < 1$ :

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0, \quad \text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2, \quad \mathbb{E}[r_t] = 0, \quad \text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

*Proof idea:*  $\sigma_t$  is  $\mathcal{F}_{t-1}$ -measurable and  $\varepsilon_t$  is independent of  $\mathcal{F}_{t-1}$ , so  $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0$  and  $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[\varepsilon_t^2] = \sigma_t^2$ . Iterated expectation gives  $\mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]] = 0$  and  $\text{Var}(r_t) = \mathbb{E}[\sigma_t^2] = \alpha_0 + \alpha_1 \text{Var}(r_{t-1})$ ; under stationarity  $\text{Var}(r_t) = \text{Var}(r_{t-1})$ , solve to get  $\alpha_0 / (1 - \alpha_1)$ .

*Why:* establishes that ARCH is mean-zero with a positive finite unconditional variance exactly when  $\alpha_1 < 1$  (necessary *and*, for ARCH(1), sufficient for weak stationarity).

### Proposition 8: ARCH/GARCH is white noise (uncorrelated but dependent)

Any ARCH( $p$ ) or GARCH( $p, q$ ) process is mean-zero and serially uncorrelated: for  $\tau \neq 0$ ,

$$\mathbb{E}[r_t] = 0, \quad \text{Cov}(r_{t+\tau}, r_t) = 0, \quad \text{hence} \quad \text{Corr}(r_{t+\tau}, r_t) = 0.$$

*Proof idea:* take  $\tau > 0$ . Since  $r_t$  is  $\mathcal{F}_{t+\tau-1}$ -measurable and  $\sigma_{t+\tau}$  is too, condition on  $\mathcal{F}_{t+\tau-1}$ :

$$\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0.$$

*Why:* the central qualitative fact — ARCH/GARCH leaves the *linear* correlation structure flat (white noise) while creating strong *nonlinear* dependence in the squares. The conditional-mean and conditional-variance modelling problems decouple.

### Theorem 9: Isserlis' theorem (Wick formula)

If  $(X_1, \dots, X_N)$  is a zero-mean multivariate normal vector and  $N$  is even,

$$\mathbb{E}[X_1 \cdots X_N] = \sum_{p \in P_N} \prod_{\{i,j\} \in p} \text{Cov}(X_i, X_j),$$

where  $P_N$  is the set of all pairings (perfect matchings) of  $\{1, \dots, N\}$ . For  $N = 4$ ,

$$\mathbb{E}[X_1 X_2 X_3 X_4] = \mathbb{E}[X_1 X_2] \mathbb{E}[X_3 X_4] + \mathbb{E}[X_1 X_3] \mathbb{E}[X_2 X_4] + \mathbb{E}[X_1 X_4] \mathbb{E}[X_2 X_3].$$

*Proof idea:* expand the Gaussian moment generating function / cumulant expansion; only pair-cumulants (covariances) survive for a Gaussian.

*Why:* gives  $\mathbb{E}[\varepsilon_t^4] = 3(\mathbb{E}[\varepsilon_t^2])^2 = 3$  for standard Gaussian innovations, the key input to the ARCH fourth-moment and kurtosis computations below.

### Proposition 10: Fourth moment and kurtosis of ARCH(1)

For ARCH(1) with Gaussian innovations, the unconditional fourth moment is finite iff  $0 \leq \alpha_1^2 < \frac{1}{3}$ , in which case

$$\mathbb{E}[r_t^4] = \frac{3\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}, \quad \kappa = \frac{\mathbb{E}[r_t^4]}{\text{Var}(r_t)^2} = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} > 3 \quad (\alpha_1 > 0).$$

*Proof idea:*  $\mathbb{E}[r_t^4 | \mathcal{F}_{t-1}] = \sigma_t^4 \mathbb{E}[\varepsilon_t^4 | \mathcal{F}_{t-1}] = 3\sigma_t^4 = 3(\alpha_0 + \alpha_1 r_{t-1}^2)^2$  (Isserlis). Take expectations, substitute  $\mathbb{E}[r_{t-1}^2] = \alpha_0/(1 - \alpha_1)$  and  $\mathbb{E}[r_{t-1}^4] = \mathbb{E}[r_t^4]$  by stationarity, and solve the linear equation; divide by  $\text{Var}(r_t)^2 = \alpha_0^2/(1 - \alpha_1)^2$  for  $\kappa$ .

*Why:* shows ARCH *generates* excess kurtosis (fat tails) endogenously, even with Gaussian  $\varepsilon_t$  — matching the heavy tails seen in returns.

### Two nested constraints

$\alpha_1 < 1$  buys a finite *variance*; the strictly tighter  $\alpha_1^2 < \frac{1}{3}$  (i.e.  $\alpha_1 < 1/\sqrt{3} \approx 0.577$ ) buys a finite *fourth* moment. As  $\alpha_1 \uparrow 1/\sqrt{3}$  the kurtosis  $\rightarrow \infty$ : more ARCH effect  $\Rightarrow$  heavier tails. The fourth moment is the relevant one whenever you want the ACF of the squares to be well-defined.

### Proposition 11: ACF of the squares of an ARCH(1)

For ARCH(1) with  $0 < \alpha_0$  and  $0 < \alpha_1 < 1/\sqrt{3}$ ,

$$\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}, \quad \tau \in \mathbb{Z},$$

i.e. the autocorrelation of the squared series decays *geometrically*.

*Proof idea:* condition on  $\mathcal{F}_{t+\tau-1}$  to peel one step:  $\mathbb{E}[r_{t+\tau}^2 r_t^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 \sigma_{t+\tau}^2 = r_t^2(\alpha_0 + \alpha_1 r_{t+\tau-1}^2)$ . Taking expectations and subtracting  $(\mathbb{E}[r_t^2])^2$  yields the recursion  $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1 \text{Cov}(r_{t+\tau-1}^2, r_t^2)$  (the constant terms cancel using  $\mathbb{E}[r_t^2] = \alpha_0/(1 - \alpha_1)$ ). Iterating gives  $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1^\tau \text{Cov}(r_{t+1}^2, r_t^2)$ , hence the stated correlation.

*Why:* explains the slowly decaying ACF of squared returns and warns that ARCH(1) imposes *purely geometric* memory in volatility — often too fast for real data, motivating GARCH.

**Proposition 12: Stationarity & variance of GARCH(p,q)**

A GARCH( $p, q$ ) process has a weakly stationary solution **iff**

$$\sum_{j=1}^p \alpha_j + \sum_{j=1}^q \beta_j < 1,$$

and then its unconditional variance is

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \sum_{j=1}^p \alpha_j - \sum_{j=1}^q \beta_j}.$$

*Proof idea:* take unconditional expectations of  $\sigma_t^2 = \alpha_0 + \sum_j \alpha_j r_{t-j}^2 + \sum_j \beta_j \sigma_{t-j}^2$  using  $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \text{Var}(r_t)$  at stationarity, then solve; positivity of the denominator forces the stated bound.

*Why:* the master condition for fitting GARCH — it must hold for the fitted parameters, and the closed-form variance gives the long-run volatility level. (More:  $\sum \alpha + \sum \beta < 1$  is sufficient for strict stationarity; for some noise laws including Gaussian, the boundary  $\sum \alpha + \sum \beta = 1$  still admits a strictly — but not weakly — stationary solution: the IGARCH case.)

**Proposition 13: Fourth-moment condition for GARCH(1,1)**

For a GARCH(1,1) with Gaussian noise, the unconditional fourth moment  $\mathbb{E}[r_t^4]$  exists **iff**

$$3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1.$$

*Proof idea:* square the recursion for  $\sigma_t^2$ , take expectations using  $\mathbb{E}[\varepsilon_t^4] = 3$  (Isserlis) and stationarity of  $\mathbb{E}[\sigma_t^4]$ ; the coefficient of  $\mathbb{E}[\sigma_t^4]$  on the right is  $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2$ , which must be  $< 1$  for a finite solution.

*Why:* the analogue of  $\alpha_1^2 < 1/3$ ; needed before quoting any formula for the kurtosis or the ACF of squared GARCH returns.

**Proposition 14: Maximum likelihood estimation (conditional Gaussian)**

Conditioning on the first observation  $r_1$  to avoid specifying a marginal law for the initial value, the conditional likelihood factorises as

$$L_c(\theta \mid r_1, \dots, r_n) = \prod_{t=2}^n f(r_t \mid r_{t-1}, \dots, r_1; \theta).$$

Under Gaussian innovations the ARCH(1) model gives  $r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \alpha_0 + \alpha_1 r_{t-1}^2)$ , so

$$L_c(\alpha_0, \alpha_1) = \prod_{t=2}^n \frac{1}{\sqrt{2\pi(\alpha_0 + \alpha_1 r_{t-1}^2)}} \exp\left(-\frac{r_t^2}{2(\alpha_0 + \alpha_1 r_{t-1}^2)}\right),$$

maximised over  $\alpha_0 > 0$ ,  $\alpha_1 \geq 0$  (numerically; GARCH is analogous with  $\sigma_t^2$  computed by its recursion).

*Proof idea:* the chain rule for densities plus conditional normality of each  $r_t$  given the past.

*Why:* the standard estimation route for ARCH/GARCH; the per-term variance  $\sigma_t^2$  is what is being fitted, not a constant  $\sigma^2$ .

### Reading the (G)ARCH likelihood

Each factor is a normal density with its *own* variance  $\sigma_t^2$ . The log-likelihood is  $-\frac{1}{2} \sum_t [\log \sigma_t^2 + r_t^2/\sigma_t^2]$  (up to constants): the fit trades off “don’t make  $\sigma_t^2$  needlessly large” (the  $\log \sigma_t^2$  penalty) against “explain big  $r_t^2$  with big  $\sigma_t^2$ ” (the  $r_t^2/\sigma_t^2$  term). That is exactly volatility tracking.

### Summary: pros & cons

**Pros:** ARCH/GARCH capture volatility clusters and time-varying conditional variance; they reproduce fat tails (large kurtosis) and large shocks; GARCH gives persistent volatility parsimoniously; and crucially they do *not* disturb the (white-noise) correlation structure, so they slot in as ARMA innovations. **Cons:** positive and negative shocks affect volatility *symmetrically* (no leverage effect); volatility dependence decays only *geometrically*, often too fast for real markets. Extensions (EGARCH, GJR-GARCH, long-memory GARCH) address these but are out of scope here.

## 10.3 Key Techniques

### Key Technique 1: Unconditional variance & stationarity of any (G)ARCH

To get the long-run variance and the stationarity condition in one move:

1. Write the  $\sigma_t^2$  recursion and take *unconditional* expectations.
2. Replace, by stationarity,  $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \text{Var}(r_t) =: v$  (this uses  $\mathbb{E}[r_{t-j}^2 | \mathcal{F}_{t-j-1}] = \sigma_{t-j}^2$  and iterated expectation, so  $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2]$ ).
3. Solve the scalar equation  $v = \alpha_0 + (\sum_j \alpha_j + \sum_j \beta_j)v$ :

$$v = \frac{\alpha_0}{1 - \sum_j \alpha_j - \sum_j \beta_j}.$$

4. Positivity ( $v > 0$ , finite)  $\Leftrightarrow \sum_j \alpha_j + \sum_j \beta_j < 1$  — the stationarity condition.

Special cases: ARCH(1) gives  $\alpha_0/(1 - \alpha_1)$ ; ARCH( $p$ ) gives  $\alpha_0/(1 - \sum_{j=1}^p \alpha_j)$ ; GARCH(1, 1) gives  $\alpha_0/(1 - \alpha_1 - \beta_1)$ .

### Key Technique 2: Proving uncorrelatedness via $\mathcal{F}_{t-1}$ -measurability

To show  $\text{Cov}(r_{t+\tau}, r_t) = 0$  for  $\tau > 0$  in any (G)ARCH:

1. Note  $\mathbb{E}[r_t] = 0$ , so  $\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t]$ .
2. Condition on  $\mathcal{F}_{t+\tau-1}$  (which contains  $r_t$  and  $\sigma_{t+\tau}$ ):

$$\mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}[\mathbb{E}[\sigma_{t+\tau} \varepsilon_{t+\tau} r_t | \mathcal{F}_{t+\tau-1}]].$$

3. Pull out the  $\mathcal{F}_{t+\tau-1}$ -measurable factors  $r_t, \sigma_{t+\tau}$ :  $= \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} | \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0$ .

The exact same template, with  $\varepsilon_{t+\tau}^2$  in place of  $\varepsilon_{t+\tau}$  (and  $\mathbb{E}[\varepsilon^2 | \mathcal{F}] = 1$ ), peels one lag off  $\text{Cov}(r_{t+\tau}^2, r_t^2)$ .

### Key Technique 3: Geometric recursion for the ACF of squares

To get  $\text{Corr}(r_{t+\tau}^2, r_t^2)$  for ARCH(1):

1. Condition on  $\mathcal{F}_{t+\tau-1}$  and use  $\mathbb{E}[\varepsilon^2 | \cdot] = 1$ ,  $\sigma_{t+\tau}^2 = \alpha_0 + \alpha_1 r_{t+\tau-1}^2$ :

$$\mathbb{E}[r_{t+\tau}^2 r_t^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 (\alpha_0 + \alpha_1 r_{t+\tau-1}^2).$$

2. Take expectations, subtract  $(\mathbb{E}[r_t^2])^2$ , and insert  $\mathbb{E}[r_t^2] = \alpha_0/(1 - \alpha_1)$ ; the non-covariance terms cancel, leaving  $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1 \text{Cov}(r_{t+\tau-1}^2, r_t^2)$ .
  3. Iterate  $\tau$  times:  $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1^\tau \text{Var}(r_t^2)$ , hence  $\text{Corr} = \alpha_1^{|\tau|}$ .
- The cancellation step is the crux; verify it explicitly (see Exercise 3).

#### Key Technique 4: Writing down the conditional-Gaussian (G)ARCH likelihood

To set up MLE:

1. Under Gaussian noise,  $r_t | \mathcal{F}_{t-1} \sim \mathcal{N}(0, \sigma_t^2)$ , with  $\sigma_t^2$  given by the model's recursion.
2. Condition on enough initial values to start the recursion (e.g.  $r_1$  for ARCH(1); for GARCH also seed  $\sigma_t^2$ , e.g. at the sample variance).
3. Form  $\log L_c = -\frac{1}{2} \sum_t [\log(2\pi\sigma_t^2) + r_t^2/\sigma_t^2]$  and maximise numerically subject to positivity and the stationarity constraint  $\sum \alpha + \sum \beta < 1$ .

#### Classic Pitfall

**Do not** confuse the two thresholds.  $\alpha_1 < 1$  is for a finite *variance*; the kurtosis / ACF-of-squares formulas require the *stronger*  $\alpha_1^2 < 1/3$  (ARCH(1)) or  $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1$  (GARCH(1,1)). Also, ARCH/GARCH are uncorrelated **but not independent**:  $\text{Corr}(r_{t+\tau}, r_t) = 0$  yet  $\text{Corr}(r_{t+\tau}^2, r_t^2) \neq 0$ . “White noise”  $\neq$  “no structure”.

## 10.4 Exercises

### Exercise 1: ARCH(2): mean, variance, stationarity [Sheet 12]

Let  $r_t = \sigma_t \varepsilon_t$  with  $\sigma_t^2 = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2$ , the  $\beta_j \geq 0$ , and  $\varepsilon_t$  a zero-mean unit-variance process independent of  $\mathcal{F}_{t-1}$ . (a) Compute  $\mathbb{E}[r_t]$  via iterated expectation. (b) Assuming stationarity, find the marginal variance  $\text{Var}(r_t)$ . (c) Give necessary conditions for stationarity in terms of the  $\beta_j$ .

#### Solution

(a) Since  $\sigma_t$  is  $\mathcal{F}_{t-1}$ -measurable and  $\varepsilon_t$  is independent of  $\mathcal{F}_{t-1}$  with  $\mathbb{E}[\varepsilon_t] = 0$ ,

$$\mathbb{E}[r_t | \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t \varepsilon_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0 \implies \mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t | \mathcal{F}_{t-1}]] = \mathbb{E}[0] = 0.$$

(b) By the law of total variance, with  $\text{Var}(\varepsilon_t) = 1$ ,

$$\text{Var}(r_t | \mathcal{F}_{t-1}) = \sigma_t^2 \text{Var}(\varepsilon_t) = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2.$$

Then, since  $\mathbb{E}[r_t | \mathcal{F}_{t-1}] = 0$  has zero variance,

$$\text{Var}(r_t) = \mathbb{E}[\text{Var}(r_t | \mathcal{F}_{t-1})] + \text{Var}(\mathbb{E}[r_t | \mathcal{F}_{t-1}]) = \mathbb{E}[\beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2] + 0.$$

Using  $\mathbb{E}[r_{t-j}^2] = \text{Var}(r_{t-j})$  (mean zero) and stationarity  $\text{Var}(r_t) = \text{Var}(r_{t-1}) = \text{Var}(r_{t-2}) =: v$ ,

$$v = \beta_0 + \beta_1 v + \beta_2 v \implies (1 - \beta_1 - \beta_2) v = \beta_0 \implies \text{Var}(r_t) = \frac{\beta_0}{1 - \beta_1 - \beta_2} \quad (\beta_1 + \beta_2 < 1).$$

(c) For the variance to be positive and finite we need  $\beta_0 > 0$  and  $\beta_1 + \beta_2 < 1$  (with  $\beta_1, \beta_2 \geq 0$ ). This is the necessary stationarity condition (the  $p = 2, q = 0$  instance of  $\sum \alpha + \sum \beta < 1$ ).

### Exercise 2: GARCH is zero-mean and uncorrelated [Sheet 12]

For a GARCH( $m, r$ ) with Gaussian noise,  $\sigma_t^2 = \alpha_0 + \sum_{j=1}^m \alpha_j r_{t-j}^2 + \sum_{j=1}^r \beta_j \sigma_{t-j}^2$ ,  $\alpha_j, \beta_j \geq 0$ , prove that  $\mathbb{E}[r_t] = 0$  and  $\text{Corr}(r_{t+\tau}, r_t) = 0$ .

#### Solution

**Zero mean.**  $\sigma_t$  is determined by the past, hence  $\mathcal{F}_{t-1}$ -measurable, and  $\varepsilon_t$  is independent of  $\mathcal{F}_{t-1}$  with  $\mathbb{E}[\varepsilon_t] = 0$ :

$$\mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t | \mathcal{F}_{t-1}]] = \mathbb{E}[\mathbb{E}[\sigma_t \varepsilon_t | \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t \mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}]] = \mathbb{E}[\sigma_t \cdot 0] = 0.$$

**Uncorrelated.** Fix  $\tau > 0$ . As  $\mathbb{E}[r_t] = 0$ ,  $\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t]$ . Both  $r_t$  and  $\sigma_{t+\tau}$  are  $\mathcal{F}_{t+\tau-1}$ -measurable, so by iterated expectation

$$\mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}[\mathbb{E}[\sigma_{t+\tau} \varepsilon_{t+\tau} r_t | \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} | \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0.$$

By symmetry the same holds for  $\tau < 0$ , and at  $\tau = 0$  the covariance is the variance. Therefore  $\text{Cov}(r_{t+\tau}, r_t) = 0$  and so  $\text{Corr}(r_{t+\tau}, r_t) = 0$  for all  $\tau \neq 0$ : GARCH is white noise.

### Exercise 3: ARCH(1): geometric ACF of the squares [Sheet 12]

For an ARCH(1) with  $0 < \alpha_0$  and  $0 < \alpha_1 < 1/\sqrt{3}$ , show  $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ .

#### Solution

Take  $\tau > 0$  and start from the covariance:

$$\text{Cov}(r_{t+\tau}^2, r_t^2) = \mathbb{E}[r_{t+\tau}^2 r_t^2] - \mathbb{E}[r_{t+\tau}^2] \mathbb{E}[r_t^2] = \mathbb{E}[r_{t+\tau}^2 r_t^2] - (\mathbb{E}[r_t^2])^2,$$

the last step by stationarity ( $\mathbb{E}[r_{t+\tau}^2] = \mathbb{E}[r_t^2]$ ). Condition on  $\mathcal{F}_{t+\tau-1}$ ; both  $r_t^2$  and  $\sigma_{t+\tau}^2$  are measurable there and  $\mathbb{E}[\varepsilon_{t+\tau}^2 | \mathcal{F}_{t+\tau-1}] = 1$ :

$$\mathbb{E}[r_{t+\tau}^2 r_t^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 \sigma_{t+\tau}^2 \mathbb{E}[\varepsilon_{t+\tau}^2 | \mathcal{F}_{t+\tau-1}] = r_t^2 (\alpha_0 + \alpha_1 r_{t+\tau-1}^2).$$

Taking expectations,

$$\mathbb{E}[r_{t+\tau}^2 r_t^2] = \alpha_0 \mathbb{E}[r_t^2] + \alpha_1 \mathbb{E}[r_{t+\tau-1}^2 r_t^2].$$

Hence, writing  $C_\tau := \text{Cov}(r_{t+\tau}^2, r_t^2)$  and using  $\mathbb{E}[r_{t+\tau-1}^2 r_t^2] = C_{\tau-1} + \mathbb{E}[r_{t+\tau-1}^2] \mathbb{E}[r_t^2] = C_{\tau-1} + (\mathbb{E}[r_t^2])^2$ ,

$$C_\tau = \alpha_1 C_{\tau-1} + \underbrace{\alpha_0 \mathbb{E}[r_t^2] + \alpha_1 (\mathbb{E}[r_t^2])^2 - (\mathbb{E}[r_t^2])^2}_{=: R}.$$

Now substitute  $\mathbb{E}[r_t^2] = \text{Var}(r_t) = \alpha_0 / (1 - \alpha_1)$ :

$$R = \alpha_0 \frac{\alpha_0}{1 - \alpha_1} - (1 - \alpha_1) \left( \frac{\alpha_0}{1 - \alpha_1} \right)^2 = \frac{\alpha_0^2}{1 - \alpha_1} - \frac{\alpha_0^2}{1 - \alpha_1} = 0.$$

So  $C_\tau = \alpha_1 C_{\tau-1}$ , and iterating  $\tau$  times from  $C_0 = \text{Var}(r_t^2)$  gives  $C_\tau = \alpha_1^\tau \text{Var}(r_t^2)$ . Therefore

$$\text{Corr}(r_{t+\tau}^2, r_t^2) = \frac{C_\tau}{\text{Var}(r_t^2)} = \alpha_1^\tau,$$

and by symmetry of the autocovariance  $\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ . (The condition  $\alpha_1^2 < 1/3$  guarantees  $\text{Var}(r_t^2) = \mathbb{E}[r_t^4] - (\mathbb{E}[r_t^2])^2 < \infty$ , so the correlation is well-defined.)

**Exercise 4: GARCH(1,1): variance and fourth-moment check** [New]

A GARCH(1, 1) is fitted to log-returns with  $\alpha_0 = 3.686 \times 10^{-6}$ ,  $\alpha_1 = 0.1713$ ,  $\beta_1 = 0.7897$  (the lecture's S&P 500 fit). (a) Verify stationarity and compute the implied unconditional variance and standard deviation (daily volatility). (b) Does a finite fourth moment exist? (c) Comment on the persistence  $\alpha_1 + \beta_1$ .

**Solution**

(a)  $\alpha_1 + \beta_1 = 0.1713 + 0.7897 = 0.9610 < 1$ , so the process is weakly stationary. The unconditional variance is

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1} = \frac{3.686 \times 10^{-6}}{1 - 0.9610} = \frac{3.686 \times 10^{-6}}{0.0390} \approx 9.45 \times 10^{-5}.$$

The implied daily volatility is  $\sqrt{9.45 \times 10^{-5}} \approx 9.72 \times 10^{-3}$ , i.e. about 0.97% per day — a sensible equity figure.

(b) Check  $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2$ :

$$3(0.1713)^2 + 2(0.1713)(0.7897) + (0.7897)^2 = 0.08803 + 0.27055 + 0.62363 = 0.9822 < 1.$$

Since  $0.9822 < 1$ , the fourth moment exists (only just): the fitted model has finite kurtosis, but the closeness to 1 signals very heavy tails.

(c)  $\alpha_1 + \beta_1 = 0.961$  is close to 1, the hallmark of financial GARCH fits: volatility shocks are highly persistent (slow geometric decay of the ACF of squares), reproducing long quiet/turbulent regimes, while the process remains stationary because the sum is still strictly below 1.

**Exercise 5: ARCH(1) parameter from a given kurtosis** [New]

An ARCH(1) with Gaussian innovations is observed to have kurtosis  $\kappa = 9$ . Find  $\alpha_1$ . What is the largest kurtosis an ARCH(1) can attain while keeping a finite fourth moment?

**Solution**

Use  $\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2}$  and set it equal to 9:

$$9(1 - 3\alpha_1^2) = 3(1 - \alpha_1^2) \Rightarrow 9 - 27\alpha_1^2 = 3 - 3\alpha_1^2 \Rightarrow 6 = 24\alpha_1^2 \Rightarrow \alpha_1^2 = \frac{1}{4} \Rightarrow \alpha_1 = \frac{1}{2}.$$

(Take the positive root; check  $\alpha_1^2 = 1/4 < 1/3$ , so the fourth moment is indeed finite.) As  $\alpha_1^2 \uparrow 1/3$  (i.e.  $\alpha_1 \uparrow 1/\sqrt{3}$ ) the denominator  $1 - 3\alpha_1^2 \downarrow 0^+$  while the numerator stays positive, so  $\kappa \rightarrow +\infty$ : there is **no** finite upper bound on the attainable kurtosis short of losing the fourth moment. Conversely  $\kappa \rightarrow 3$  as  $\alpha_1 \rightarrow 0$  (back to constant variance), confirming  $\kappa > 3$  for all  $\alpha_1 > 0$ .

**Exercise 6: ARCH(1) one-step variance forecast & predictive band** [New]

For an ARCH(1) with  $\alpha_0 = 0.0001$ ,  $\alpha_1 = 0.4$ , suppose the most recent return is  $r_t = 0.03$ . (a) Give the one-step-ahead conditional variance  $\sigma_{t+1}^2$  and a 95% predictive interval for  $r_{t+1}$  under Gaussian innovations. (b) Compare with the unconditional ( $\pm 1.96\sigma$ ) band. (c) Is this model fourth-moment stationary, and what is its kurtosis?

**Solution**

(a)  $\sigma_{t+1}^2 = \alpha_0 + \alpha_1 r_t^2 = 0.0001 + 0.4(0.03)^2 = 0.0001 + 0.4(0.0009) = 0.0001 + 0.00036 = 0.00046$ . Thus  $\sigma_{t+1} = \sqrt{0.00046} \approx 0.02145$  and, since  $r_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(0, \sigma_{t+1}^2)$ , the conditional 95%

interval is

$$0 \pm 1.96 \sigma_{t+1} = \pm 1.96(0.02145) \approx \pm 0.0421.$$

(b) The unconditional variance is  $\text{Var}(r_t) = \alpha_0/(1 - \alpha_1) = 0.0001/0.6 \approx 1.667 \times 10^{-4}$ , so  $\sigma \approx 0.01291$  and the unconditional band is  $\pm 1.96(0.01291) \approx \pm 0.0253$ . After the large move  $r_t = 0.03$  the conditional band ( $\pm 0.042$ ) is markedly *wider* than the unconditional one ( $\pm 0.025$ ): the model has raised its volatility forecast — exactly the clustering behaviour we want.

(c) Variance stationarity:  $\alpha_1 = 0.4 < 1$  holds. Fourth moment:  $\alpha_1^2 = 0.16 < 1/3 \approx 0.333$  holds, so the fourth moment (hence kurtosis) is finite, with

$$\kappa = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} = \frac{3(1 - 0.16)}{1 - 0.48} = \frac{3(0.84)}{0.52} = \frac{2.52}{0.52} \approx 4.85 > 3.$$

#### Exercise 7: GARCH(1,1) as an ARCH( $\infty$ ) [New]

Explain — qualitatively, then with the recursion algebra — why a GARCH(1,1) can mimic an ARCH( $\infty$ ) with geometrically decaying coefficients, and why this is more parsimonious than ARCH( $p$ ).

#### Solution

**Algebra.** Iterate the GARCH(1,1) variance recursion  $\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta_1 \sigma_{t-1}^2$  by repeatedly substituting  $\sigma_{t-1}^2$ :

$$\sigma_t^2 = \alpha_0 \sum_{k=0}^{\infty} \beta_1^k + \alpha_1 \sum_{k=0}^{\infty} \beta_1^k r_{t-1-k}^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{k=0}^{\infty} \beta_1^k r_{t-1-k}^2,$$

valid when  $0 \leq \beta_1 < 1$  (so the geometric series converges). This is an ARCH( $\infty$ ) with intercept  $\alpha_0/(1 - \beta_1)$  and coefficients  $\alpha_1 \beta_1^k$  on  $r_{t-1-k}^2$  that decay *geometrically* in the lag.

**Parsimony.** Reproducing the same slowly decaying dependence with a finite ARCH( $p$ ) would need many free  $\alpha_j$ 's (large  $p$ ), whereas GARCH(1,1) encodes the entire geometric tail with just *three* parameters ( $\alpha_0, \alpha_1, \beta_1$ ). This is exactly the “feed past variance back in” idea:  $\beta_1$  sets the persistence/decay rate,  $\alpha_1$  the immediate shock impact. It also explains the geometric ACF-of-squares decay carried over from the ARCH(1) case.

## 11 Linear Time-Invariant (LTI) Filters

A (*digital*) *filter* is a map that turns one sequence into another,  $\{Y_t\} = \mathcal{L}[\{X_t\}]$ , with index set  $\mathbb{Z}$ . The filters that matter for time series are the **linear time-invariant** (LTI) ones: they treat all time points alike and act linearly. Two “test inputs” completely characterise such a filter — a single impulse (giving the *impulse response*  $h$ ) and a pure complex wave (giving the *transfer function*  $H$ ). Their power is twofold: every LTI filter is a *convolution* with  $h$ , and in the frequency domain it acts *diagonally*, multiplying the spectral density by  $|H(f)|^2$ . This single fact,  $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$ , is the engine behind the closed-form spectral density of any ARMA process. Throughout we allow time series to be *complex-valued* and assume the (possibly infinite) sums involved converge, the standard sufficient condition being  $h \in \ell^1$ .

## 11.1 Definitions

### Definition 1: Complex-valued time series & its autocovariance

A **complex-valued** time series can be written  $X_t = U_t + iV_t$  with  $\{U_t\}, \{V_t\}$  real-valued, and has mean  $\mathbb{E}[X_t] = \mathbb{E}[U_t] + i\mathbb{E}[V_t]$ . Its **autocovariance** uses a *conjugate* on the second factor:

$$\gamma_\tau^{(X)} = \text{Cov}(X_{t+\tau}, X_t) = \mathbb{E}[(X_{t+\tau} - \mu)(\overline{X_t - \mu})].$$

This reduces to the usual real ACVS when  $X_t \in \mathbb{R}$ . (Stationarity is required jointly of the real and imaginary parts.) Throughout we drop the index set:  $\{X_t\} = \{X_t\}_{t \in \mathbb{Z}}$ .

### Definition 2: Digital filter & output notation

A **digital filter**  $\mathcal{L}$  maps an input sequence to an output sequence,  $\{Y_t\} = \mathcal{L}[\{X_t\}]$ . To name a single output entry we put the time index *after* the bracket,

$$Y_u = \mathcal{L}[\{X_t\}]_u, \quad u \in \mathbb{Z}.$$

The  $t$  in  $\{X_t\}$  is a *dummy* variable; the  $u \in \mathbb{Z}$  after the bracket has a definite meaning (the output time).

### Recall: the backshift operator

The **backshift**  $B$  satisfies  $B[\{X_t\}]_u = X_{u-1}$  for every  $u \in \mathbb{Z}$ . Iterating,  $B^u[\{X_t\}]_s = X_{s-u}$  for all  $s \in \mathbb{Z}$  (for  $u < 0$  this applies  $B^{-1}$  a total of  $-u$  times).  $B$  is itself the prototypical LTI filter.

### Definition 3: Linear time-invariant (LTI) filter

A digital filter  $\mathcal{L}$  is **linear time-invariant** if for all sequences  $\{X_t\}, \{Y_t\}$  and all  $\alpha \in \mathbb{C}$ :

1. **Linearity:**  $\mathcal{L}[\{\alpha X_t + Y_t\}] = \alpha \mathcal{L}[\{X_t\}] + \mathcal{L}[\{Y_t\}]$ ;
2. **Time invariance:**  $\mathcal{L}[B[\{X_t\}]] = B[\mathcal{L}[\{X_t\}]]$ , i.e.  $\mathcal{L}$  commutes with the backshift,  $\mathcal{L}B = B\mathcal{L}$ .

Time invariance for the single shift  $B$  already implies it for every shift:  $\mathcal{L}[B^u\{X_t\}] = B^u\mathcal{L}[\{X_t\}]$  for all  $u \in \mathbb{Z}$ . (The continuous-time analogue is usually stated with  $B^u$ , but the weaker one-step condition suffices.)

### Example 4: First LTI filters

- (i) The backshift  $B$  itself is LTI.
- (ii) For fixed  $\phi \in \mathbb{R}$ , the one-tap filter  $\mathcal{L}[\{X_t\}]_u = X_u + \phi X_{u-1}$  is LTI: linearity is immediate from

$$\mathcal{L}[\{\alpha X_t + Y_t\}]_u = \alpha(X_u + \phi X_{u-1}) + (Y_u + \phi Y_{u-1}) = \alpha \mathcal{L}[\{X_t\}]_u + \mathcal{L}[\{Y_t\}]_u,$$

and time invariance is checked the same way. This is exactly  $\mathcal{L} = I + \phi B$ .

### Definition 5: Impulse sequence & impulse response

For  $m \in \mathbb{Z}$  the **impulse sequence**  $\{\delta_{t,m}\}$  is  $\delta_{t,m} = 1$  if  $t = m$  and 0 otherwise. The **impulse response** of an LTI filter  $\mathcal{L}$  is the sequence  $\{h_m\}$  defined by

$$h_m = \mathcal{L}[\{\delta_{t,-m}\}]_0, \quad m \in \mathbb{Z},$$

where the subscript 0 is the *output time index*. Informally  $h_m$  is the response, read off at time 0, to a unit shock placed at time  $-m$ . A common sufficient regularity condition is  $h \in \ell^1$  (then  $\|h\|_\infty \leq \|h\|_1 < \infty$ ).

### Definition 6: Complex wave & transfer function

For  $f \in \mathbb{R}$  the **complex wave** is the sequence  $\{\xi_{t,f}\}$  with  $\xi_{t,f} = e^{2\pi i f t}$ . The **transfer function** of an LTI filter  $\mathcal{L}$  that can be applied to these waves is

$$H(f) = \mathcal{L}[\{\xi_{t,f}\}]_0, \quad f \in \mathbb{R}.$$

It encodes the frequency-domain behaviour of  $\mathcal{L}$ : how each pure frequency is re-weighted in amplitude and shifted in phase.

### Recall: mean-square equality

For random variables  $U, V \in L^2$  we write  $U \stackrel{\text{ms}}{=} V$  to mean  $\mathbb{E}[|U - V|^2] = 0$  (equality in mean square). This is the natural notion of equality for the stochastic infinite sums (convolutions, spectral integrals) appearing below.

## 11.2 Theorems & Propositions

### Proposition 7: Linear combination of LTI filters is LTI

If  $\mathcal{L}_1, \mathcal{L}_2$  are LTI and  $\alpha \in \mathbb{C}$ , then  $\mathcal{L} = \alpha\mathcal{L}_1 + \mathcal{L}_2$  is LTI.

*Proof idea:* apply linearity of  $\mathcal{L}_1, \mathcal{L}_2$  to  $\mathcal{L}[\beta\{X_t\} + \{Y_t\}]$  and regroup; for time invariance use that each  $\mathcal{L}_j$  commutes with  $B$  and that  $B$  is linear, so  $\mathcal{L}[B\{X_t\}] = B[\mathcal{L}\{X_t\}]$ . (Full computation in Exercise 1.)

*Why:* it makes the LTI filters a *vector space* (over  $\mathbb{C}$ ); in particular any polynomial in  $B$ , e.g.  $\Phi(B) = \sum_j \phi_j B^j$ , is LTI.

### Proposition 8: Cascade of LTI filters is LTI

If  $\mathcal{L}_1, \mathcal{L}_2$  are LTI, then the **cascade**  $\mathcal{L} = \mathcal{L}_1\mathcal{L}_2$ , defined by  $\mathcal{L}[\{X_t\}] = \mathcal{L}_1[\mathcal{L}_2[\{X_t\}]]$ , is LTI.

*Proof idea:* linearity is the composition of two linear maps; time invariance follows by pushing  $B$  through both filters,  $\mathcal{L}_1[\mathcal{L}_2[B\{X_t\}]] = \mathcal{L}_1[B[\mathcal{L}_2\{X_t\}]] = B[\mathcal{L}_1[\mathcal{L}_2\{X_t\}]]$ . (Exercise 2.)

*Why:* filters can be *chained* and the family is closed under composition; combined with the previous proposition, all of {polynomials and power series in  $B$ } are LTI.

### Intuition / Why it matters

LTI filters are exactly the maps that “do the same thing at every time and respect superposition”. Commuting with  $B$  is time invariance: shifting the input then filtering gives the same as filtering then shifting. Both closure properties (linear combinations, cascades) are inherited directly from those two structural rules.

### Theorem 9: LTI $\Leftrightarrow$ convolution with the impulse response

Under suitable convergence assumptions, an LTI filter with impulse response  $h$  acts as a **convolution**:

$$\mathcal{L}[\{X_t\}]_u \stackrel{\text{ms}}{=} \sum_{m \in \mathbb{Z}} h_{u-m} X_m, \quad u \in \mathbb{Z}.$$

Conversely, any filter defined by such a convolution (with, say,  $h \in \ell^1$ ) is LTI. The impulse response gives the convolution weights.

*Proof idea (deterministic  $\ell^1$  version, Exercise 3):* write  $X^{(N)} = \sum_{|m| \leq N} X_m \delta_{.,m}$ ; by continuity in  $\ell^1$ ,  $\mathcal{L}[\{X_t\}]_u = \lim_N \mathcal{L}[X^{(N)}]_u$ . Linearity turns each finite truncation into a finite sum, and time invariance gives  $\mathcal{L}[\{\delta_{t,m}\}]_u = \mathcal{L}[\{\delta_{t,m-u}\}]_0 = h_{u-m}$ ; absolute convergence ( $h \in \ell^1$ ,  $X \in \ell^1$ ) lets the partial sums pass to the full series. The converse checks linearity and time invariance

directly on the convolution (re-indexing  $m' = m - 1$  for the shift).

*Why:* this is the master representation — it reduces *any* LTI filter to a single object, the weights  $h$ , and underlies every spectral computation below.

### Intuition / Why it matters

Two complementary pictures of a time series, and an LTI filter acts simply on both “building blocks”:

$$X_t = \sum_{m \in \mathbb{Z}} X_m \delta_{m,t} \quad (\text{sum of impulses}), \quad X_t \stackrel{\text{ms}}{=} \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f) \quad (\text{sum of waves}).$$

Applying  $\mathcal{L}$  gives, respectively,  $\mathcal{L}[\{X_t\}]_u \stackrel{\text{ms}}{=} \sum_m X_m \mathcal{L}[\{\delta_{m,t}\}]_u$  (the impulse-response view) and  $\mathcal{L}[\{X_t\}]_u \stackrel{\text{ms}}{=} \mu H(0) + \int_{-1/2}^{1/2} H(f) e^{2\pi i f u} dZ(f)$  (the transfer-function view).

### Theorem 10: Complex waves are eigensequences; $H(f)$ is the eigenvalue

Let  $\mathcal{L}$  be an LTI filter applicable to the complex waves  $\{\xi_{t,f}\}$ . Then each wave is an **eigensequence**: for every  $f \in \mathbb{R}$ ,

$$\mathcal{L}[\{\xi_{t,f}\}] = H(f) \{\xi_{t,f}\},$$

i.e.  $\mathcal{L}$  only rescales amplitude and shifts phase while preserving the frequency.

*Proof idea:* since  $B^{-u}[\{\xi_{t,f}\}]_\tau = \xi_{\tau+u,f} = e^{2\pi i u f} \xi_{\tau,f}$ , we have  $B^{-u}[\{\xi_{t,f}\}] = e^{2\pi i u f} \{\xi_{t,f}\}$ . Then for any  $u$ ,

$$\mathcal{L}[\{\xi_{t,f}\}]_u = B^u[\mathcal{L}[B^{-u}\{\xi_{t,f}\}]]_u = \mathcal{L}[e^{2\pi i f u} \{\xi_{t,f}\}]_0 = e^{2\pi i f u} \mathcal{L}[\{\xi_{t,f}\}]_0 = \xi_{u,f} H(f),$$

using time invariance, the shift identity, linearity and the definition of  $H$ .

*Why:* it diagonalises every LTI filter in the frequency basis — filtering becomes *pointwise multiplication* by  $H(f)$ , the source of all clean frequency-domain formulas.

### Theorem 11: Transfer function = Fourier transform of $h$

If the impulse response satisfies  $h \in \ell^1$ , the transfer function is the (discrete) Fourier transform of  $h$ :

$$H(f) = \sum_{m \in \mathbb{Z}} h_m e^{-2\pi i f m}, \quad f \in \mathbb{R}.$$

*Proof idea:* feed the wave  $\xi_{t,f}$  into the convolution representation:  $H(f) = \mathcal{L}[\{\xi_{t,f}\}]_0 = \sum_m h_{-m} e^{2\pi i f m} = \sum_m h_m e^{-2\pi i f m}$  (re-index  $m \mapsto -m$ );  $h \in \ell^1$  makes the sum absolutely convergent. (Exercise.)

*Why:* the practical recipe — read off  $h$  from the filter, then sum the series to get  $H$  (and conversely, recover  $h$  as Fourier coefficients of  $H$ ).

### Theorem 12: Filtering preserves stationarity

Let  $\{X_t\}$  be stationary and  $\mathcal{L}$  an LTI filter with  $h \in \ell^1$ . Then  $\{Y_t\} = \mathcal{L}[\{X_t\}]$  is stationary.

*Proof idea:* by the convolution theorem  $Y_t \stackrel{\text{ms}}{=} \sum_m h_m X_{t-m}$ . By Fubini,  $\mathbb{E}[Y_t] = \sum_m h_m \mathbb{E}[X_{t-m}] = \mu_X \sum_m h_m$  is  $t$ -free; the covariance  $\gamma_\tau^{(Y)} = \sum_{m,s} h_m \bar{h}_s \gamma_{\tau-m+s}^{(X)}$  depends only on  $\tau$ ; and  $\gamma_0^{(Y)} \leq \gamma_0^{(X)} \|h\|_1^2 < \infty$ , so all is finite.

*Why:* it licenses speaking of  $\mathcal{S}_Y$  at all — only a stationary output has a spectral density.

### Theorem 13: Output spectrum: $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$

Let  $\{X_t\}$  be stationary with  $\gamma^{(X)} \in \ell^1$ , and  $\mathcal{L}$  an LTI filter with  $h \in \ell^1$  and transfer function  $H$ . If  $\{Y_t\} = \mathcal{L}[\{X_t\}]$  then

$$\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f), \quad f \in \mathbb{R}.$$

*Proof idea:* from the covariance formula above,  $\gamma_\tau^{(Y)} = \sum_u w_u \gamma_{\tau-u}^{(X)}$  where  $w_u = \sum_s h_{u+s} \bar{h}_s$  (set  $u = m - s$ ). With  $\tilde{h}_t = \bar{h}_{-t}$  one has  $w = h * \tilde{h} \in \ell^1$ , so the convolution theorem gives  $\mathcal{S}_Y = W \mathcal{S}_X$  with  $W(f) = H(f) \tilde{H}(f) = H(f) \overline{H(f)} = |H(f)|^2$ , since  $\tilde{H}(f) = \overline{H(f)}$ .

*Why:* the workhorse identity. It converts *any* filtering question into multiplying the input spectrum by a gain  $|H(f)|^2$  — and, applied to  $\Phi(B)$  and  $\Theta(B)$ , it yields the ARMA spectral density below.

### Lemma 14: Transfer function of a polynomial filter $\Phi(B)$

For a degree- $p$  polynomial  $\Phi(z) = \sum_{j=0}^p \phi_j z^j$ , the filter  $\Phi(B)$  is LTI with transfer function

$$H(f) = \Phi(e^{-2\pi i f}), \quad f \in \mathbb{R}.$$

*Proof idea:*  $\Phi(B)$  is a linear combination of the LTI filters  $B^j$ , hence LTI, with impulse response  $h_m = \phi_m$  for  $0 \leq m \leq p$  and 0 otherwise. Then  $H(f) = \sum_m h_m e^{-2\pi i f m} = \sum_{j=0}^p \phi_j e^{-2\pi i f j} = \Phi(e^{-2\pi i f})$ .

*Why:* it turns the operator  $\Phi(B)$  into a concrete function of  $f$ ; the bridge from time-domain difference equations to the spectrum.

### Theorem 15: ARMA spectral density

For a stationary ARMA( $p, q$ ) process  $\Phi(B)X_t = \Theta(B)\varepsilon_t$  with  $\varepsilon_t$  white noise of variance  $\sigma^2$ ,

$$\mathcal{S}_X(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2} = \sigma^2 \left| \frac{\sum_{j=0}^q \theta_j e^{-2\pi i f j}}{\sum_{j=0}^p \phi_j e^{-2\pi i f j}} \right|^2.$$

*Proof idea:* write  $\Phi(B)[\{X_t\}] = \Theta(B)[\{\varepsilon_t\}]$ . Applying the output-spectrum theorem to each side and the polynomial-filter lemma,  $|\Phi(e^{-2\pi i f})|^2 \mathcal{S}_X(f) = |\Theta(e^{-2\pi i f})|^2 \mathcal{S}_\varepsilon(f)$  with  $\mathcal{S}_\varepsilon(f) = \sigma^2$ ; divide, legal because stationarity forbids zeros of  $\Phi$  on the unit circle, so  $\Phi(e^{-2\pi i f}) \neq 0$ .

*Why:* ARMA processes have *no* closed-form ACVS but a clean closed-form spectral density — the practical way to read off their frequency content.

### Classic Pitfall

The conjugate sits on the *second* factor in the complex ACVS,  $\text{Cov}(X_{t+\tau}, X_t) = \mathbb{E}[(X_{t+\tau} - \mu)(\overline{X_t - \mu})]$ . Forgetting it breaks Hermitian symmetry. Also: the gain is  $|H(f)|^2$ , *not*  $H(f)$  or  $H(f)^2$  — the  $\mathcal{S}$  multiplier is always the squared modulus, which is real and nonnegative as a spectral density must be.

## 11.3 Key Techniques

### Key Technique 1: Spectral density of a filtered process

To get  $\mathcal{S}_Y$  when  $\{Y_t\} = \mathcal{L}[\{X_t\}]$  for an LTI filter:

1. **Identify the impulse response  $h$**  (the convolution weights), or write  $\mathcal{L}$  as a polynomial  $\Phi(B)$ .

2. **Form the transfer function**  $H(f) = \sum_m h_m e^{-2\pi i f m}$  (for  $\Phi(B)$  this is  $\Phi(e^{-2\pi i f})$ ).
  3. **Multiply by the squared gain:**  $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$ .
- Handy simplification:  $|1 - e^{-2\pi i f}|^2 = 2 - 2 \cos(2\pi f) = 4 \sin^2(\pi f)$ .

#### Key Technique 2: Transfer function from an impulse response (and back)

**Forward** ( $h \rightarrow H$ ): sum the Fourier series  $H(f) = \sum_m h_m e^{-2\pi i f m}$ ; for a *finite* filter this is a finite trigonometric polynomial. **Backward** ( $H \rightarrow h$ ): the weights are the Fourier coefficients,  $h_m = \int_{-1/2}^{1/2} H(f) e^{2\pi i f m} df$ . To read  $h$  straight off a convolution filter, recall  $\mathcal{L}[\{X_t\}]_u = \sum_m h_{u-m} X_m = \sum_k h_k X_{u-k}$ , so the coefficient of  $X_{u-k}$  is  $h_k$ .

#### Key Technique 3: Composing filters (cascades) in the frequency domain

Filtering is multiplication by  $H$  on each frequency, so a cascade  $\mathcal{L} = \mathcal{L}_1 \mathcal{L}_2$  has transfer function  $H(f) = H_1(f) H_2(f)$  and squared gain  $|H_1(f)|^2 |H_2(f)|^2$ . Likewise convolution in time corresponds to *product* in frequency:  $h^{(1)} * h^{(2)} \leftrightarrow H_1 H_2$ . This is why ARMA decomposes as “MA part on top, AR part on the bottom”.

#### Key Technique 4: Proving a filter is LTI

Check the two axioms on a generic input:

1. **Linearity:** expand  $\mathcal{L}[\{\alpha X_t + Y_t\}]_u$  and match  $\alpha \mathcal{L}[\{X_t\}]_u + \mathcal{L}[\{Y_t\}]_u$ .
2. **Time invariance:** show  $\mathcal{L}[B\{X_t\}]_u = B[\mathcal{L}\{X_t\}]_u = \mathcal{L}[\{X_t\}]_{u-1}$ , typically by re-indexing the sum ( $m' = m - 1$ ); absolute convergence ( $h \in \ell^1$ ) justifies the re-indexing.

Shortcut: any linear combination or cascade of known LTI filters is automatically LTI — and any filter expressible as a convolution with  $h \in \ell^1$  is LTI.

## 11.4 Exercises

### Exercise 1: Linear combination of LTI filters [Sheet 13]

Let  $\mathcal{L}_1, \mathcal{L}_2$  be LTI and  $\alpha \in \mathbb{C}$ . Show  $\mathcal{L} = \alpha \mathcal{L}_1 + \mathcal{L}_2$  is LTI.

#### Solution

**Linearity.** For sequences  $\{X_t\}, \{Y_t\}$  and  $\beta \in \mathbb{C}$ , using linearity of  $\mathcal{L}_1, \mathcal{L}_2$ ,

$$\begin{aligned} \mathcal{L}[\beta\{X_t\} + \{Y_t\}] &= \alpha \mathcal{L}_1[\beta\{X_t\} + \{Y_t\}] + \mathcal{L}_2[\beta\{X_t\} + \{Y_t\}] \\ &= \alpha(\beta \mathcal{L}_1[\{X_t\}] + \mathcal{L}_1[\{Y_t\}]) + (\beta \mathcal{L}_2[\{X_t\}] + \mathcal{L}_2[\{Y_t\}]) \\ &= \beta(\alpha \mathcal{L}_1[\{X_t\}] + \mathcal{L}_2[\{X_t\}]) + (\alpha \mathcal{L}_1[\{Y_t\}] + \mathcal{L}_2[\{Y_t\}]) \\ &= \beta \mathcal{L}[\{X_t\}] + \mathcal{L}[\{Y_t\}]. \end{aligned}$$

**Time invariance.** Using that each  $\mathcal{L}_j$  commutes with  $B$  and that  $B$  is linear,

$$\begin{aligned} \mathcal{L}[B[\{X_t\}]] &= \alpha \mathcal{L}_1[B[\{X_t\}]] + \mathcal{L}_2[B[\{X_t\}]] = \alpha B[\mathcal{L}_1[\{X_t\}]] + B[\mathcal{L}_2[\{X_t\}]] \\ &= B[\alpha \mathcal{L}_1[\{X_t\}] + \mathcal{L}_2[\{X_t\}]] = B[\mathcal{L}[\{X_t\}]]. \end{aligned}$$

Both axioms hold, so  $\mathcal{L}$  is LTI. □

### Exercise 2: Cascade of LTI filters [Sheet 13]

Let  $\mathcal{L}_1, \mathcal{L}_2$  be LTI. Show the cascade  $\mathcal{L} = \mathcal{L}_1 \mathcal{L}_2$ ,  $\mathcal{L}[\{X_t\}] = \mathcal{L}_1[\mathcal{L}_2[\{X_t\}]]$ , is LTI.

### Solution

**Linearity.** For  $\alpha \in \mathbb{C}$ ,

$$\begin{aligned}\mathcal{L}[\alpha\{X_t\} + \{Y_t\}] &= \mathcal{L}_1[\mathcal{L}_2[\alpha\{X_t\} + \{Y_t\}]] = \mathcal{L}_1[\alpha\mathcal{L}_2[\{X_t\}] + \mathcal{L}_2[\{Y_t\}]] \\ &= \alpha\mathcal{L}_1[\mathcal{L}_2[\{X_t\}]] + \mathcal{L}_1[\mathcal{L}_2[\{Y_t\}]] = \alpha\mathcal{L}[\{X_t\}] + \mathcal{L}[\{Y_t\}].\end{aligned}$$

**Time invariance.** Push  $B$  through each filter in turn:

$$\mathcal{L}[B[\{X_t\}]] = \mathcal{L}_1[\mathcal{L}_2[B[\{X_t\}]]] = \mathcal{L}_1[B[\mathcal{L}_2[\{X_t\}]]] = B[\mathcal{L}_1[\mathcal{L}_2[\{X_t\}]]] = B[\mathcal{L}[\{X_t\}]].$$

Hence  $\mathcal{L}$  is LTI. (Note the order is preserved; for general operators  $\mathcal{L}_1\mathcal{L}_2 \neq \mathcal{L}_2\mathcal{L}_1$ , though for polynomials in  $B$  they do commute.)  $\square$

### Exercise 3: LTI on $\ell^1 \Leftrightarrow$ convolution [Sheet 13]

Let  $\mathcal{L} : \ell^1 \rightarrow \mathbb{C}^{\mathbb{Z}}$  be a digital filter such that for each fixed  $u$  the map  $\{X_t\} \mapsto \mathcal{L}[\{X_t\}]_u$  is continuous in the  $\ell^1$  norm. Define  $h_m = \mathcal{L}[\{\delta_{t,-m}\}]_0$  and assume  $h \in \ell^1$ . Prove  $\mathcal{L}$  is LTI on  $\ell^1$  iff  $\mathcal{L}[\{X_t\}]_u = \sum_{m \in \mathbb{Z}} h_{u-m} X_m$  for all  $\{X_t\} \in \ell^1$  and  $u \in \mathbb{Z}$ . Justify all limit/sum interchanges.

### Solution

( $\Rightarrow$ ) Assume  $\mathcal{L}$  is LTI on  $\ell^1$ . Truncate:  $X_t^{(N)} = \sum_{|m| \leq N} X_m \delta_{t,m}$ . Since  $\{X_t\} \in \ell^1$ ,  $\|X^{(N)} - X\|_1 \rightarrow 0$ , so by continuity of  $\{X_t\} \mapsto \mathcal{L}[\{X_t\}]_u$ ,

$$\mathcal{L}[\{X_t\}]_u = \lim_{N \rightarrow \infty} \mathcal{L}[\{X_t^{(N)}\}]_u.$$

For each  $N$ , *linearity* (finite sum) gives  $\mathcal{L}[\{X_t^{(N)}\}]_u = \sum_{|m| \leq N} X_m \mathcal{L}[\{\delta_{t,m}\}]_u$ , and *time invariance* gives  $\mathcal{L}[\{\delta_{t,m}\}]_u = \mathcal{L}[\{\delta_{t,m-u}\}]_0 = h_{u-m}$ , so

$$\mathcal{L}[\{X_t^{(N)}\}]_u = \sum_{|m| \leq N} h_{u-m} X_m.$$

Since  $h \in \ell^1$  we have  $\|h\|_\infty \leq \|h\|_1 < \infty$ , hence  $\sum_{m \in \mathbb{Z}} |h_{u-m} X_m| \leq \|h\|_\infty \sum_m |X_m| < \infty$ : the series converges absolutely, the partial sums converge to it, and  $\mathcal{L}[\{X_t\}]_u = \sum_{m \in \mathbb{Z}} h_{u-m} X_m$ .

( $\Leftarrow$ ) Suppose  $\mathcal{L}[\{X_t\}]_u = \sum_m h_{u-m} X_m$  with  $h \in \ell^1$ . Absolute convergence holds because  $\sum_m |h_{u-m} X_m| \leq \|h\|_\infty \|X\|_1 \leq \|h\|_1 \|X\|_1$ . *Linearity*: for  $\alpha \in \mathbb{C}$ ,

$$\begin{aligned}\mathcal{L}[\alpha\{X_t\} + \{Y_t\}]_u &= \sum_m h_{u-m} (\alpha X_m + Y_m) \\ &= \alpha \sum_m h_{u-m} X_m + \sum_m h_{u-m} Y_m = \alpha \mathcal{L}[\{X_t\}]_u + \mathcal{L}[\{Y_t\}]_u.\end{aligned}$$

*Time invariance*: absolute convergence allows reindexing  $m' = m - 1$ ,

$$\mathcal{L}[B[\{X_t\}]]_u = \sum_m h_{u-m} X_{m-1} = \sum_{m'} h_{u-1-m'} X_{m'} = \mathcal{L}[\{X_t\}]_{u-1} = B[\mathcal{L}[\{X_t\}]]_u.$$

Hence  $\mathcal{L}$  is LTI on  $\ell^1$ .  $\square$

### Exercise 4: Tapered DFT as convolution with $dZ$ [Sheet 13]

Let  $\{X_t\}$  be stationary, mean zero, with spectral representation  $X_t = \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f)$ . Observe  $X_1, \dots, X_n$  and define the *tapered DFT*  $J_h(f) = \sum_{t=1}^n h_t X_t e^{-2\pi i f t}$ , with  $\|h_t\|_2^2 = 1$ .

Show  $J_h(f) = \int_{-1/2}^{1/2} H(f - f') dZ(f')$ , where  $H(f) = \sum_{t=1}^n h_t e^{-2\pi i f t}$ .

### Solution

Substitute the spectral representation and interchange the finite sum with the integral:

$$\begin{aligned} J_h(f) &= \sum_{t=1}^n h_t X_t e^{-2\pi i f t} = \sum_{t=1}^n h_t \left( \int_{-1/2}^{1/2} e^{2\pi i f' t} dZ(f') \right) e^{-2\pi i f t} \\ &= \int_{-1/2}^{1/2} \left( \sum_{t=1}^n h_t e^{2\pi i (f' - f) t} \right) dZ(f') = \int_{-1/2}^{1/2} H(f - f') dZ(f'), \end{aligned}$$

since  $\sum_{t=1}^n h_t e^{2\pi i (f' - f) t} = \sum_{t=1}^n h_t e^{-2\pi i (f - f') t} = H(f - f')$ . Thus the tapered DFT is the spectral increment  $dZ$  smeared by the taper's transfer function  $H$  centred at  $f$ .  $\square$

### Exercise 5: Spectral density of the differenced process [Sheet 13]

Let  $\{X_t\}$  be stationary and  $\{Y_t\} = (I - B)[\{X_t\}]$ , i.e.  $Y_t = X_t - X_{t-1}$ . Is  $\{Y_t\}$  stationary? If so, express  $\mathcal{S}_Y$  in terms of  $\mathcal{S}_X$ .

### Solution

$I - B$  is LTI with impulse response  $h_0 = 1$ ,  $h_1 = -1$  (others 0), so  $h \in \ell^1$  and  $\{Y_t\}$  is stationary (filtering preserves stationarity). Its transfer function is

$$H(f) = h_0 + h_1 e^{-2\pi i f} = 1 - e^{-2\pi i f}.$$

Therefore

$$\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f) = |1 - e^{-2\pi i f}|^2 \mathcal{S}_X(f).$$

Simplify the gain:  $|1 - e^{-2\pi i f}|^2 = (1 - \cos 2\pi f)^2 + \sin^2 2\pi f = 2 - 2 \cos(2\pi f) = 4 \sin^2(\pi f)$ . Hence

$$\mathcal{S}_Y(f) = 4 \sin^2(\pi f) \mathcal{S}_X(f).$$

The differencing filter *kills*  $f = 0$  (since  $H(0) = 0$ ) and amplifies high frequencies, as expected of a high-pass operation.  $\square$

### Exercise 6: Time reversibility of AR(p) [Sheet 13]

In an AR(p) process  $X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t$  (Gaussian noise, variance  $\sigma_\varepsilon^2$ , mean zero), set  $Y_t = X_{-t}$ . Show  $\{Y_t\}$  is an AR(p) with the *same* parameters, driven by white noise with the same distribution as  $\varepsilon$ .

### Solution

Put  $\phi_0 = -1$  and define  $\tilde{\nu}_t = \sum_{j=0}^p \phi_j Y_{t-j} = -Y_t + \sum_{j=1}^p \phi_j Y_{t-j}$ , i.e.  $Y_t = \sum_{j=1}^p \phi_j Y_{t-j} - \tilde{\nu}_t$ ; we must show  $\{\tilde{\nu}_t\}$  is white noise  $\sim$  that of  $\varepsilon$  (the sign is immaterial for noise). It is Gaussian (a linear combination of Gaussians) with  $\mathbb{E}[\tilde{\nu}_t] = 0$  since  $\mathbb{E}[X_t] = 0$ .

*Spectral matching.* For every  $\tau$ ,

$$\gamma_\tau^{(Y)} = \mathbb{E}[Y_t Y_{t+\tau}] = \mathbb{E}[X_{-t} X_{-t-\tau}] = \gamma_{-\tau}^{(X)} = \gamma_\tau^{(X)},$$

so  $\mathcal{S}_Y(f) = \mathcal{S}_X(f)$  for all  $f$ . Now  $\{\tilde{\nu}_t\} = \Phi(B)[\{Y_t\}]$  is an LTI filtering of  $\{Y_t\}$ , so by the polynomial-filter lemma and the output-spectrum theorem,

$$\mathcal{S}_{\tilde{\nu}}(f) = |\Phi(e^{-2\pi i f})|^2 \mathcal{S}_Y(f) = |\Phi(e^{-2\pi i f})|^2 \mathcal{S}_X(f).$$

But the AR spectral density (the  $q = 0$  ARMA case) gives  $\mathcal{S}_X(f) = \sigma_\varepsilon^2 / |\Phi(e^{-2\pi if})|^2$ , hence

$$\mathcal{S}_\nu(f) = |\Phi(e^{-2\pi if})|^2 \cdot \frac{\sigma_\varepsilon^2}{|\Phi(e^{-2\pi if})|^2} = \sigma_\varepsilon^2.$$

A constant spectral density means  $\{\tilde{\nu}_t\}$  is white noise of variance  $\sigma_\varepsilon^2$ ; being Gaussian and mean-zero it has the same distribution as  $\{\varepsilon_t\}$ . Therefore the reversed process is AR( $p$ ) with identical coefficients.  $\square$

#### Exercise 7: Spectral density of an MA(1) via a filter [New]

Let  $\varepsilon_t$  be white noise of variance  $\sigma^2$  and  $X_t = \varepsilon_t - \theta\varepsilon_{t-1}$ . Using the LTI machinery, find  $\mathcal{S}_X(f)$  and check it against the direct ACVS.

#### Solution

Here  $X_t = \Theta(B)\varepsilon_t$  with  $\Theta(z) = 1 - \theta z$ , an LTI filter with impulse response  $h_0 = 1$ ,  $h_1 = -\theta$ . Its transfer function is  $H(f) = 1 - \theta e^{-2\pi if}$ , so

$$\mathcal{S}_X(f) = |H(f)|^2 \mathcal{S}_\varepsilon(f) = \sigma^2 |1 - \theta e^{-2\pi if}|^2.$$

Expand:

$$|1 - \theta e^{-2\pi if}|^2 = (1 - \theta e^{-2\pi if})(1 - \theta e^{2\pi if}) = 1 + \theta^2 - 2\theta \cos(2\pi f),$$

giving  $\mathcal{S}_X(f) = \sigma^2(1 + \theta^2 - 2\theta \cos(2\pi f))$ .

*Check via ACVS.* For this MA(1),  $\gamma_0 = (1 + \theta^2)\sigma^2$ ,  $\gamma_{\pm 1} = -\theta\sigma^2$ , else 0. Then  $\mathcal{S}_X(f) = \sum_\tau \gamma_\tau e^{-2\pi if\tau} = (1 + \theta^2)\sigma^2 - \theta\sigma^2(e^{-2\pi if} + e^{2\pi if}) = \sigma^2(1 + \theta^2 - 2\theta \cos 2\pi f)$ , identical.  $\square$

#### Exercise 8: Transfer function and gain of a two-sided moving average [New]

Let  $\mathcal{L}[\{X_t\}]_u = \frac{1}{3}(X_{u-1} + X_u + X_{u+1})$  (a centred 3-point average). Find the impulse response, the transfer function, and the squared gain  $|H(f)|^2$ . What does it do to  $f = 0$  versus  $f = \frac{1}{3}$ ?

#### Solution

Writing  $\mathcal{L}[\{X_t\}]_u = \sum_m h_{u-m}X_m = \sum_k h_k X_{u-k}$ , the coefficients of  $X_{u-1}, X_u, X_{u+1}$  are  $h_1, h_0, h_{-1}$ , so  $h_{-1} = h_0 = h_1 = \frac{1}{3}$  (others 0). Then

$$H(f) = \sum_m h_m e^{-2\pi ifm} = \frac{1}{3}(e^{2\pi if} + 1 + e^{-2\pi if}) = \frac{1}{3}(1 + 2\cos(2\pi f)).$$

$H$  is real (the filter is symmetric, zero phase), so

$$|H(f)|^2 = \frac{1}{9}(1 + 2\cos 2\pi f)^2.$$

At  $f = 0$ :  $H(0) = 1$ , gain 1 (the mean / DC component passes untouched). At  $f = \frac{1}{3}$ :  $\cos(2\pi/3) = -\frac{1}{2}$ , so  $H = \frac{1}{3}(1 - 1) = 0$  — that frequency is annihilated. The 3-point average is a low-pass smoother with a null at  $f = \frac{1}{3}$ .  $\square$

#### Exercise 9: Spectral density of an AR(1) [New]

Let  $X_t = \phi X_{t-1} + \varepsilon_t$  with  $|\phi| < 1$  and  $\varepsilon$  white noise of variance  $\sigma^2$ . Use the polynomial-filter lemma to obtain  $\mathcal{S}_X(f)$  and state where it peaks for  $\phi > 0$  and for  $\phi < 0$ .

### Solution

Rewrite as  $\Phi(B)X_t = \varepsilon_t$  with  $\Phi(z) = 1 - \phi z$ . By the polynomial-filter lemma the AR filter has transfer function  $\Phi(e^{-2\pi i f}) = 1 - \phi e^{-2\pi i f}$ , so the output-spectrum theorem applied to  $\Phi(B)[\{X_t\}] = \{\varepsilon_t\}$  gives  $|1 - \phi e^{-2\pi i f}|^2 \mathcal{S}_X(f) = \sigma^2$ , hence

$$\mathcal{S}_X(f) = \frac{\sigma^2}{|1 - \phi e^{-2\pi i f}|^2} = \frac{\sigma^2}{1 + \phi^2 - 2\phi \cos(2\pi f)}.$$

The denominator is smallest where  $\phi \cos(2\pi f)$  is largest. For  $\phi > 0$  this is  $f = 0$  (spectrum peaks at low frequency — slowly varying, persistent series); for  $\phi < 0$  it is  $f = \frac{1}{2}$  (peak at high frequency — rapidly oscillating series). Stationarity ( $|\phi| < 1$ ) keeps the denominator  $\geq (1 - |\phi|)^2 > 0$ , so  $\mathcal{S}_X$  is finite everywhere.  $\square$

### Exercise 10: Cascade gain of difference then smooth [New]

A series is first differenced,  $\{V_t\} = (I - B)[\{X_t\}]$ , then run through the 3-point average  $\mathcal{L}_2$  of Exercise 8, giving  $\{Y_t\} = \mathcal{L}_2[\{V_t\}]$ . Find  $\mathcal{S}_Y$  in terms of  $\mathcal{S}_X$ .

### Solution

The whole operation is the cascade  $\mathcal{L} = \mathcal{L}_2(I - B)$ , which is LTI with transfer function the *product* of the two transfer functions:

$$H(f) = \underbrace{(1 - e^{-2\pi i f})}_{\text{difference}} \cdot \underbrace{\frac{1}{3}(1 + 2 \cos 2\pi f)}_{\text{3-point average}}.$$

The squared gains multiply, so using  $|1 - e^{-2\pi i f}|^2 = 4 \sin^2(\pi f)$ ,

$$\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f) = 4 \sin^2(\pi f) \cdot \frac{1}{9} (1 + 2 \cos 2\pi f)^2 \mathcal{S}_X(f).$$

This both removes the  $f = 0$  component (the difference) and nulls  $f = \frac{1}{3}$  (the average) — a band-suppressing combination.  $\square$

## 12 Exam Strategy Checklist

The exam (cf. the mock) is **closed-book, no calculator**, and made of a few multi-part questions mixing *stated definitions*, *short proofs*, and *calculations*. The recurring tasks are: write down a definition; compute an ACVS/ACF; find an  $\text{MA}(\infty)$  representation; check causality/invertibility; derive a spectral density; forecast and give the error; prove (joint) second-order stationarity; and read cross-spectra. This checklist maps a *question cue* to the *tool* you deploy.



| When the question says...                                 | Deploy...                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “State the definition of (weak/strict) stationarity”      | The three moment conditions (mean const, finite const variance, covariance a function of $\tau$ only) / equality of all finite-dim. distributions under shift.                                                                                            |
| “Is $\{X_t\}$ (weakly) stationary?”                       | Check the 3 conditions in order; stop at first failure. Watch for undefined moments and a residual $t$ -dependence in the covariance.                                                                                                                     |
| “Compute the ACVS/ACF of an MA( $q$ )”                    | $\gamma_\tau = \sigma^2 \sum_j \theta_j \theta_{j+ \tau }$ for $ \tau  \leq q$ , else 0. ACF <i>cuts off</i> after lag $q$ .                                                                                                                              |
| “Compute the ACVS of an AR/ARMA”                          | Multiply the defining equation by $X_{t-\tau}$ , take expectations $\Rightarrow$ a <b>difference equation</b> $\gamma_\tau = \sum \phi_j \gamma_{\tau-j}$ ; solve with initial conditions (handle small $\tau$ separately because of the MA/noise terms). |
| “Find $\psi_j$ in $X_t = \sum \psi_j \varepsilon_{t-j}$ ” | Write $X_t = \Theta(B)/\Phi(B) \varepsilon_t$ ; expand by partial fractions / $1/(1-gB) = \sum g^k B^k$ , or match coefficients in $\Phi(B) \sum \psi_j B^j = \Theta(B)$ .                                                                                |
| “Is it causal/stationary? invertible?”                    | <b>Roots of <math>\Phi(z)</math> outside the unit circle</b> $\Rightarrow$ causal/stationary; <b>roots of <math>\Theta(z)</math> outside</b> $\Rightarrow$ invertible. (AR always invertible; MA always stationary.)                                      |
| “Find the PACF”                                           | Durbin–Levinson recursion on the ACF; for AR( $p$ ), $\alpha_\tau = 0$ for $\tau > p$ . PACF <i>cuts off</i> after $p$ .                                                                                                                                  |
| “Identify a model from ACF/PACF plots”                    | ACF cuts at $q$ & PACF tails $\Rightarrow$ MA( $q$ ); PACF cuts at $p$ & ACF tails $\Rightarrow$ AR( $p$ ); both tail $\Rightarrow$ ARMA; slow ACF decay $\Rightarrow$ difference (non-stationary).                                                       |
| “Estimate AR parameters”                                  | Yule–Walker $\hat{\phi} = \hat{\Gamma}^{-1} \hat{\gamma}$ , $\hat{\sigma}^2 = \hat{\gamma}_0 - \sum \hat{\phi}_j \hat{\gamma}_j$ ; or (forward/backward) LS.                                                                                              |
| “Estimate MA/ARMA by LS”                                  | Reconstruct innovations recursively ( $\varepsilon_0 = 0$ ), minimise $\sum \hat{\varepsilon}_t^2$ .                                                                                                                                                      |
| “Remove a trend / make stationary”                        | Difference: $\nabla^q$ kills a degree- $(q-1)$ polynomial trend; $\nabla_{(s)}$ removes period- $s$ seasonality.                                                                                                                                          |
| “Forecast $X_{n+h}$ (causal ARMA)”                        | $X_{n+h}^n = \sum_{j \geq h} \psi_j \varepsilon_{n+h-j}$ (future $\varepsilon \equiv 0$ ); MSE $= \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$ . Or the recursive truncated recipe.                                                                                |
| “Forecast an ARIMA”                                       | Forecast the differenced ARMA $Z = \nabla^d X$ , then un-difference (cumulate). $P_{n+h}^n \rightarrow \infty$ .                                                                                                                                          |
| “Give a prediction interval”                              | $X_{n+h}^n \pm z_{1-\alpha/2} \sqrt{\text{Var}(e_n(h))}$ (Gaussian).                                                                                                                                                                                      |
| “Find the spectral density”                               | $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau}$ ; for ARMA use $\mathcal{S}(f) = \sigma^2  \Theta(e^{-2\pi i f}) ^2 /  \Phi(e^{-2\pi i f}) ^2$ .                                                                                              |
| “Spectrum of a filtered series $Y = \mathcal{L}X$ ”       | $\mathcal{S}_Y(f) =  H(f) ^2 \mathcal{S}_X(f)$ , $H(f) = \sum_m h_m e^{-2\pi i f m}$ .                                                                                                                                                                    |
| “Periodogram expectation/variance”                        | $\mathbb{E}[\hat{\mathcal{S}}^{(p)}] = \mathcal{F}_n * \mathcal{S}$ (biased, blurring); $\text{Var} \rightarrow \mathfrak{S}(\mathbb{H})^2$ (does not vanish). Taper for bias, multi-taper for variance.                                                  |
| “Is this $\gamma_\tau$ a valid ACVS?”                     | Compute its SDF; it is valid <i>iff</i> $\mathcal{S}(f) \geq 0$ for                                                                                                                                                                                       |

### Key Technique 1: The six moves you will almost certainly need

1. **Stationarity check** (3 conditions).
2. **ACVS via the white-noise filter trick** (only equal noise indices survive) or via the **difference equation** (multiply by  $X_{t-\tau}$ , take expectations).
3. **Root test** for causality/invertibility/stability.
4.  $\Theta/\Phi$  **expansion** for  $\psi$ -weights (partial fractions or coefficient matching).
5. **Recursive forecasting** and its MSE.
6. **Spectral density** from ACVS, from a filter, or from ARMA polynomials.

### Key Technique 2: Exam hygiene

- State which *theorem/definition* you invoke (examiners reward naming the tool: “by causality  $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$ ”; “by Herglotz / the prediction equations / Isserlis”).
- No calculator: keep fractions exact; factor polynomials by hand ( $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ) and report  $|z|$  vs 1.
- For “ $|z| > 1$ ” with complex roots, compare  $|z|^2 = (\Re z)^2 + (\Im z)^2$  to 1 — no need to take the square root.
- Always sanity-check:  $\gamma_0 \geq 0$ ,  $|\rho_\tau| \leq 1$ , MA(1) has  $|\rho_1| \leq \frac{1}{2}$ , an SDF must be  $\geq 0$  and real, a spectral *matrix* must be Hermitian.

## 13 Classic Pitfalls to Avoid

### 🔊 Classic Pitfall

**Roots inside vs outside the unit circle.** Stationarity/causality requires the roots of  $\Phi(z)$  to lie *outside*  $|z| = 1$  (equivalently  $\Phi(z) \neq 0$  for  $|z| \leq 1$ ). Students routinely flip this. The *characteristic* root condition (“ $|z| > 1$ ”) is the opposite of the companion-eigenvalue condition (“ $|\lambda| < 1$ ”): both say the same thing, so do not mix the two phrasings in one argument.

### 🔊 Classic Pitfall

**Strict vs weak stationarity.** They do not imply each other. Strict but not weak: i.i.d. Cauchy (no finite variance). Weak but not strict: a process with matched first two moments but non-Gaussian higher structure. They coincide *only* for Gaussian processes.

### 🔊 Classic Pitfall

**Lag goes in the FIRST argument (multivariate).**  $\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t)$ , so  $\Gamma_{-\tau} = \Gamma_\tau^\top$  (not  $\Gamma_\tau$ ). Cross-covariance sequences are *not* symmetric in  $\tau$ :  $\gamma_\tau^{(j,k)} = \gamma_{-\tau}^{(k,j)}$ . In the univariate real case this distinction is invisible, which is exactly why it trips people in the multivariate case.

### 🔊 Classic Pitfall

**ACF cut-off, PACF cut-off — which is which.** MA( $q$ ): ACF cuts off after  $q$  (PACF tails). AR( $p$ ): PACF cuts off after  $p$  (ACF tails). ARMA: both tail off. Swapping these is the single most common identification error.

### 🔊 Classic Pitfall

**Identifiability of MA / sign of  $\theta_0$ .** Fixing  $\theta_0 = -1$  is not cosmetic: without it, scaling  $(\theta_j, \sigma^2) \mapsto (c\theta_j, c^{-2}\sigma^2)$  gives the *same* ACVS, so the model is unidentifiable. Likewise a non-

invertible MA and its invertible “twin” (root  $g$  vs  $1/g$ ) share the same ACF — pick the invertible one.

#### 🔊 Classic Pitfall

**Biased is not bad.** The divide-by- $n$  ACVS estimator  $\hat{\gamma}_\tau$  is biased yet positive semi-definite and has *smaller* MSE than the unbiased  $\tilde{\gamma}_\tau$ . Unbiasedness is not the objective; positive semi-definiteness and low MSE are. Do not “correct” the estimator on autopilot.

#### 🔊 Classic Pitfall

**The periodogram is not consistent.** Its variance tends to  $\mathcal{S}(f)^2$  and does *not* shrink with  $n$  — more data makes it wigglier, not smoother. Reducing bias needs *tapering*; reducing variance needs *smoothing/multitapering* ( $\text{Var} \approx \mathcal{S}(f)^2/K$ ).

#### 🔊 Classic Pitfall

**Forecasts of stationary vs integrated processes.** A stationary ARMA forecast reverts to the mean and its error variance saturates at  $\gamma_0$ . An *integrated* (ARIMA,  $d \geq 1$ ) forecast does *not* revert and its error variance  $\rightarrow \infty$ . Quoting a finite long-horizon variance for a random walk is wrong.

#### 🔊 Classic Pitfall

**Setting future innovations to zero (only).** In the  $\psi$ -weight forecast, set  $\varepsilon_{n+1}, \dots, \varepsilon_{n+h} = 0$  but keep the *observed past* innovations as residuals; in the recursive recipe, replace future *observations* by their forecasts, future errors by 0, and past errors by residuals — do not zero the past errors.

#### 🔊 Classic Pitfall

**ARCH/GARCH are uncorrelated but dependent.**  $\text{Corr}(r_{t+\tau}, r_t) = 0$ , yet  $\text{Corr}(r_{t+\tau}^2, r_t^2) \neq 0$  (geometric for ARCH(1):  $\alpha_1^{|\tau|}$ ). “Uncorrelated”  $\neq$  “independent” here. Also check *moment* conditions ( $\alpha_1 < 1$  for variance,  $\alpha_1^2 < 1/3$  for finite kurtosis under Gaussian noise) separately from stationarity.

#### 🔊 Classic Pitfall

**Ljung–Box degrees of freedom.** Use  $\chi_{m-p-q}^2$ , subtracting the number of fitted ARMA parameters — not  $\chi_m^2$ . Forgetting the  $-(p+q)$  shifts the critical value and can flip the conclusion.

#### 🔊 Classic Pitfall

**Angular vs ordinary frequency.** This course uses ordinary frequency  $f$  (period  $1/f$ ), so the SDF integrates over  $[-\frac{1}{2}, \frac{1}{2}]$  and  $\mathcal{S}(f) = \sum \gamma_\tau e^{-2\pi i f \tau}$ . Switching to  $\omega = 2\pi f$  scatters stray  $2\pi$  factors — keep one convention.

#### 🔊 Classic Pitfall

**A valid ACVS must be positive semi-definite.** A plausible-looking sequence (e.g.  $\gamma_\tau = 1$  for  $|\tau| \leq K$ , else 0) need *not* be a real ACVS: its SDF can go negative. Always test via  $\mathcal{S}(f) \geq 0$  before claiming a sequence is an autocovariance.

You are ready when you can do every **Exercise** in this document unaided, name the tool behind every **Theorem**, and avoid every **Pitfall** above. Good luck!